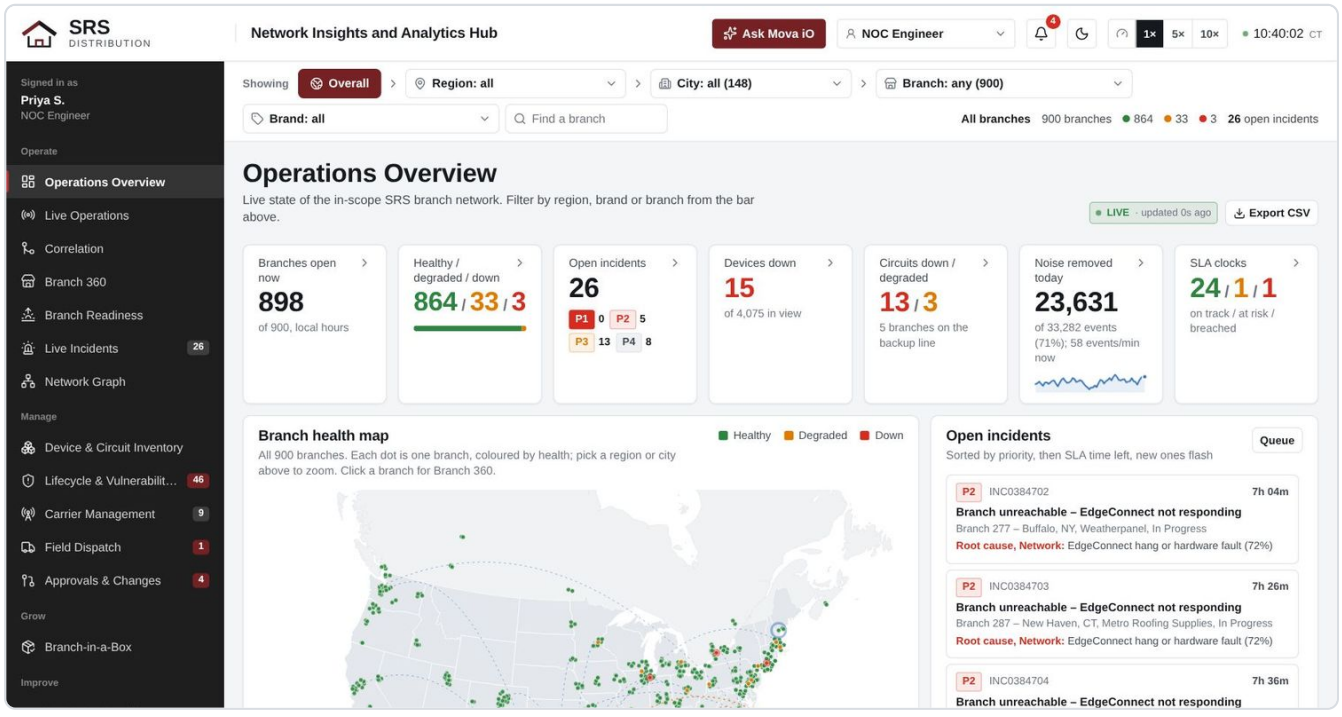
SRS DISTRIBUTION

# SRS Network Insights and Analytics Hub

The complete guide to the console: every screen, every visual, every demo




**Managed Network Services** · Movate for SRS Distribution

Version 2.0 · October 2026

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# How to use this guide

This guide explains every screen of the SRS Network Insights and Analytics Hub: each tile, chart, table, tab and button, what it measures, how to read it and what to do about it. It is written for anyone who uses or presents the console, from a NOC engineer on shift to a branch manager or an executive sponsor.

The guide is in four parts:

1. **Getting started:** what the console is, how the screen is laid out, the common way every screen is read, and where the data comes from.
2. **Screen by screen:** one chapter for each of the 19 console screens, in the order of the left menu, then the live demos page and the role dashboards, which are pages of their own. Every visual has a picture and the same five points: what it shows, its parameters, how to read it, good and bad, and what to do.
3. **The live demos:** all 33 demos, what each one proves, its control mode and where it comes from in the proposal documents.
4. **Presenting and reference:** a suggested walkthrough, answers to common questions and a glossary.

The console runs on realistic sample data that moves every two seconds, so numbers in the pictures will differ slightly from what you see on screen. The structure, the thresholds and the rules are the same.

# Part I. Getting started

## 1. The console at a glance

The Network Insights and Analytics Hub is the working console for SRS Distribution's managed network: 900 branches across the United States and Canada, 4,075 managed devices and 1,800 WAN and internet circuits. It brings the network tools together on one screen and shows what Mova iO, the AI layer, is doing about each problem.

| Estate | Count | Notes |
|---|---|---|
| Branches | 900 | Nine regions, from Southeast to Canada; each branch keeps its local brand name |
| SD-WAN appliances | 933 | HPE Aruba EdgeConnect EC-XS; larger branches have an HA pair. End of sale 31 Jan 2026, supported until 31 Jan 2031; new units are EdgeConnect 10104 |
| Switches | 1,856 | Aruba 2530 (728, end of support in 2027), CX 6100, 6200M and 6300; 1,268 are standalone (outside Aruba Central) |
| Access points | 1,286 | Indoor AP-635 and outdoor AP-565 and AP-585 for yards; 812 are standalone |
| Circuits | 1,800 | A primary and a backup path per branch, from AT&T, Lumen, Comcast, Spectrum, Verizon, Cox and others |

The tools behind the console:

| Tool | Role in the console |
|---|---|
| HPE Aruba EdgeConnect Orchestrator | SD-WAN paths, tunnels, loss, latency and jitter for every branch |
| Aruba Central | Switch, access point and client health; firmware; config groups. It manages 1,062 switches and APs; the other 2,080 are standalone and monitored by OpsRamp only |
| HPE OpsRamp | Monitoring and event management: SNMP, syslog, config backups, correlation |
| New Relic | Synthetic checks of the payment and ERP paths from each branch |
| ServiceNow | Incidents, changes, problems, CMDB and the service catalog |
| Mova iO | AI agents that correlate, diagnose, run approved runbooks, verify and record |
| Carrier portals | Carrier cases, loopback tests, maintenance calendars and outage maps |

Scope is the network only: SD-WAN, circuits, switching, Wi-Fi, network services such as DHCP, configuration, firmware and the network paths to the data centre and cloud. Applications, end-user devices and building systems are outside it.

# 2. The screen layout

Every screen has the same frame: a header across the top, the menu on the left, a filter bar under the header and the working area in the middle.

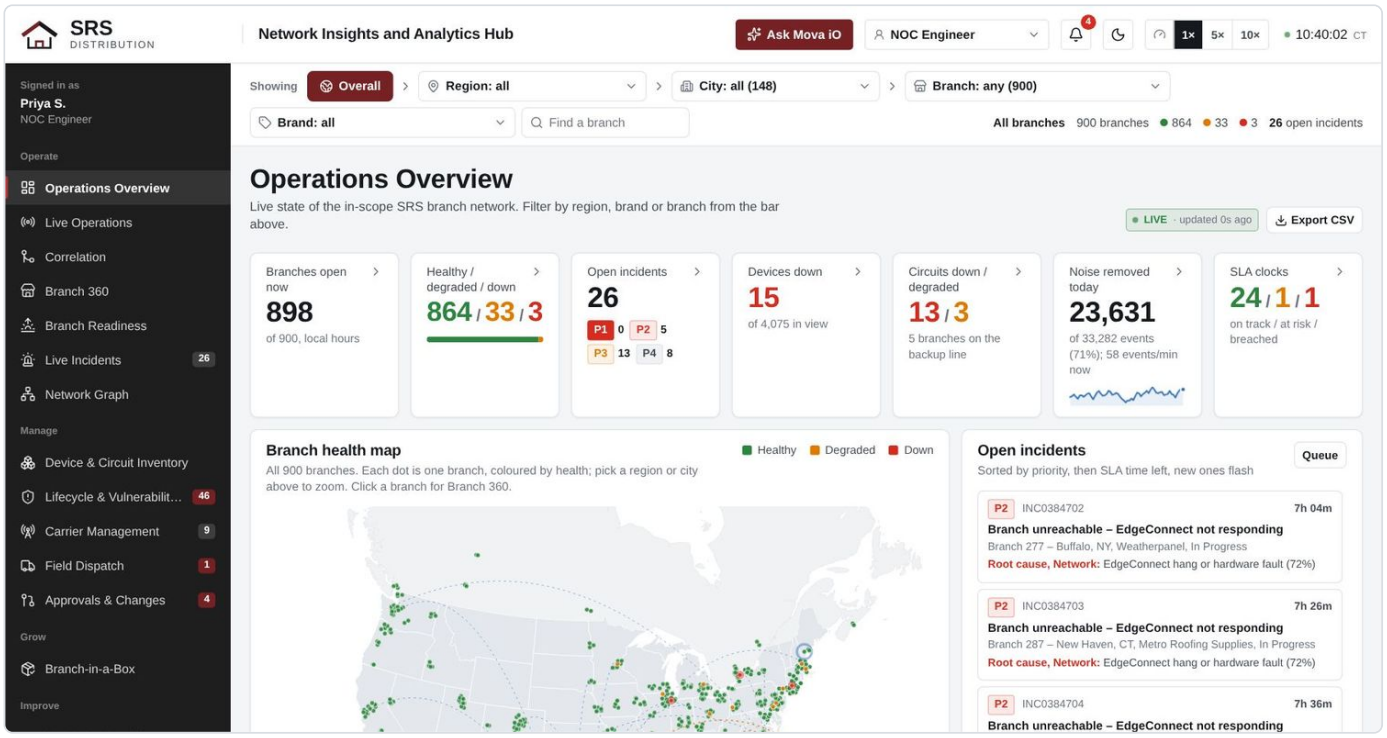



**Figure 1.** The Operations Overview with the header, left menu and filter bar

## Header




**Figure 2.** Header

| Element | What it does |
|---|---|
| SRS mark and "Network Insights and Analytics Hub" | The console name. The live demos (chapter 24) and the role dashboards (chapter 25) are separate pages with their own links; the console does not link to them. |
| Tool sync (wide screens only) | OpsRamp, Aruba Central, ServiceNow and Mova iO, each with a green dot and the seconds since the last sync. A dot that is not green means that feed is late. |
| Ask Mova iO | Opens the question panel on the right. |
| Role selector | Switches the viewing role. The role sets the landing page, the suggested questions and some defaults. |
| Bell | Notifications. A red number shows how many are new. Short pop-up messages (for example a new P1, or a change of role) appear at the top right under the header; each has a close button and clears itself after a few seconds. |
| Moon or sun | Switches between light and dark mode. |

| Element | What it does |
|---|---|
| Speed (1×, 5×, 10×) | How fast live data refreshes: every 2 seconds at 1×, every 0.4 seconds at 5× and every 0.2 seconds at 10×. Charts, numbers and streams all move at the chosen speed. Use 5× or 10× to show a lot of activity in a short meeting. |
| Clock | Current time in Central Time (CT), which the whole console uses. |

## Left menu

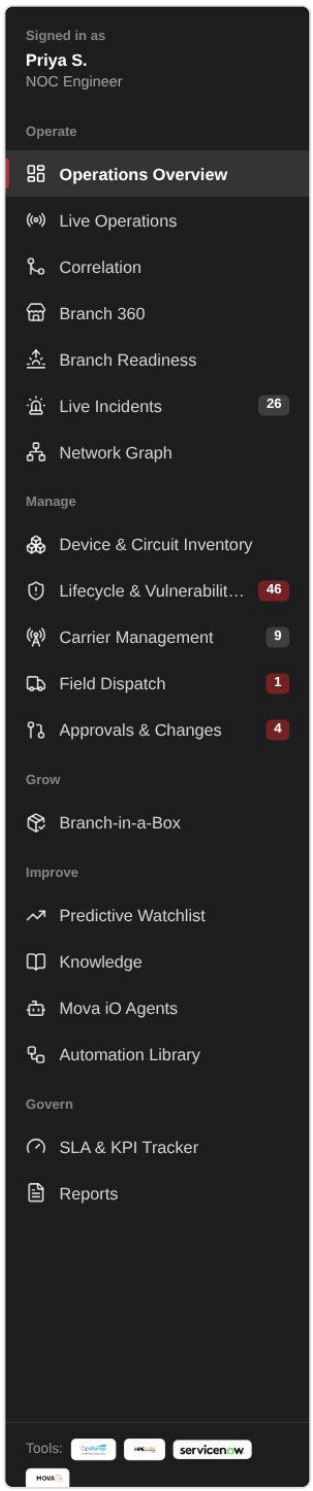



**Figure 3.** Left menu

The menu groups the 19 screens by what you are doing:

| Group | Screens |
|---|---|
| Operate | Operations Overview, Live Operations, Correlation, Branch 360, Branch Readiness, Live Incidents, Network Graph |

| Group | Screens |
|---|---|
| Manage | Device & Circuit Inventory, Lifecycle & Vulnerabilities, Carrier Management, Field Dispatch, Approvals & Changes |
| Grow | Branch-in-a-Box |
| Improve | Predictive Watchlist, Knowledge, Mova iO Agents, Automation Library |
| Govern | SLA & KPI Tracker, Reports |

The console opens on the Operations Overview. A number next to a screen is a live count: open incidents, open carrier cases. A maroon number means something is waiting for a person: devices with a critical vulnerability, unassigned work orders or approvals pending. "Signed in as" at the top shows the person and role in use. **Collapse** at the bottom shrinks the menu to icons.

## Filter bar




**Figure 4.** Filter bar

The filter bar sets which branches every screen shows. It works left to right:

1. **Overall** resets to all 900 branches.
2. **Region** narrows to one of nine regions; the count in brackets is its branches.
3. **City** lists only the cities in the chosen region.
4. **Branch** lists the branches in that city. Picking one on the Overview or Branch 360 opens that branch's Branch 360.
5. **Brand** narrows to one of the SRS family brands.
6. **Find a branch** searches by name, brand, city, state or branch ID.

The right end of the bar always summarises the current view: its name, the number of branches, how many are green, amber and red, and the open incidents among them. The filter stays in place as you move between screens.

## Ask Mova iO

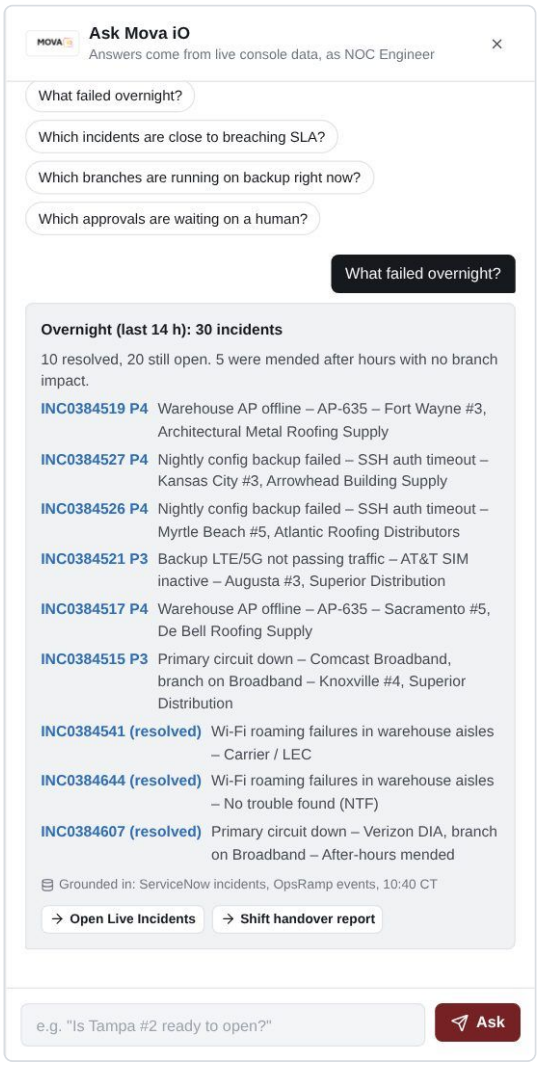



**Figure 5.** Ask Mova iO panel

The ask panel answers questions from the live console data. Each role gets four suggested questions, and you can type your own, such as "Is Tampa #2 ready to open?". Every answer shows its lines, names you can click to open the incident or branch, the source it was grounded in with the time, and buttons that jump to the right screen.

## Notifications

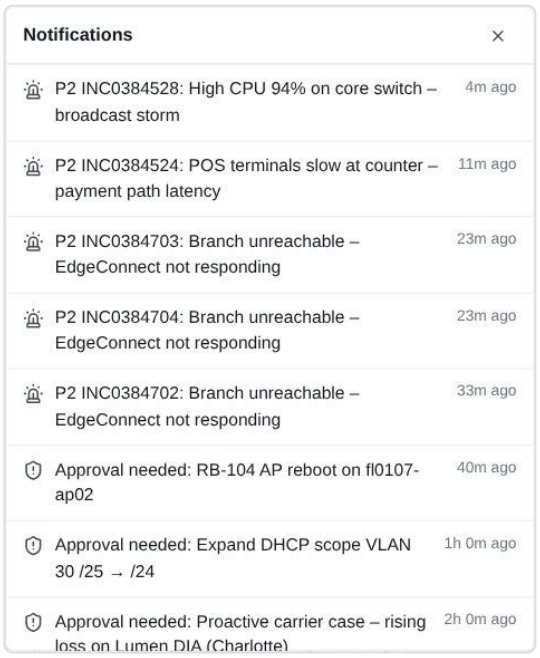



**Figure 6.** Notifications

The bell collects things that need a person: approvals waiting, new P1 or P2 incidents, breach risks and demo results. Click an entry to go to it.

## Roles

The role selector changes the landing page and the suggested questions, and it decides who may act. **Movate roles carry out actions. SRS roles see everything, live, and escalate to Movate**; they cannot trigger an action.

| Role | Organisation | Person in the sample | Lands on | Typical questions |
|---|---|---|---|---|
| NOC Engineer | Movate | Priya S. | Live Incidents | What failed overnight? Which incidents are close to breaching? |
| Network Engineer L3 | Movate | Marcus L. | Approvals & Changes | Which carriers breached this week? Where is config drift? |
| StoreOps Lead | SRS | Jenna H. | Branch Readiness | Which branches are not ready to open? |
| Branch Manager | SRS | Rick D. | Operations Overview, for one branch | Is my branch ready to open? When is the technician arriving? |
| Field Technician | SRS field technical services | Carlos M. | Field Dispatch (My jobs today, with the onsite work pack for each job) | What are my jobs today? Which spares are in my depot? |
| M&A Integration Lead | SRS | Alicia R. | Branch-in-a-Box | Status of this weekend's cutovers? Which projects are red? |

| Role | Organisation | Person in the sample | Lands on | Typical questions |
|---|---|---|---|---|
| Security Lead | SRS | Omar N. | Approvals & Changes, firmware tab | Which devices are below firmware policy? Who approved changes today? |

The Branch Manager role locks the console to one branch and uses plain language for it.

## SRS sees, Movate acts: Escalate to Movate

For an SRS role, a dark strip under the filter bar says so, and every control that would do something (acknowledge, assign, change priority or state, approve, run a runbook, raise a case or a work order, resolve, roll back, restock, re-run checks, start a project) is greyed out with the reason. In its place is **Escalate to Movate**: in the strip, in the incident record, in the root-cause view, in the work pack and at the top of every drill-down level. The one exception is the field technician, who records the onsite work in the work pack Movate sends (steps, photos, readings, findings).

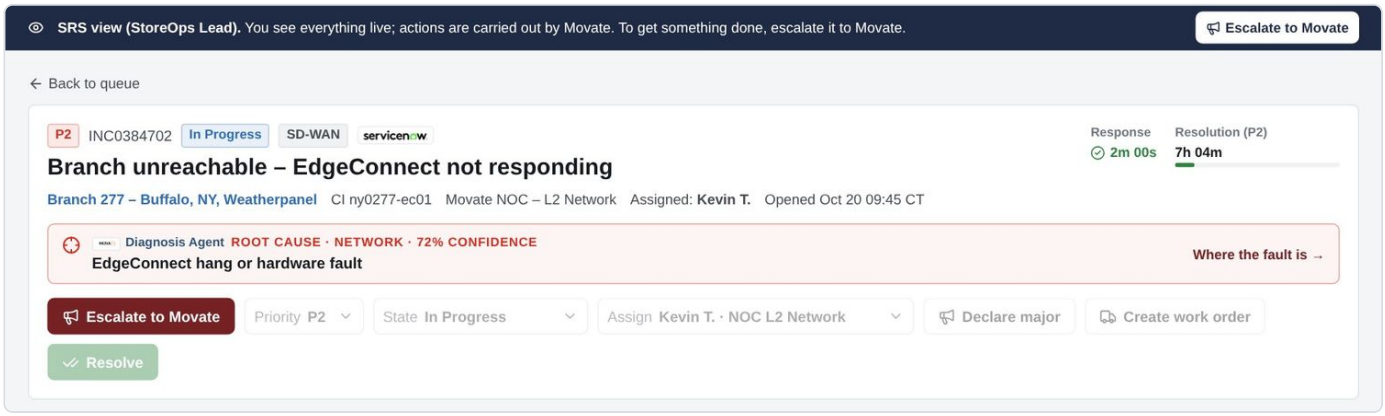



**Figure 7.** SRS view: the actions are greyed out and Escalate to Movate is the way to get something done

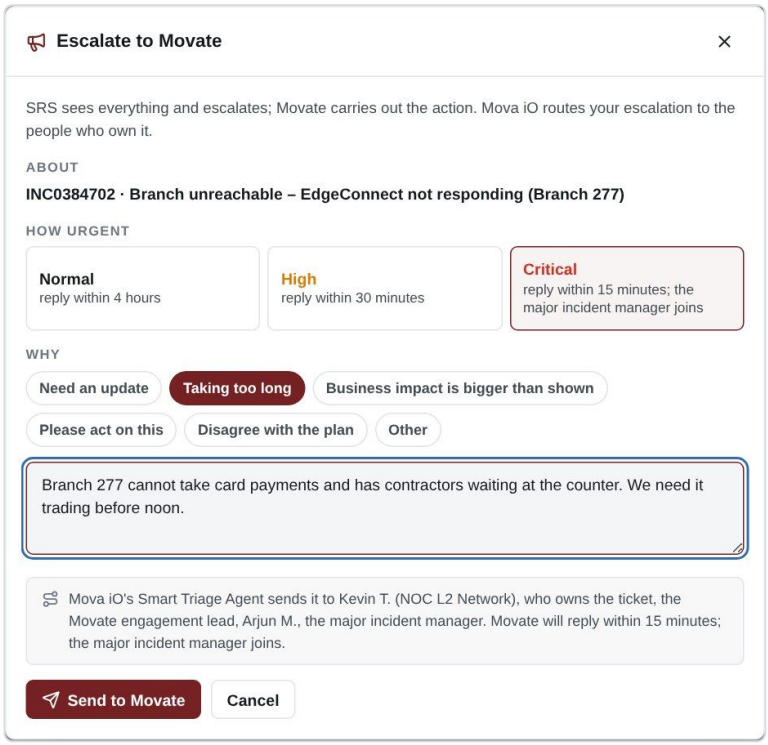



**Figure 8.** Escalate to Movate: what it is about, how urgent, why, and who Mova iO sends it to

| Part | What it does |
|---|---|
| About | Filled in from where you are: the incident, the branch, the work order, the change or the screen |
| How urgent | **Normal** (reply within 4 hours), **High** (within 30 minutes) or **Critical** (within 15 minutes; the major incident manager joins) |
| Why | Need an update, Taking too long, Business impact is bigger than shown, Please act on this, Disagree with the plan, or Other, with a message |
| Routing | Mova iO's Smart Triage Agent sends it to the ticket owner and the Movate engagement lead, and to the major incident manager when critical. The owner acknowledges and sets the time of the next update |
| Your escalations | Each one with its status: sent, acknowledged (by whom and the next update time) or answered |

Movate sees each escalation on the incident record, the root-cause view and the work pack, and in the notifications, and replies from there. The reply reaches the SRS person in their notifications.

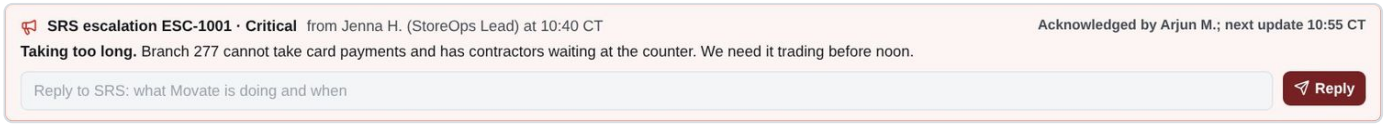



**Figure 9.** What Movate sees on the incident, with the reply to SRS

The role dashboards are SRS leaders' views: the same rule applies, with **Escalate to Movate** in the header and in every drill-down level. Escalations sent from the dashboards reach the console in the same browser.

## Live updates

The console refreshes every two seconds at normal speed, faster when the header speed control is set to 5× or 10×. A green **LIVE** label means the card is updating; click it on a screen to pause updates while you explain something, and click again to resume. New rows slide in at the top of streams, changed numbers flash briefly, and "updated 1s ago" shows the age of the last reading. Live panels at the top of many screens show the stream for that screen, with a small chart and the latest events. Branch-in-a-Box is a project board, so it does not refresh live: its numbers change when a project moves forward, not every few seconds.

# 3. How to read any screen

The same conventions apply on every screen.

## Colours

| Colour | Means | Used for |
|---|---|---|
| Green | Healthy, passed, on track | Branches up, checks passed, SLA clocks with time left |
| Amber | Degraded, warning, at risk | A branch on its backup path, a check that passed with a warning, a clock past 75% |
| Red | Down, failed, breached | A branch with no path, a failed check, a breached SLA |
| Blue | Information and data series | Chart lines and bars that are measurements, not judgements |
| Maroon (SRS brand) | Actions and navigation | Buttons, the active menu item, badges that need a person |

Maroon is never used for health, so a red item always means a problem.

Severity on alerts and events follows the ITOM convention: **Critical** (red), **Major** (orange), **Minor** (yellow) and **Info** (blue).

## Priorities and SLA clocks

| Priority | When it is used | Respond | Resolve |
|---|---|---|---|
| P1 | Several branches down or a critical service lost | 15 min | 4 h |
| P2 | One branch with both paths down, or a significant impact at one branch | 15 min | 8 h |
| P3 | One path down while the branch trades on its backup, or degradation with a workaround | 30 min | 24 h |
| P4 | Minor issues and planned work | 30 min | 96 h |

An SLA clock shows the time left to resolve. It turns amber when more than 75% of the time is used and red when it is breached. "On track / at risk / breached" triples are always read in that colour order.

## Numbers and small charts

- A triple such as **872 / 25 / 3** is healthy / degraded / down, in green, amber and red.
- A sparkline is the recent history of the number next to it; the last point is now.
- Where a note says "square-root scale", small values are drawn larger than their share so they stay visible. Read the number, not the bar length.
- Percentages for availability use two or three decimals because the targets are 99.9% and above.

## Drill down to any level

Every number in the console opens the detail behind it. A tile you can click shows a small arrow (›) next to its label and a pointer when you hover. In a tile that shows several numbers, such as **864 / 33 / 3** for healthy, degraded and down, each number opens its own list: click the red **3** to see the three branches that are down.

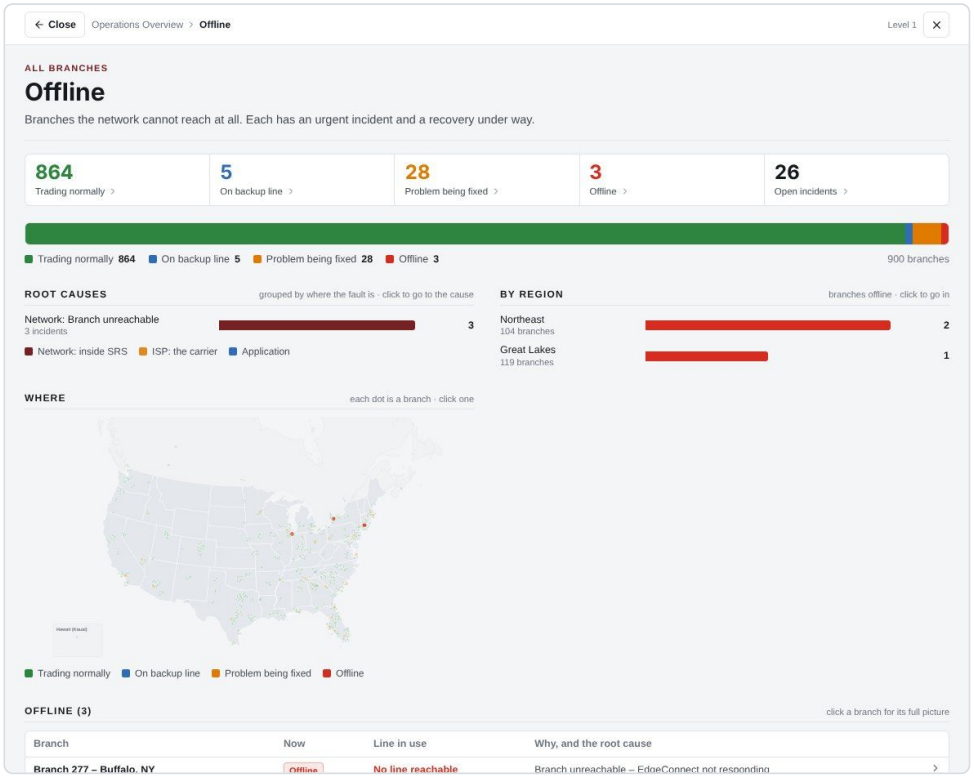



**Figure 10.** Clicking the 3 branches down on the Operations Overview opens the list of those branches

The detail opens in a panel from the right, over the screen you are on. Each level has the same shape: a headline in plain words, a row of numbers, two or more charts and a list, and every number, bar, map dot and row opens the next level down:

| From | You can go to |
|---|---|
| A tile or count | The list behind it: branches, incidents, internet lines, devices, changes, decisions or service levels |
| A list of branches | A region, a state, a city, any branch, and the root causes behind the problems in the list |
| A branch | Its lines, devices, opening checks, incidents, technician visits, carrier repairs, decisions and changes, and its root cause |
| An internet line | Its readings for the last 24 hours, its uptime against the carrier's promise, its carrier and its carrier cases |
| An incident | Its timeline, deadline, owner and fix, and its root cause |

A breadcrumb at the top shows the path; **Back** or any part of the breadcrumb steps out a level and **Escape** closes the panel. The filter bar carries into the panel: with a region, city, brand or one branch selected, every list shows only that part of the estate. Buttons in the panel hand over to the console's own screens when you need them: **Open Branch 360**, **Open the incident record**, or the screen that owns the list (for example **Open Carrier Management**).

## Going to the root cause

Wherever a branch, line or incident has a problem, the root cause is one click away: the red **Root cause** line on an incident in the Operations Overview, the **See the root cause** banner at the top of a branch, the root cause under each row of a list, and **Where the fault is** in the incident record.

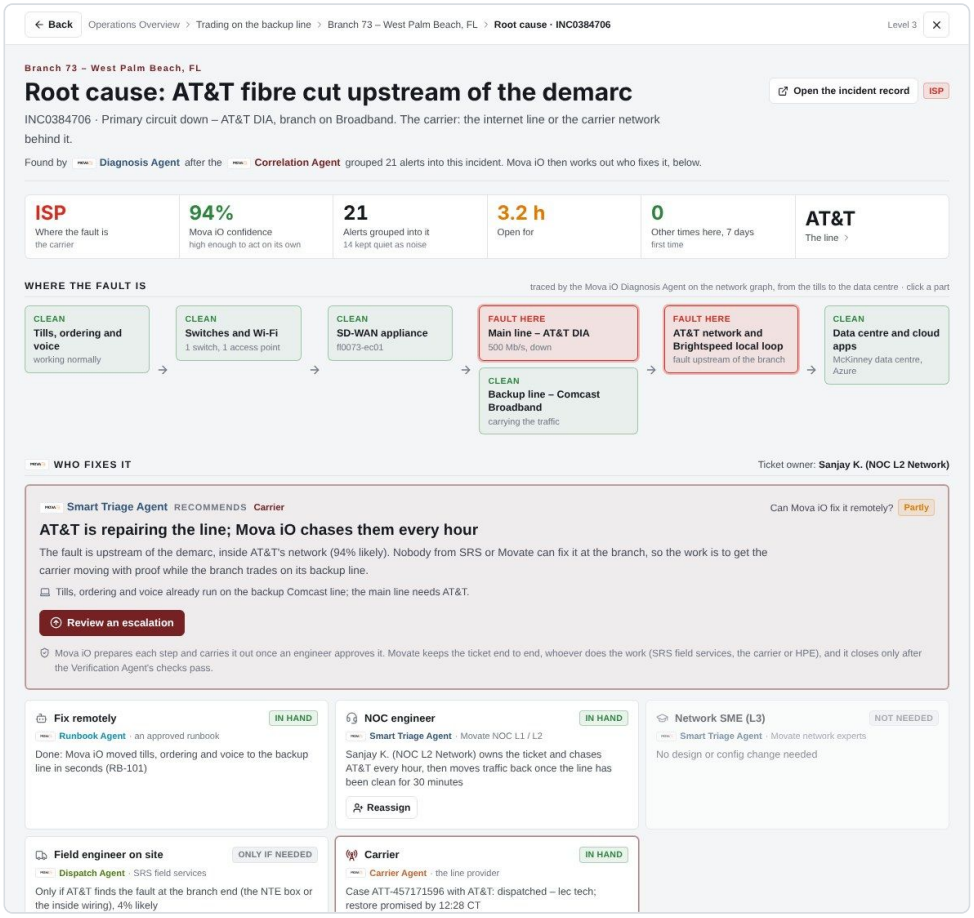



**Figure 11.** The root cause of the fibre cut at Branch 73

| Part of the root-cause view | What it shows |
|---|---|
| Headline | The root cause in one line, the incident, and where the fault is: **Network** (inside SRS: the branch network or the SD-WAN appliance), **ISP** (the carrier: the line or the carrier network behind it) or **Application** (the application or the data centre) |
| Numbers | Where the fault is, Mova iO's confidence (80% or more means it may act on its own), how many alerts were grouped into the incident, how long it has been open or took, whether it has happened at this branch before in 7 days, and the line, device or change at fault (each opens its own detail) |
| Where the fault is | The path from what the branch feels (tills, ordering and voice) through the switches and Wi-Fi, the SD-WAN appliance, the main and backup lines and the carrier network to the data centre. The part at fault is red (**Fault here**), a degraded part amber, the rest green; after the fix the parts show **Fixed**. Click any part to open it |
| Network, ISP or application | Each layer checked, with the cause marked and the evidence that rules the others out |
| Ranked causes | Every cause Mova iO considered, with its confidence and evidence |

| Part of the root-cause view | What it shows |
|---|---|
| How it was found | The first alerts and Mova iO's grouping and diagnosis steps, in time order |
| Fix and prevention | The runbook or fix and the checks before closing; the carrier case; the change behind it, if a change caused it; more faults from the same carrier in the same region (a wider carrier problem is likely); repeats at this branch; and the problem record for a permanent fix |

When a change caused the problem, the root cause names the change and the change's own page lists the incidents it caused.

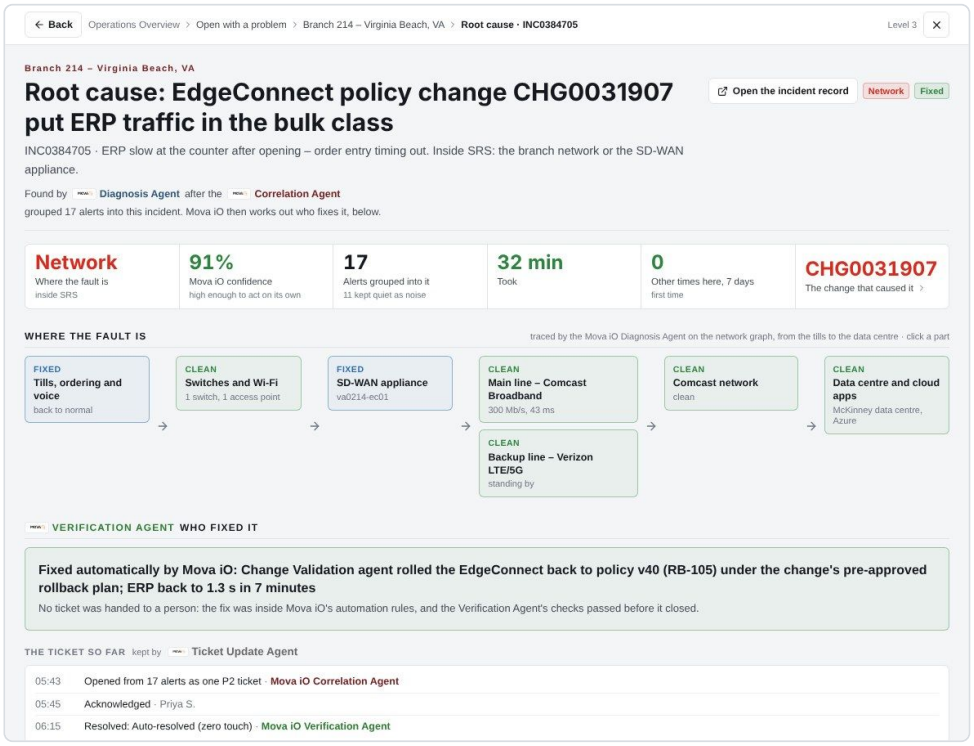



**Figure 12.** A root cause that is a change: CHG0031907 at Branch 214

A list of branches with problems (down, degraded, on the backup line or not ready to open) starts with **Root causes**: the problems grouped by where the fault is and by kind, so you can see at once whether they share a cause. A group with one incident opens its root cause; a larger group opens the incidents.

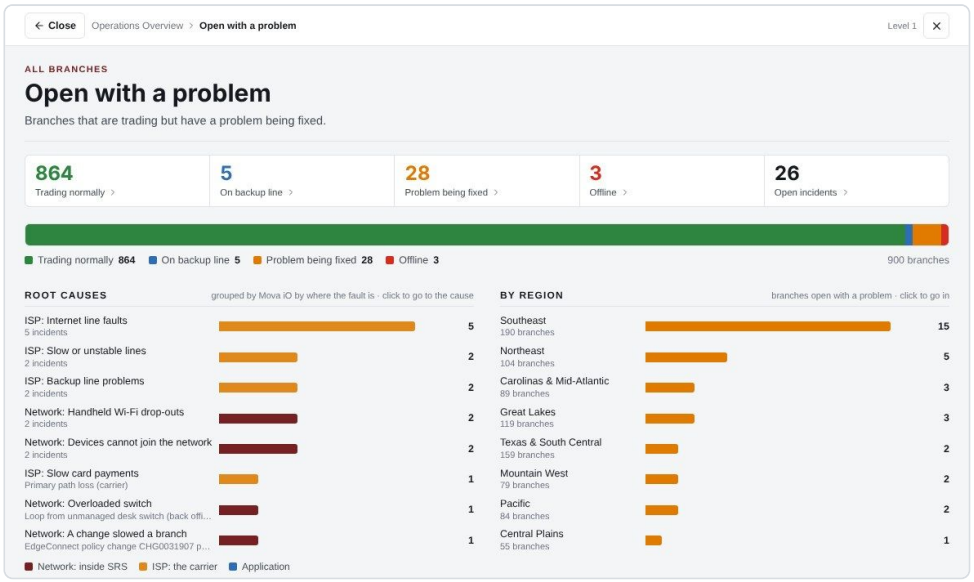



**Figure 13.** Root causes for the branches open with a problem

## Mova iO: who fixes it

Everything in the root-cause view comes from Mova iO. The **Correlation Agent** groups the alerts into one incident, the **Diagnosis Agent** finds the root cause and traces it on the network graph, and under the path the **Smart Triage Agent** works out who fixes it and whether it can be fixed remotely. The **Runbook**, **Dispatch**, **Carrier**, **Ticket Update** and **Change Validation** agents then prepare each step: the runbook, the field visit, the carrier case, the hand-off or the security request. Where the rules allow, Mova iO acts on its own; otherwise an engineer reviews what it has prepared and approves it (the only human step), and the agent carries it out. The view also opens from the **Next** line on each open incident in the Operations Overview and from **Who fixes it** in the incident record.

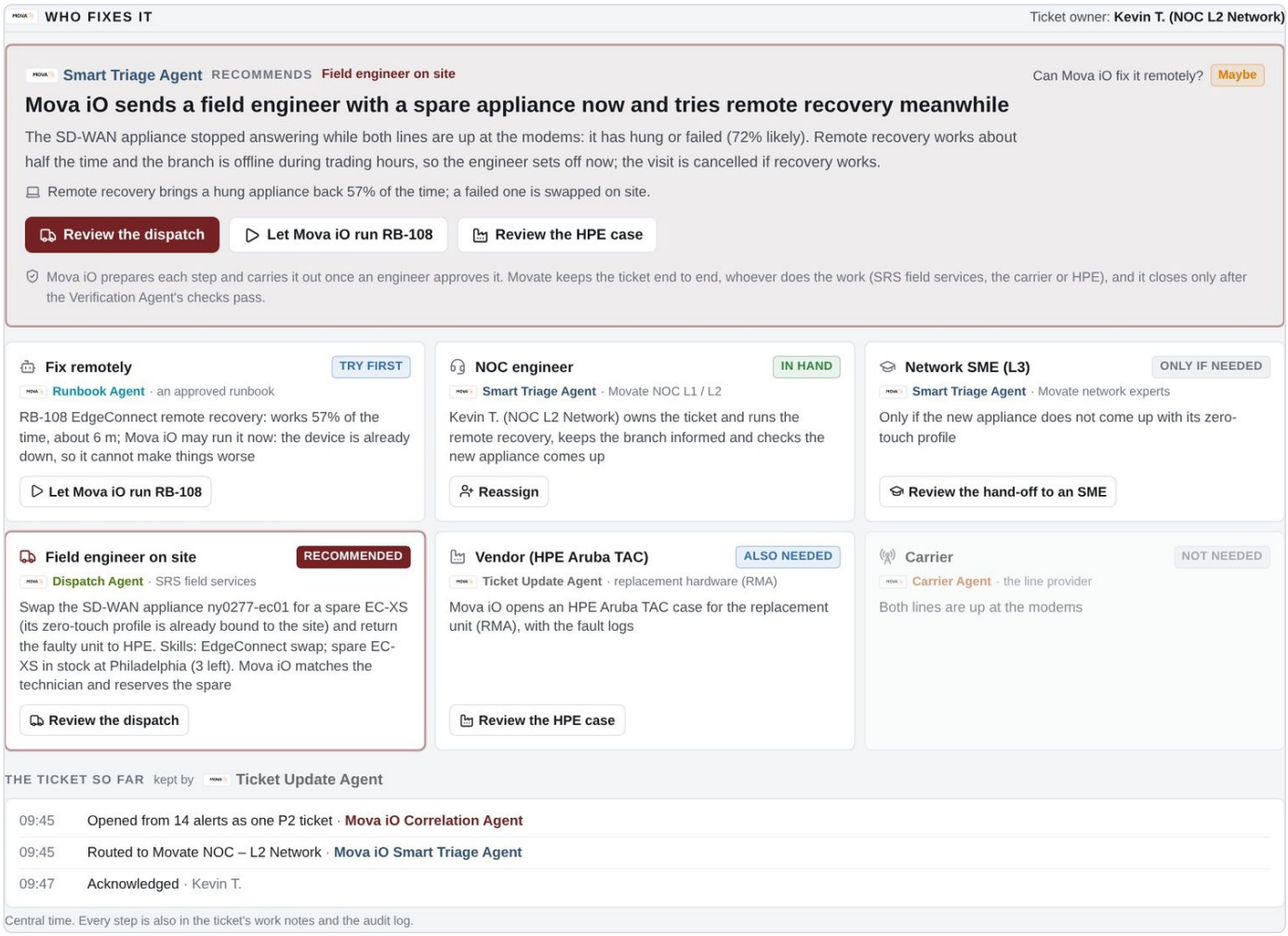



**Figure 14.** Mova iO: who fixes it, for a branch whose SD-WAN appliance stopped answering

| Part | What it shows |
|---|---|
| Smart Triage Agent recommends | The recommended route, what happens next in one line, why in plain words, and **Can Mova iO fix it remotely?**: Yes, Partly, Maybe or No |
| Buttons | What Mova iO has prepared, ready for review, for example **Review the dispatch**, **Let Mova iO run RB-108** and **Review the HPE case** |
| Routes | One card for each way the fault can be fixed, with the Mova iO agent behind it, marked Recommended, Try first, Also needed, In hand, Tried (did not fix it), Only if needed or Not needed |
| The ticket so far | Kept by the Ticket Update Agent: opened, routed, acknowledged, each hand-off, field visit, carrier case and HPE case, with the agent that did it and who approved it, in Central time |

Movate keeps the ticket end to end, whoever does the work, and it closes only after the Verification Agent's checks pass.

| Route | Mova iO agent | Who does the work | When |
|---|---|---|---|
| Fix remotely | Runbook Agent | Mova iO itself, with an approved runbook | The fix is a setting, a restart or a traffic move the rules allow, for example re-running a failed backup, a port bounce, or moving tills to the backup line |
| NOC engineer | Smart Triage Agent | Movate NOC L1 (24x7 analysts) or L2 (network engineers) | Remote fault-finding and fixes, chasing the carrier, talking the branch through a simple step |
| Network SME (L3) | Smart Triage Agent | US-based L3 engineers: SD-WAN and routing (Marcus L.), Wi-Fi and radio (Lina K.), change and config (Dana W.) | The fix is a design change: a Wi-Fi radio plan, a policy or routing change, a risky rollback |
| Field engineer on site | Dispatch Agent | SRS field technical services, directed by the Movate NOC | Something physical: a failed appliance, an optic, a power cable, work at height in the yard |
| Vendor (HPE Aruba TAC) | Ticket Update Agent | HPE | A replacement unit under the support contract (RMA), or a software fault |
| Carrier | Carrier Agent | The line provider | The fault is in the carrier network, or a backup SIM is blocked on the carrier account |
| Application support | Smart Triage Agent | SRS application teams | The network and lines are clean and the application is at fault |
| Security | Change Validation Agent | SRS Security Operations | A change made with no change record |
| Major incident manager | Smart Triage Agent | Movate NOC | Any P1 |

How Mova iO routes the common root causes:

| Root cause | Can Mova iO fix it remotely? | Mova iO's next step |
|---|---|---|
| Carrier fault on the main line | Partly: it has already moved traffic to the backup line | Carrier case with the evidence; NOC L2 chases it |
| Slow or lossy line at the carrier | Partly | Move tills, ordering and voice to the backup line (RB-101), then a carrier case |
| Backup SIM blocked on the carrier account | No | Ask the carrier account team to re-activate it; no visit |
| Access point with no power | Maybe: a port bounce fixes a hung port | Field engineer for the cable or the AP; port bounce first |
| Errors on a switch uplink | Partly: the bad link is taken out of service | Field engineer swaps the optic or patch lead |
| SD-WAN appliance not answering | Maybe: remote recovery works about half the time | Field engineer with a spare appliance now, remote recovery meanwhile, HPE case for the replacement |
| Wi-Fi drop-outs in the aisles | Partly | Hand to the Wi-Fi SME to re-plan channels and power |
| Address pool full | Yes | NOC L2 standard change |

| Root cause | Can Mova iO fix it remotely? | Mova iO's next step |
|---|---|---|
| Unrecorded config change | Yes | Roll back once approved; security reviews the change |
| Failed config backup | Yes | Mova iO re-runs it on its own |
| Loop from a desk switch | Partly: loop protection blocks it | The NOC talks the branch through unplugging it |

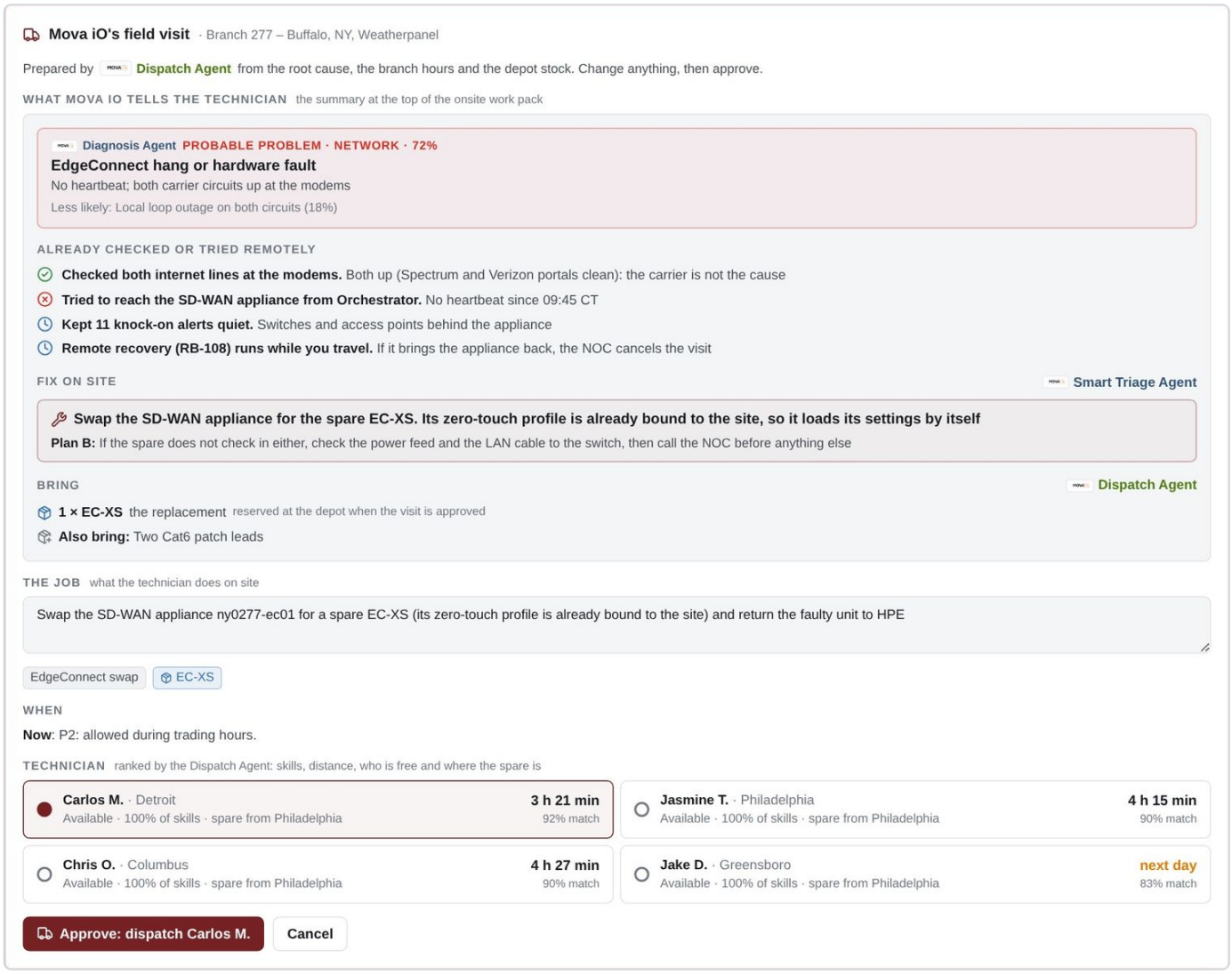



**Figure 15.** The field visit the Dispatch Agent prepared, ready to approve

| What Mova iO prepares | Agent | What happens when an engineer approves |
|---|---|---|
| A hand-off | Smart Triage and Ticket Update | The team and the person (least busy first, best-fit specialist marked) and a hand-off package with the root cause, the evidence and what was tried. The agent moves the ticket and records it in the work notes and the audit log |

| What Mova iO prepares | Agent | What happens when an engineer approves |
|---|---|---|
| A field visit | Dispatch | The job, skills and spare part from the root cause; the visit **Now** for P1 and P2 or a device already down, otherwise after the branch closes (with **Go now anyway**); the best four technicians ranked by skills, distance, who is free and a depot in the region with the spare. The agent books the work order in Field Dispatch, reserves the spare and sends the technician the onsite work pack (chapter 15), which opens with Mova iO's summary: the probable problem, what was already done remotely, the fix on site and what to bring |
| A carrier case or escalation | Carrier | The case with an hour of loss and delay readings, path tests, counters and timestamps; or an escalation to the carrier duty manager, then the account manager, then the service director, with the reason. The agent opens or escalates it and chases it every hour |
| An HPE case | Ticket Update | The serial, the support status, the fault logs and what was tried, for a replacement unit. The agent opens the case with HPE Aruba TAC |
| A runbook | Runbook | Opens the incident record on the Runbook tab, where the gate rule decides whether Mova iO runs it on its own or needs approval |
| A security request | Change Validation | Brings security in on an unrecorded change without moving the ticket |

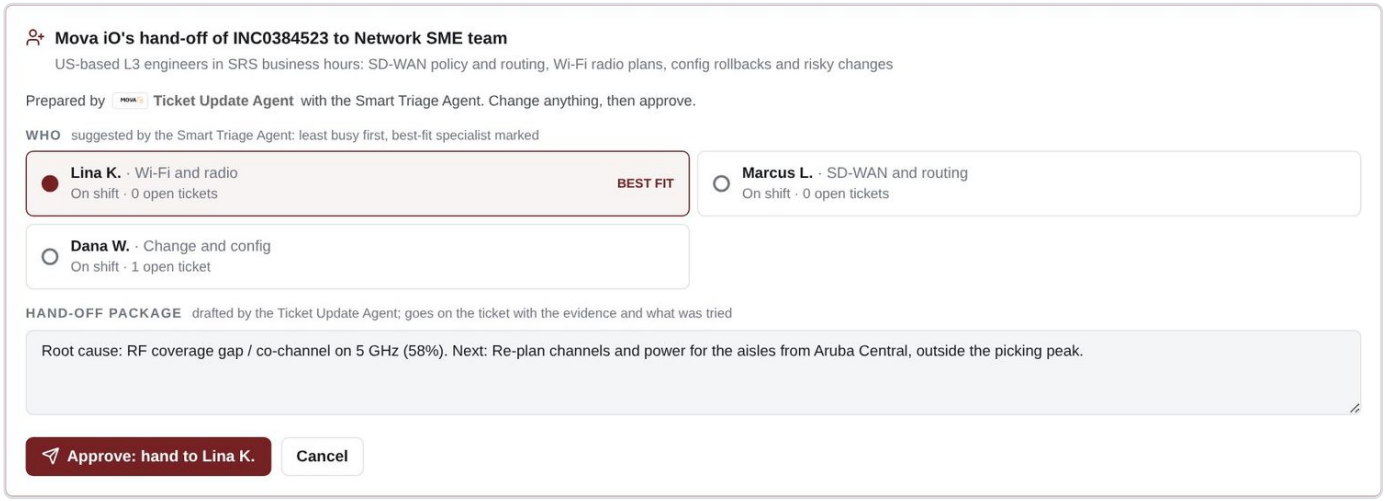



**Figure 16.** The hand-off Mova iO prepared for the Wi-Fi expert

The NOC Engineer and Network Engineer L3 roles approve. The Security Lead approves security requests. Other roles see what Mova iO recommends with the buttons greyed out, and the role dashboards show it read-only.

## Time

All times are in Central Time (CT) unless a card says otherwise. Branch opening and closing times use each branch's local time, which matters on Branch Readiness, where branches open in waves across five time zones.

## Confidence and the gate rule

Every Mova iO decision carries **evidence** (why it thinks so), a **confidence** from 0 to 100%, a **rollback** (how to undo it) and a **verification** (how it checks the fix worked). A person must approve before anything changes when any of these is true:

- confidence is below 80%;
- the action changes a device at a branch that is open for trading;
- a freeze window covers the device or region.

Some actions are never autonomous, whatever the confidence: firmware and OS upgrades, routing and security policy changes, QoS changes, device reloads, rollbacks on hub devices and any action on a device without a confirmed alternate path.

## The three control modes

| Mode | Catalog tier | What happens | A person's role |
|---|---|---|---|
| Autonomous | Act | Mova iO acts on its own inside a policy SRS and Movate set and audit | Over the loop: reviews outcomes and the policy |
| Human-on-the-loop | Act and notify | Mova iO acts and tells a named person at once; the person can stop it during a stop window (5 minutes in production, 10 seconds in demos) | On the loop: watches and can stop |
| Human-in-the-loop | Stage for approval | Mova iO prepares the plan, evidence and rollback, then waits | In the loop: nothing changes until they approve |

New actions start at the Stage tier. They move up only on evidence: a record of clean runs, proposed by the Catalog Steward and approved by SRS.

## Network terms in plain words

Every reading in the console is a network measurement. This section explains each term once: what it is, what is normal at SRS, and what it means for the branch and its customers. The glossary in chapter 29 repeats the short definitions.

### How a branch connects

| Term | What it is | At SRS | Why the business cares |
|---|---|---|---|
| LAN (branch network) | The switches, access points and cabling inside the building | 1 to 6 switches and 1 to 3 APs per branch | Counters, office PCs, phones and warehouse handhelds all plug into it |
| WAN | The lines that connect the branch to the data centre, Azure and the internet | Two per branch | ERP, pricing, card payments and calls all cross the WAN |
| Line, circuit | One WAN connection bought from a carrier | 1,800 in total | A line down means the branch relies on its other line |
| Primary line, backup line | The main line the branch normally uses, and the second line kept for failures | Every branch has both | With one line left there is no spare if it also fails |

| Term | What it is | At SRS | Why the business cares |
|---|---|---|---|
| MPLS | A private line inside a carrier's network, steady but expensive | 22% of primaries (AT&T, Lumen, Bell) | Very steady latency; slow to install |
| DIA (dedicated internet access) | A business internet line with guaranteed speed | 30% of primaries | Steady, with a carrier repair commitment |
| Broadband | A cable or fibre internet line shared with other customers | 48% of primaries, many backups | Cheaper and fast, but busier at peak times |
| LTE/5G | A cellular line from a router with a SIM | Backup at 55% of branches | Works when wires are cut, but adds 25 to 40 ms and is metered |
| Carrier, LEC | The company that sells the line; the local exchange carrier that owns the last mile to the building | AT&T, Lumen, Comcast, Spectrum, Verizon, Cox, Bell, Rogers | The carrier case and its repair time decide how long a line stays down |
| Demarc, NTE | The point where the carrier's equipment ends and the branch's begins | One per line | Proving a fault is on the carrier's side of the demarc gets it fixed faster |
| SD-WAN | Software that uses both lines together and picks the best line for each application | On every branch | Payments and ERP keep working when one line is poor |
| EdgeConnect | The HPE Aruba SD-WAN appliance at each branch (model EC-XS; EdgeConnect 10104 for new and replacement units) | 933 at 900 branches | It measures both lines every second and moves traffic between them |
| HA pair | Two EdgeConnects at one branch, one taking over if the other fails | 33 larger branches | A hardware fault does not take the branch down |
| Orchestrator | The central system that configures every EdgeConnect and collects its statistics | One for the estate | The source of the latency, jitter, loss and tunnel readings |
| Tunnel | An encrypted path over one line from the branch EdgeConnect to the data centre | Two per appliance, 1,866 in total | A tunnel down means one line cannot carry business traffic |
| Overlay, business intent policy | Rules that say which line each kind of traffic should use (payments, ERP, voice, guest) | Set centrally in Orchestrator | Critical traffic gets the best line; guest Wi-Fi never competes with payments |
| Steering, failover, failback | Moving traffic to the other line; doing it automatically when a line fails; moving it back once the main line is stable | Failover takes about 4 seconds | Customers at the counter do not notice a failover |
| Bandwidth | The capacity of a line, in megabits per second (Mb/s) | 20 to 1,000 Mb/s primaries, 150 Mb/s LTE | A line with too little bandwidth fills up at peak times |
| WAN use | The share of a line's bandwidth in use right now | Usually 10 to 60% | Above 85% traffic queues, which raises latency and loss for everyone |

### Line quality and what the business feels

| Term | What it is | Normal at SRS | Warning | What the business feels |
|---|---|---|---|---|
| Latency (to the data centre) | Time for a request to reach the McKinney data centre and come back, in milliseconds | 7 to 15 ms in Texas, 35 to 50 ms on the coasts, about 90 ms in Hawaii; LTE adds 25 to 40 ms; 32 ms on average | Normal plus 25 ms; critical at normal plus 60 ms | ERP, pricing and order screens respond slowly |
| Jitter | How much the delay varies from one packet to the next | 0.5 to 3 ms on wired lines, 5 to 12 ms on LTE | Above 20 ms; critical above 30 ms | Choppy calls; card terminals time out more often |
| Packet loss | The share of data that never arrives and must be sent again | Under 0.2% | Above 1%; critical above 2% | Card payments retry, calls break up, ERP sessions drop |
| Degraded line | A line that works but with high latency, jitter or loss | A handful at any time | Any degraded primary at an open branch | The branch trades, but slowly, until traffic moves to the other line |
| Payment path round trip | A synthetic test payment from the counter VLAN (PCI segment) to the payment processor and back | 100 to 200 ms | Above 400 ms | Card payments are slow at the counter |
| ERP login | A synthetic login to the ERP from the branch | About 1 second | Above 2 seconds | Staff wait on every order and stock check |
| Voice call quality (MOS) | A score from 1 to 4.4 for how a call sounds, worked out from jitter, loss and delay | 4.2 to 4.4 | Below 3.6 | Customers and branches struggle to hear each other |
| Wi-Fi experience | Aruba Central's score out of 100 for the Wi-Fi inside the branch: signal, connection success and roaming | 90 to 97 | Below 80; critical below 65 | Handhelds and laptops in the warehouse and office struggle. It does not depend on the WAN: Wi-Fi can be good while a WAN line is poor |
| p95 | The 95th percentile: 95% of measurements are at or below this value | – | – | Shows what most users get, without being skewed by one bad sample |
| Synthetic probe | A scripted test run every minute from each EdgeConnect: payment path, ERP login, DNS, DHCP | 5,400 checks every 10 minutes | A failed probe | Finds problems before a customer does |

### Inside the branch

| Term | What it is | Why the business cares |
|---|---|---|
| Switch, port, uplink | The box that connects wired devices; each socket is a port; the uplink connects one switch to another or to the EdgeConnect | An uplink fault cuts off everything behind it |
| PoE (Power over Ethernet) | Power delivered to APs, phones and cameras through the network cable | A switch out of PoE budget switches devices off |

| Term | What it is | Why the business cares |
|---|---|---|
| CRC errors | Corrupted frames on a link, usually a failing cable or optic | Slow or dropping connections; often worse over days |
| Optic, SFP | The plug-in module for a fibre link | A cheap part that is a common cause of uplink faults |
| VLAN | A separate network segment on the same switches: 10 corporate, 20 payments (PCI), 30 warehouse Wi-Fi, 40 voice, 50 cameras, 60 guest | Keeps payments, staff and guests apart |
| PCI segment | VLAN 20, the isolated network for card terminals, with no routes to other VLANs | Required for card payment compliance |
| Access point (AP), SSID | The Wi-Fi radio; the network name it broadcasts, such as SRS-SCAN for warehouse handhelds | Handhelds, laptops and yard devices depend on it |
| Roaming | A handheld moving from one AP to the next as staff walk the aisles | Failures stop scanning and picking mid-task |
| Channel utilisation | How busy an AP's radio channel is | Above about 70% Wi-Fi slows for everyone on that AP |
| DHCP, scope | The service that gives each device an address; the pool of addresses for a VLAN | A full scope means new devices cannot connect |
| Broadcast storm, loop | Traffic copied endlessly round a cabling loop, often from an unmanaged desk switch | The whole branch network slows or stops |
| BPDU guard | A switch setting that shuts a port when another switch is plugged into it | Stops loops before they become storms |
| OOB (out-of-band) console | A cellular console path to devices when the WAN is down | Engineers can still reach a branch that has lost both lines |

### Monitoring, events and incidents

| Term | What it is | At SRS |
|---|---|---|
| Telemetry, poll, trap, syslog | Readings collected from devices; a regular query (SNMP poll); a message a device sends when something happens (trap); a device's log lines (syslog) | OpsRamp polls every device every minute |
| Event | Any single message or threshold crossing from a tool | About 50 a minute across the estate |
| Noise removal | Dropping duplicates, flaps and information-only events before they become alerts | About 70% of events (KPI at least 50%) |
| Duplicate, flap, hold-down | The same event repeated; a line or port that goes down and up quickly; the wait (for example 30 to 40 seconds) before acting, in case it recovers | Flaps that recover in the hold-down never become tickets |
| Alert | An event worth acting on, left after noise removal | Most clear on their own |
| Auto-cleared | An alert that recovered by itself and was checked by ping and SNMP | About 70% of alerts |
| Digest | A daily summary of warnings that do not need a ticket | Read by the morning shift |

| Term | What it is | At SRS |
|---|---|---|
| Correlation, correlation group | Grouping related alerts by topology, time, change, carrier and similarity into one incident | One incident per branch per event |
| Incident | A ServiceNow record of a problem, with a priority and SLA clocks | 20 to 40 a day |
| Confidence | How sure Mova iO is about a diagnosis or action, 0 to 100% | Below 80% a person decides |
| Root cause | The underlying fault behind an incident, for example a carrier fault upstream of the demarc | Recorded in the RCA |
| Problem record | A ServiceNow record for the cause behind repeat incidents | Opened after three incidents in 30 days |
| Repeat offender | A circuit with three or more tickets in 90 days | Gets a carrier review |
| NTF (no trouble found) | A carrier closing a ticket without finding a fault | 23% of tickets in the last six months |

### Devices, firmware and lifecycle

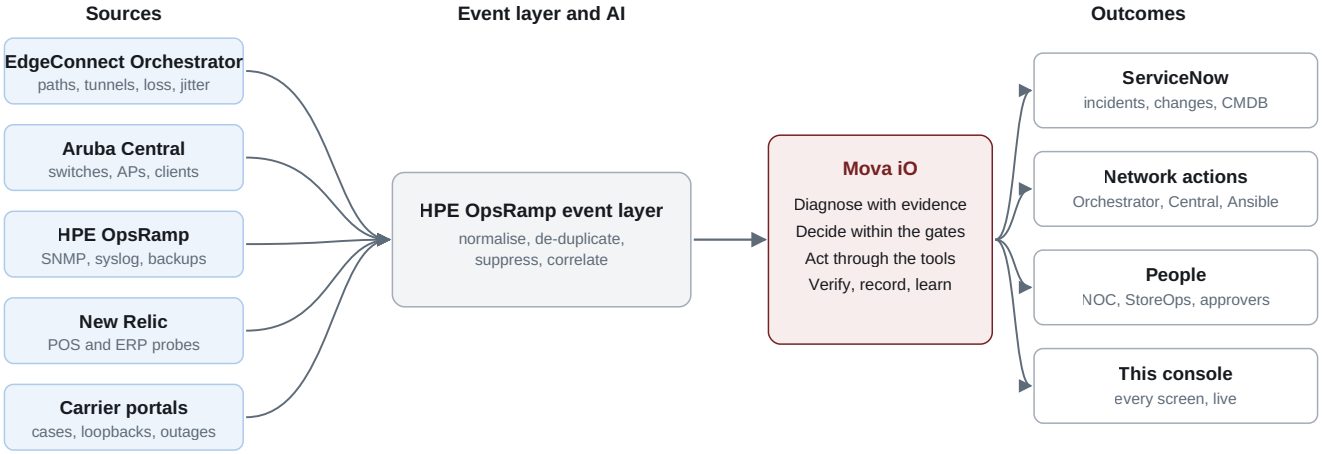
| Term | What it is | At SRS |
|---|---|---|
| Aruba Central, standalone | HPE Aruba's cloud management for switches and APs; a standalone device is managed one by one and only monitored by OpsRamp | 1,062 in Central, 2,080 standalone |
| Firmware, release, N, N-1, N-2 | The device software; the target release (N), one behind (N-1) and two behind (N-2) | Policy: EdgeConnects at N or N-1, others no older than N-2 |
| Config backup | A nightly copy of each device's configuration | 02:00 CT every night |
| Golden config, drift, Network-as-Code | The approved configuration kept as files in a repository; any difference on a device is drift; changes go through reviewed merge requests | Drift is checked every night |
| Advisory, CVE, CVSS | A vendor security bulletin; the public ID of a vulnerability; its severity score from 0 to 10 (9.0 and above is critical) | Each advisory is matched to the firmware every device runs |
| Patch, canary, ring, wave | A fixed release; a small first group upgraded and checked before the rest; the groups that follow; one batch of upgrades after branch close | Canary first, then waves by region |
| End of sale | The last day the vendor sells a model; it is still supported | EC-XS: 31 Jan 2026. Aruba 2530: 31 Jan 2022 (48-port), 31 Mar 2022 (24-port) |
| End of support | The last day of vendor support and security fixes; after it the device should be replaced | EC-XS: 31 Jan 2031. Aruba 2530: 31 Jan 2027 (48-port), 31 Mar 2027 (24-port). Other models: not announced |
| Limited lifetime warranty | HPE's warranty for Aruba switches and APs, which runs until five years after the model's end of sale | Switches and APs show "Lifetime" |
| Support contract | The yearly HPE support agreement that covers the EdgeConnects, which have a 1-year warranty | Renewed every year |

### Service measures and automation

| Term | What it is | At SRS |
|---|---|---|
| SLA, respond, resolve | The agreed time to start working on an incident and to fix it | P1: respond in 15 minutes, resolve in 4 hours |
| MSL, ESL, KPI | Minimum service level (below it a service credit applies); expected service level (the target); key performance indicator | All from Annexure B |
| Availability | The share of time a service works | Branches online: target 99.95% |
| MTTR | Mean time to restore service | Tracked per priority |
| Error budget, burn rate | The failure a target allows over 30 days (99.9% allows about 43 minutes); how fast it is being used compared with an even spend | Burning fast holds non-essential changes |
| Runbook | An approved, tested automation for one kind of fix, numbered RB-1xx | 10 in the library |
| Act, Act and notify, Stage | The three catalog tiers: acts on its own; acts and tells a person at once; prepares and waits for approval | New runbooks start at Stage |
| Gate rule | The rule that sends a decision to a person: confidence below 80%, a branch open for trading, or a freeze | Applies to every action |
| Verification, rollback | The checks that prove a fix worked; the prepared way to undo it | Every action has both |
| Mova iO agent | One of Movate's 13 AI agents, each with one job: noise reduction, correlation, diagnosis, runbooks, carrier cases and others | Every action lands in the audit log |

# 4. Where the data comes from

Five monitoring sources feed one event layer. Mova iO correlates the events, decides what to do within the gate rules, acts through the tools and writes the record to ServiceNow.




Data flow: five sources feed one event layer; Mova iO decides within the gate rules and acts through the tools; every step is recorded in ServiceNow and the audit log.

| Source | What it sends | How often |
|---|---|---|
| EdgeConnect Orchestrator | Path and tunnel health per appliance: loss, latency, jitter, failover | Stats every 30 s, alarms live |
| Aruba Central | Switch, AP and client health, PoE, radios, roaming, firmware | Every minute, events live |
| HPE OpsRamp | SNMP and syslog from every device, config backups | Polls every minute, traps live |
| New Relic | Synthetic payment-path and ERP checks from each branch | Every minute |
| Carrier portals | Ticket status, loopback results, maintenance windows, outage maps | On change |

Typical ticket triggers are a device unreachable for three polls, an SD-WAN tunnel down for more than two minutes, packet loss above 2% for ten minutes, sustained latency or jitter above the path's limit, recurring interface errors and a failed synthetic check.

# Part II. Screen by screen

## 5. Operations Overview

The Operations Overview answers "is the network all right, and if not, where?" in one screen. It covers every branch in the current filter. The header buttons are the **LIVE** label (click to pause), **Run a scenario**, which replays one of five scripted events into the console, and **Export**, which downloads the branch health list as a CSV file.

## Headline tiles




**Figure 17.** Headline tiles

| Tile | What it counts | Good | Look closer when | Click |
|---|---|---|---|---|
| Branches open now | Branches inside their local trading hours, out of the branches in view | Follows the clock: low at night, near 900 in the day | Not applicable | The branches open now, with their status and line |
| Healthy / degraded / down | Branches by health: green has both lines and its devices up; amber is trading but impaired, for example on its backup line or with an unstable main line; red has no working line and cannot trade | Green above 97%, red at 0 | Any red, or amber rising | The tile opens every branch; each number opens its own list: the healthy, the degraded or the branches down |
| Open incidents | All open incidents in view, with the split by priority. "+N new" flashes when incidents arrive | No P1 | Any P1 (tile turns red), or P2 count rising | Opens the open incidents; each priority chip opens that priority. **Open Live Incidents** in the panel goes to the queue |
| Devices down | EdgeConnects, switches and APs not answering, out of all devices in view. Every device at a branch that is down counts, because none of them can be reached | Single digits | A jump, which usually means a branch is down | The devices down, by model and region |
| Circuits down / degraded | WAN lines (circuits) down (red) or degraded (amber), primary and backup together, and how many branches are trading on their backup line. A branch that is down counts both its lines | Low and steady | Several on one carrier or in one area | The tile opens every line down or degraded; the red and amber numbers open the lines down or degraded; the line about the backup opens the branches on their backup line |
| Noise removed today | Events removed before they became alerts: duplicates, flaps that recovered within the hold-down and information-only messages. Shown with all events since midnight, the share removed and events per minute now | Above 50% (KPI); usually near 70% | The share drops, so engineers see more noise | Opens Correlation |
| SLA clocks | Open incidents on track, at risk (more than 75% of resolve time used) and breached | No breached clocks | Any amber or red | The tile opens the open incidents; each number opens the incidents on track, at risk or past their deadline |

Every list opens to any branch, line, device or incident, and to its root cause (chapter 3).

## Branch health map




**Figure 18.** Branch health map

**What it shows.** All 900 branches on a map of the United States and Canada, one dot per branch, coloured by health. Dashed moving lines are live traffic from the busiest branches to the three hosting points: the Primary DC, Azure South Central and Azure East (black diamonds).

| Parameter | Meaning |
|---|---|
| Dot colour | Green healthy, amber degraded, red down. Red dots pulse |
| Faded dots | Branches outside the current filter, kept for context |
| Flow line colour | Blue normal traffic, amber from a degraded branch, red from a branch that is down |
| Zoom | Picking a region or city in the filter bar zooms the map to it |

**How to read it.** Look for clusters, not single dots. A cluster of amber or red in one metro usually means a carrier or weather event, which the Correlation screen will show as one major incident. **What to do:** hover a dot for the branch name and state; click it to open Branch 360.

## Open incidents

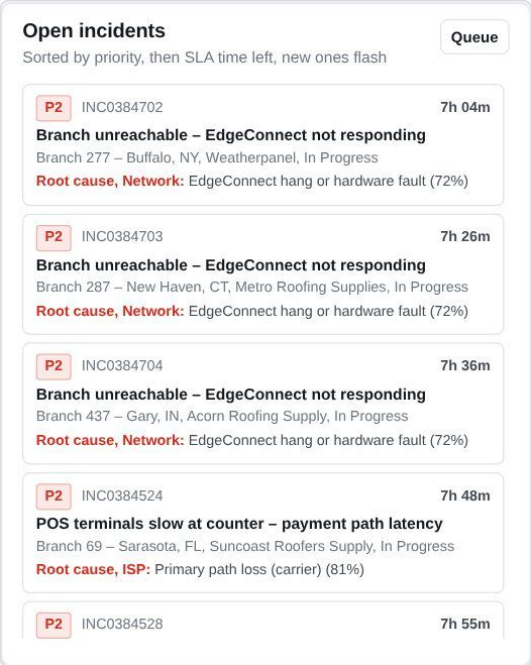



**Figure 19.** Open incidents

The 14 most urgent open incidents, sorted by priority and then by SLA time left. Each row shows the priority, incident number, MAJOR or NEW flags, time left to resolve (amber past 75% used, red "breached"), the title and the branch, brand and state, and in red the root cause: where the fault is (network, ISP or application), the cause and its confidence. Under it, Mova iO's **Next** says who fixes it and what happens next. New incidents flash. **Queue** opens Live Incidents; clicking a row opens that incident; clicking the root cause opens the root-cause view, and clicking **Next** opens it at **Mova iO: who fixes it** (chapter 3).

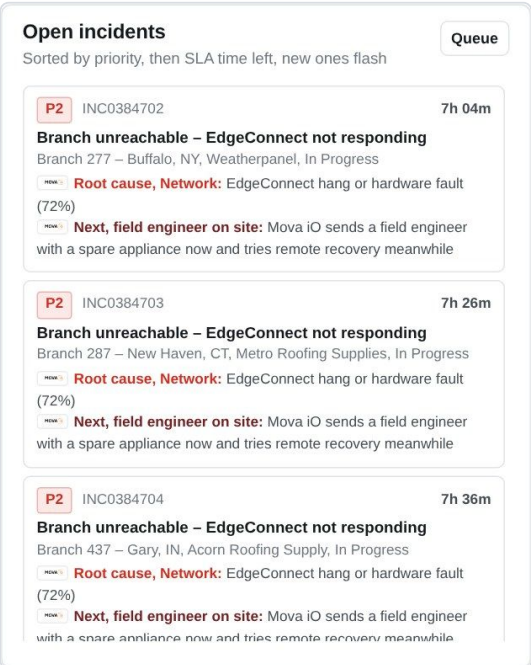



**Figure 20.** Open incidents with the root cause and the next step

## Service levels and error budgets

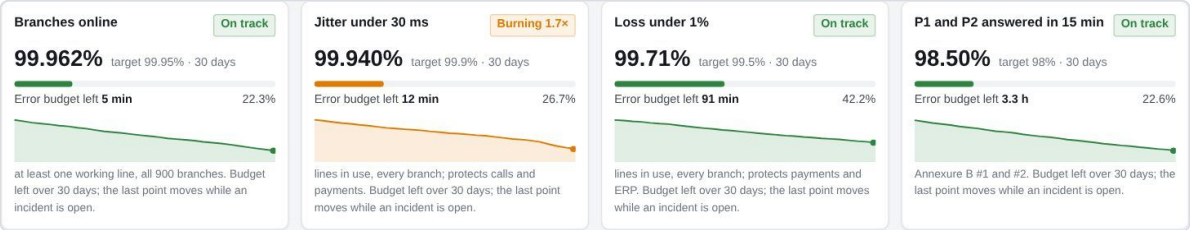



**Figure 21.** Error budgets

Four service-level objectives over 30 days: **Branches online** (at least one working line), **Jitter under 30 ms** on the lines in use (protects calls and card payments), **Loss under 1%** on the lines in use (protects payments and ERP) and **P1/P2 answered in 15 minutes**.

| Parameter | Meaning |
|---|---|
| Large figure | Achieved level over the last 30 days, against the target shown beside it |
| Badge | **On track**, or **Burning N×** when the budget is being used faster than an even spend (more than 1.5 times) |
| Bar and percentage | Share of the 30-day error budget still left |
| Error budget left | The same budget as time: minutes or hours of failure still allowed this month |
| Line | Budget left over the 30 days; the last point moves while an incident is open |

**How to read it.** The error budget is the failure the target allows. A 99.9% target over 30 days allows about 43 minutes. Green means plenty left; amber means burning fast; red means less than 20% left. **What it means for change:** when a budget burns fast, non-essential changes in that area wait (see the error budget demo in Part III).

## Incident outlook, next 12 hours




**Figure 22.** Incident outlook

**What it shows.** New incidents per hour for the last 24 hours (solid) and the forecast for the next 12 hours (dashed), with a shaded band for the 80% likely range. Above the chart: expected new incidents, the busiest hour and the weather risk.

**How to read it.** The forecast uses the hourly pattern, the weather outlook and open changes. Evening hours are busier because most incidents are worked after branches close. If the actual line runs above the band, something unusual is happening. **What to do:** plan the evening shift and surge cover from the busiest hour.

## Prevention in flight

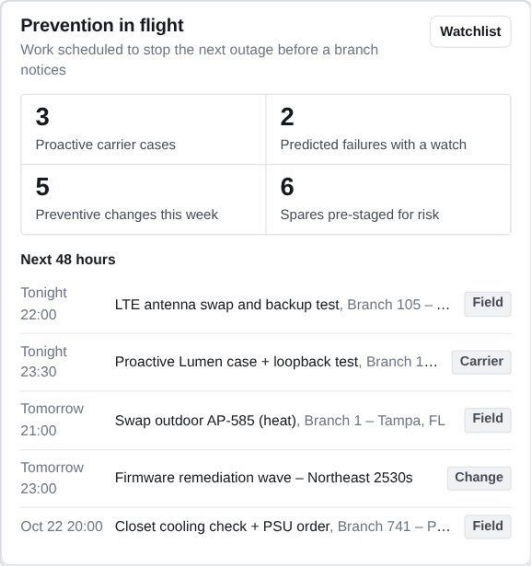



**Figure 23.** Prevention in flight

Work already booked to stop the next outage before a branch notices: proactive carrier cases, predicted failures under watch, preventive changes this week and spares pre-staged for risk, then a list of the next 48 hours with time, task, branch and owner (Field, Carrier or Change). **Watchlist** opens the Predictive Watchlist.

## Event flow




**Figure 24.** Event flow

| Parameter | Meaning |
|---|---|
| Raw / min | Events per minute arriving from all monitoring tools (grey line) |
| After correlation | What is left once duplicates, flaps and children of an open incident are removed (blue line) |
| Noise removed | 1 minus after/raw. The KPI asks for at least 50%; the console usually runs near 70% |
| Tabs | Live 5 minutes, 1 hour, 24 hours |

**How to read it.** The gap between the lines is the work Mova iO saves. A sudden rise in raw events with a flat blue line is a storm being contained. A rising blue line means more real events, which should match new incidents.

## WAN quality – live

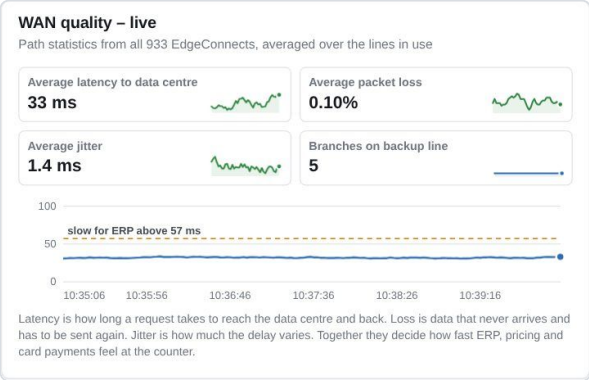



**Figure 25.** WAN quality

The quality of the lines every branch is using right now, averaged across all 933 EdgeConnects from Orchestrator path statistics and refreshed with the live data:

| Tile | What it is | Normal at SRS | Look closer when | What the business feels |
|---|---|---|---|---|
| Average latency to data centre | Time for a request to travel from the branch to the McKinney data centre and back | About 32 ms | More than 10 ms above normal | ERP, pricing and order screens respond more slowly |
| Average packet loss | Share of data that never arrives and must be sent again | Under 0.2% | Above 0.5% on average | Card payments retry, calls break up |
| Average jitter | How much the delay varies from one packet to the next | 1 to 3 ms | Above 10 ms on average | Voice becomes choppy; payment terminals time out more often |
| Branches on backup line | Branches whose main line is down, or whose traffic was moved off it | Under 15 | Above 20, or a cluster in one area | Trading continues, but LTE is slower and there is no spare line |

The chart tracks average latency with a dashed line at normal plus 25 ms, the point where staff start to notice ERP screens slowing. A rise across many branches at once usually means a carrier or regional problem; one branch alone shows on Top branches and Branch 360.

## Live event stream

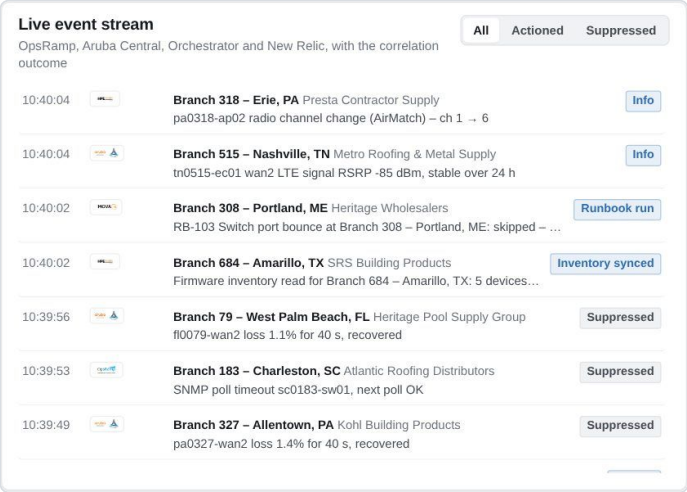



**Figure 26.** Live event stream

Every event in view as it arrives: time, source tool, branch and brand, the message and the outcome. Outcome badges: **OK** (a probe passed), **Suppressed** or **Duplicate** (grey), **Watching** (amber), **Auto-resolved** or **Recovered** (green), **New incident** (red), and others such as **Accepted** (a knowledge article accepted by an engineer). Tabs filter to **All**, **Actioned** (events that led to something) or **Suppressed**. Click a row to open its incident, or the branch if there is none.

## Top branches

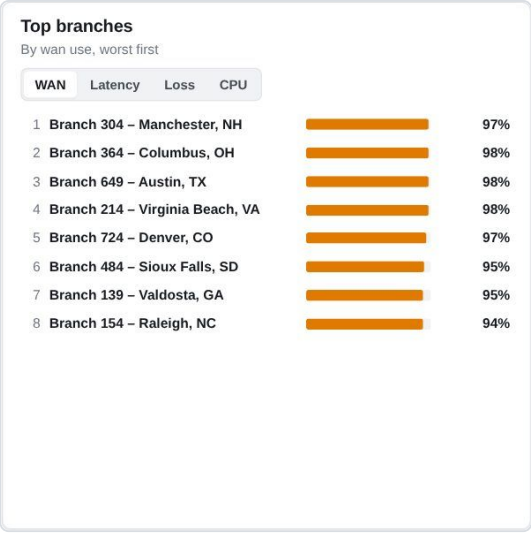
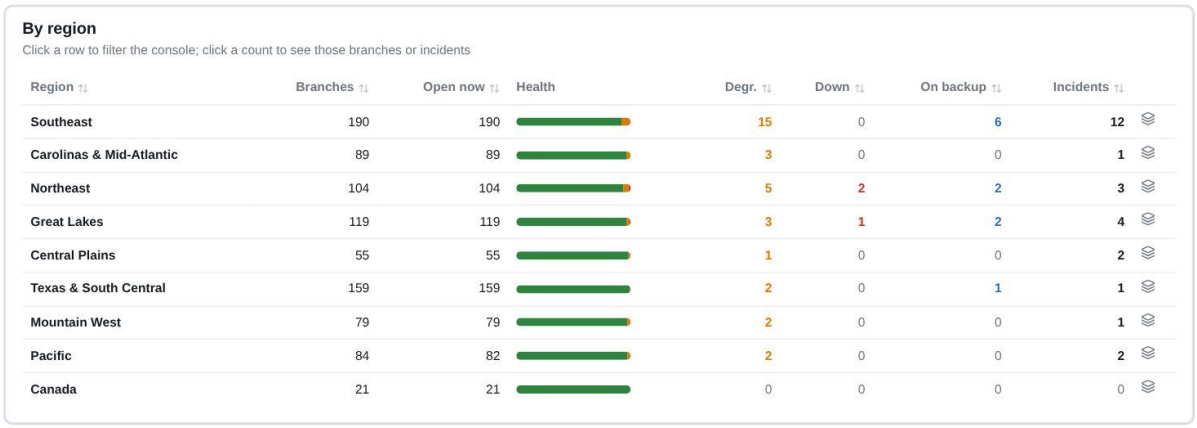



**Figure 27.** Top branches

The eight worst branches on one measure, chosen with the tabs: **WAN** use (%, amber from 80%), **Latency** (ms, amber from 80 ms), **Loss** (%, amber from 1%) or EdgeConnect **CPU** (%, amber from 75%). Use it to find the branch that is about to become an incident. Click a row for Branch 360.

## By region



By region

Click a row to filter the console; click a count to see those branches or incidents

| Region | Branches | Open now | Health | Degr. | Down | On backup | Incidents |
|---|---|---|---|---|---|---|---|
| Southeast | 190 | 190 | | 15 | 0 | 6 | 12 |
| Carolinas & Mid-Atlantic | 89 | 89 | | 3 | 0 | 0 | 1 |
| Northeast | 104 | 104 | | 5 | 2 | 2 | 3 |
| Great Lakes | 119 | 119 | | 3 | 1 | 2 | 4 |
| Central Plains | 55 | 55 | | 1 | 0 | 0 | 2 |
| Texas & South Central | 159 | 159 | | 2 | 0 | 1 | 1 |
| Mountain West | 79 | 79 | | 2 | 0 | 0 | 1 |
| Pacific | 84 | 82 | | 2 | 0 | 0 | 2 |
| Canada | 21 | 21 | | 0 | 0 | 0 | 0 |

**Figure 28.** By region

One row per region: branches, open now, a health bar (green, amber, red), degraded, down, on backup and incidents. Click a row to filter the whole console to that region.

## Alerts per minute – live

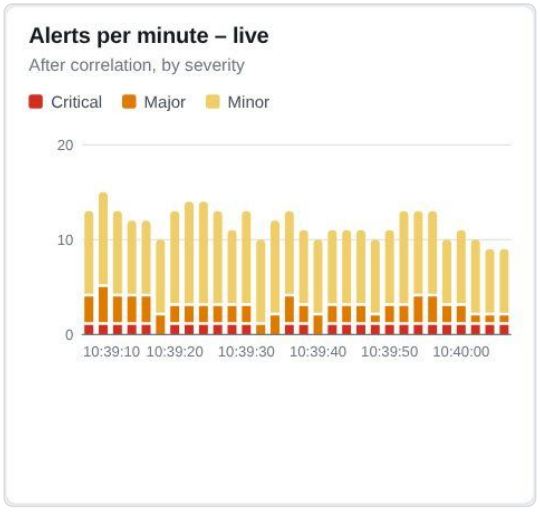



**Figure 29.** Alerts per minute

Alerts per minute after correlation for the last 30 minutes, stacked by severity: Critical (red), Major (orange) and Minor (yellow). A tall red bar is the moment to look at the event stream.

## Circuits down or degraded

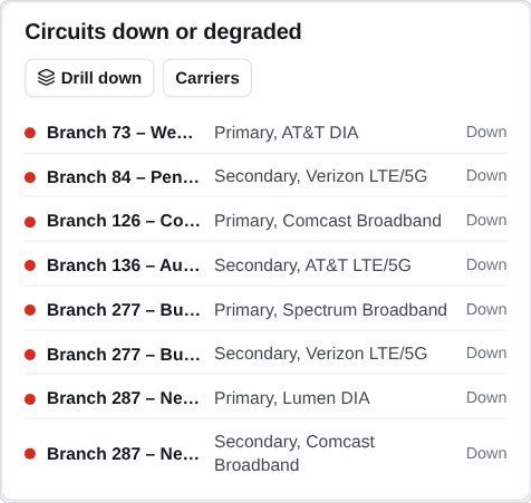



**Figure 30.** Circuits down or degraded

Every circuit not fully up: branch, primary or secondary, carrier and type, and Down (red dot) or Degraded (amber dot). A branch listed twice has lost both paths. **Carriers** opens Carrier Management; a row opens the branch.

## Ticket mix – last 6 months

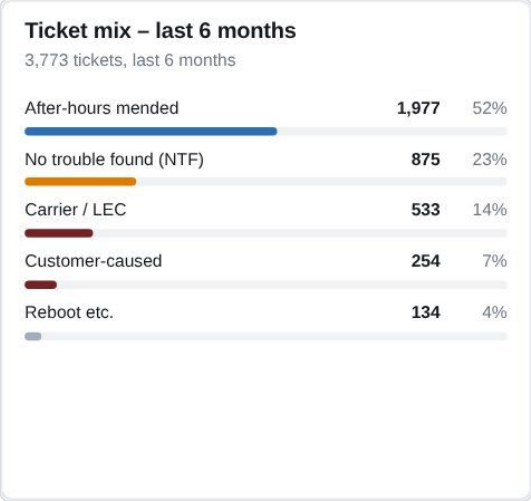



**Figure 31.** Ticket mix

The 3,773 network tickets of the last six months by type: after-hours mended (52%), no trouble found (23%), carrier or LEC (14%), customer-caused (7%) and reboots (4%). This is the baseline the automation targets: after-hours fixes and no-trouble-found tickets are where Mova iO saves the most.

## Unified timeline

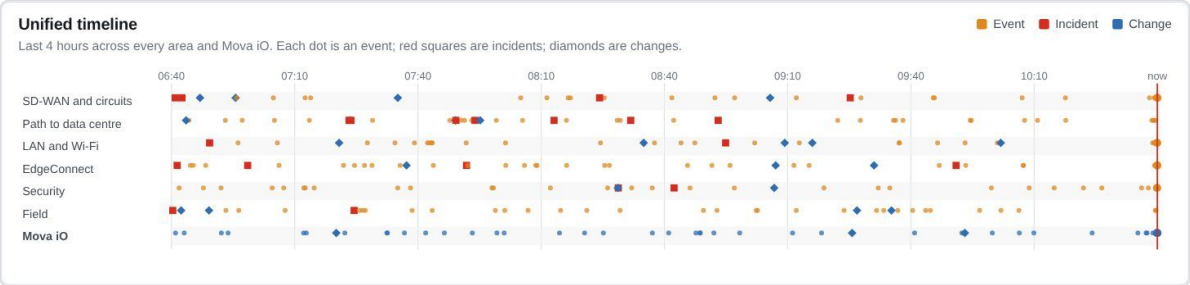



**Figure 32.** Unified timeline

The last four hours on one axis, one lane per area (SD-WAN and circuits, path to data centre, LAN and Wi-Fi, EdgeConnect, security, field) plus a lane for Mova iO. Dots are events (larger when new), red squares are incidents and blue diamonds are changes; the red line is now. Use it to see whether an incident followed a change, or whether one area's events lead another's.

# 6. Live Operations

Live Operations is the "watch it happen" screen: the network as it runs, line by line, with alerts and automation as they arrive, for every branch in the current filter. Every tile says what its number means for trading. Use it on a wall screen or when something is unfolding.

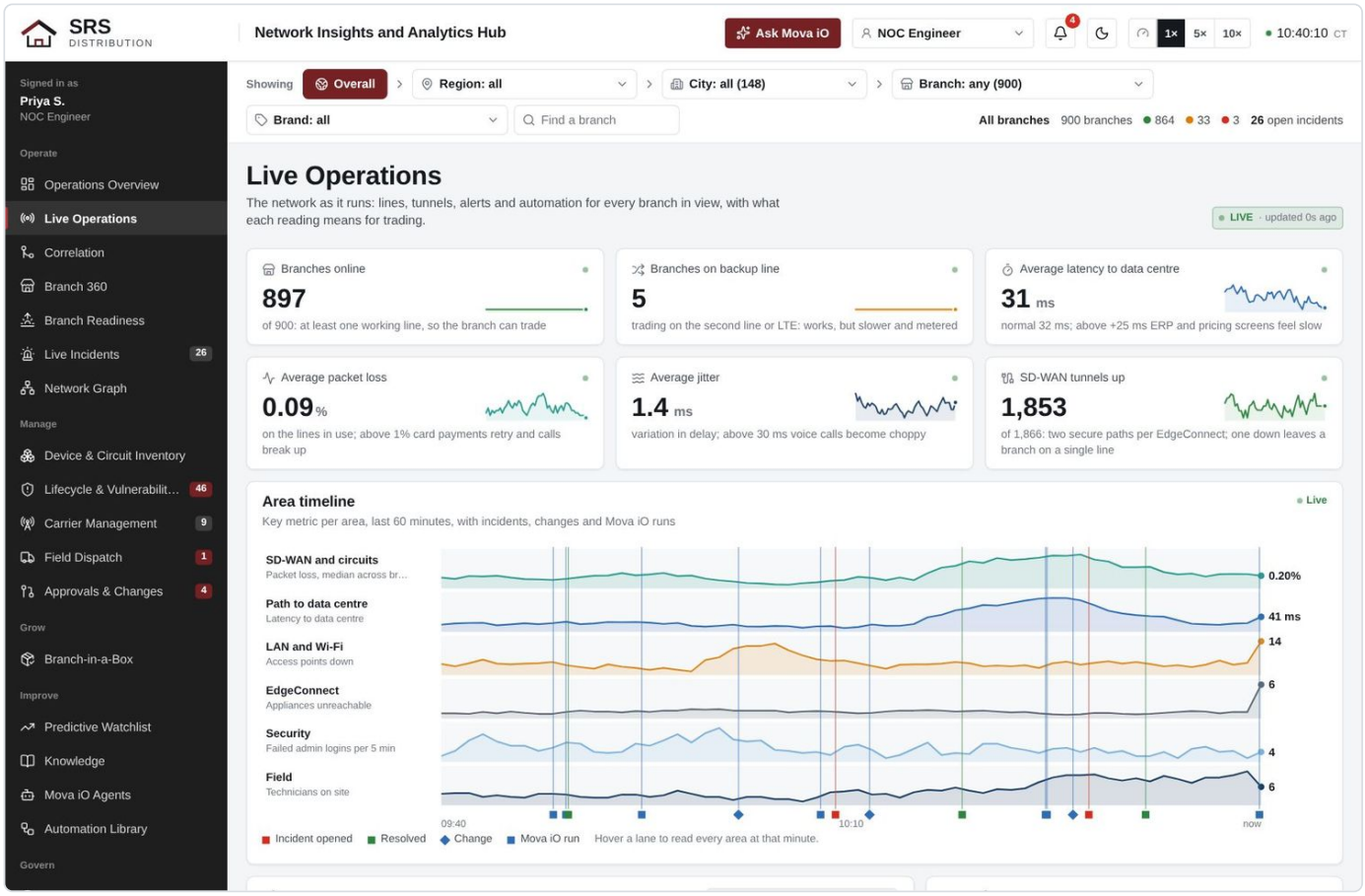



**Figure 33.** Live Operations

## Live tiles

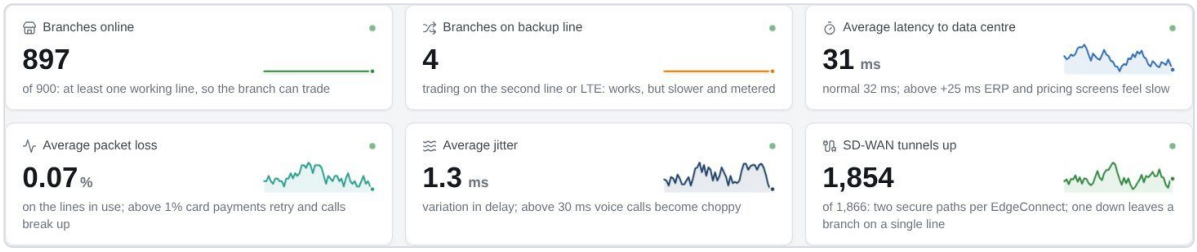



**Figure 34.** Live tiles

| Tile | What it is | Normal at SRS | What it means for the business |
|---|---|---|---|
| Branches online | Branches with at least one working WAN line | 897 to 900 of 900 | A branch online can take orders and card payments. A branch that drops off cannot trade until a line returns |

| Tile | What it is | Normal at SRS | What it means for the business |
|---|---|---|---|
| Branches on backup line | Branches whose main line is down, or whose traffic was moved off it, running on their second line (broadband or LTE/5G) | Under 15 | Trading continues; LTE is slower and metered, and there is no spare line if it fails too |
| Average latency to data centre | Round trip from each branch to the McKinney data centre on the line in use, averaged. The subtitle shows the normal value | About 32 ms: 7 to 15 ms in Texas, 35 to 50 ms on the coasts, about 90 ms in Hawaii | Above normal plus 25 ms, ERP, pricing and order entry feel slow |
| Average packet loss | Data lost on the lines in use, averaged | Under 0.2% | Above 1% card payments retry and calls break up; above 2% ERP sessions drop |
| Average jitter | Variation in delay on the lines in use, averaged | 1 to 3 ms | Above 30 ms voice calls become choppy and payment terminals time out more |
| SD-WAN tunnels up | Encrypted paths from each EdgeConnect to the data centre, one per line, out of 1,866 | 1,866 minus the lines that are down | Each tunnel down leaves a branch on a single line |

Each tile has a sparkline of its recent history; the green dot shows it is live. Click a tile for the detail behind it: **Branches online** opens the branches online, **Branches on backup line** the branches on their backup line, **Average latency** the branches slowest to the data centre first, **Average packet loss** and **Average jitter** the lines down or degraded now, and **SD-WAN tunnels up** the lines down (each down line takes its tunnel with it). The numbers follow the header speed control (1×, 5× or 10×).

## Area timeline

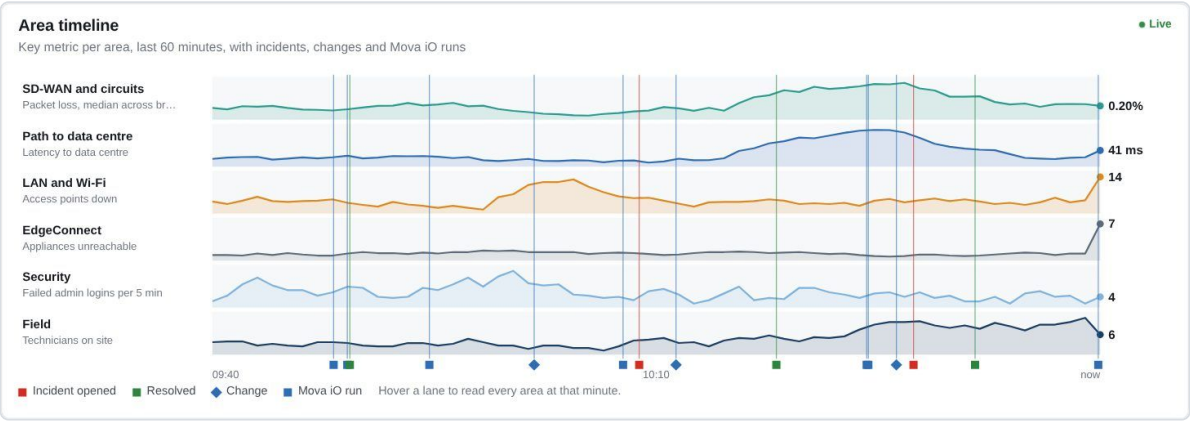



**Figure 35.** Area timeline

**What it shows.** One lane per area for the last 60 minutes, each with its key metric and the value now at the right.

| Lane | Metric |
|---|---|
| SD-WAN and circuits | Packet loss, median across branches (%) |
| Path to data centre | Latency to the data centre (ms) |
| LAN and Wi-Fi | Access points down |

| Lane | Metric |
|---|---|
| EdgeConnect | Appliances unreachable |
| Security | Failed admin logins per 5 minutes |
| Field | Technicians on site |

Markers along the bottom: red squares for incidents opened, green for resolved, blue diamonds for changes and dark blue for Mova iO runs, each with a thin line up through the lanes. **How to read it.** Each lane is scaled to its own range, so read the shape, not the height between lanes. A bump in one lane followed by bumps in others shows cause and effect, for example a loss rise on SD-WAN followed by higher latency on the path to the data centre. Hover anywhere to read every lane at that minute.

## Live event stream

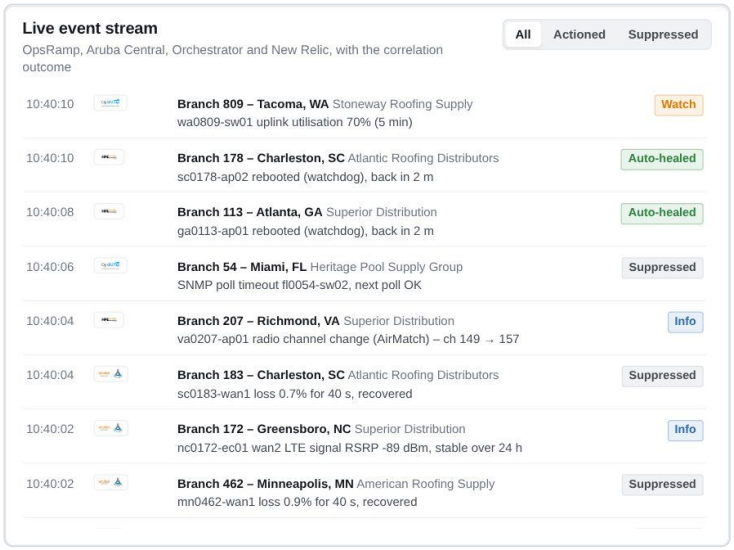



**Figure 36.** Live event stream

The same stream as on the Overview (chapter 5): every event with its source, branch, message and outcome badge, filtered by **All**, **Actioned** or **Suppressed**.

## Mova iO agents and Needs attention

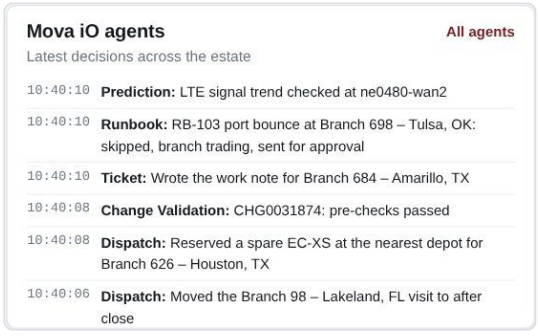



**Figure 37.** Latest Mova iO decisions

The last eight decisions by any agent, with time and agent name; **All agents** opens the Mova iO Agents screen.

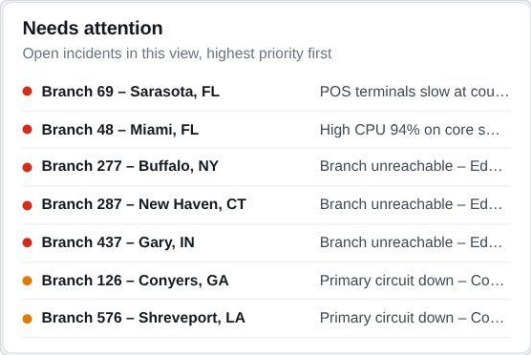



**Figure 38.** Needs attention

Open incidents in view, highest priority first: a red dot for P1 and P2, amber for P3 and P4, then the branch and title. Click to open the incident.

## Alerts by area and hour

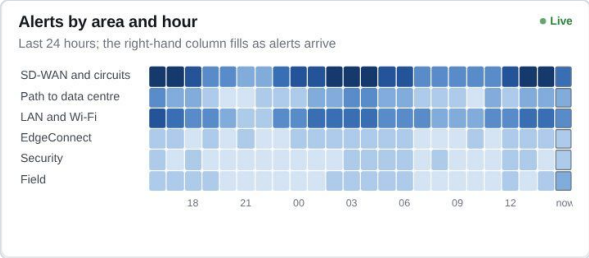



**Figure 39.** Alerts by area and hour

A heat map of alerts by area (rows) and hour for the last 24 hours (columns). Darker blue means more alerts; the outlined right-hand column is the current hour and fills as alerts arrive. Read it for patterns: SD-WAN alerts usually peak in the evening, when carriers do maintenance and branches are closed.

## Alerts per minute – live

The same severity bars as on the Overview: Critical, Major and Minor after correlation, last 30 minutes.

## Tools

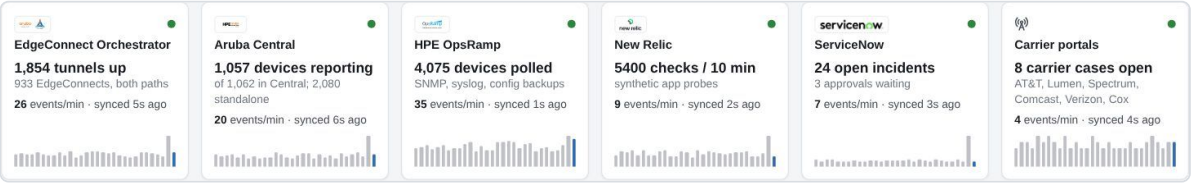



**Figure 40.** Tool status strip

One tile per source system: EdgeConnect Orchestrator, Aruba Central, HPE OpsRamp, New Relic, ServiceNow and the carrier portals.

| Parameter | Meaning |
|---|---|
| Headline | What the tool holds now: tunnels up, devices reporting, devices polled, synthetic checks every 10 minutes, open incidents and approvals, open carrier cases |

| Parameter | Meaning |
|---|---|
| Events/ min | Events arriving from that tool this minute |
| Synced | Seconds since the last sync |
| Bars | Events per minute for the last half hour; the blue bar is now |

A tool whose bars drop to nothing, or whose sync time grows, is a monitoring gap, not a quiet network.

## Branch heartbeat

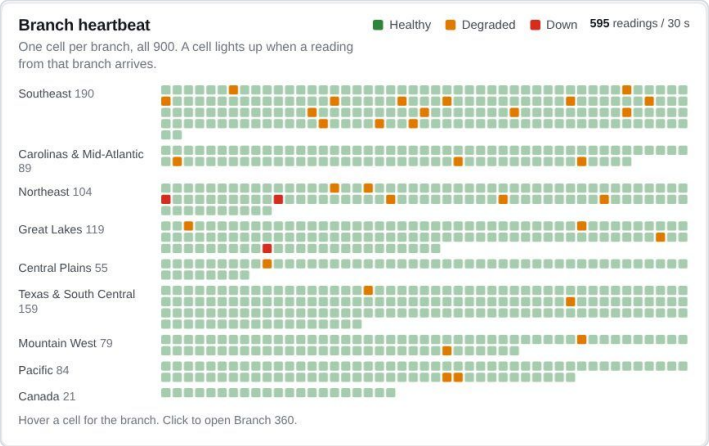



**Figure 41.** Branch heartbeat

**What it shows.** One small square per branch, all 900, grouped by region. The colour is health; a square lights up with an outline when a reading from that branch arrives. The counter shows readings per 30 seconds.

**How to read it.** A healthy estate twinkles evenly. A square that stays dim for a long time is a branch that has stopped reporting, which matters even if it is still green. Hover for the branch; click for Branch 360.

## Branch health score distribution




**Figure 42.** Health score distribution

A histogram of the branch health score, from 30 to 100, for every branch in view, with the average at top right. Bars from 80 up are green-blue (good), 60 to 79 amber (warning) and below 60 red (critical). The dashed line marks "good from 80". Height uses a square-root scale so the few critical branches stay visible. The score combines path health, device health, incidents and recent events; Branch 360 shows how one branch's score is built.

### Top branches

The eight worst branches by WAN use, latency, loss or EdgeConnect CPU, as on the Overview.

# 7. Correlation

Correlation shows how tens of thousands of events become a handful of incidents: one ServiceNow incident per branch per event. Every screen uses the same pipeline:

1. **Events** arrive from Orchestrator, Aruba Central, OpsRamp and New Relic: traps, syslog lines and threshold crossings, about 50 a minute across the estate.
2. **Noise removal** drops duplicates, flaps that recover within the hold-down and information-only messages, usually about 70% of events.
3. What is left are **alerts**. Most clear on their own (**auto-cleared**), warnings go to the morning **digest**, and the rest are **correlated** into incidents.
4. **Incidents** go to ServiceNow, one per branch per event, each with all its alerts as evidence.

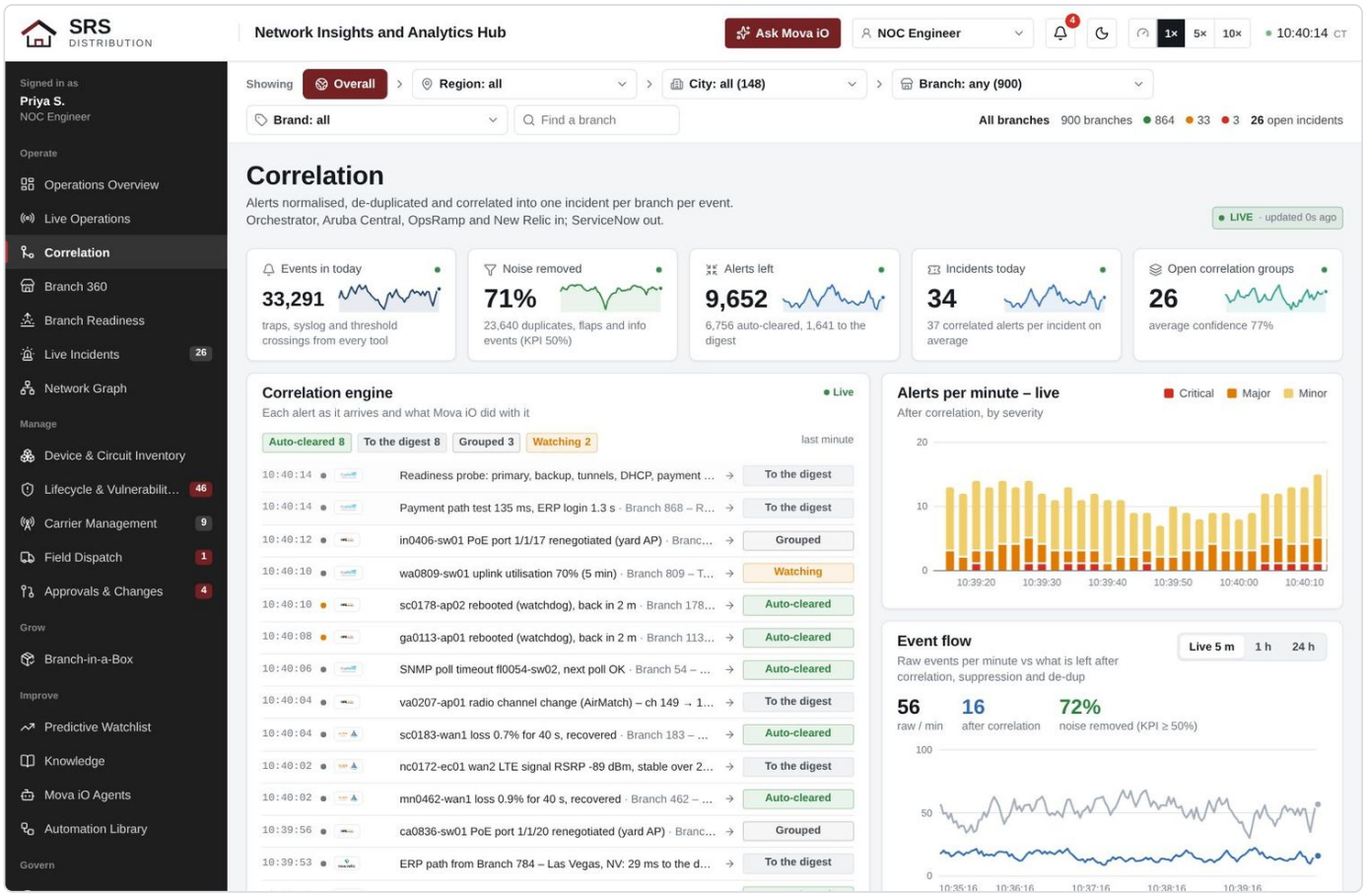



**Figure 43.** Correlation

## Live tiles




**Figure 44.** Correlation tiles

| Tile | Meaning |
|---|---|
| Events in today | All events received since midnight from every tool |
| Noise removed | Share of today's events removed before they became alerts, with the count (KPI at least 50%) |
| Alerts left | Alerts after noise removal, with how many cleared on their own and how many went to the digest |
| Incidents today | Incidents created today, and the average number of correlated alerts in each |
| Open correlation groups | Open incidents, each a group of correlated alerts, with their average confidence |

The same numbers appear on the Overview (**Noise removed today**) and in the funnel below, so the screens always agree. **Incidents today** opens the incidents opened since midnight Central Time, and **Open correlation groups** opens the open incidents, each with its root cause.

## Correlation engine

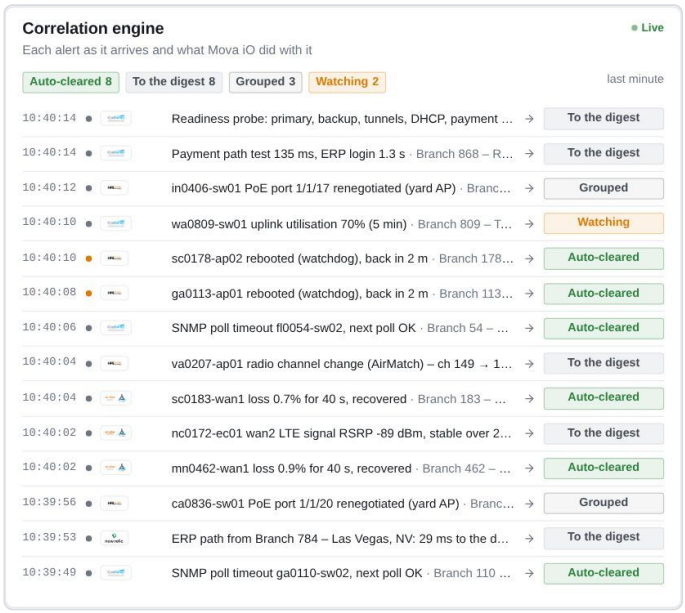



**Figure 45.** Correlation engine

**What it shows.** Each alert as it arrives (time, severity dot, source tool, message and branch) and, after the arrow, what Mova iO did with it. The counters above count the last minute.

| Outcome | Meaning |
|---|---|
| Auto-cleared | A transient: checked by ping and SNMP and cleared after the hold-down (for example a 40-second flap) |
| To the digest | An informational warning, bundled into the daily digest instead of a ticket |
| Grouped (into nnnn) | Linked to an open incident as one more piece of its evidence |
| Watching | Held for a re-check before deciding |

Click a row to open the incident it was grouped into, or the branch.

## How alerts were grouped

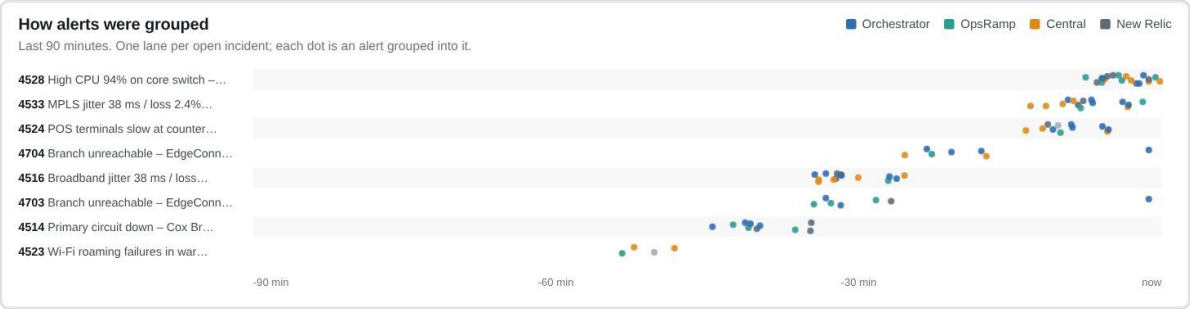



**Figure 46.** How alerts were grouped

One lane per open incident for the last 90 minutes, labelled with the last four digits of the incident and its title. Each dot is an alert grouped into it, coloured by source: Orchestrator, OpsRamp, Central or New Relic. A lane with many colours is an incident confirmed by several tools, which is why its confidence is high. A lane with one dot is a single-signal incident; check it is not a noisy device. Click a lane to open the incident.

## From events to incidents




**Figure 47.** From events to incidents

A funnel of today's counts since 00:00 CT:

| Stage | Meaning |
|---|---|
| Events received | Every trap, syslog line and threshold crossing from the four tools |
| Alerts | Left after noise removal |
| Auto-cleared | Alerts that recovered by themselves: checked by ping and SNMP after the hold-down |
| Rolled to digest | Warnings bundled into the morning digest instead of tickets |
| Correlated | Alerts grouped by topology, time, change and carrier into incidents |
| Incidents | ServiceNow incidents, one per branch per event |

Bar length uses a square-root scale so the small stages stay visible; read the numbers. The two ratios to watch are events to alerts (noise removed) and correlated alerts per incident.

## Alert flow, today

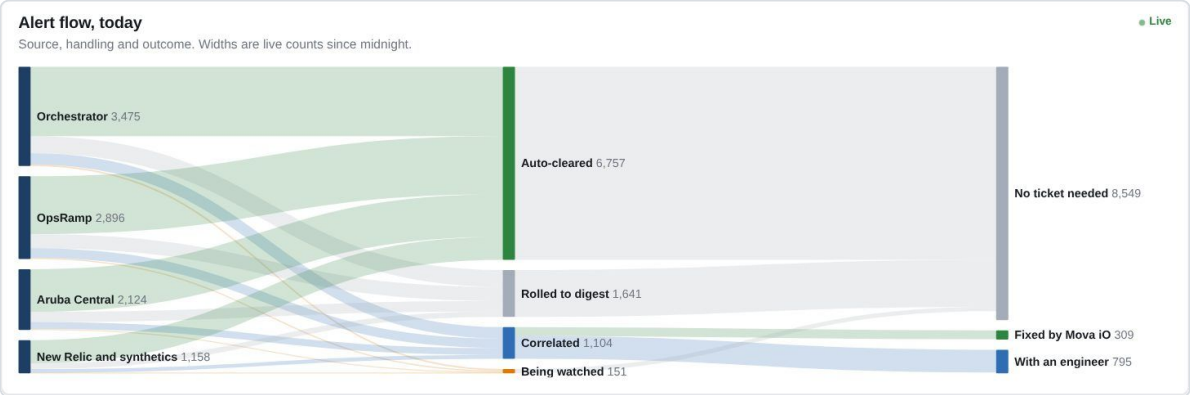



**Figure 48.** Alert flow

The same day as a flow from left to right. Left: source tools with their alert counts. Middle: how alerts were handled (auto-cleared, rolled to digest, correlated, being watched). Right: the outcome (no ticket needed, fixed by Mova iO, with an engineer). Band widths are live counts since midnight. Use it to explain in one picture why engineers see only a small share of what the tools raise.

## Alerts by area and hour

The 24-hour heat map from Live Operations (chapter 6).

## Area-to-area correlations

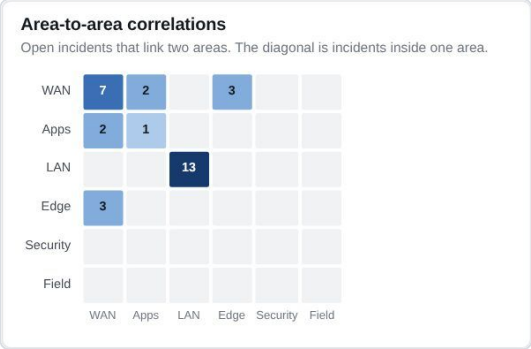



**Figure 49.** Area-to-area correlations

A grid of open incidents by the area where they started (rows) and the area they affect (columns): WAN, Apps (the path to payments and ERP), LAN, Edge (EdgeConnect), Security and Field. The diagonal counts incidents that stay inside one area; cells off the diagonal count incidents that link two areas, such as a WAN problem slowing the POS path. Darker cells hold more incidents.

## Correlation methods

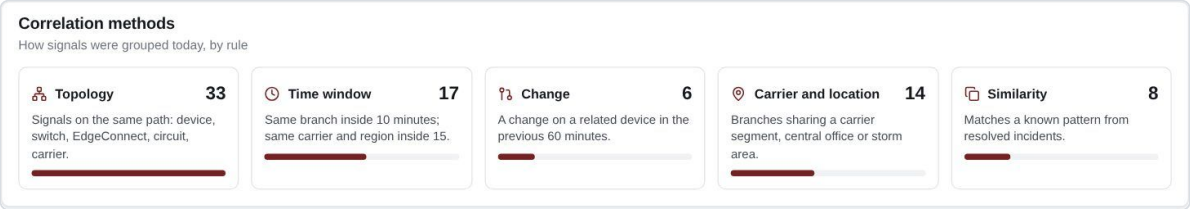



**Figure 50.** Correlation methods

How today's signals were grouped, by rule, with a count and a bar for each:

| Method | Rule |
|---|---|
| Topology | Signals on the same path: device, switch, EdgeConnect, circuit, carrier |
| Time window | Same branch inside 10 minutes; same carrier and region inside 15 |
| Change | A change on a related device in the previous 60 minutes |
| Carrier and location | Branches sharing a carrier segment, central office or storm area |
| Similarity | Matches a known pattern from resolved incidents |

Topology does most of the work. A rising share for "Change" means changes are causing incidents and the change process needs attention.

# 8. Branch 360

Branch 360 is everything about one branch on one screen: its live metrics, health, risk, who is working on it, its circuits, applications, devices, vulnerabilities and history. Open it by clicking any branch anywhere in the console, by picking a branch in the filter bar, or from **Branch 360** at the end of a demo.

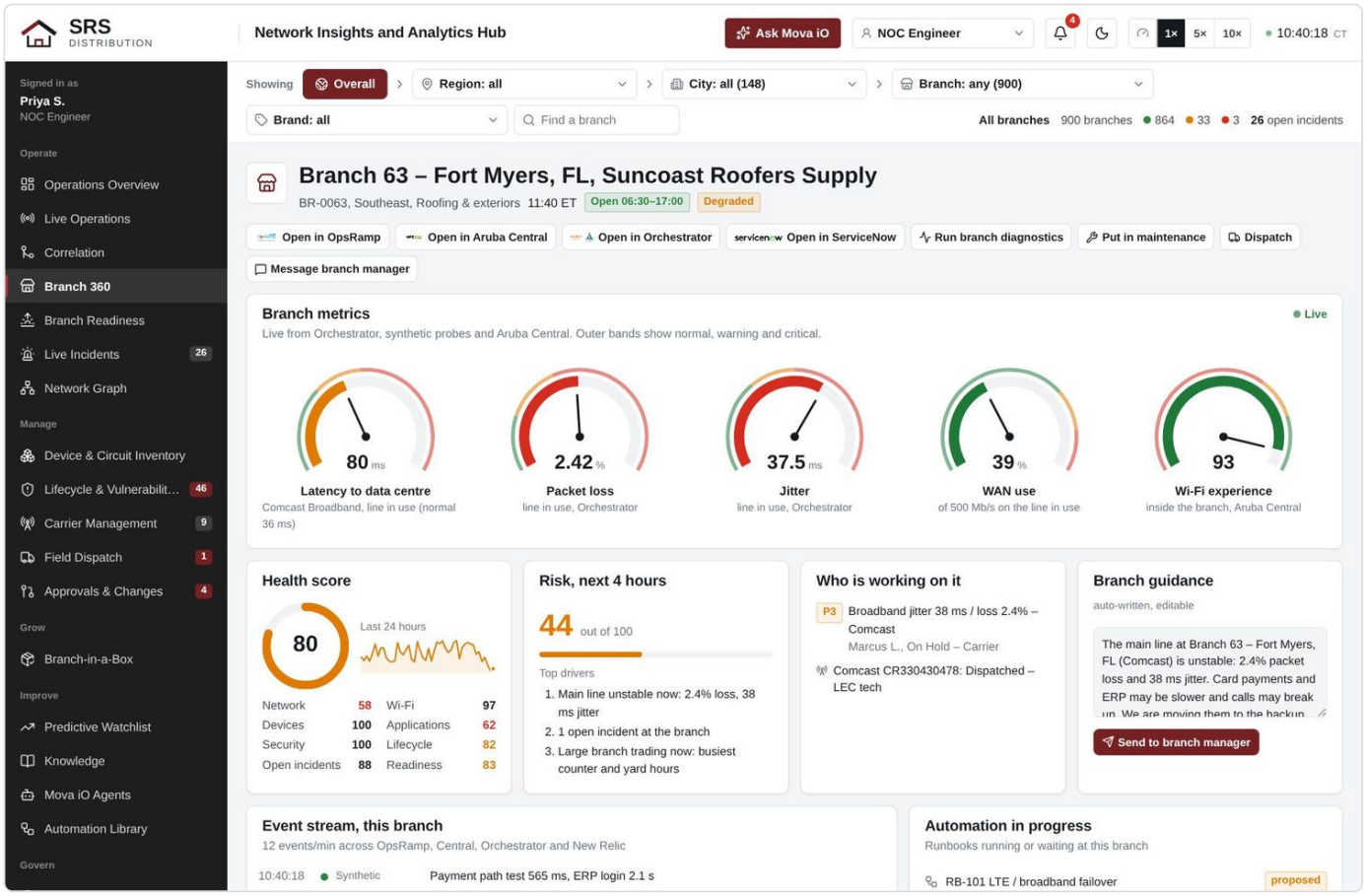



**Figure 51.** Branch 360

## Header and actions

The header gives the branch name and brand, branch ID, city, state, region and line of business, the local time, the opening hours (green when open now) and badges such as **Degraded** or **On backup path**.

| Button | What it does |
|---|---|
| Open in OpsRamp, Aruba Central, Orchestrator, ServiceNow | Opens the same branch in that tool |
| Run branch diagnostics | Runs the read-only checks: reachability, both lines, tunnels, payment and ERP probes, switch and AP health |
| Put in maintenance | Holds alerts for planned work so they do not page anyone; recorded in the audit log |
| Dispatch | Creates a field work order for the branch |
| Message branch manager | Sends the guidance text below to the branch manager |

## Branch metrics




**Figure 52.** Branch metrics

Five live dials. The outer band of each dial shows the normal (green), warning (amber) and critical (red) ranges; the needle and inner arc show the value now. The first four describe the WAN line the branch is using at this moment; the fifth describes Wi-Fi inside the building.

| Dial | What it is | Source | Warning | Critical | What the business feels |
|---|---|---|---|---|---|
| Latency to data centre (ms) | Round trip from this branch to the McKinney data centre on the line in use. The label shows the line's normal value, which depends on distance and line type | Orchestrator | normal plus 25 ms | normal plus 60 ms | ERP, pricing and order screens slow down |
| Packet loss (%) | Data lost on the line in use | Orchestrator | above 1% | above 2% | Card payments retry, calls break up, ERP sessions can drop |
| Jitter (ms) | How much the delay varies on the line in use | Orchestrator | above 20 ms | above 30 ms | Choppy calls; payment terminals time out more often |
| WAN use (%) | Share of the line's bandwidth in use (for example of 500 Mb/s) | Orchestrator | above 70% | above 85% | A full line queues traffic, which raises latency and loss for everyone |
| Wi-Fi experience (score out of 100) | Aruba Central's score for the Wi-Fi inside the branch: signal, connection success and roaming for every client | Aruba Central | below 80 | below 65 | Handhelds and laptops in the office, counter and warehouse struggle |

**Why the readings move together.** When a line is congested or faulty, latency, jitter and loss rise at the same time; the dials are drawn from that one line, so they always tell the same story. Wi-Fi is measured inside the building and does not depend on the WAN, so a branch can show a poor WAN line and good Wi-Fi at the same time. That is normal: staff can connect, but anything that crosses the WAN, such as ERP and payments, is affected.

**Normal values.** A branch in Texas normally sees 7 to 15 ms to the data centre, one on the east or west coast 35 to 50 ms, Hawaii about 90 ms; an LTE/5G backup adds 25 to 40 ms. Jitter is normally 0.5 to 3 ms on wired lines and 5 to 12 ms on LTE; loss is normally under 0.2%.

## Health score, risk, owner and guidance




**Figure 53.** Health score

**Health score** is a number from 0 to 100 with its last 24 hours, built from eight parts: Network, Wi-Fi, Devices, Applications, Security, Lifecycle, Open incidents and Readiness. The ring is green from 85, amber from 70 and red below 70; each part is scored the same way, with amber and red parts highlighted. The lowest parts explain a low score; for example Network drops when the main line is unstable, Applications when card payments, ERP or calls are affected, Security when a device has an unpatched critical vulnerability, Lifecycle when devices are past support.




**Figure 54.** Risk, next 4 hours

**Risk, next 4 hours** is the chance of a business-impacting problem soon, out of 100, with the top drivers listed in order, for example the main line unstable now, trading on the backup line with no spare, a backup line not passing traffic, a repeat-offender circuit, open incidents, unpatched critical vulnerabilities, a regional storm, or a large branch at its busiest time. Below 25 is low (green), 25 to 49 medium (amber), 50 and above high (red).

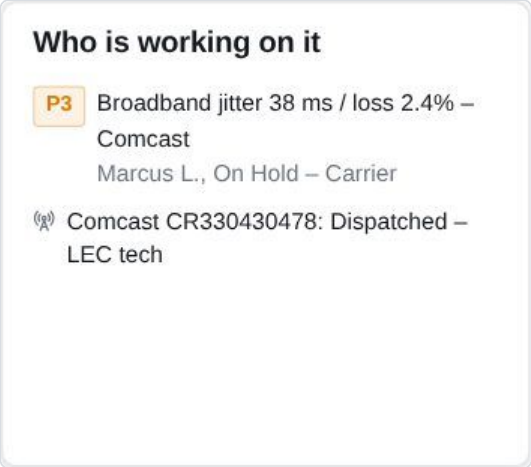



**Figure 55.** Who is working on it

**Who is working on it** lists each open incident with its priority, title, assignee and state, and any carrier case with its number and status. Nothing listed means nothing is open.

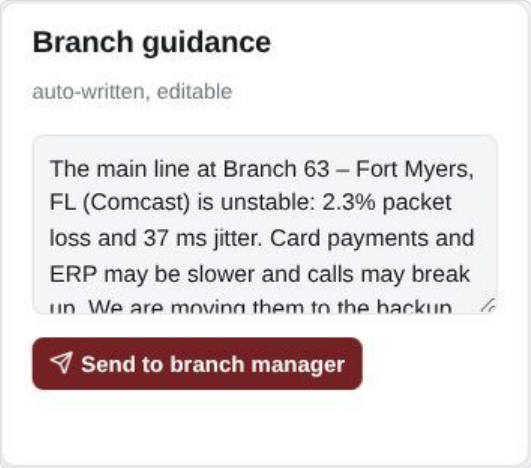



**Figure 56.** Branch guidance

**Branch guidance** is a plain-language note for the branch manager, written by Mova iO from the branch's state and matching the dials: down on both lines, running on its backup line, the main line unstable (with the loss and jitter, what staff may notice and that traffic is being moved), or the incidents being worked. It is editable; **Send to branch manager** sends it and records it.

## Event stream and automation




**Figure 57.** Event stream for this branch

Every reading and event from this branch as it arrives, with time, source and message, and the events per minute in the subtitle. Use it to watch a fix land in real time.

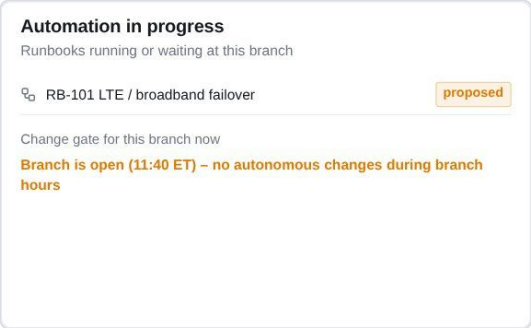



**Figure 58.** Automation in progress

Runbooks running, proposed or waiting at this branch, with their state, and the **change gate for this branch now**: whether autonomous changes are allowed at this moment, for example "Branch is open – no autonomous changes during branch hours".

## Connectivity, applications and network today

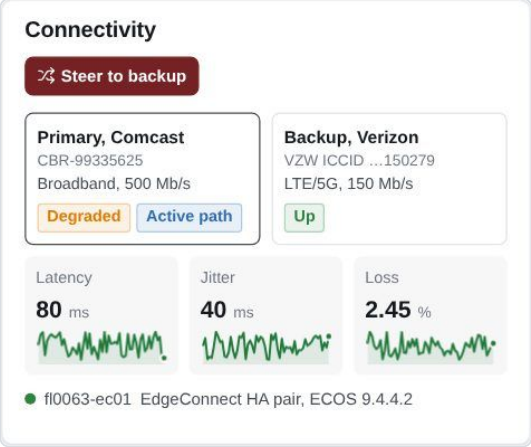



**Figure 59.** Connectivity

**Connectivity** shows both circuits side by side: primary and backup, carrier, circuit ID, type and bandwidth, and status (Up, Degraded or Down). The active path has a dark border, with its live latency, jitter and loss. Below are the EdgeConnect name, whether it is a single unit or an HA pair, and its ECOS release. **Steer to backup** moves card payments, ERP and voice to the backup line (an SD-WAN override in Orchestrator); at an open branch it follows the gate rules.

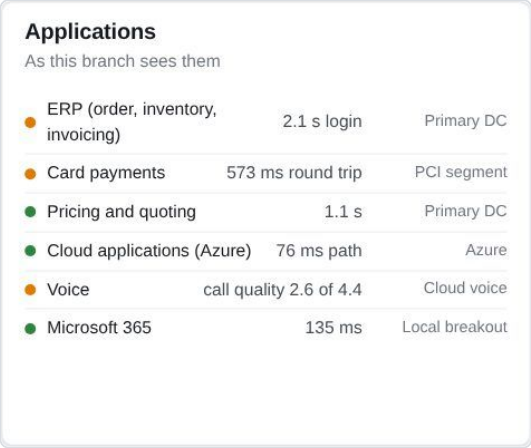



**Figure 60.** Applications

**Applications** shows what the business feels, worked out from the line in use: ERP login time (normally about 1 s), card payment round trip from the counter to the payment processor (normally 100 to 200 ms; slow above 400 ms), pricing and quoting, the path to Azure, voice call quality on the 1 to 4.4 scale (good above 4.0, poor below 3.6) and Microsoft 365, each with where it is hosted. An amber dot means that service is slower than it should be at this branch.

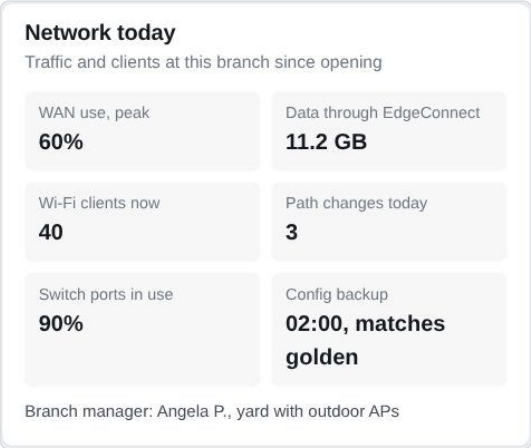



**Figure 61.** Network today

**Network today** gives the day since opening: peak WAN use, data through the EdgeConnect, Wi-Fi clients now, path changes, switch ports in use, the last config backup and whether it matches the golden config, and the branch manager's name.

## LAN and Wi-Fi

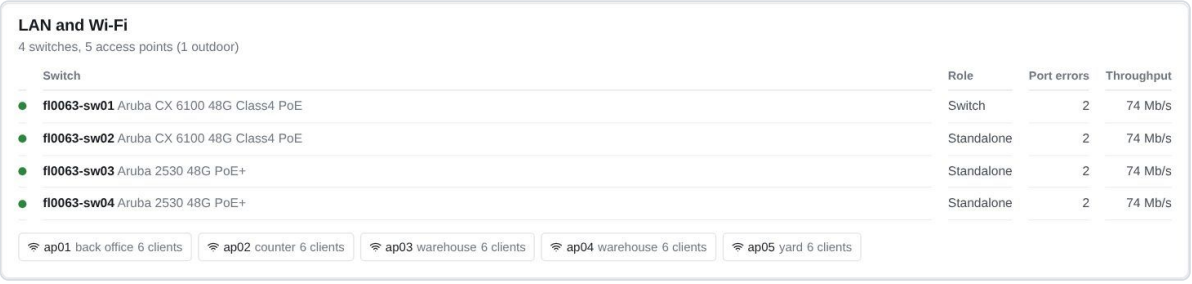



**Figure 62.** LAN and Wi-Fi

Every switch with model, role (Switch or Standalone), port errors and throughput, then every access point with its zone (back office, counter, warehouse, yard) and clients. Port errors above zero, or an AP with no clients during trading hours, are worth a look.

## Dependencies

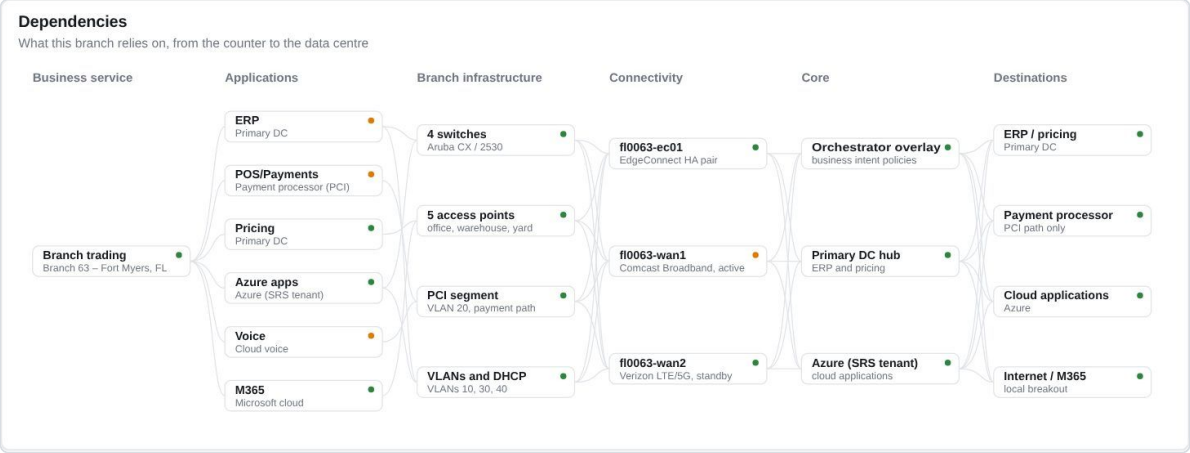
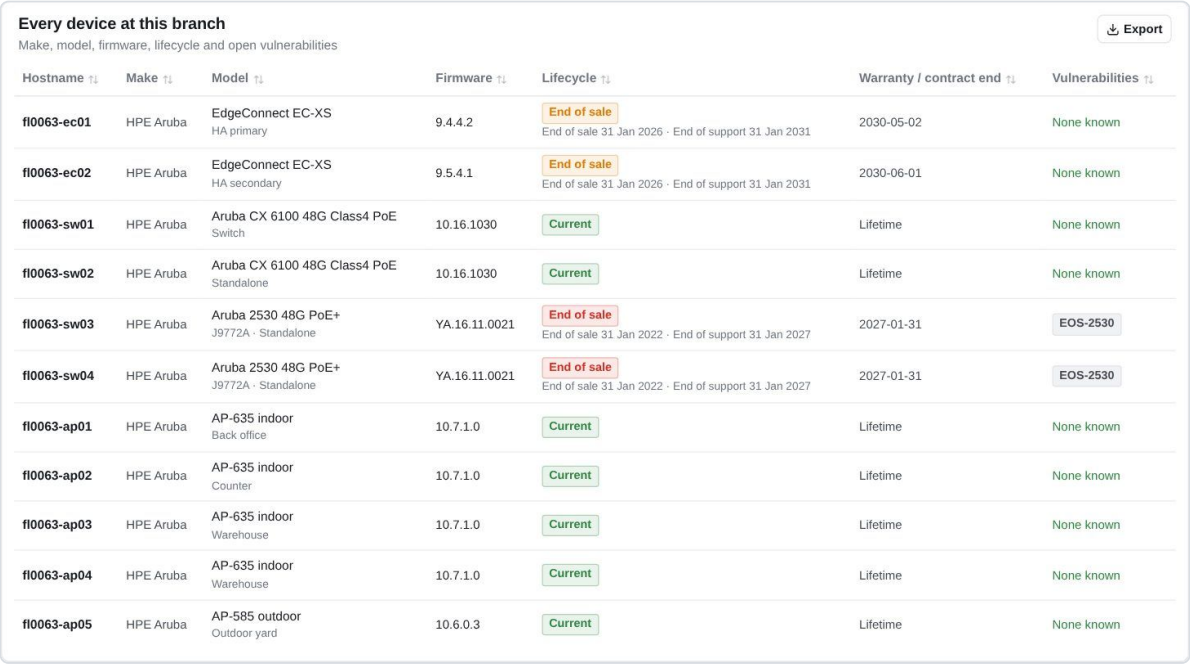



**Figure 63.** Dependencies

**What it shows.** Everything the branch relies on, from left to right: the business service (branch trading), the applications, the branch infrastructure (switches, APs, PCI segment, client VLANs and DHCP), connectivity (EdgeConnect, both circuits, the Orchestrator overlay), the core and the destinations (Primary DC, payment processor, Azure, cloud voice, Microsoft cloud). Each box has a status dot.

**How to read it.** Follow a red or amber dot to the right to see what it affects, or to the left to see what it depends on. A degraded circuit with healthy applications to its right means the backup path is doing its job.

## Tab: Devices and lifecycle




| Hostname | Make | Model | Firmware | Lifecycle | Warranty / contract end | Vulnerabilities |
|---|---|---|---|---|---|---|
| fl0063-ec01 | HPE Aruba | EdgeConnect EC-XS<br>HA primary | 9.4.4.2 | End of sale<br>End of sale 31 Jan 2026 · End of support 31 Jan 2031 | 2030-05-02 | None known |
| fl0063-ec02 | HPE Aruba | EdgeConnect EC-XS<br>HA secondary | 9.5.4.1 | End of sale<br>End of sale 31 Jan 2026 · End of support 31 Jan 2031 | 2030-06-01 | None known |
| fl0063-sw01 | HPE Aruba | Aruba CX 6100 48G Class4 PoE<br>Switch | 10.16.1030 | Current | Lifetime | None known |
| fl0063-sw02 | HPE Aruba | Aruba CX 6100 48G Class4 PoE<br>Standalone | 10.16.1030 | Current | Lifetime | None known |
| fl0063-sw03 | HPE Aruba | Aruba 2530 48G PoE+<br>J9772A · Standalone | YA.16.11.0021 | End of sale<br>End of sale 31 Jan 2022 · End of support 31 Jan 2027 | 2027-01-31 | EOS-2530 |
| fl0063-sw04 | HPE Aruba | Aruba 2530 48G PoE+<br>J9772A · Standalone | YA.16.11.0021 | End of sale<br>End of sale 31 Jan 2022 · End of support 31 Jan 2027 | 2027-01-31 | EOS-2530 |
| fl0063-ap01 | HPE Aruba | AP-635 indoor<br>Back office | 10.7.1.0 | Current | Lifetime | None known |
| fl0063-ap02 | HPE Aruba | AP-635 indoor<br>Counter | 10.7.1.0 | Current | Lifetime | None known |
| fl0063-ap03 | HPE Aruba | AP-635 indoor<br>Warehouse | 10.7.1.0 | Current | Lifetime | None known |
| fl0063-ap04 | HPE Aruba | AP-635 indoor<br>Warehouse | 10.7.1.0 | Current | Lifetime | None known |
| fl0063-ap05 | HPE Aruba | AP-585 outdoor<br>Outdoor yard | 10.6.0.3 | Current | Lifetime | None known |

**Figure 64.** Every device at this branch

**Every device at this branch** lists each device: hostname, make, model and role, firmware, lifecycle (Current, End of sale or End of support, with both dates under the badge), warranty or support contract end (Lifetime for switches and APs) and open vulnerabilities with their CVSS score. **Export** downloads the list.

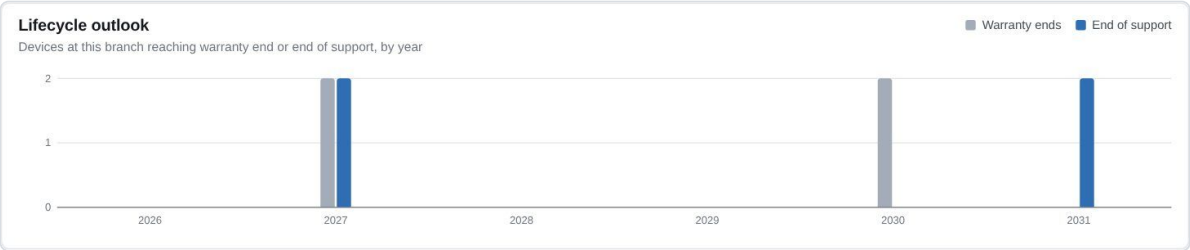



**Figure 65.** Lifecycle outlook

**Lifecycle outlook** counts the branch's devices whose warranty or support contract ends, or whose end of support falls, in each year, to plan a refresh.

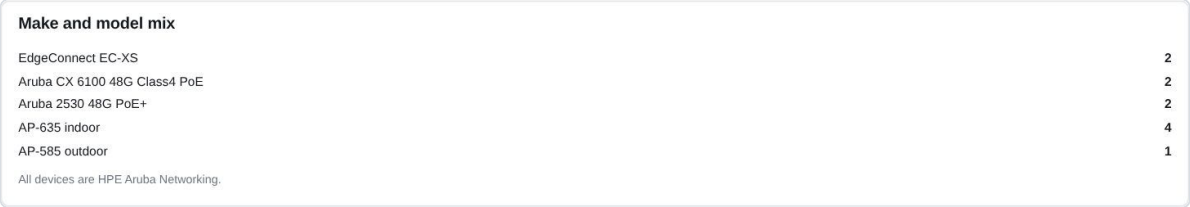



**Figure 66.** Make and model mix

**Make and model mix** counts devices by model.

## Tab: Vulnerabilities




**Figure 67.** Vulnerabilities at this branch

Each finding matched from HPE Aruba Networking security bulletins against the running firmware of this branch's devices: severity and CVSS score, title, CVE and bulletin number with date, attack vector, the affected devices here, the fix and the mitigation available now. **Schedule fix after close** creates the change for after trading hours; **Apply mitigation** applies the interim protection (for example restricting management access) and follows the gate rules. Lifecycle risks, such as switches past end of sale, are listed the same way.

## Tab: Open work and history

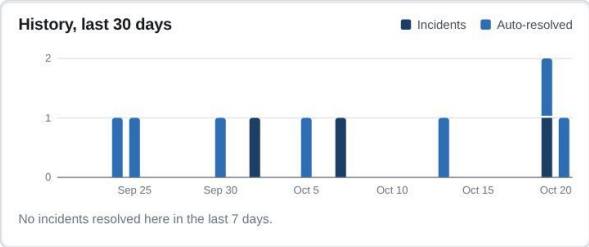



**Figure 68.** History, last 30 days

Incidents per day for 30 days, with those that were auto-resolved.




**Figure 69.** Open work across every team

Everything open for this branch in one list: incidents with time left, carrier cases, work orders and changes.

# 9. Branch Readiness

Branch Readiness answers "will every branch open on time with a working network?". At 05:30 local time, Mova iO runs ten network checks at every branch. Problems are fixed overnight where the rules allow, or given an owner before the doors open. **Run check now** re-runs the checks for the branches in view; **Export** downloads the results.

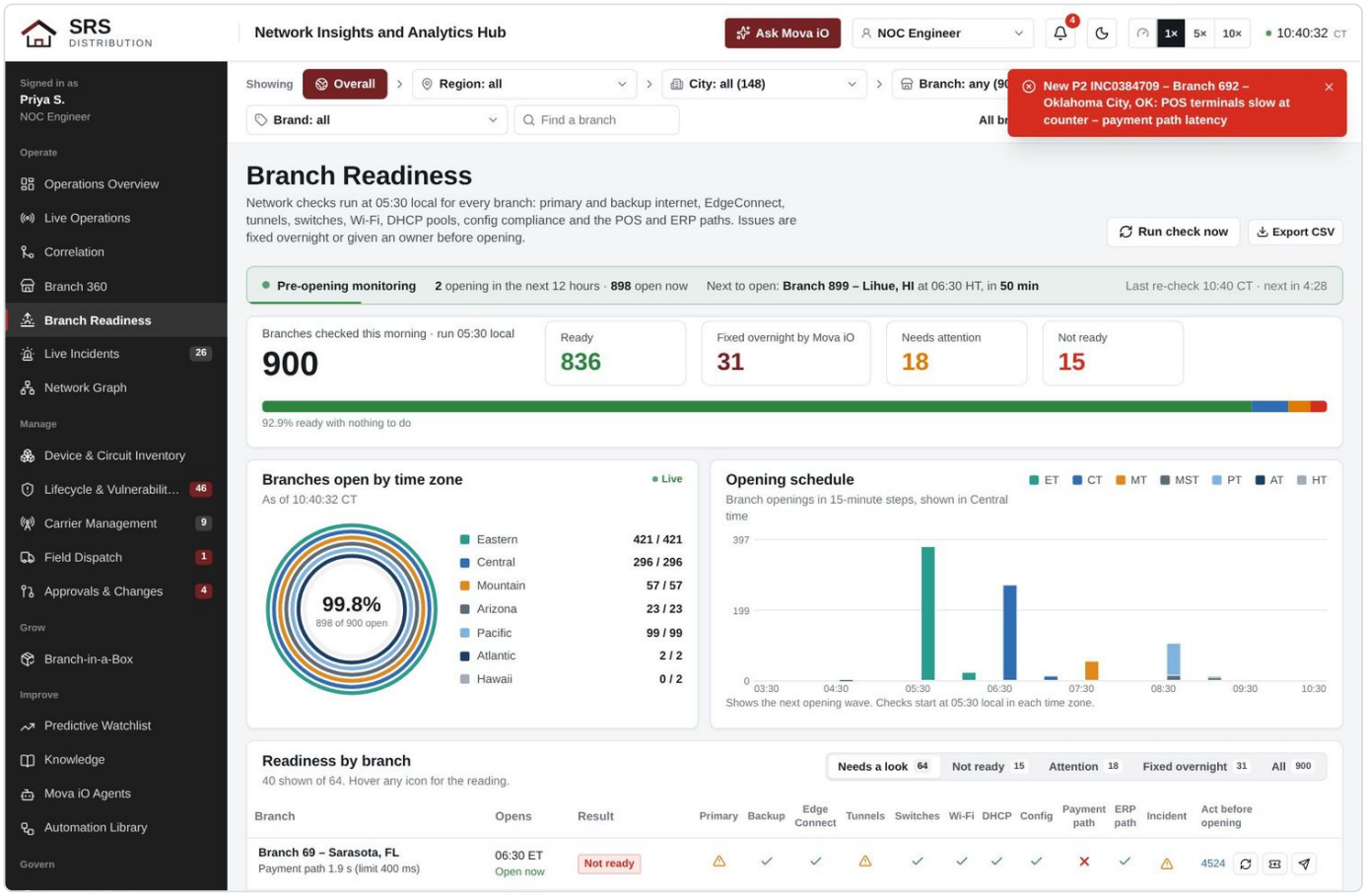



**Figure 70.** Branch Readiness

## This morning's result




**Figure 71.** Readiness summary

The bar splits every branch checked this morning into four results:

| Result | Meaning | Colour |
|---|---|---|
| Ready | All ten checks passed | Green |
| Fixed overnight by Mova iO | A check failed overnight and an approved runbook fixed it before 05:30, with no engineer woken | Blue |

| Result | Meaning | Colour |
|---|---|---|
| Needs attention | The branch can trade, but something is degraded (for example the backup line is down while the primary is clean) | Amber |
| Not ready | A check that stops normal trading failed (for example no working line, or the payment path over its limit) | Red |

The headline "ready with nothing to do" is Ready plus Fixed overnight as a share of all branches.

## The ten checks

| Check | Passes when | Source |
|---|---|---|
| Primary | The primary internet line is up and clean | Orchestrator |
| Backup | The backup line is up; LTE backups pass their weekly failover test | Orchestrator |
| EdgeConnect | The EdgeConnect is reachable, its HA state is healthy and its firmware is in policy | Orchestrator |
| Tunnels | Overlay tunnels to the data centre and Azure are up on the active path | Orchestrator |
| Switches | All switches are up, uplinks clean, no port errors | Aruba Central |
| Wi-Fi | All access points are up in office, warehouse and yard | Aruba Central |
| DHCP | Address pools are below 85% on every branch VLAN | OpsRamp |
| Config | Last night's config backup was taken and there is no drift from the golden config | OpsRamp |
| Payment path | A test payment from the counter VLAN (PCI segment) reaches the payment processor and back in under 400 ms | Synthetic probe |
| ERP path | The path to the data centre is within its latency and loss limits | Synthetic probe |

A failed check on the primary line, the EdgeConnect, the tunnels or the payment path makes the branch **Not ready** and raises a P2; other failures raise a P3. A backup line down while the primary is clean is a warning, not a failure.

## Branches open by time zone and the opening schedule




**Figure 72.** Branches open by time zone

How many branches are open now in each time zone (Eastern, Central, Mountain, Arizona, Pacific, Atlantic, Hawaii), with the total share open. Read it with the clock: at 06:00 CT the Eastern wave is open and the Pacific wave is still hours away.

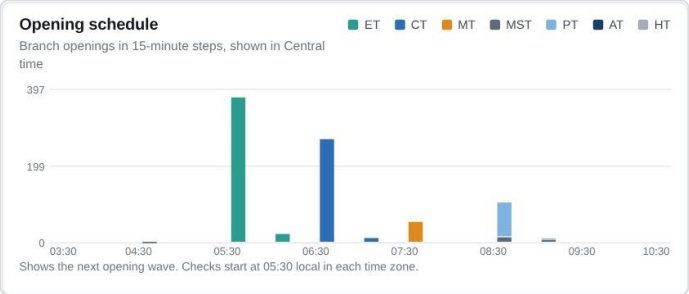



**Figure 73.** Opening schedule

Branch openings in 15-minute steps, shown in Central Time and coloured by time zone. Checks start at 05:30 local in each zone, so the NOC works the problems in waves: east first, west last.

## Readiness by branch

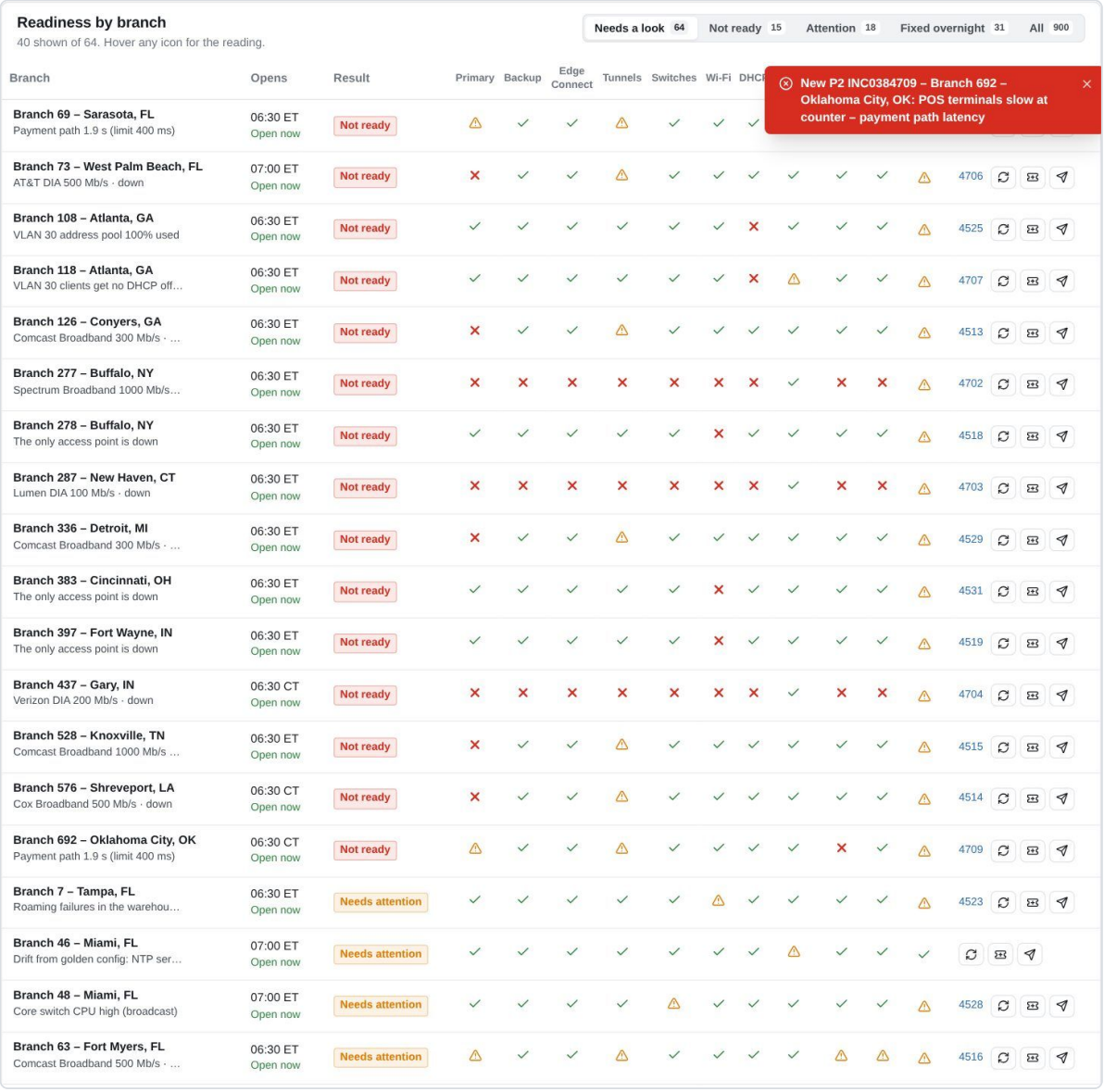



**Figure 74.** Readiness by branch

**What it shows.** One row per branch with a problem, worst first. The tabs choose the rows: **Needs a look** (everything not plainly ready), **Not ready**, **Attention**, **Fixed overnight** and **All**.

| Column | Meaning |
|---|---|
| Branch | Name, with the first failing reading under it |
| Opens | Local opening time and "Open now" once trading |
| Result | Ready, Fixed overnight, Needs attention or Not ready |
| Ten check columns | A green tick, amber warning, red cross or blue wrench (fixed overnight). Hover an icon for the reading |
| Incident | The incident raised for the failure, if any |
| Act before opening | **Re-run** the checks or **Raise** an incident; "No action" when ready |

**Show 40 more** loads further rows. Click a branch name for Branch 360.

## Pre-opening checks

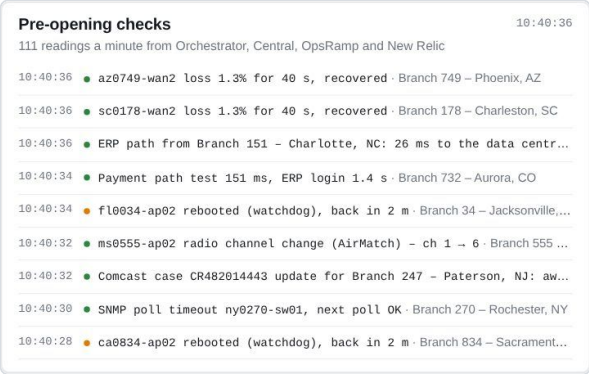



**Figure 75.** Pre-opening checks

The live stream of readiness readings: probes of the primary and backup lines, tunnels, DHCP and the payment and ERP paths, with the branch and the result. The subtitle shows readings per minute.

## Fixed overnight

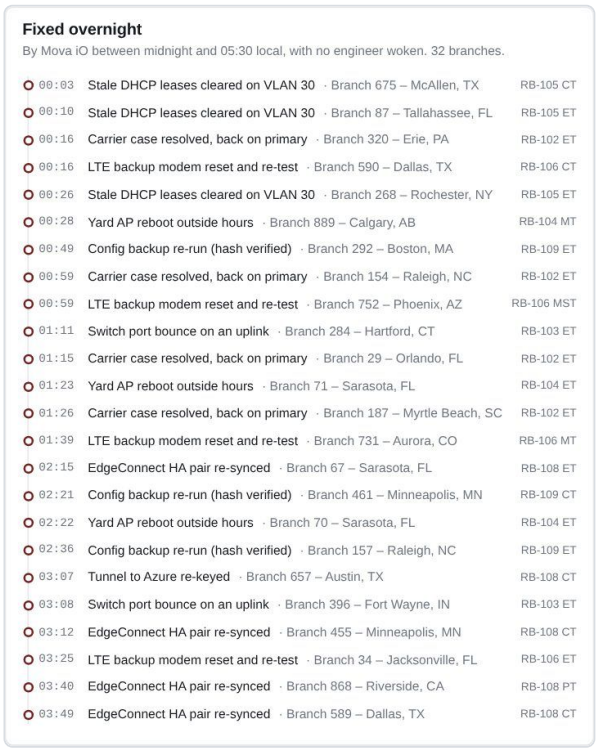



**Figure 76.** Fixed overnight

What Mova iO fixed between midnight and 05:30 local, with no engineer woken: time, what was fixed, branch, the runbook used and the time zone. Typical fixes are an LTE modem reset and re-test (RB-106), a yard AP reboot outside hours (RB-104), a config backup re-run (RB-109), stale DHCP leases cleared (RB-105), a tunnel to Azure re-keyed (RB-108) and a return to the primary after a carrier fix (RB-102). These count as Ready at opening.

## Ready at opening, 14 days




**Figure 77.** Ready at opening, 14 days

The share of branches that passed every check at opening, each day for 14 days, against the 96% target line. The note explains any dip, for example a carrier outage that sent a group of branches to LTE.

## Tomorrow's opening risk

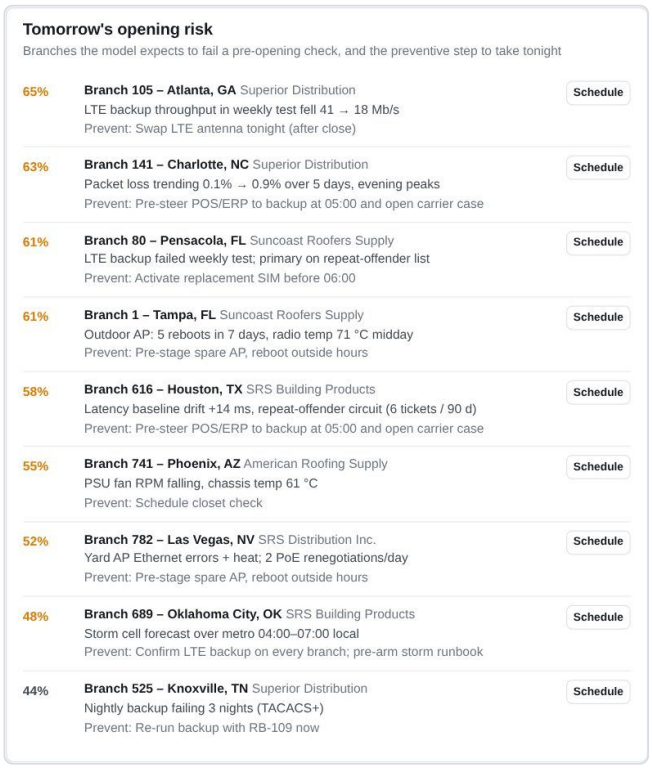
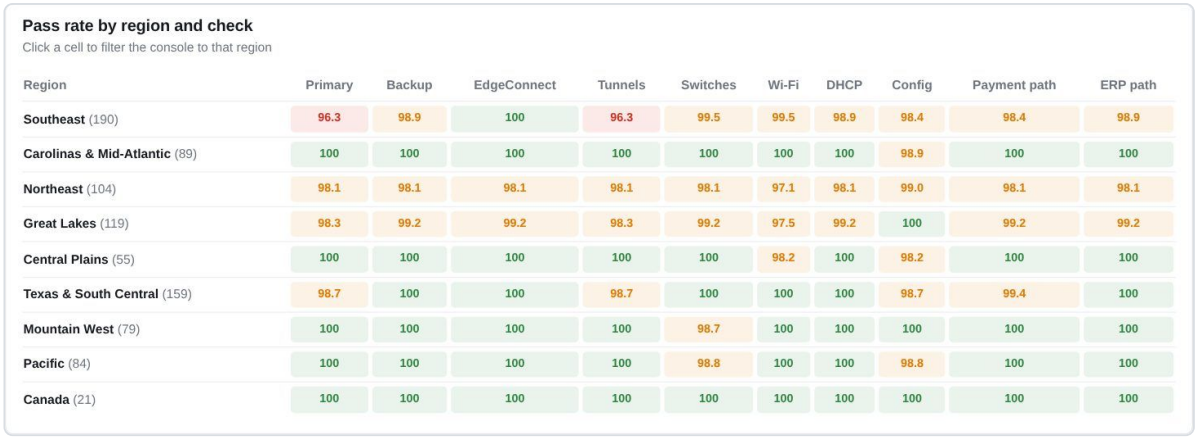



**Figure 78.** Tomorrow's opening risk

Branches the model expects to fail a check tomorrow morning, with the probability, the reason and the preventive step to take tonight. **Schedule** books that step. This is where tomorrow's "Not ready" branches are prevented.

## Pass rate by region and check




| Region | Primary | Backup | EdgeConnect | Tunnels | Switches | Wi-Fi | DHCP | Config | Payment path | ERP path |
|---|---|---|---|---|---|---|---|---|---|---|
| **Southeast** (190) | 96.3 | 98.9 | 100 | 96.3 | 99.5 | 99.5 | 98.9 | 98.4 | 98.4 | 98.9 |
| **Carolinas & Mid-Atlantic** (89) | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 98.9 | 100 | 100 |
| **Northeast** (104) | 98.1 | 98.1 | 98.1 | 98.1 | 98.1 | 97.1 | 98.1 | 99.0 | 98.1 | 98.1 |
| **Great Lakes** (119) | 98.3 | 99.2 | 99.2 | 98.3 | 99.2 | 97.5 | 99.2 | 100 | 99.2 | 99.2 |
| **Central Plains** (55) | 100 | 100 | 100 | 100 | 100 | 98.2 | 100 | 98.2 | 100 | 100 |
| **Texas & South Central** (159) | 98.7 | 100 | 100 | 98.7 | 100 | 100 | 100 | 98.7 | 99.4 | 100 |
| **Mountain West** (79) | 100 | 100 | 100 | 100 | 98.7 | 100 | 100 | 100 | 100 | 100 |
| **Pacific** (84) | 100 | 100 | 100 | 100 | 98.8 | 100 | 100 | 98.8 | 100 | 100 |
| **Canada** (21) | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 | 100 |

**Figure 79.** Pass rate by region and check

The share of branches passing each check, by region. Green cells are 100%, amber 97% or more, red below 97%. A red column across many regions points at a common cause, such as a carrier or a template; a red row points at one region. Click a cell to filter the console to that region.

# 10. Live Incidents

Live Incidents is the NOC queue: every open incident with its SLA clocks, and the full record of each one. It is the NOC Engineer's landing page.

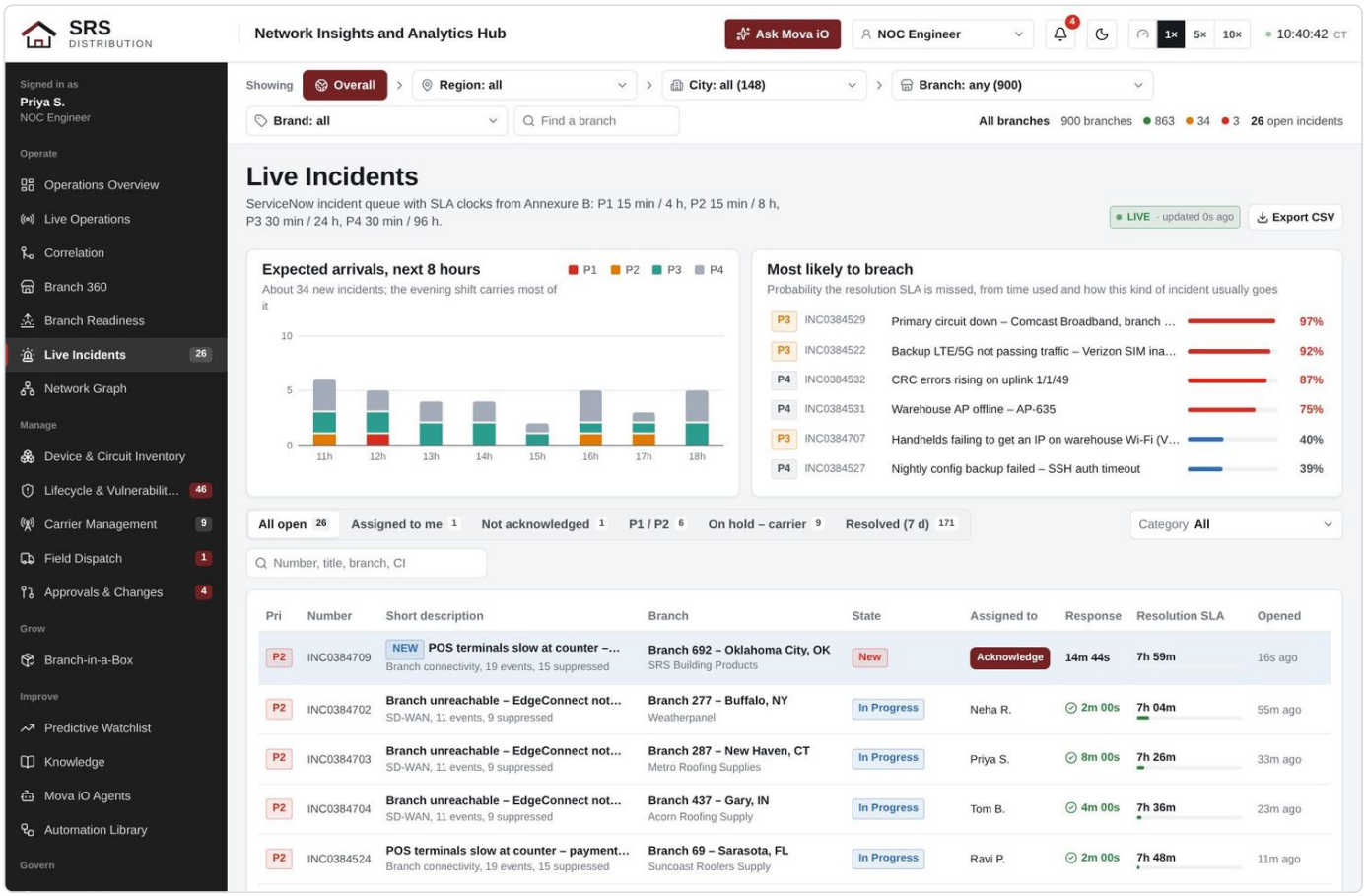



**Figure 80.** Live Incidents

## Expected arrivals and breach risk

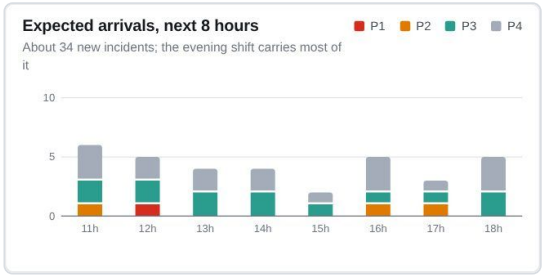



**Figure 81.** Expected arrivals, next 8 hours

New incidents expected per hour for the next eight hours, stacked by priority (P1 red, P2 orange, P3 blue, P4 grey), with the total in the subtitle. Use it to staff the next shift.

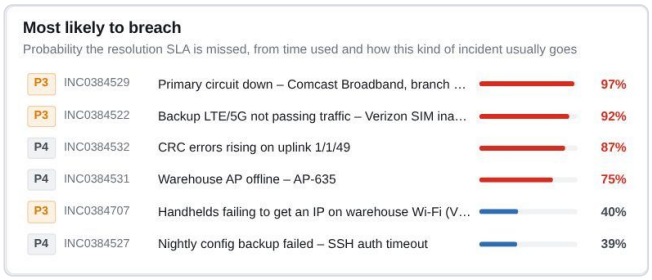
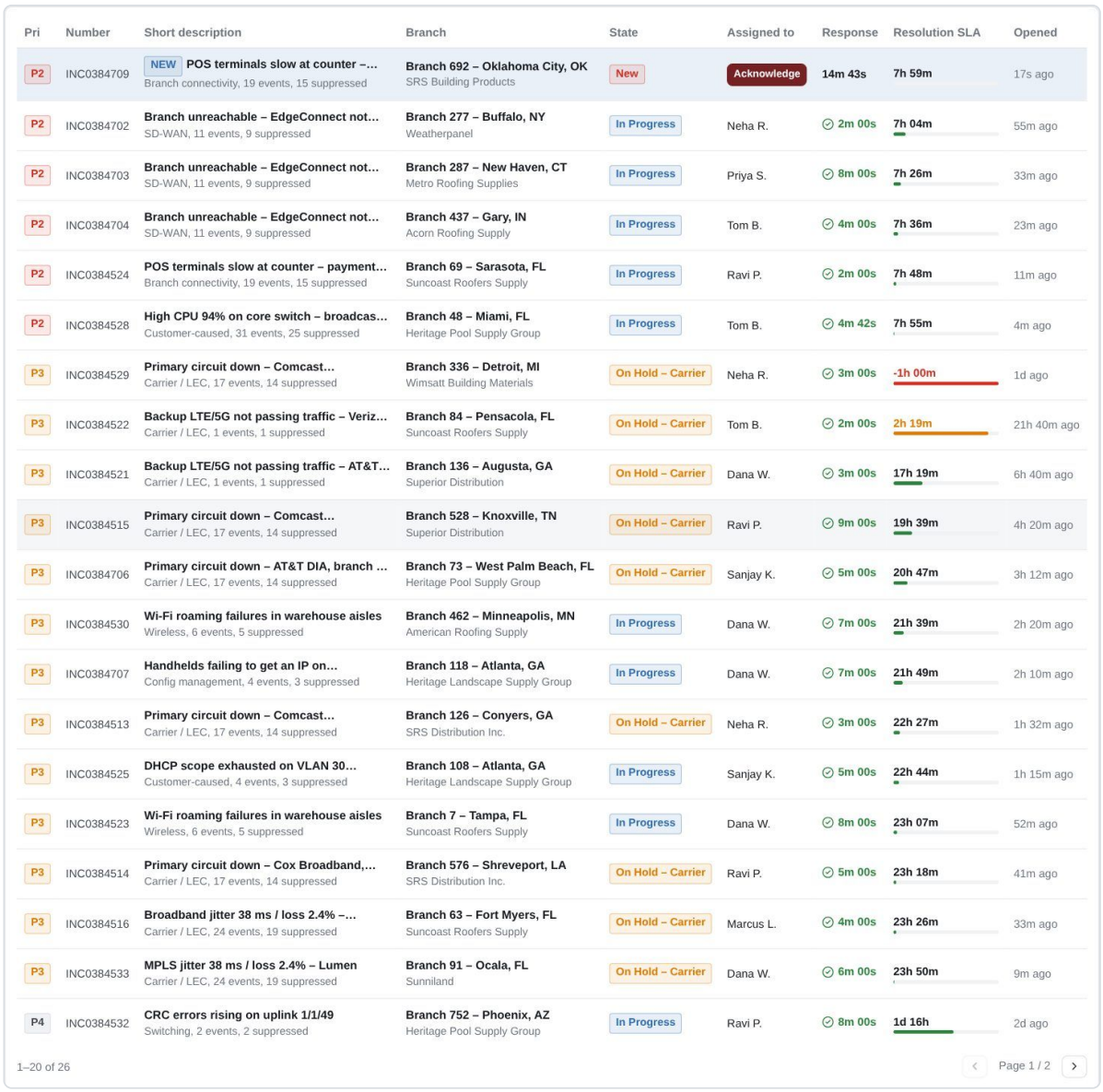



**Figure 82.** Most likely to breach

Open incidents ranked by the chance their resolution SLA is missed, worked out from the time already used and how this kind of incident usually goes. Anything above 70% needs action now: escalate, chase the carrier or dispatch.

## The queue



| Pri | Number | Short description | Branch | State | Assigned to | Response | Resolution SLA | Opened |
|---|---|---|---|---|---|---|---|---|
| P2 | INC0384709 | NEW POS terminals slow at counter –… Branch connectivity, 19 events, 15 suppressed | Branch 692 – Oklahoma City, OK SRS Building Products | New | Acknowledge | 14m 43s | 7h 59m | 17s ago |
| P2 | INC0384702 | Branch unreachable – EdgeConnect not… SD-WAN, 11 events, 9 suppressed | Branch 277 – Buffalo, NY Weatherpanel | In Progress | Neha R. | 2m 00s | 7h 04m | 55m ago |
| P2 | INC0384703 | Branch unreachable – EdgeConnect not… SD-WAN, 11 events, 9 suppressed | Branch 287 – New Haven, CT Metro Roofing Supplies | In Progress | Priya S. | 8m 00s | 7h 26m | 33m ago |
| P2 | INC0384704 | Branch unreachable – EdgeConnect not… SD-WAN, 11 events, 9 suppressed | Branch 437 – Gary, IN Acorn Roofing Supply | In Progress | Tom B. | 4m 00s | 7h 36m | 23m ago |
| P2 | INC0384524 | POS terminals slow at counter – payment… Branch connectivity, 19 events, 15 suppressed | Branch 69 – Sarasota, FL Suncoast Roofers Supply | In Progress | Ravi P. | 2m 00s | 7h 48m | 11m ago |
| P2 | INC0384528 | High CPU 94% on core switch – broadcas… Customer-caused, 31 events, 25 suppressed | Branch 48 – Miami, FL Heritage Pool Supply Group | In Progress | Tom B. | 4m 42s | 7h 55m | 4m ago |
| P3 | INC0384529 | Primary circuit down – Comcast… Carrier / LEC, 17 events, 14 suppressed | Branch 336 – Detroit, MI Wimsatt Building Materials | On Hold – Carrier | Neha R. | 3m 00s | -1h 00m | 1d ago |
| P3 | INC0384522 | Backup LTE/5G not passing traffic – Veriz… Carrier / LEC, 1 events, 1 suppressed | Branch 84 – Pensacola, FL Suncoast Roofers Supply | On Hold – Carrier | Tom B. | 2m 00s | 2h 19m | 21h 40m ago |
| P3 | INC0384521 | Backup LTE/5G not passing traffic – AT&T… Carrier / LEC, 1 events, 1 suppressed | Branch 136 – Augusta, GA Superior Distribution | On Hold – Carrier | Dana W. | 3m 00s | 17h 19m | 6h 40m ago |
| P3 | INC0384515 | Primary circuit down – Comcast… Carrier / LEC, 17 events, 14 suppressed | Branch 528 – Knoxville, TN Superior Distribution | On Hold – Carrier | Ravi P. | 9m 00s | 19h 39m | 4h 20m ago |
| P3 | INC0384706 | Primary circuit down – AT&T DIA, branch … Carrier / LEC, 17 events, 14 suppressed | Branch 73 – West Palm Beach, FL Heritage Pool Supply Group | On Hold – Carrier | Sanjay K. | 5m 00s | 20h 47m | 3h 12m ago |
| P3 | INC0384530 | Wi-Fi roaming failures in warehouse aisles Wireless, 6 events, 5 suppressed | Branch 462 – Minneapolis, MN American Roofing Supply | In Progress | Dana W. | 7m 00s | 21h 39m | 2h 20m ago |
| P3 | INC0384707 | Handhelds failing to get an IP on… Config management, 4 events, 3 suppressed | Branch 118 – Atlanta, GA Heritage Landscape Supply Group | In Progress | Dana W. | 7m 00s | 21h 49m | 2h 10m ago |
| P3 | INC0384513 | Primary circuit down – Comcast… Carrier / LEC, 17 events, 14 suppressed | Branch 126 – Conyers, GA SRS Distribution Inc. | On Hold – Carrier | Neha R. | 3m 00s | 22h 27m | 1h 32m ago |
| P3 | INC0384525 | DHCP scope exhausted on VLAN 30… Customer-caused, 4 events, 3 suppressed | Branch 108 – Atlanta, GA Heritage Landscape Supply Group | In Progress | Sanjay K. | 5m 00s | 22h 44m | 1h 15m ago |
| P3 | INC0384523 | Wi-Fi roaming failures in warehouse aisles Wireless, 6 events, 5 suppressed | Branch 7 – Tampa, FL Suncoast Roofers Supply | In Progress | Dana W. | 8m 00s | 23h 07m | 52m ago |
| P3 | INC0384514 | Primary circuit down – Cox Broadband,… Carrier / LEC, 17 events, 14 suppressed | Branch 576 – Shreveport, LA SRS Distribution Inc. | On Hold – Carrier | Ravi P. | 5m 00s | 23h 18m | 41m ago |
| P3 | INC0384516 | Broadband jitter 38 ms / loss 2.4% –… Carrier / LEC, 24 events, 19 suppressed | Branch 63 – Fort Myers, FL Suncoast Roofers Supply | On Hold – Carrier | Marcus L. | 4m 00s | 23h 26m | 33m ago |
| P3 | INC0384533 | MPLS jitter 38 ms / loss 2.4% – Lumen Carrier / LEC, 24 events, 19 suppressed | Branch 91 – Ocala, FL Sunniland | On Hold – Carrier | Dana W. | 6m 00s | 23h 50m | 9m ago |
| P4 | INC0384532 | CRC errors rising on uplink 1/1/49 Switching, 2 events, 2 suppressed | Branch 752 – Phoenix, AZ Heritage Pool Supply Group | In Progress | Ravi P. | 8m 00s | 1d 16h | 2d ago |

1–20 of 26 Page 1 / 2

**Figure 83.** Incident queue

The tabs filter the queue: **All open**, **Assigned to me**, **Not acknowledged**, **P1 / P2**, **On hold – carrier** and **Resolved (7 d)**.

| Column | Meaning |
|---|---|
| Pri | P1 to P4 |
| Number | The ServiceNow incident number |
| Short description | Title, then category, the number of events correlated into it and how many were suppressed |
| Branch | Branch and brand |
| State | New, In Progress, On Hold – Carrier, Resolved |
| Assigned to | The engineer or agent, or an **Acknowledge** button |
| Response | Time left to acknowledge (15 or 30 minutes by priority), or the time taken |
| Resolution SLA | Time left to resolve; amber past 75% used; negative and red once breached; "Met" when resolved in time |
| Opened | How long ago |

**Scenarios** replays a scripted event into the queue; **Export CSV** downloads it. Click a row to open the incident.

## When incidents happen

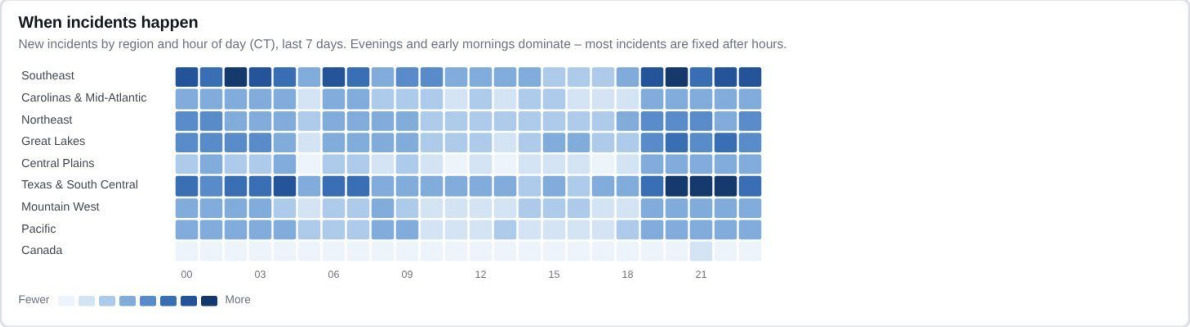



**Figure 84.** When incidents happen

New incidents by region (rows) and hour of day in CT (columns) over the last seven days; darker means more. Evenings and early mornings dominate because most fixes are made after branches close and carriers work at night.

## The incident record

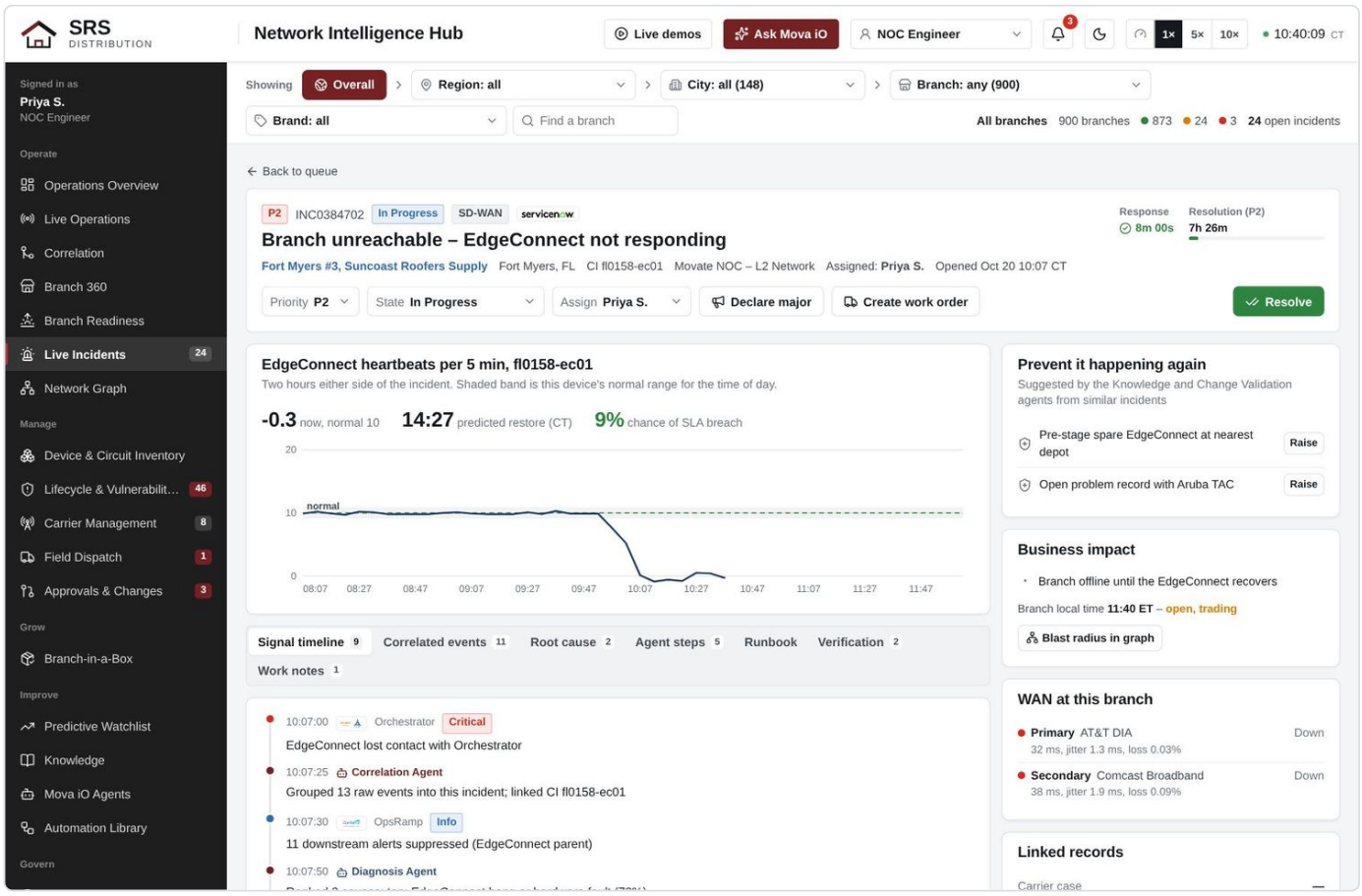



**Figure 85.** Incident detail

The top of the record gives the priority, number, state, category, the ServiceNow link, the title, branch, city, CI, assignment group, assignee and opening time, with the response and resolution clocks on the right.

| Action | What it does |
|---|---|
| Escalate to Movate | SRS roles only: sends an escalation about this incident to the ticket owner (chapter 3); the actions below are Movate's |
| Acknowledge & take | Stops the response clock and assigns the incident to you |
| Priority, State, Assign | Change them; every change is in the audit log. **Assign** lists the people by team (NOC L1, NOC L2, major incident, Network SME), and the group follows the person |
| Declare major | Makes it a major incident with a bridge and leadership updates |
| Create work order | Opens a field visit with skills and parts |
| Resolve | Closes with a close code once verification passes |

Under the actions:

- **Device chart:** the key metric of the CI for two hours either side of the incident, with the device's normal band for that time of day, the value now, the predicted restore time and the chance of an SLA breach.

- **Who fixes it:** what Mova iO's Smart Triage Agent recommends (for example field engineer on site), what happens next, whether Mova iO can fix it remotely and the routes already in hand. **Review what Mova iO has prepared** opens the root-cause view at **Mova iO: who fixes it** (chapter 3), where an engineer approves the hand-off, the field visit, the carrier or HPE case, or the runbook.

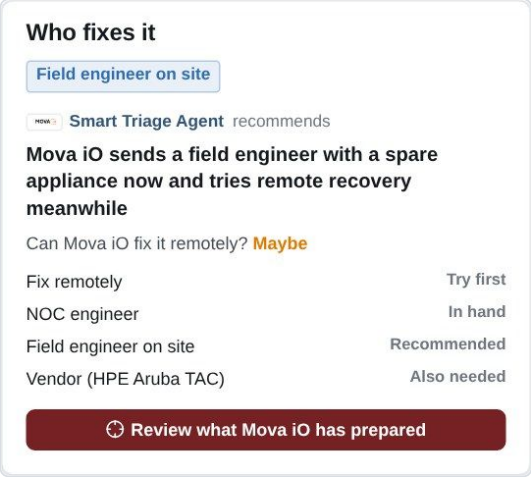



**Figure 86.** Who fixes it in the incident record

- **Prevent it happening again:** fixes suggested by the Knowledge and Change Validation agents from similar incidents; **Raise** creates the change or task.
- **Business impact:** what the branch feels and whether it is open and trading; **Blast radius in graph** opens the Network Graph on this CI.
- **WAN at this branch:** both circuits with latency, jitter, loss and status.
- **Linked records:** carrier case, work order, change and problem.

## The record tabs

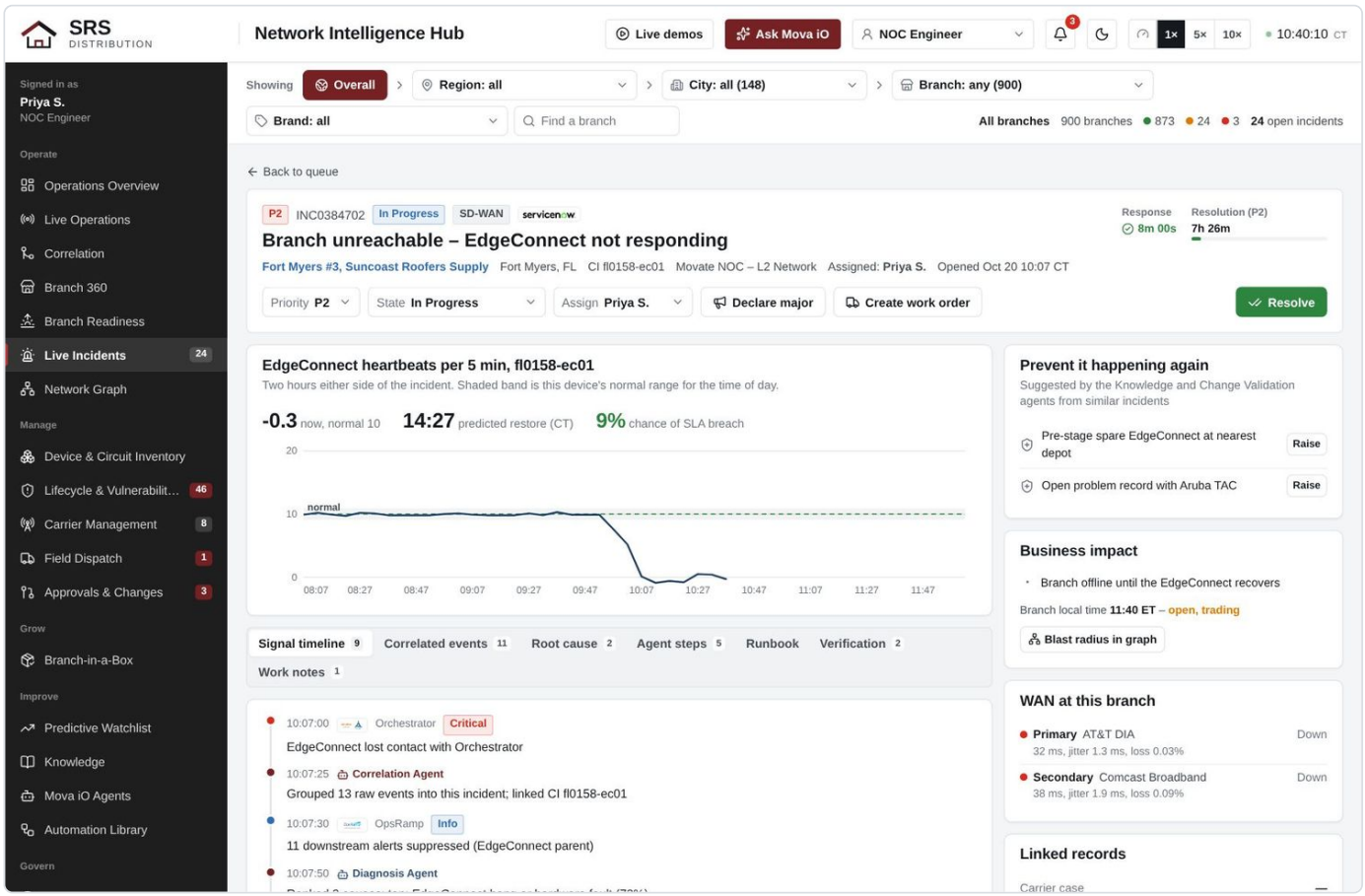



**Figure 87.** Signal timeline

| Tab | What it holds |
|---|---|
| Signal timeline | Every signal, agent step and note in time order, with source and severity |
| Correlated events | All raw events grouped into this incident and which were suppressed |
| Root cause | Ranked causes, each with its confidence and the evidence for it |
| Agent steps | What each Mova iO agent did, in order |
| Runbook | The recommended runbook, its steps, autonomy level, gate result, and **Run**, **Send to approvals queue** or **Rollback** |
| Verification | The checks that must pass before the incident can close, with values |
| Work notes | Notes from engineers and agents; **Post** adds one |

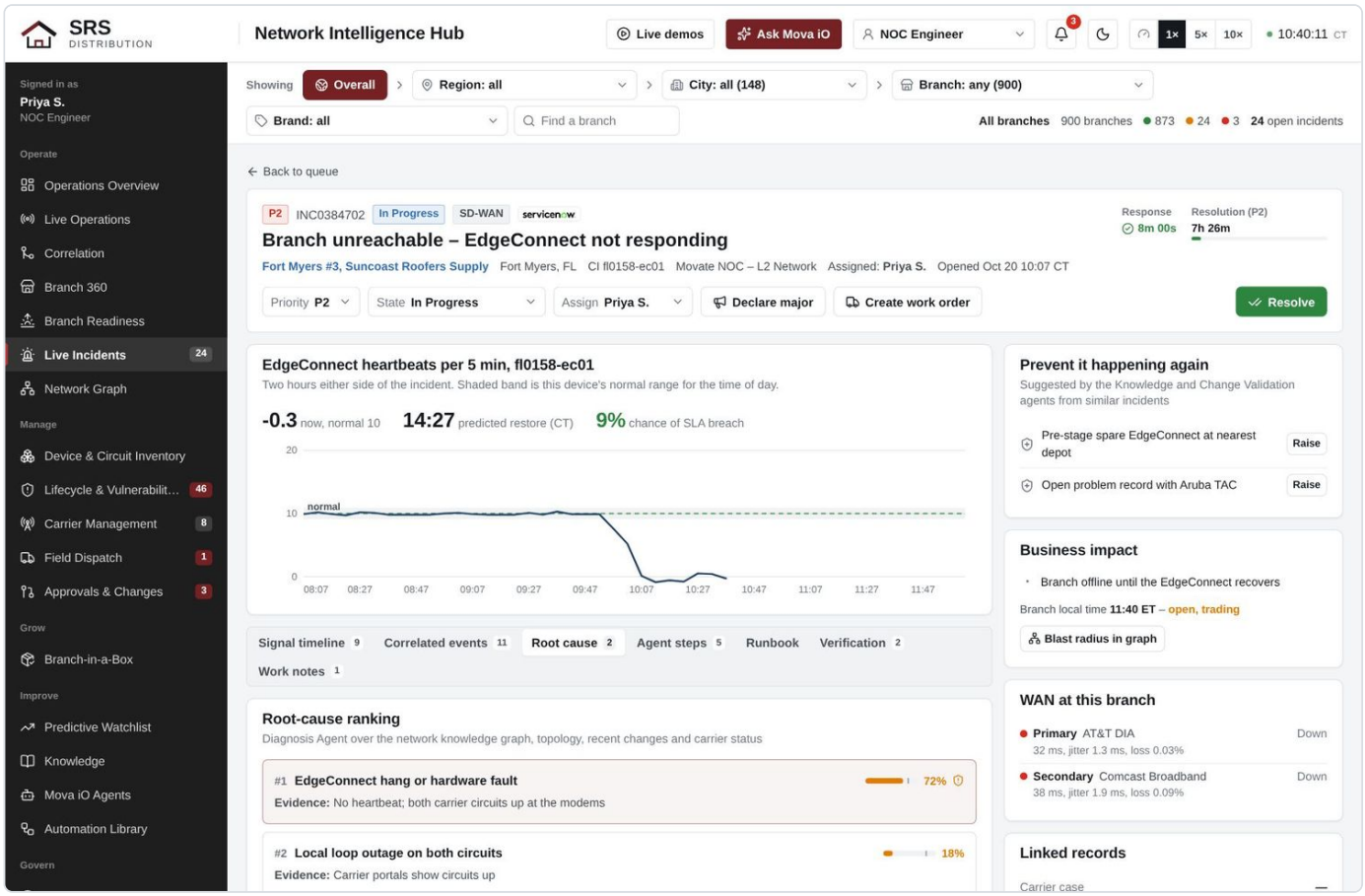



**Figure 88.** Root cause tab

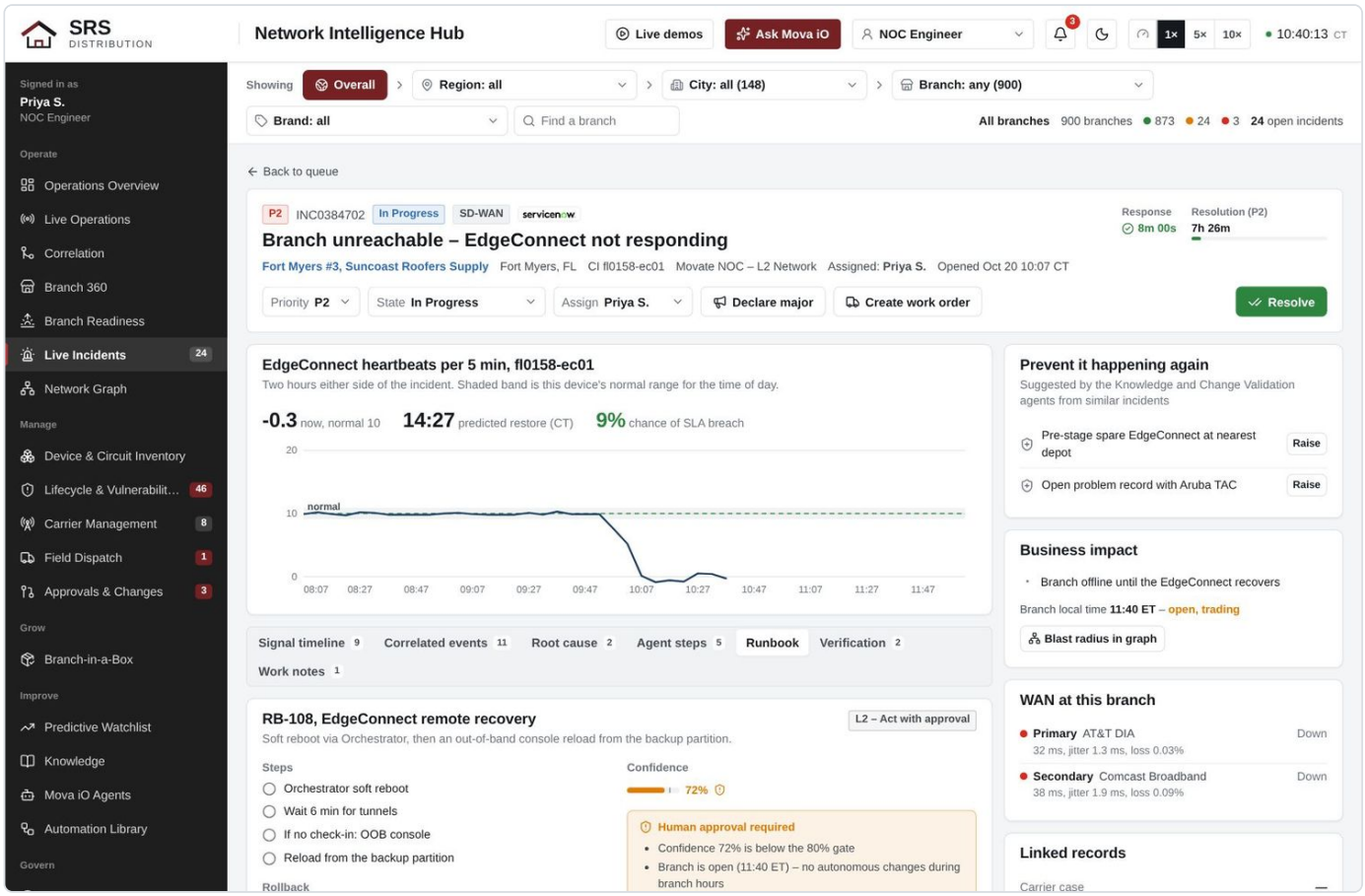



**Figure 89.** Runbook tab

**How to read the root cause.** The top cause carries the confidence that drives the gate rule: 80% or more allows the runbook to run within its tier, below 80% sends it to a person. The lower causes show what was considered and why it is less likely.

# 11. Network Graph

The Network Graph shows how each branch reaches the business: its LAN and EdgeConnect, both WAN lines, the carrier networks, the hosting points and the applications. It is built from OpsRamp, Orchestrator and the ServiceNow CMDB. Click anything to see what depends on it.

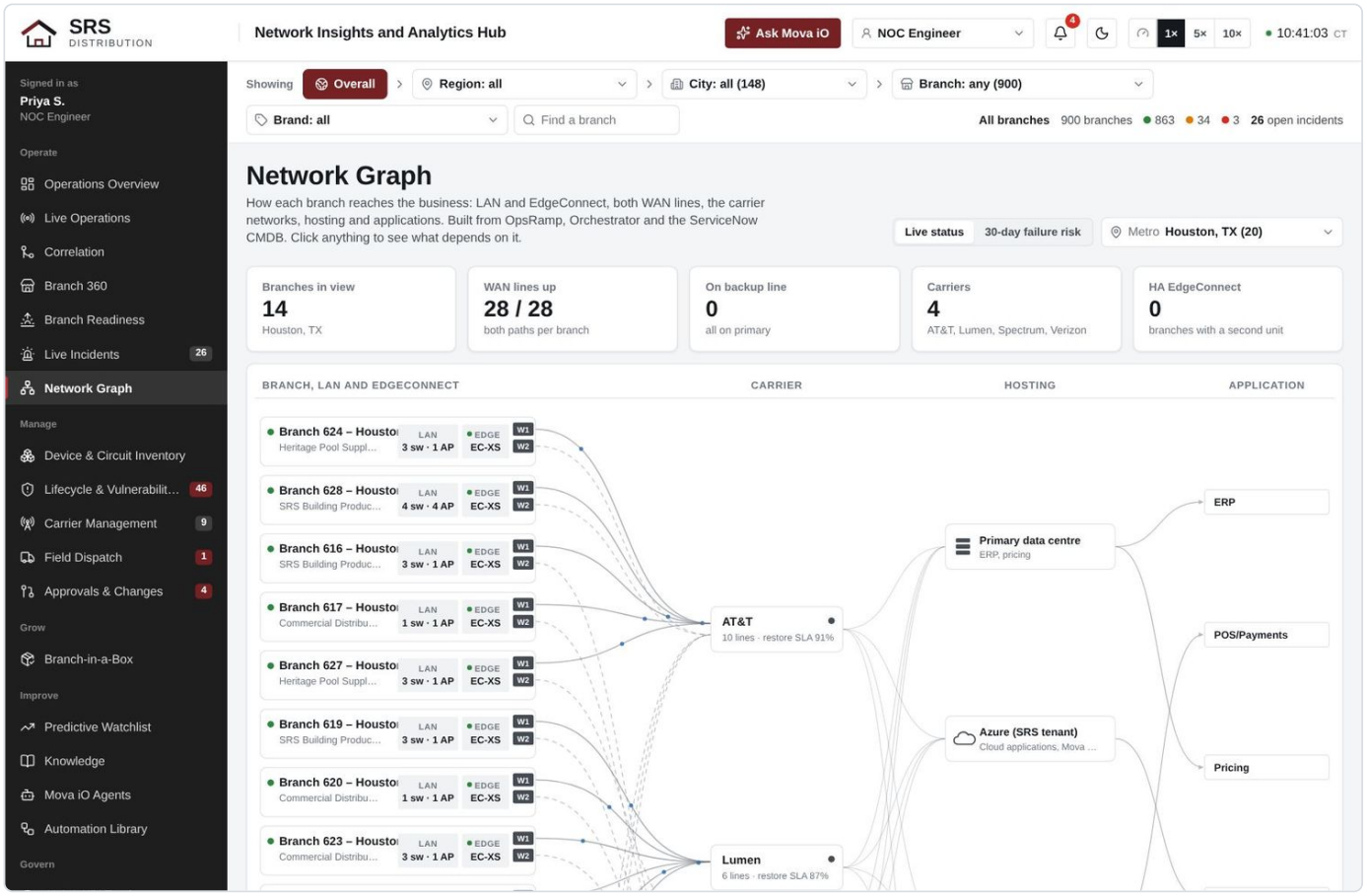



**Figure 90.** Network Graph

## Controls

| Control | What it does |
|---|---|
| Live status / 30-day failure risk | Colours the graph by health now, or by each element's predicted chance of failing in the next 30 days |
| Metro | Picks the metro area to draw, with its branch count. The graph draws up to the first 14 branches in that metro; the filter bar narrows further |

## Tiles

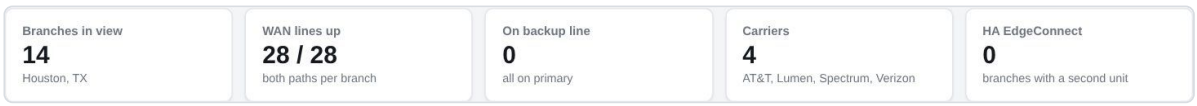



**Figure 91.** Graph tiles

| Tile | Meaning |
|---|---|
| Branches in view | Branches drawn, and the metro |
| WAN lines up | Lines up out of all lines drawn (two per branch) |
| On backup line | Branches running on their second line now |
| Carriers | Carriers serving these branches |
| HA EdgeConnect | Branches with a second EdgeConnect unit |

## Reading the graph

The graph runs left to right in four columns.

| Column | Nodes | What to look at |
|---|---|---|
| Branch, LAN and EdgeConnect | One card per branch: name, brand, the LAN (switches and APs), the EdgeConnect model, and the two WAN ports W1 and W2 | The dot by the name is the branch health; W1 and W2 turn amber or red when a line is degraded or down |
| Carrier | One node per carrier, with the number of lines and its restore SLA over six months | A carrier with many lines and a low restore SLA is a concentration risk |
| Hosting | Primary data centre (ERP, pricing) and Azure (SRS tenant: cloud applications and Mova iO) | – |
| Application | ERP, POS and payments, pricing and the other business services | – |

Solid lines are primary circuits, dashed lines are backups. Moving dots on a line show traffic. Branches are sorted by carrier so lines do not cross more than they must, and each carrier sits next to the branches it serves.

## What depends on it




**Figure 92.** What depends on it

Click a carrier to see which branches would go down (both lines on that carrier) and which would fail over (only one line on it). Click a branch's LAN, EdgeConnect or a W1 or W2 port to see what depends on that element. This is the **blast radius**: the set of branches and services affected if that one element fails. It is the same blast radius Mova iO checks before any change.

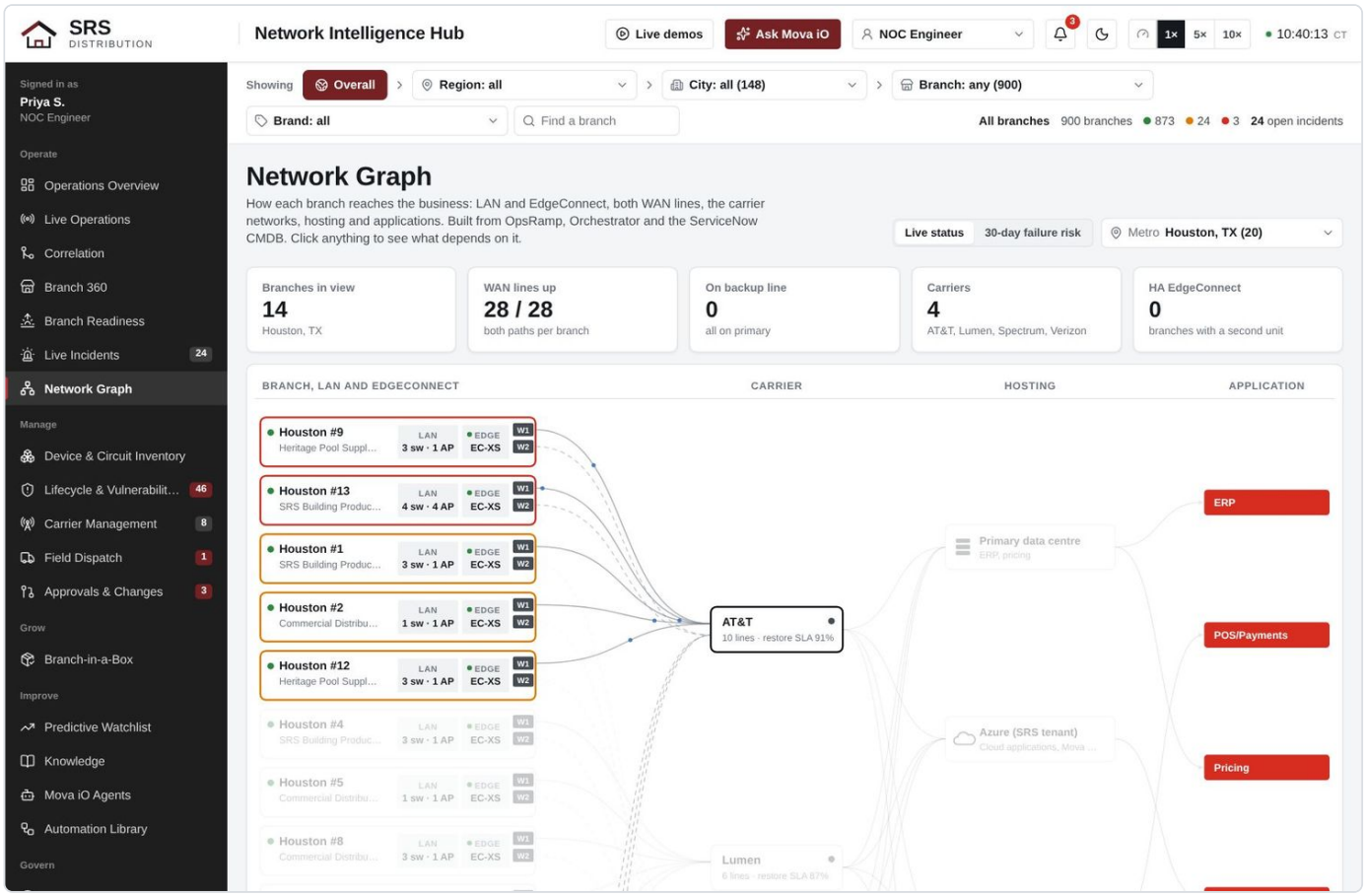



**Figure 93.** Blast radius after clicking a carrier

## Line detail

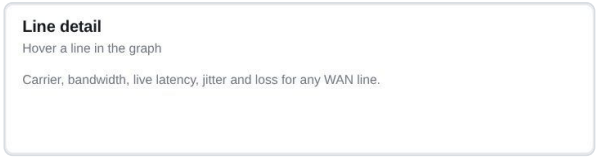



**Figure 94.** Line detail

Hover any line for its carrier, bandwidth and live latency, jitter and loss.

## 30-day failure risk

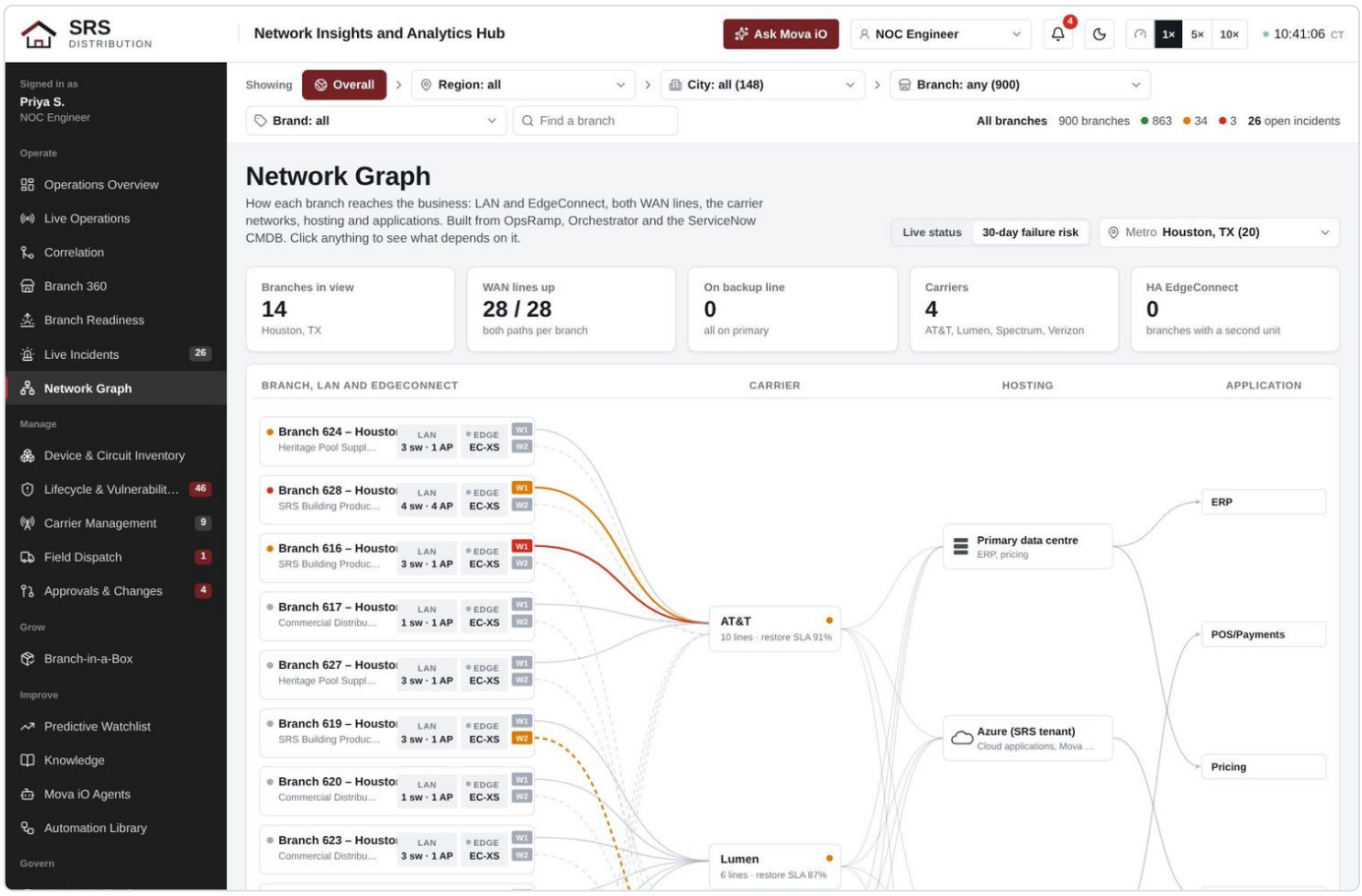



**Figure 95.** 30-day failure risk

The same graph coloured by predicted risk instead of health. Use it before a change window or a busy season: the red elements are where to raise a proactive case, pre-stage a spare or test a failover first.

## Network activity – live

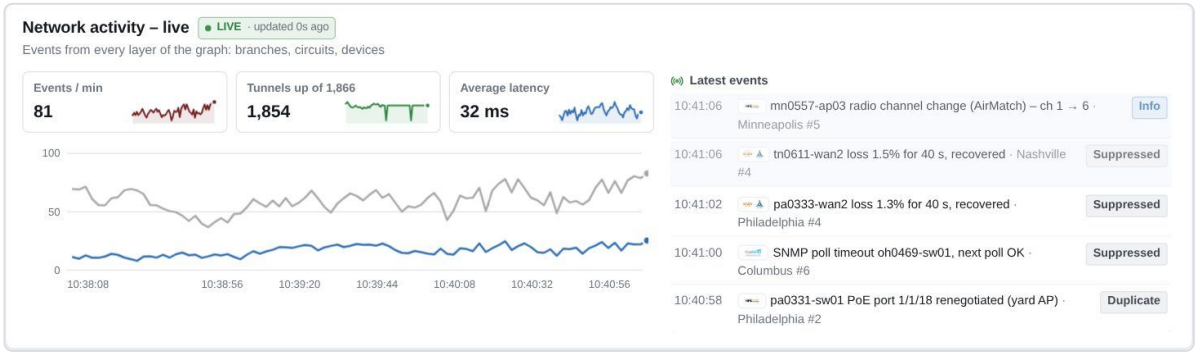



**Figure 96.** Network activity

The live panel for this screen: events per minute from every layer of the graph, tunnels up out of 1,866 and the average latency to the data centre, with a chart and the latest events. Every screen in Manage, Grow and Improve has a live panel like this one at the top, showing the stream that matters for that screen.

# 12. Device & Circuit Inventory

The inventory is the CMDB view of every managed device and circuit: model, firmware, config backup, warranty and circuit details. It has three tabs: **Devices**, **Circuits** and **Reconciliation**.

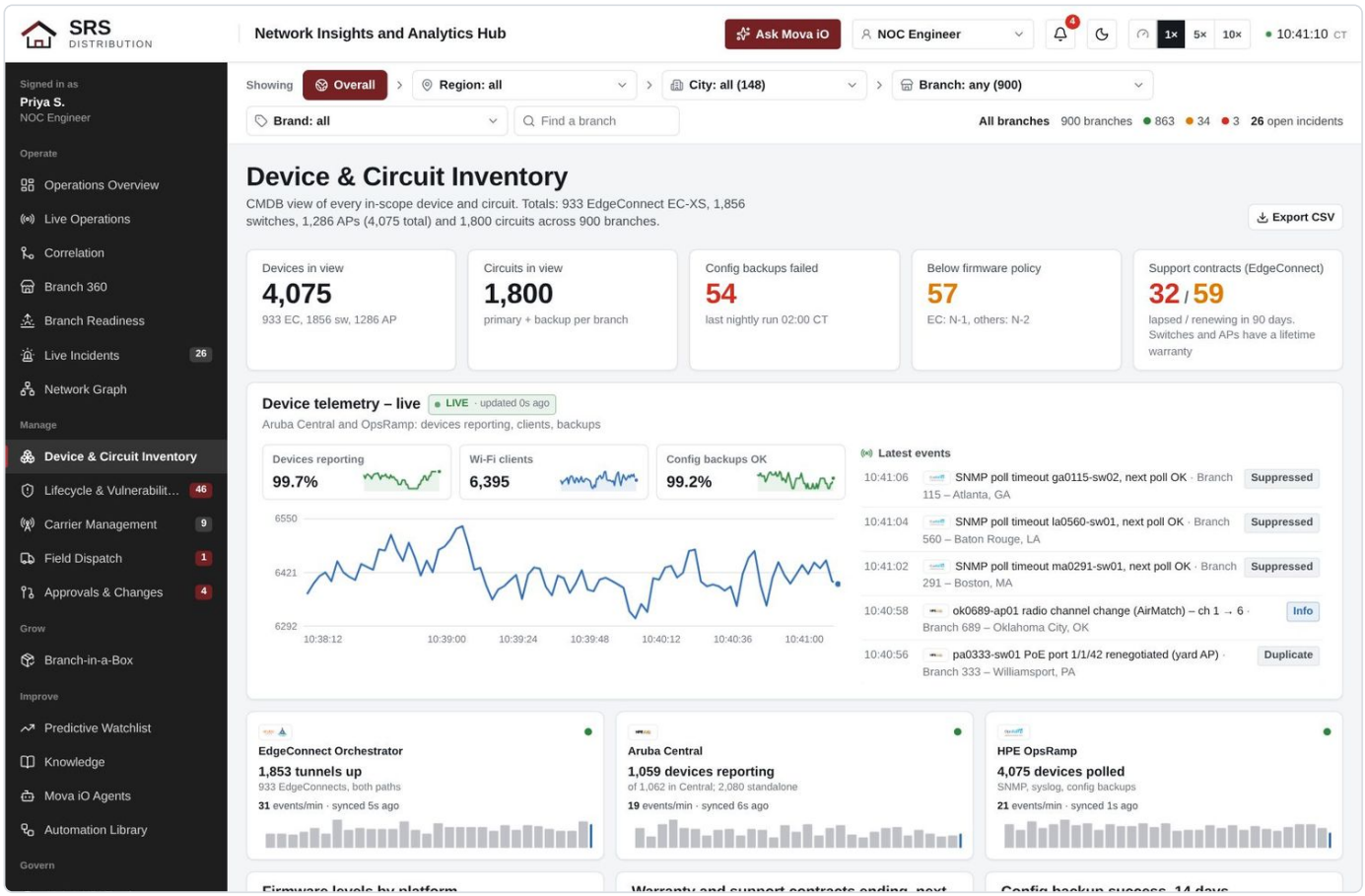



**Figure 97.** Device & Circuit Inventory

## Tiles

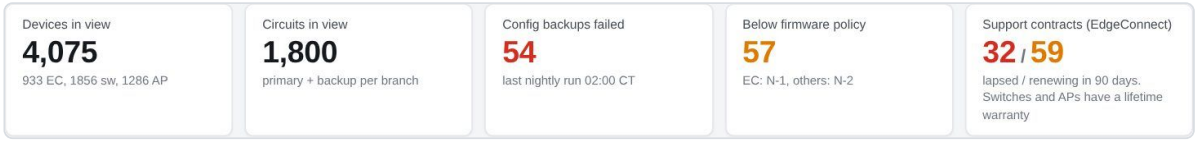



**Figure 98.** Inventory tiles

| Tile | Meaning | Look closer when |
|---|---|---|
| Devices in view | All managed devices, split into EdgeConnects, switches and APs | The count does not match the contract (see Reconciliation) |
| Circuits in view | Primary and backup lines, two per branch | – |
| Config backups failed | Devices whose last nightly backup (02:00 CT) failed | More than about 2% of devices with backups |

| Tile | Meaning | Look closer when |
|---|---|---|
| Below firmware policy | Devices further behind than the policy allows: EdgeConnects more than one release behind (N-1), others more than two (N-2) | Rising after a wave |
| Support contracts (EdgeConnect) | EdgeConnect support contracts that have lapsed, and those renewing within 90 days. Switches and APs carry HPE's limited lifetime warranty, which runs until 5 years after the model's end of sale, so they have no yearly date | Any lapsed contract |

## Device telemetry – live

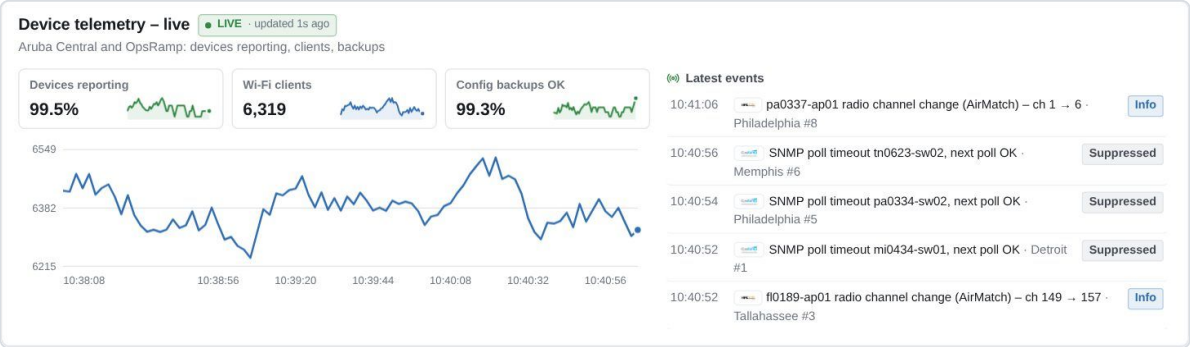



**Figure 99.** Device telemetry

Devices reporting (%), Wi-Fi clients connected and config backups OK (%), with a chart of clients and the latest device events.

## Firmware levels by platform

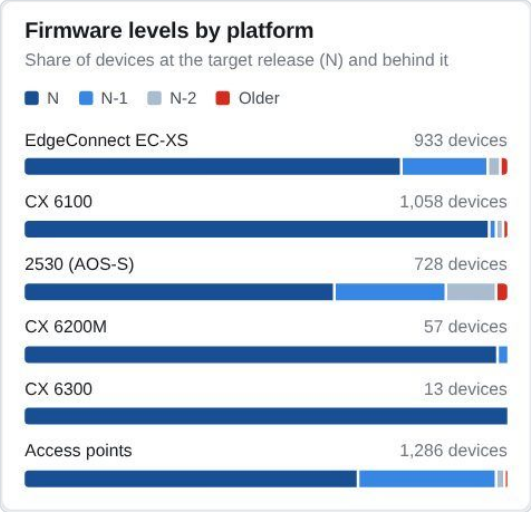



**Figure 100.** Firmware levels by platform

For each platform (EdgeConnect EC-XS, CX 6100, 2530, CX 6200M, CX 6300 and access points), a bar split into the share of devices on the target release (N), one behind (N-1), two behind (N-2) and older. More of the bar in the darkest shade (N) is better. A large "Older" share on one platform means it missed waves; on the 2530 it reflects the end-of-sale model.

## Warranty and support contracts ending, next 8 quarters

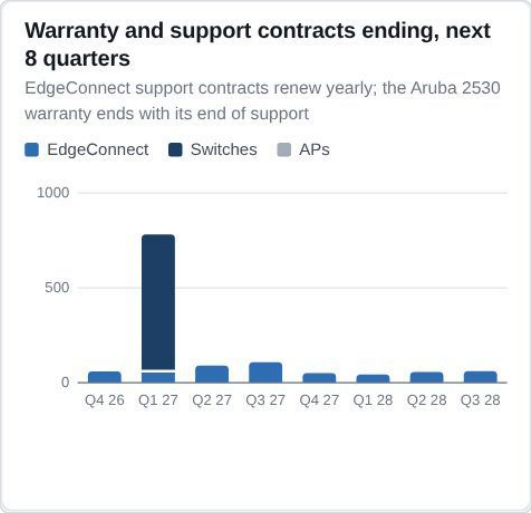



**Figure 101.** Warranty and support contracts

Dates ending in each of the next eight quarters, stacked by EdgeConnect, switches and APs. EdgeConnects have a 1-year hardware warranty and are covered by an HPE support contract renewed every year, so their bars are contract renewals. The Aruba 2530 warranty ends with its end of support in 2027. Current switches and APs have a lifetime warranty and do not appear.

## Config backup success, 14 days

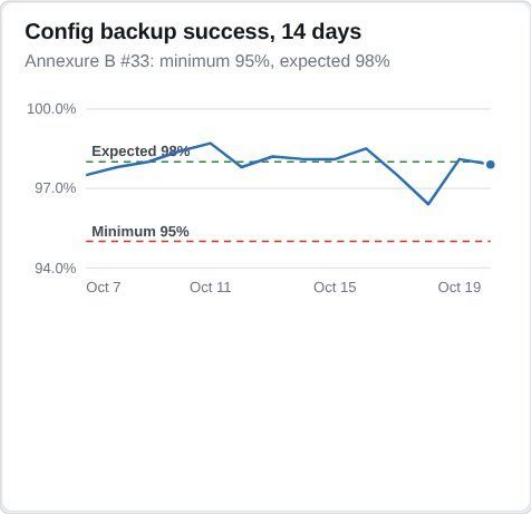



**Figure 102.** Config backup success

Daily backup success for 14 days against the two lines from service level #33: minimum 95% and expected 98%. Below 95% in a month triggers a service credit.

## Hardware likely to fail in 30 days

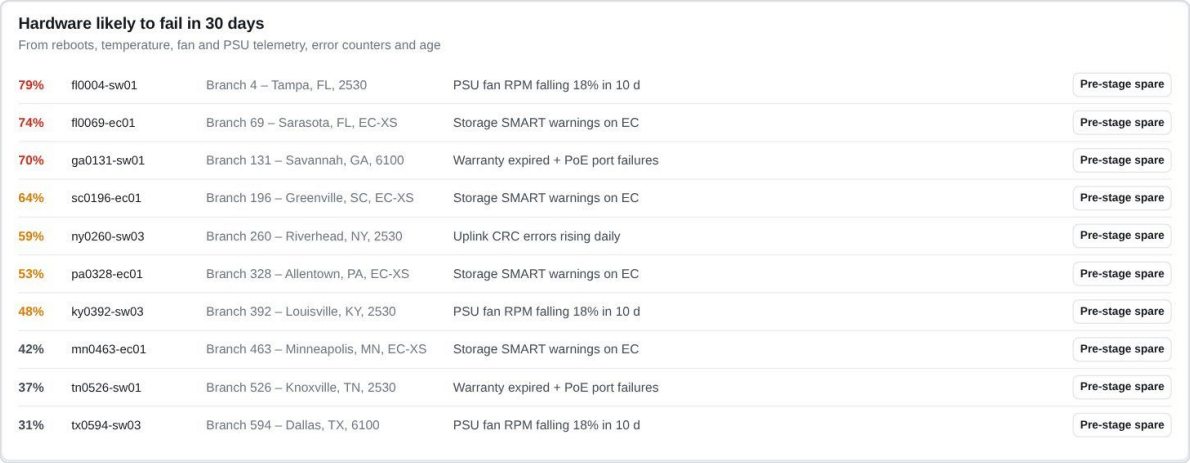



**Figure 103.** Hardware likely to fail

Devices ranked by their chance of failing in 30 days, worked out from reboots, temperature, fan and power-supply telemetry, error counters and age. Each row shows the device, branch and model, the signal behind the prediction, and **Pre-stage spare**, which reserves a spare at the nearest depot.

## Devices tab

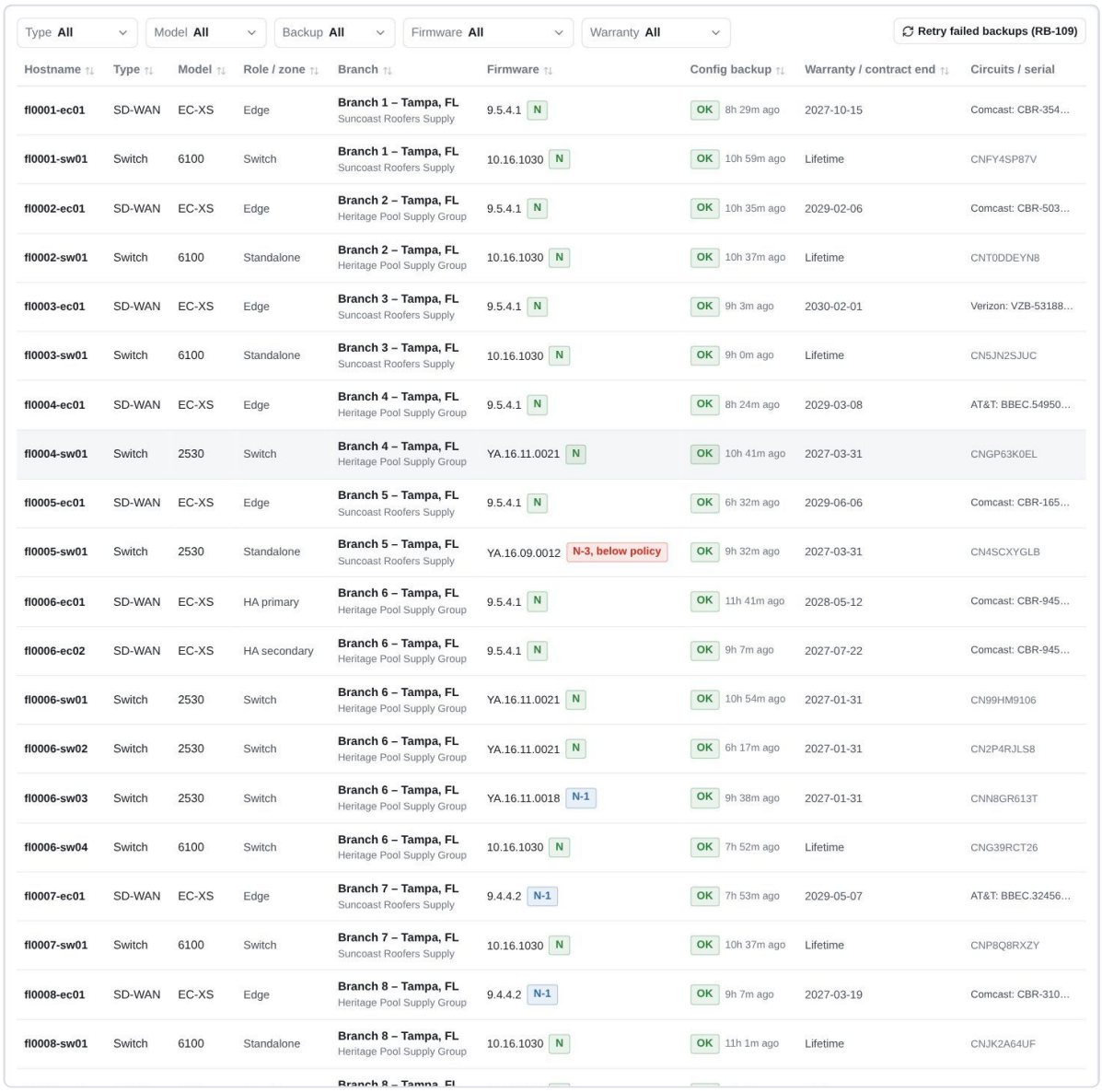


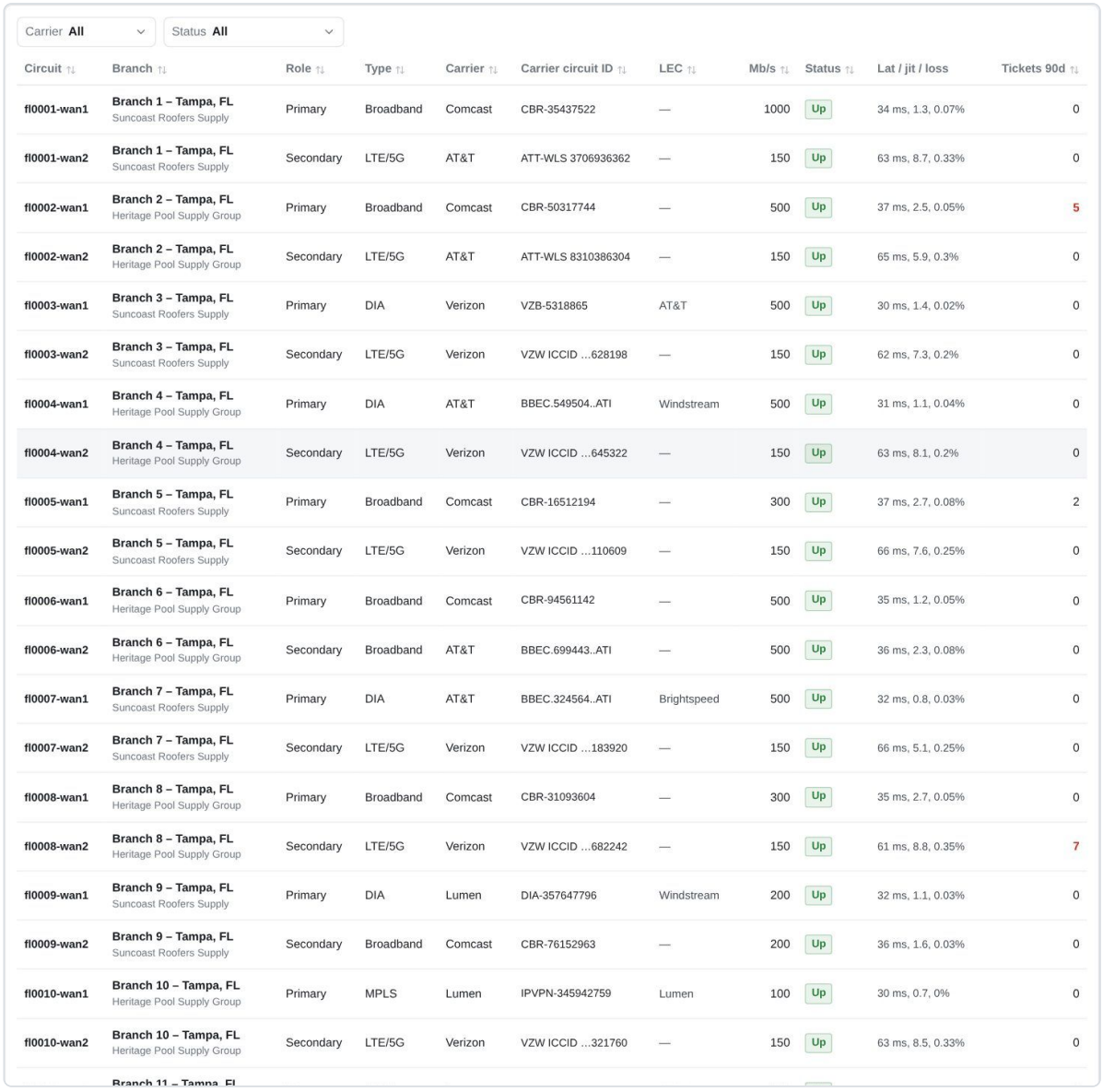

**Figure 104.** Device table

Filters at the top: type, model, backup state (OK, Failed, Stale, N/A), firmware (below policy) and warranty or contract (ending within 90 days, lapsed). **Retry failed backups (RB-109)** re-runs every failed backup in view.

| Column | Meaning |
|---|---|
| Hostname | Device name, built from state, branch number and device type |
| Type, model, role or zone | For example SD-WAN EC-XS Edge, Switch CX 6100, AP-635 warehouse |
| Branch | Branch and brand |
| Firmware | Running release and its level: N, N-1, N-2 or older |
| Config backup | OK, Failed or Stale, with the age of the last good backup |

| Column | Meaning |
|---|---|
| Warranty / contract end | Date for EdgeConnect support contracts and the Aruba 2530; Lifetime for current switches and APs. Amber within 90 days, red when lapsed |
| Circuits or serial | Circuits for an EdgeConnect, serial number for others |

## Circuits tab




| Circuit | Branch | Role | Type | Carrier | Carrier circuit ID | LEC | Mb/s | Status | Lat / jit / loss | Tickets 90d |
|---|---|---|---|---|---|---|---|---|---|---|
| fl0001-wan1 | Branch 1 – Tampa, FL<br>Suncoast Roofers Supply | Primary | Broadband | Comcast | CBR-35437522 | — | 1000 | Up | 34 ms, 1.3, 0.07% | 0 |
| fl0001-wan2 | Branch 1 – Tampa, FL<br>Suncoast Roofers Supply | Secondary | LTE/5G | AT&T | ATT-WLS 3706936362 | — | 150 | Up | 63 ms, 8.7, 0.33% | 0 |
| fl0002-wan1 | Branch 2 – Tampa, FL<br>Heritage Pool Supply Group | Primary | Broadband | Comcast | CBR-50317744 | — | 500 | Up | 37 ms, 2.5, 0.05% | 5 |
| fl0002-wan2 | Branch 2 – Tampa, FL<br>Heritage Pool Supply Group | Secondary | LTE/5G | AT&T | ATT-WLS 8310386304 | — | 150 | Up | 65 ms, 5.9, 0.3% | 0 |
| fl0003-wan1 | Branch 3 – Tampa, FL<br>Suncoast Roofers Supply | Primary | DIA | Verizon | VZB-5318865 | AT&T | 500 | Up | 30 ms, 1.4, 0.02% | 0 |
| fl0003-wan2 | Branch 3 – Tampa, FL<br>Suncoast Roofers Supply | Secondary | LTE/5G | Verizon | VZW ICCID …628198 | — | 150 | Up | 62 ms, 7.3, 0.2% | 0 |
| fl0004-wan1 | Branch 4 – Tampa, FL<br>Heritage Pool Supply Group | Primary | DIA | AT&T | BBEC.549504..ATI | Windstream | 500 | Up | 31 ms, 1.1, 0.04% | 0 |
| fl0004-wan2 | Branch 4 – Tampa, FL<br>Heritage Pool Supply Group | Secondary | LTE/5G | Verizon | VZW ICCID …645322 | — | 150 | Up | 63 ms, 8.1, 0.2% | 0 |
| fl0005-wan1 | Branch 5 – Tampa, FL<br>Suncoast Roofers Supply | Primary | Broadband | Comcast | CBR-16512194 | — | 300 | Up | 37 ms, 2.7, 0.08% | 2 |
| fl0005-wan2 | Branch 5 – Tampa, FL<br>Suncoast Roofers Supply | Secondary | LTE/5G | Verizon | VZW ICCID …110609 | — | 150 | Up | 66 ms, 7.6, 0.25% | 0 |
| fl0006-wan1 | Branch 6 – Tampa, FL<br>Heritage Pool Supply Group | Primary | Broadband | Comcast | CBR-94561142 | — | 500 | Up | 35 ms, 1.2, 0.05% | 0 |
| fl0006-wan2 | Branch 6 – Tampa, FL<br>Heritage Pool Supply Group | Secondary | Broadband | AT&T | BBEC.699443..ATI | — | 500 | Up | 36 ms, 2.3, 0.08% | 0 |
| fl0007-wan1 | Branch 7 – Tampa, FL<br>Suncoast Roofers Supply | Primary | DIA | AT&T | BBEC.324564..ATI | Brightspeed | 500 | Up | 32 ms, 0.8, 0.03% | 0 |
| fl0007-wan2 | Branch 7 – Tampa, FL<br>Suncoast Roofers Supply | Secondary | LTE/5G | Verizon | VZW ICCID …183920 | — | 150 | Up | 66 ms, 5.1, 0.25% | 0 |
| fl0008-wan1 | Branch 8 – Tampa, FL<br>Heritage Pool Supply Group | Primary | Broadband | Comcast | CBR-31093604 | — | 300 | Up | 35 ms, 2.7, 0.05% | 0 |
| fl0008-wan2 | Branch 8 – Tampa, FL<br>Heritage Pool Supply Group | Secondary | LTE/5G | Verizon | VZW ICCID …682242 | — | 150 | Up | 61 ms, 8.8, 0.35% | 7 |
| fl0009-wan1 | Branch 9 – Tampa, FL<br>Suncoast Roofers Supply | Primary | DIA | Lumen | DIA-357647796 | Windstream | 200 | Up | 32 ms, 1.1, 0.03% | 0 |
| fl0009-wan2 | Branch 9 – Tampa, FL<br>Suncoast Roofers Supply | Secondary | Broadband | Comcast | CBR-76152963 | — | 200 | Up | 36 ms, 1.6, 0.03% | 0 |
| fl0010-wan1 | Branch 10 – Tampa, FL<br>Heritage Pool Supply Group | Primary | MPLS | Lumen | IPVPN-345942759 | Lumen | 100 | Up | 30 ms, 0.7, 0% | 0 |
| fl0010-wan2 | Branch 10 – Tampa, FL<br>Heritage Pool Supply Group | Secondary | LTE/5G | Verizon | VZW ICCID …321760 | — | 150 | Up | 63 ms, 8.5, 0.33% | 0 |
| | Branch 11 – Tampa, FL | | | | | | | | | |

**Figure 105.** Circuit table

Filters: carrier, status and **Repeat offenders** (circuits with repeated tickets). Columns: circuit, branch, role (primary or secondary), type (DIA, broadband, MPLS, LTE/5G), carrier, carrier circuit ID, LEC, bandwidth, status, live latency, jitter and loss, and tickets in 90 days.

## Reconciliation tab

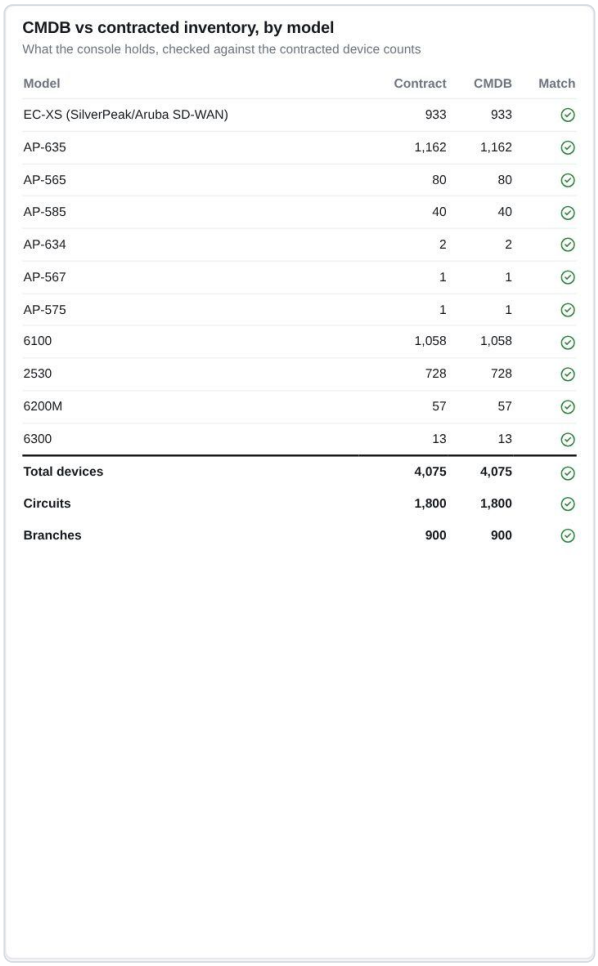
**Contracted inventory**

HPE Aruba Networking pivot – 3,142 devices + 933 EC-XS = 4,075; 1,800 circuits; 900 branch locations

| | |
|---|---|
| **Access Point** | **474** |
| AP-565-US | 24 |
| AP-567-US | 1 |
| AP-575-US | 1 |
| AP-585-US | 20 |
| AP-635-US | 428 |
| **Access Point; Standalone** | **812** |
| AP-565-US | 56 |
| AP-585-US | 20 |
| AP-634-US | 2 |
| AP-635-US | 734 |
| **Switch** | **524** |
| 2530 | 47 |
| 6100 | 473 |
| 6300 | 2 |
| 6200M | 2 |
| **Switch; Stack (Conductor)** | **17** |
| 6300 | 5 |
| 6200M | 12 |
| **Switch; Stack (Member)** | **30** |
| 6200M | 30 |
| **Switch; Stack (Standby)** | **17** |
| 6300 | 5 |
| 6200M | 12 |
| **Switch; Standalone** | **1,268** |
| 2530 | 681 |
| 6100 | 585 |
| 6300 | 1 |
| 6200M | 1 |
| **Total HPE Aruba Networking** | **3,142** |
| **SilverPeak/Aruba SD-WAN (EC-XS)** | **933** |
| **Total SD-WAN, switches and APs** | **4,075** |
| **Total circuits** | **1,800** |

**Figure 106.** Contracted inventory

**Contracted inventory** is the device list from the contract, by role and model: 3,142 HPE Aruba Networking devices plus 933 EC-XS for 4,075 in total, 1,800 circuits and 900 branch locations. It shows how many access points and switches are managed in Aruba Central and how many are standalone: 1,268 switches and 812 APs, 2,080 devices or 51% of the estate, are standalone. Standalone devices are monitored by OpsRamp (SNMP and syslog) but cannot be configured or upgraded from Central; demo 25 brings them under Central. The device panel shows **Managed by** for each device.



**CMDB vs contracted inventory, by model**

What the console holds, checked against the contracted device counts

| Model | Contract | CMDB | Match |
|---|---|---|---|
| EC-XS (SilverPeak/Aruba SD-WAN) | 933 | 933 | ✓ |
| AP-635 | 1,162 | 1,162 | ✓ |
| AP-565 | 80 | 80 | ✓ |
| AP-585 | 40 | 40 | ✓ |
| AP-634 | 2 | 2 | ✓ |
| AP-567 | 1 | 1 | ✓ |
| AP-575 | 1 | 1 | ✓ |
| 6100 | 1,058 | 1,058 | ✓ |
| 2530 | 728 | 728 | ✓ |
| 6200M | 57 | 57 | ✓ |
| 6300 | 13 | 13 | ✓ |
| **Total devices** | **4,075** | **4,075** | ✓ |
| **Circuits** | **1,800** | **1,800** | ✓ |
| **Branches** | **900** | **900** | ✓ |

**Figure 107.** CMDB versus contract

**CMDB vs contracted inventory, by model** checks what the console holds against the contracted counts for every model, circuits and branches. A tick means they match; a mismatch is a CMDB clean-up task.

# 13. Lifecycle & Vulnerabilities

This screen matches every device's running firmware against HPE Aruba Networking security bulletins and lifecycle notices, and plans the fixes. It has four tabs: **Advisories**, **Models and lifecycle**, **Exposure by region** and **Most exposed branches**.

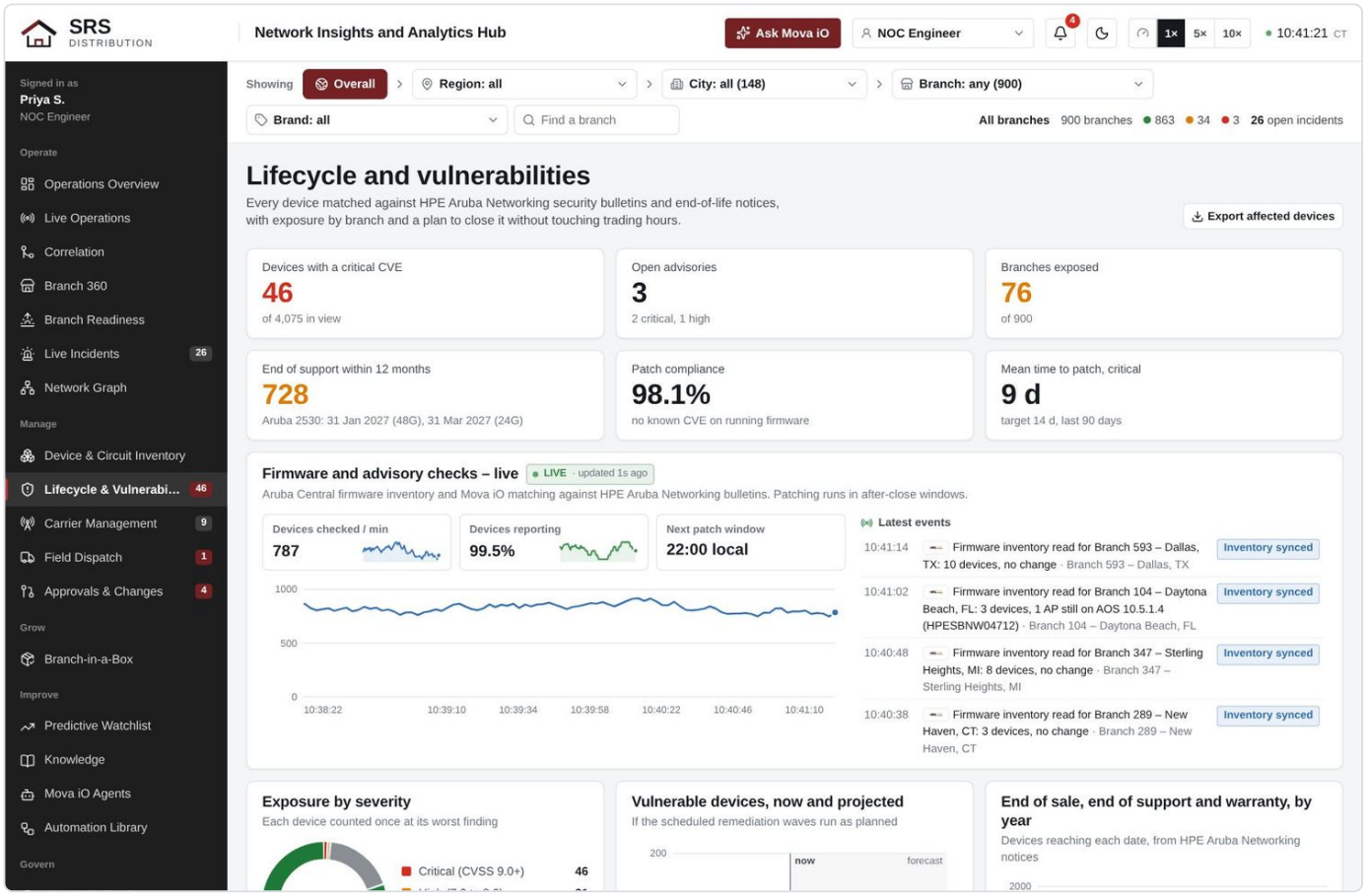



**Figure 108.** Lifecycle & Vulnerabilities

## Tiles

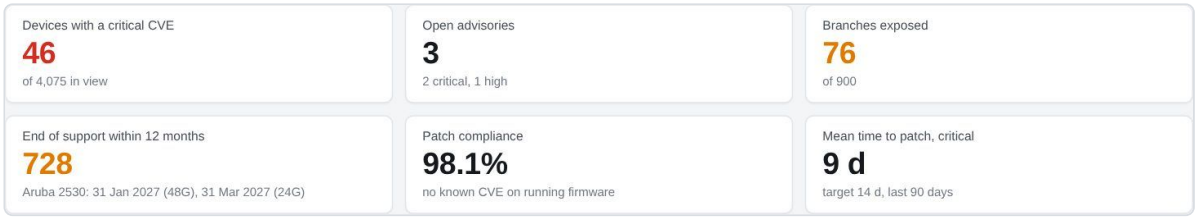



**Figure 109.** Vulnerability tiles

| Tile | Meaning | Target |
|---|---|---|
| Devices with a critical CVE | Devices whose running firmware has a CVSS 9.0+ vulnerability | Falling with each wave |
| Open advisories | Advisories that affect at least one device, by severity | – |

| Tile | Meaning | Target |
|---|---|---|
| Branches exposed | Branches with at least one affected device | – |
| End of support within 12 months | Devices whose end of support falls in the next 12 months: the 728 Aruba 2530 switches (31 Jan 2027 for the 48-port, 31 Mar 2027 for the 24-port) | Falling with the refresh |
| Patch compliance | Share of devices with no known CVE on their running firmware | Rising |
| Mean time to patch, critical | Average days from advisory to patched, last 90 days | 14 days |

## Firmware and advisory checks – live

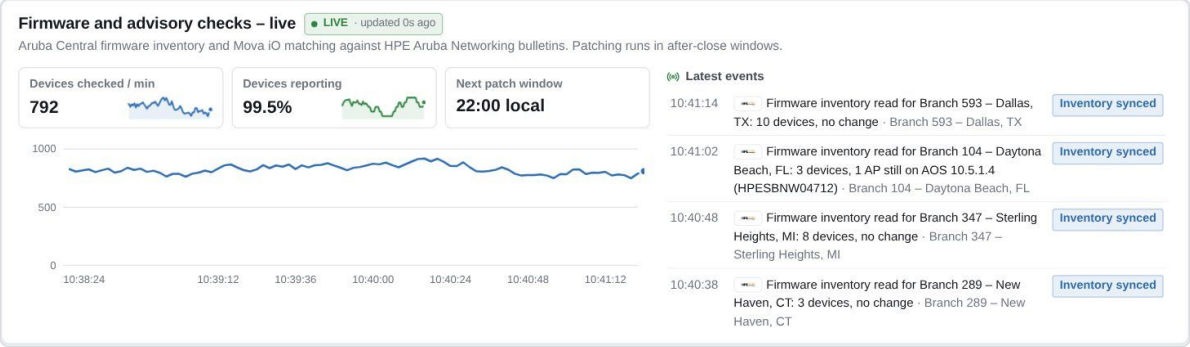



**Figure 110.** Firmware and advisory checks

Devices checked per minute, devices reporting and the next patch window (22:00 local, after close), with the latest inventory and matching events.

## Exposure by severity

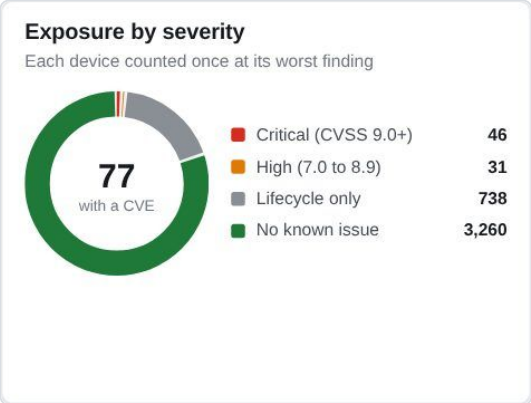



**Figure 111.** Exposure by severity

Every device counted once at its worst finding: critical (CVSS 9.0+), high (7.0 to 8.9), lifecycle only (no CVE, but near end of support or on an older software release) and no known issue.

## Vulnerable devices, now and projected

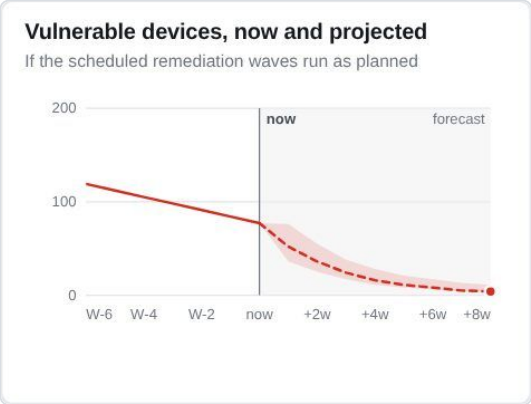



**Figure 112.** Vulnerable devices

Vulnerable devices for the last six weeks and, after "now", the forecast for six weeks if the scheduled remediation waves run as planned. A forecast that does not fall means waves are missing.

## End of sale, end of support and warranty, by year

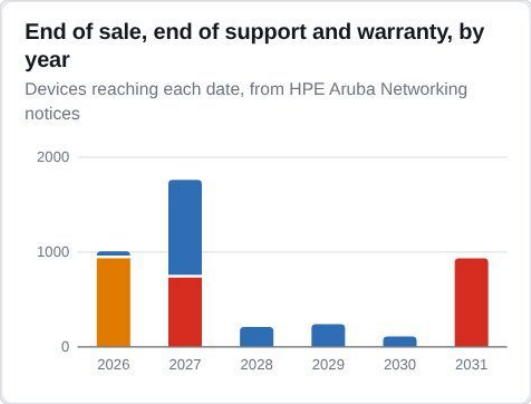



**Figure 113.** End of sale, end of support and warranty

Devices reaching each date in each year, from HPE Aruba Networking notices: **end of sale** (the last day the model can be ordered), **end of support** (the last day of vendor support and security fixes) and warranty or contract end. The 2026 end-of-sale bar is the 933 EdgeConnect EC-XS; the 2027 end-of-support bar is the 728 Aruba 2530 switches; the 2031 bar is the EC-XS end of support.

## Advisory cards




**Figure 114.** Advisory card

One card per advisory:

| Field | Meaning |
|---|---|
| Severity and CVSS | Critical, High or Lifecycle, with the score |
| Bulletin and date | HPE Aruba Networking bulletin number and publication date |
| Title and CVEs | What the flaw is and its CVE numbers |
| Product | Which platform and software is affected |
| Fixed in | The release that fixes it |
| Mitigation until patched | What can be done now without a reboot |
| Devices and branches affected | Counts in the current view |
| Status | Not yet scheduled, waves scheduled, mitigated or patched |

**Plan patch waves** creates region-by-region changes after branch close, canary first. **Mitigate now** applies the policy mitigation on every branch; it is a change with no client impact, so it can run sooner. **Devices** lists the affected devices. Lifecycle cards offer **Plan refresh** instead, using the Branch-in-a-Box standard kit.

## Models and lifecycle tab

| Model | Software | Devices | Lifecycle | End of sale | End of support | With open CVE | Warranty or contract lapsed | Replacement |
|---|---|---|---|---|---|---|---|---|
| **Aruba 2530 48G PoE+** J9772A<br>HPE Aruba Networking, Aruba 2530 (AOS-S) | AOS-S | 551 | End of sale | **31 Jan 2022** | **31 Jan 2027**<br>3 months left | 0 | 0 | CX 6100 48G Class4 PoE |
| **Aruba 2530 24G PoE+** J9773A<br>HPE Aruba Networking, Aruba 2530 (AOS-S) | AOS-S | 177 | End of sale | **31 Mar 2022** | **31 Mar 2027**<br>5 months left | 0 | 0 | CX 6100 24G Class4 PoE |
| **Aruba CX 6100 48G Class4 PoE**<br>HPE Aruba Networking, Aruba CX 6100 | AOS-CX | 1,058 | Current | Not announced | Not announced | 26 | 0 | — |
| **Aruba CX 6300M 48G Class4 PoE**<br>HPE Aruba Networking, Aruba CX 6300 | AOS-CX | 13 | Current | Not announced | Not announced | 0 | 0 | — |
| **EdgeConnect EC-XS** JM962A<br>HPE Aruba Networking, EdgeConnect SD-WAN | ECOS | 933 | End of sale | **31 Jan 2026** | **31 Jan 2031**<br>51 months left | 31 | 32 | EdgeConnect 10104 |
| **Aruba CX 6200M 48G Class4 PoE**<br>HPE Aruba Networking, Aruba CX 6200M | AOS-CX | 57 | Current | Not announced | Not announced | 1 | 0 | — |
| **AP-635 indoor**<br>HPE Aruba Networking, 630 Series (Wi-Fi 6E) | AOS 10 | 1,162 | Current | Not announced | Not announced | 18 | 0 | — |
| **AP-634 indoor**<br>HPE Aruba Networking, 630 Series (Wi-Fi 6E) | AOS 10 | 2 | Current | Not announced | Not announced | 1 | 0 | — |
| **AP-565 outdoor**<br>HPE Aruba Networking, 560 Series outdoor | AOS 10 | 80 | Current | Not announced | Not announced | 0 | 0 | — |
| **AP-567 outdoor**<br>HPE Aruba Networking, 560 Series outdoor | AOS 10 | 1 | Current | Not announced | Not announced | 0 | 0 | — |
| **AP-575 outdoor**<br>HPE Aruba Networking, 570 Series outdoor | AOS 10 | 1 | Current | Not announced | Not announced | 0 | 0 | — |
| **AP-585 outdoor**<br>HPE Aruba Networking, 580 Series outdoor | AOS 10 | 40 | Current | Not announced | Not announced | 0 | 0 | — |

Sources: HPE Aruba Networking end-of-sale notice for the 2530 24/48-port Gigabit switches: 48G PoE+ (J9772A) end of sale 31 Jan 2022, end of support 31 Jan 2027; 24G PoE+ (J9773A) end of sale 31 Mar 2022, end of support 31 Mar 2027. HPE Aruba Networking EdgeConnect EC-XS (JM962A): end of sale 31 Jan 2026, end of support 31 Jan 2031 (EdgeConnect lifecycle policy: support ends 5 years after end of sale). No end of sale has been announced for the CX 6100, 6200M and 6300M switches or the AP-634, AP-635, AP-565, AP-567, AP-575 and AP-585 access points.

The 728 Aruba 2530 switches need replacing before their end of support; refresh waves use the Branch-in-a-Box kit. EC-XS units stay supported until 31 Jan 2031; new and replacement units are EdgeConnect 10104.

**Figure 115.** Models and lifecycle

One row per model, with the Aruba 2530 split into its 48-port and 24-port SKUs because their dates differ:

| Column | Meaning |
|---|---|
| Lifecycle | **Current** (no end of sale announced) or **End of sale** (no longer sold, still supported). Red when end of support is within 12 months |
| End of sale | The last day HPE Aruba Networking sold the model, or "Not announced" |
| End of support | The last day of vendor support and security fixes, with the months left, or "Not announced" |
| With open CVE | Devices of this model with an open vulnerability |
| Warranty or contract lapsed | EdgeConnect support contracts not renewed; zero for lifetime-warranty models |
| Replacement | The model it is replaced with |

The dates come from HPE Aruba Networking notices:

| Model | End of sale | End of support | Replacement |
|---|---|---|---|
| Aruba 2530 48G PoE+ (J9772A) | 31 Jan 2022 | 31 Jan 2027 | CX 6100 48G Class4 PoE |
| Aruba 2530 24G PoE+ (J9773A) | 31 Mar 2022 | 31 Mar 2027 | CX 6100 24G Class4 PoE |
| EdgeConnect EC-XS (JM962A) | 31 Jan 2026 | 31 Jan 2031 | EdgeConnect 10104 |
| CX 6100, CX 6200M, CX 6300M | Not announced | Not announced | – |
| AP-634, AP-635, AP-565, AP-567, AP-575, AP-585 | Not announced | Not announced | – |

HPE's lifecycle policy ends support five years after end of sale. The RFP gives the 2530 date as 1 Feb 2027, the day after the 48-port end of support. EC-XS units already installed stay supported until 2031; new and replacement units are EdgeConnect 10104.

## Exposure by region tab

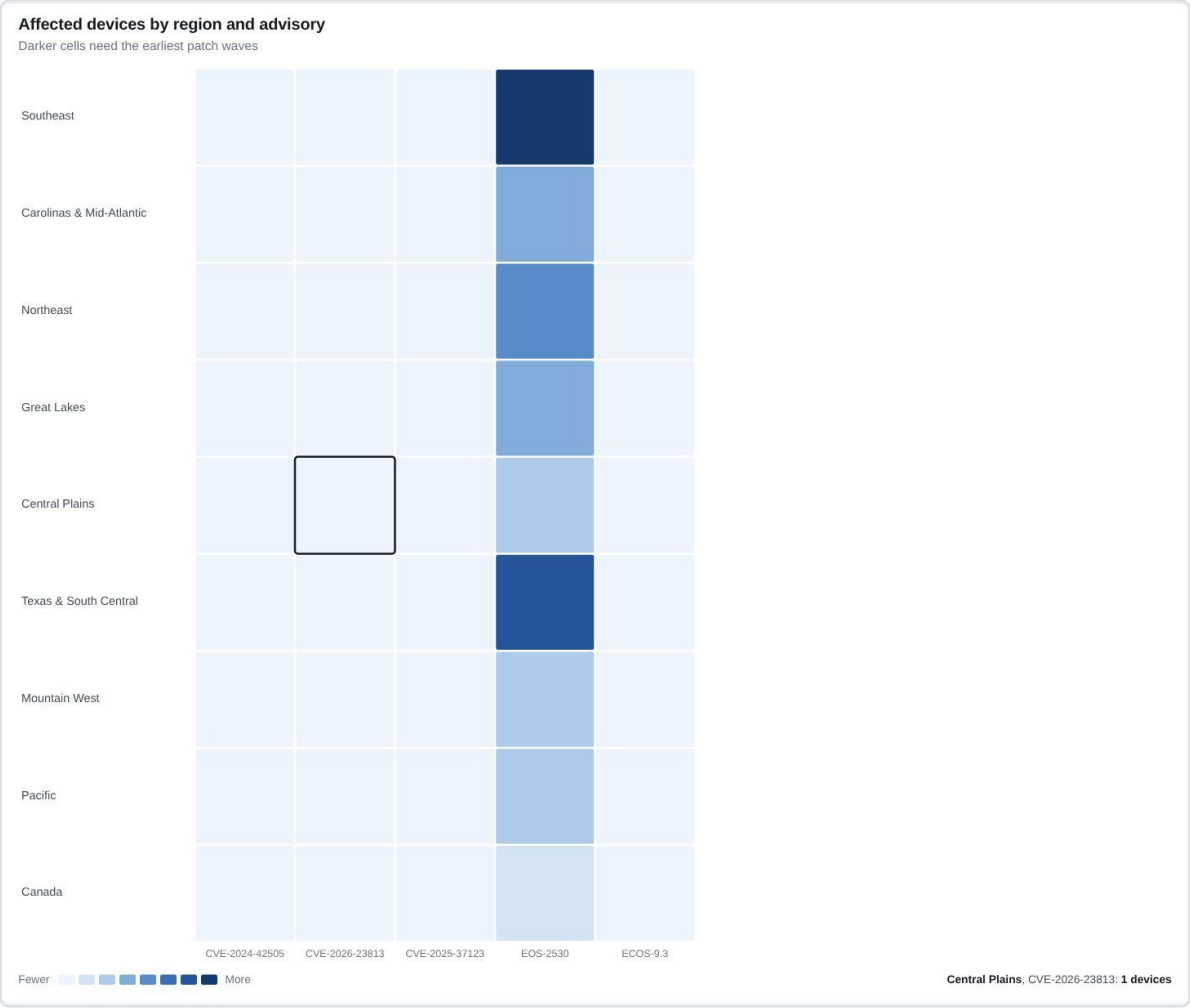



**Figure 116.** Affected devices by region and advisory

A heat map of affected devices by region (rows) and advisory (columns). Darker cells need the earliest patch waves. Hover a cell for the count.

## Most exposed branches tab

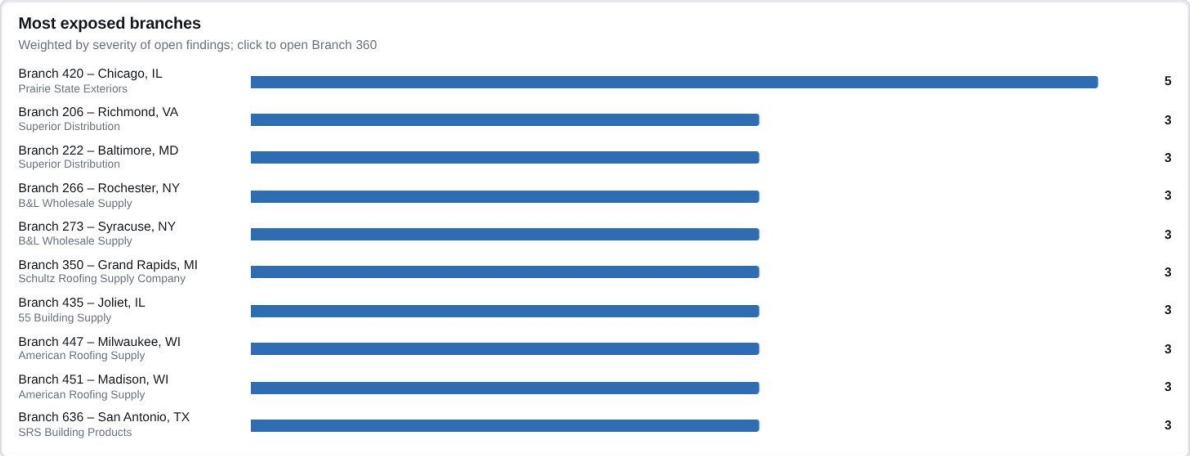



**Figure 117.** Most exposed branches

Branches ranked by a score weighted by the severity of their open findings. Click a branch to open its Branch 360 vulnerability tab.

# 14. Carrier Management

Carrier Management tracks every carrier case from opening to close, scores the carriers and drives down "no trouble found" (NTF) closures. Its tabs are **Open cases**, **Closed (7 d)**, **Scorecard**, **Repeat offenders** and **NTF tracking**.

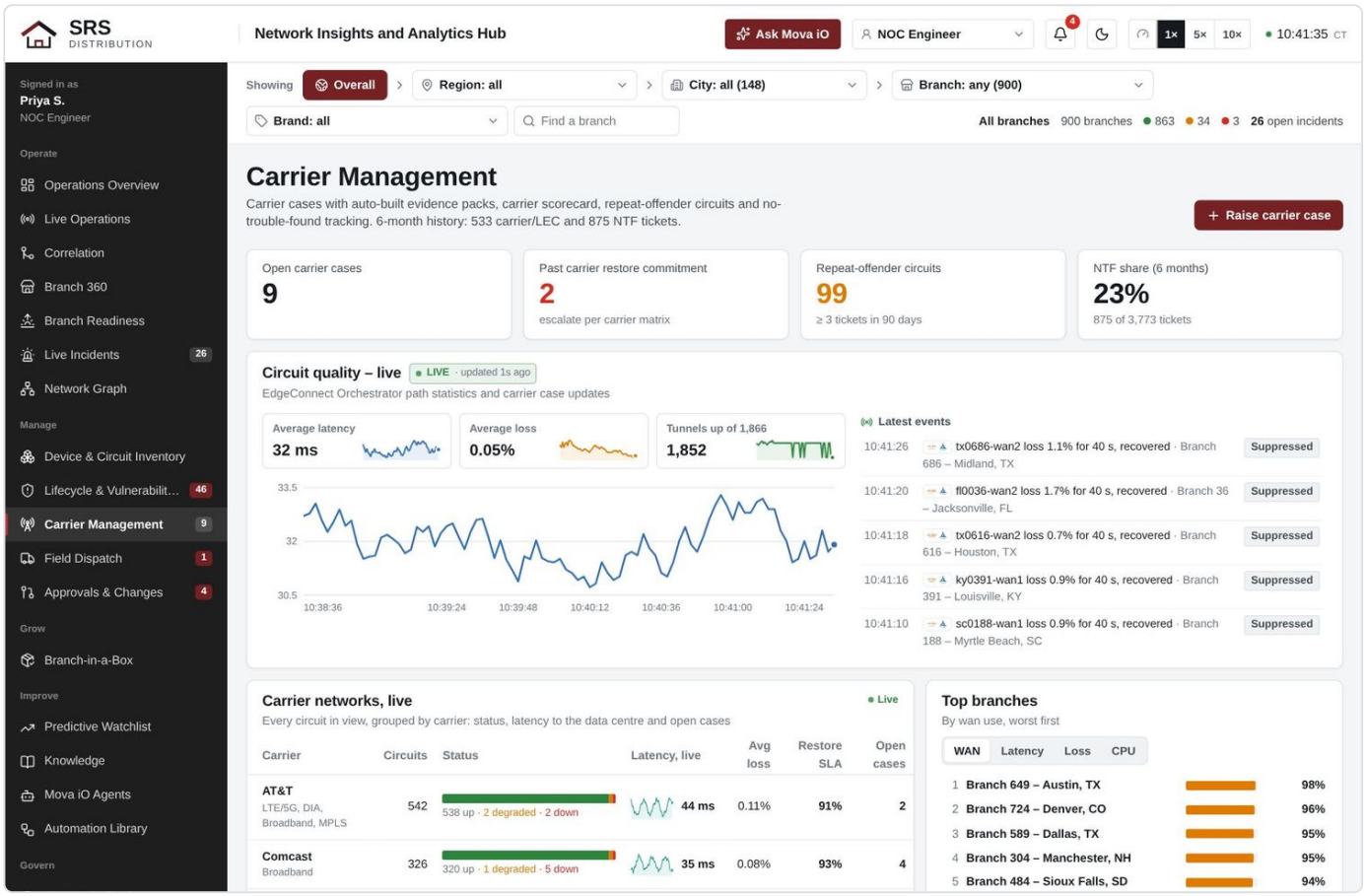



**Figure 118.** Carrier Management

## Tiles




**Figure 119.** Carrier tiles

| Tile | Meaning | Act when |
|---|---|---|
| Open carrier cases | Cases open with any carrier | – |
| Past carrier restore commitment | Open cases past the time the carrier committed to | Above zero: escalate per the carrier matrix |
| Repeat-offender circuits | Circuits with three or more tickets in 90 days | Each one should have a problem record |
| NTF share (6 months) | Tickets the carrier closed as no trouble found: 875 of 3,773 | The aim is to bring this down with evidence |

## Circuit quality – live




**Figure 120.** Circuit quality

Average latency to the data centre, average loss and tunnels up, with a chart and the latest path events from Orchestrator and carrier case updates. Short loss bursts that recover are marked Suppressed: they are watched, not ticketed.

## Carrier networks, live

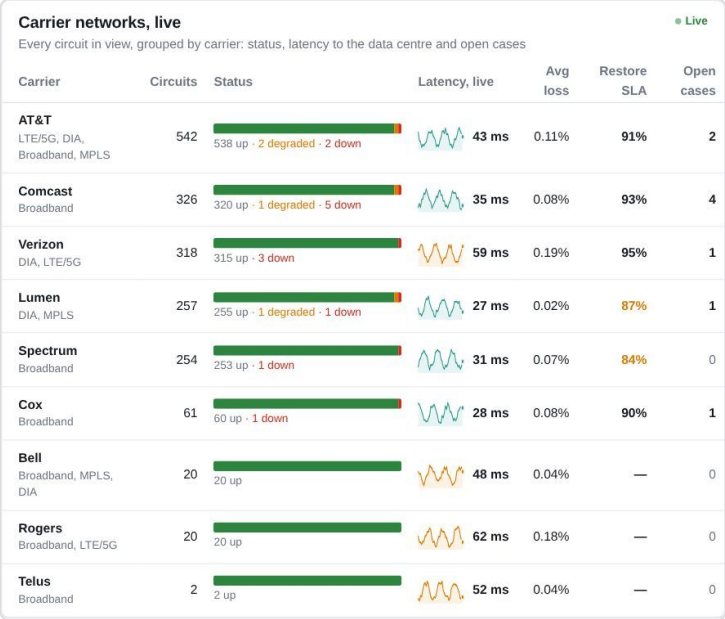



**Figure 121.** Carrier networks

One row per carrier for every circuit in view:

| Column | Meaning |
|---|---|
| Carrier | Name and the circuit types it provides (DIA, broadband, MPLS, LTE/5G) |
| Circuits | How many lines in view |
| Status | A bar of up (green), degraded (amber) and down (red), with counts |
| Latency, live | Average latency to the data centre with its recent trend; amber above 45 ms |
| Avg loss | Average packet loss |

| Column | Meaning |
|---|---|
| Restore SLA | Share of carrier faults restored within the carrier's commitment over six months; amber below 90% |
| Open cases | Click to see them |

## Circuits likely to fail this week




**Figure 122.** Circuits likely to fail

Circuits the prediction model expects to fail, with the probability, the time window, the carrier, the signal behind it and **Raise case**. A proactive case raised while the branch is still on its primary gives the carrier time to fix it before anyone notices.

## Carrier restore time, 6 months

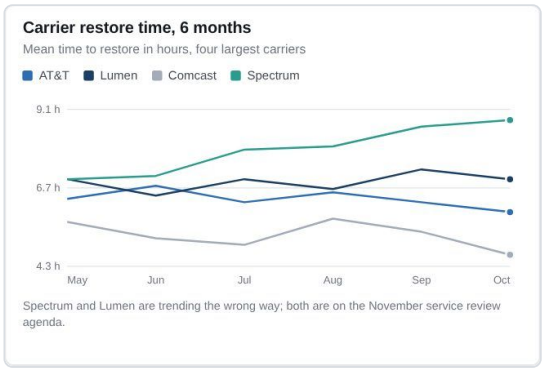
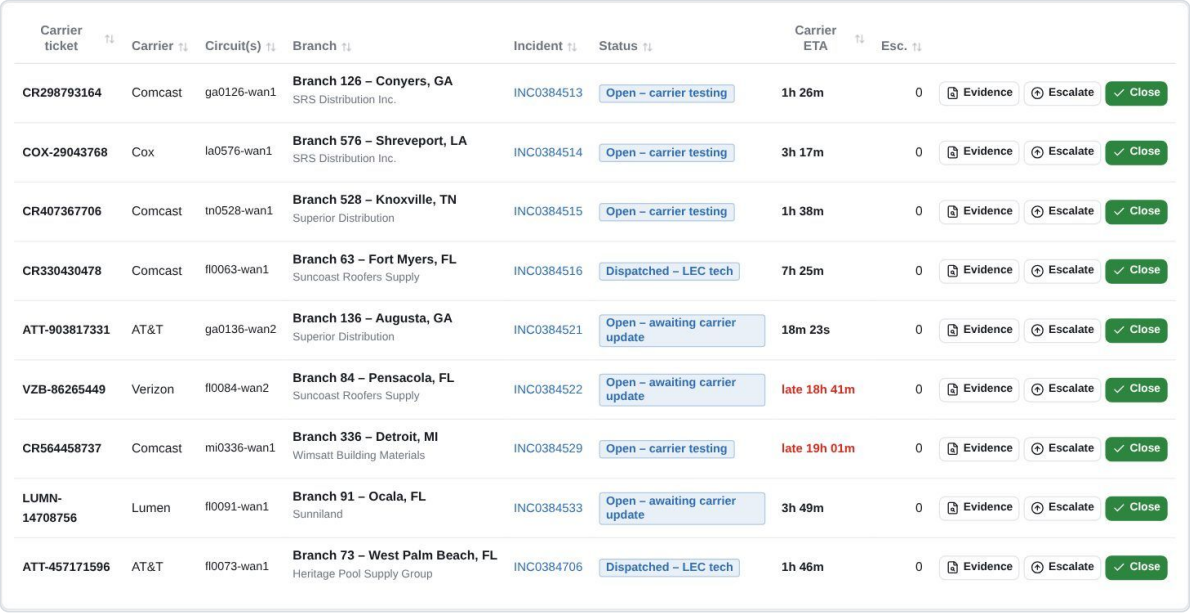



**Figure 123.** Carrier restore time

Mean time to restore in hours for the four largest carriers, month by month, with a note on which carriers are trending the wrong way and are on the next service review agenda.

## Open cases



| Carrier ticket | Carrier | Circuit(s) | Branch | Incident | Status | Carrier ETA | Esc. | |
|---|---|---|---|---|---|---|---|---|
| CR298793164 | Comcast | ga0126-wan1 | Branch 126 – Conyers, GA<br>SRS Distribution Inc. | INC0384513 | Open – carrier testing | 1h 26m | 0 | Evidence Escalate Close |
| COX-29043768 | Cox | la0576-wan1 | Branch 576 – Shreveport, LA<br>SRS Distribution Inc. | INC0384514 | Open – carrier testing | 3h 17m | 0 | Evidence Escalate Close |
| CR407367706 | Comcast | tn0528-wan1 | Branch 528 – Knoxville, TN<br>Superior Distribution | INC0384515 | Open – carrier testing | 1h 38m | 0 | Evidence Escalate Close |
| CR330430478 | Comcast | fl0063-wan1 | Branch 63 – Fort Myers, FL<br>Suncoast Roofers Supply | INC0384516 | Dispatched – LEC tech | 7h 25m | 0 | Evidence Escalate Close |
| ATT-903817331 | AT&T | ga0136-wan2 | Branch 136 – Augusta, GA<br>Superior Distribution | INC0384521 | Open – awaiting carrier update | 18m 23s | 0 | Evidence Escalate Close |
| VZB-86265449 | Verizon | fl0084-wan2 | Branch 84 – Pensacola, FL<br>Suncoast Roofers Supply | INC0384522 | Open – awaiting carrier update | late 18h 41m | 0 | Evidence Escalate Close |
| CR564458737 | Comcast | mi0336-wan1 | Branch 336 – Detroit, MI<br>Wimsatt Building Materials | INC0384529 | Open – carrier testing | late 19h 01m | 0 | Evidence Escalate Close |
| LUMN-14708756 | Lumen | fl0091-wan1 | Branch 91 – Ocala, FL<br>Sunniland | INC0384533 | Open – awaiting carrier update | 3h 49m | 0 | Evidence Escalate Close |
| ATT-457171596 | AT&T | fl0073-wan1 | Branch 73 – West Palm Beach, FL<br>Heritage Pool Supply Group | INC0384706 | Dispatched – LEC tech | 1h 46m | 0 | Evidence Escalate Close |

**Figure 124.** Open carrier cases

| Column | Meaning |
|---|---|
| Carrier ticket | The carrier's case number |
| Carrier, circuit(s), branch | Which line; a master case lists several circuits |
| Incident | The linked ServiceNow incident |
| Status | For example Open – carrier testing, Dispatched – LEC tech, Open – proactive |
| Carrier ETA | Time left to the carrier's restore commitment |
| Esc. | Escalation level reached |

**Evidence** shows the evidence pack attached (loss and jitter graphs, MTR traces, loopback results, timestamps). **Escalate** moves the case up the carrier's escalation matrix. **Close** closes it with a close code. The **Closed (7 d)** tab lists recent cases with their close code: carrier fault fixed, or NTF.

## Scorecard tab

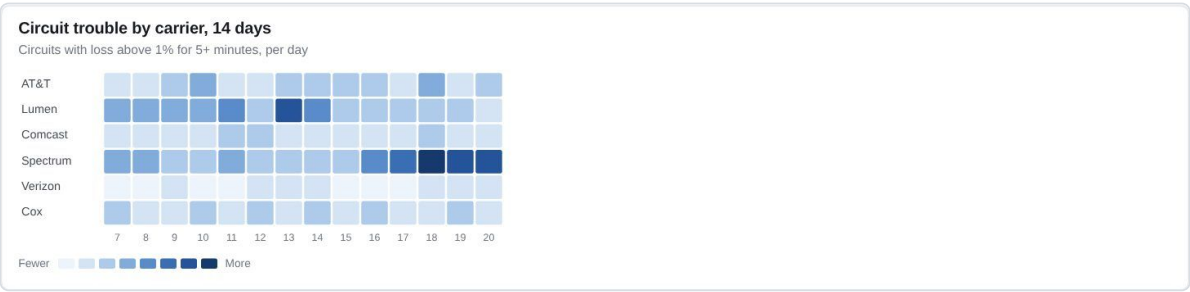



**Figure 125.** Circuit trouble by carrier

**Circuit trouble by carrier, 14 days:** circuits with loss above 1% for five minutes or more, per carrier per day. Darker cells mean more circuits in trouble; a dark run across days is a carrier-side problem.

**Carrier scorecard – last 6 months**

533 carrier/LEC and 875 no-trouble-found tickets, by carrier

| Carrier | Circuits | Carrier faults | NTF | Faults / 100 circuits | MTTR (h) | Restore commit met | Repeat ckts |
|---|---|---|---|---|---|---|---|
| AT&T | 542 | 148 | 231 | 27.3 | 6.1 | 91% | 9 |
| Lumen | 257 | 131 | 214 | 51 | 7.4 | 87% | 11 |
| Comcast | 326 | 97 | 168 | 29.8 | 5.2 | 93% | 6 |
| Spectrum | 254 | 86 | 152 | 33.9 | 8.3 | 84% | 10 |
| Verizon | 318 | 41 | 61 | 12.9 | 4.6 | 95% | 3 |
| Cox | 61 | 22 | 37 | 36.1 | 6.8 | 90% | 2 |
| Canadian carriers | 42 | 8 | 12 | 19 | 5.9 | 94% | 0 |

**Figure 126.** Carrier scorecard

**Carrier scorecard, last 6 months:** circuits, carrier faults, NTF tickets, faults per 100 circuits (the fair comparison between large and small carriers), mean time to restore in hours, restore commitment met and repeat-offender circuits. Use it in the monthly carrier review.

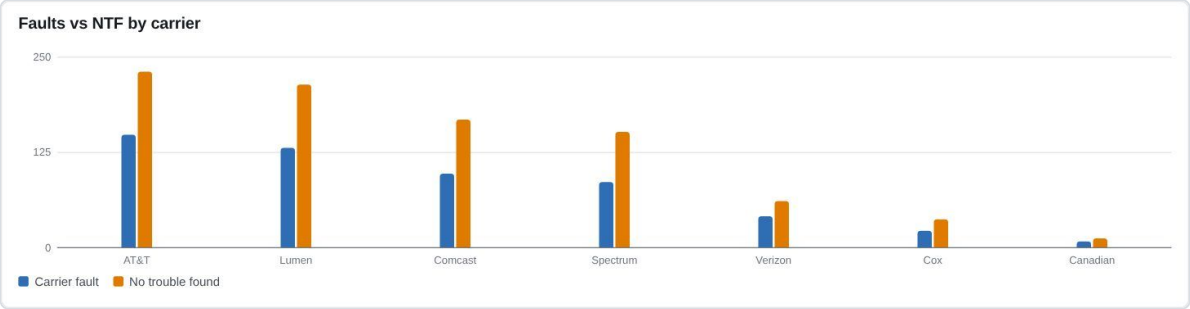



**Figure 127.** Faults versus NTF

**Faults vs NTF by carrier:** carrier faults and NTF tickets side by side. A carrier with more NTF than faults is closing tickets it should be fixing.

## Repeat offenders tab

| Circuit | Branch | Carrier | Type | LEC | Tickets 90d | NTF 90d | |
|---|---|---|---|---|---|---|---|
| fl0043-wan1 | **Branch 43 – Miami, FL**<br>Suncoast Roofers Supply | AT&T | DIA | Frontier | 8 | 1 | Problem record Carrier review |
| il0414-wan1 | **Branch 414 – Chicago, IL**<br>Prairie State Exteriors | Comcast | Broadband | — | 8 | 2 | Problem record Carrier review |
| fl0008-wan2 | **Branch 8 – Tampa, FL**<br>Heritage Pool Supply Group | Verizon | LTE/5G | — | 7 | 0 | Problem record Carrier review |
| fl0070-wan2 | **Branch 70 – Sarasota, FL**<br>Suncoast Roofers Supply | AT&T | LTE/5G | — | 7 | 1 | Problem record Carrier review |
| ny0276-wan2 | **Branch 276 – Buffalo, NY**<br>Weatherpanel | Verizon | LTE/5G | — | 7 | 0 | Problem record Carrier review |
| oh0384-wan1 | **Branch 384 – Cincinnati, OH**<br>Marsh Building Products | Spectrum | Broadband | — | 7 | 1 | Problem record Carrier review |
| mo0495-wan1 | **Branch 495 – St. Louis, MO**<br>Arrowhead Building Supply | Spectrum | Broadband | — | 7 | 2 | Problem record Carrier review |
| tn0527-wan2 | **Branch 527 – Knoxville, TN**<br>Superior Distribution | AT&T | Broadband | — | 7 | 1 | Problem record Carrier review |
| co0719-wan1 | **Branch 719 – Denver, CO**<br>American Roofing Supply | Comcast | Broadband | — | 7 | 3 | Problem record Carrier review |
| az0741-wan1 | **Branch 741 – Phoenix, AZ**<br>American Roofing Supply | AT&T | DIA | Lumen | 7 | 1 | Problem record Carrier review |
| wa0819-wan1 | **Branch 819 – Bellingham, WA**<br>CB Wholesale | Lumen | DIA | Brightspeed | 7 | 0 | Problem record Carrier review |
| on0881-wan1 | **Branch 881 – Toronto, ON**<br>SRS Canada (illustrative) | Rogers | Broadband | — | 7 | 0 | Problem record Carrier review |
| ga0107-wan1 | **Branch 107 – Atlanta, GA**<br>Superior Distribution | AT&T | DIA | Windstream | 6 | 3 | Problem record Carrier review |
| nc0142-wan1 | **Branch 142 – Charlotte, NC**<br>Heritage Pool Supply Group | Lumen | MPLS | Frontier | 6 | 2 | Problem record Carrier review |
| fl0002-wan1 | **Branch 2 – Tampa, FL**<br>Heritage Pool Supply Group | Comcast | Broadband | — | 5 | 2 | Problem record Carrier review |
| fl0052-wan2 | **Branch 52 – Miami, FL**<br>Heritage Pool Supply Group | Comcast | Broadband | — | 5 | 1 | Problem record Carrier review |
| fl0079-wan2 | **Branch 79 – West Palm Beach, FL**<br>Heritage Pool Supply Group | Comcast | Broadband | — | 5 | 1 | Problem record Carrier review |
| ga0114-wan2 | **Branch 114 – Atlanta, GA**<br>Heritage Landscape Supply … | Verizon | LTE/5G | — | 5 | 0 | Problem record Carrier review |
| sc0188-wan2 | **Branch 188 – Myrtle Beach, SC**<br>Atlantic Roofing Distributors | AT&T | LTE/5G | — | 5 | 0 | Problem record Carrier review |
| pa0232-wan1 | **Branch 232 – Philadelphia, PA**<br>Marvic Supply | Lumen | DIA | AT&T | 5 | 3 | Problem record Carrier review |
| ct0286-wan1 | **Branch 286 – Hartford, CT**<br>Metro Roofing Supplies | Comcast | Broadband | — | 5 | 0 | Problem record Carrier review |

**Figure 128.** Repeat offenders

Circuits with three or more tickets in 90 days: circuit, branch, carrier, type, LEC, tickets and NTF in 90 days. **Problem record** opens a problem in ServiceNow; **Carrier review** asks the carrier's account team for a chronic-circuit review instead of another repair ticket.

## NTF tracking tab

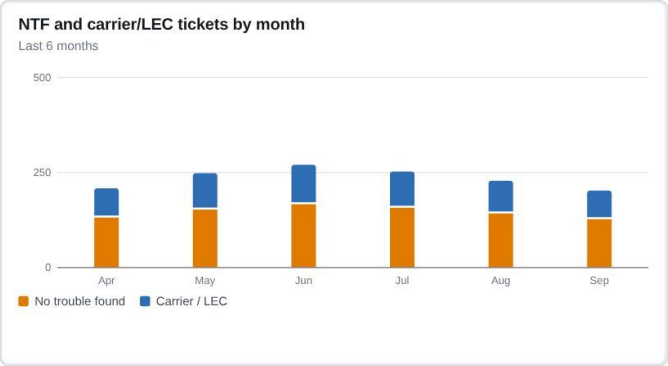



**Figure 129.** NTF and carrier tickets by month

NTF and carrier/LEC tickets per month for six months. The target is a falling NTF bar.

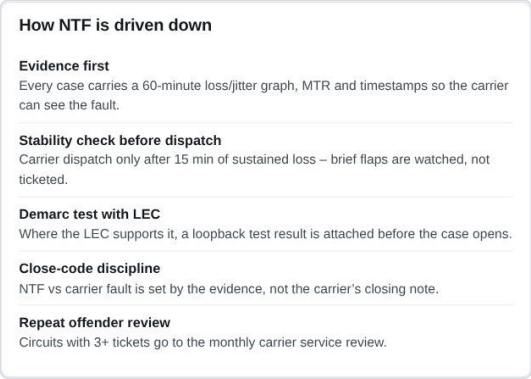



**Figure 130.** How NTF is driven down

The four rules that bring NTF down:

1. **Evidence first:** every case carries a 60-minute loss and jitter graph, an MTR trace and timestamps, so the carrier can see the fault.
2. **Stability check before dispatch:** the carrier is dispatched only after 15 minutes of sustained loss; brief flaps are watched, not ticketed.
3. **Demarc test with the LEC:** where the LEC supports it, a loopback result is attached before the case opens.
4. **Close-code discipline:** NTF or carrier fault is set by the evidence, not by the carrier's closing note.

# 15. Field Dispatch

Field Dispatch matches work orders to technicians by skill, depot spares and branch hours, and tracks every visit. Its tabs are **Work orders**, **Technicians** and **Depot spares**. It is the Field Technician's landing page. Remote fixes come first; a visit is booked only when the fix needs hands on site.

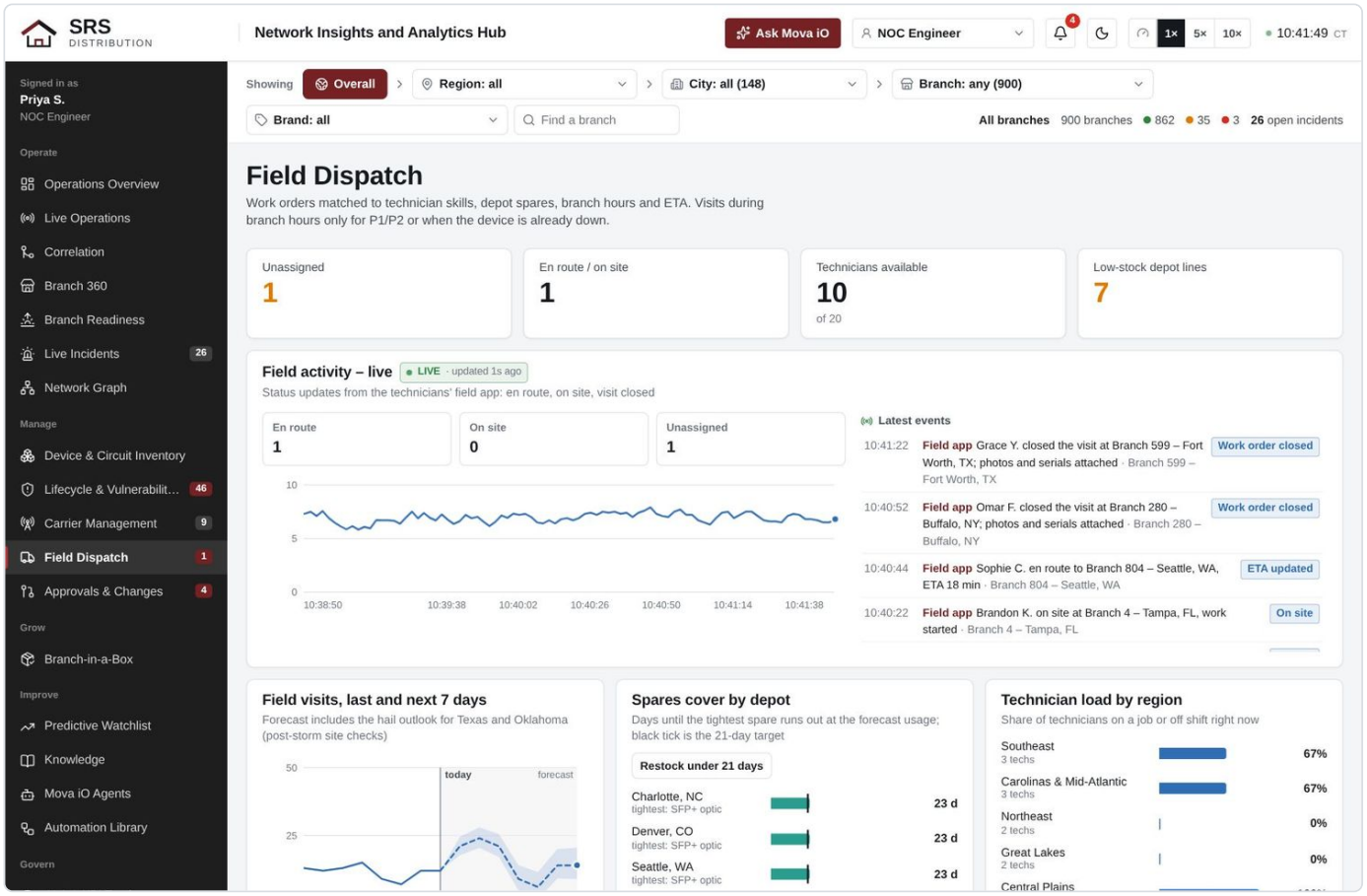



**Figure 131.** Field Dispatch

## Tiles

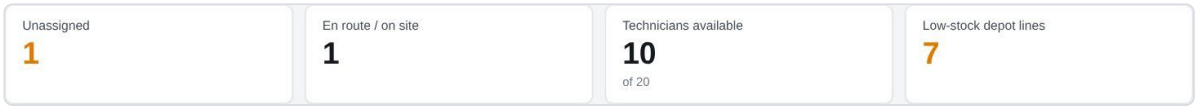



**Figure 132.** Dispatch tiles

| Tile | Meaning |
|---|---|
| Unassigned | Work orders with no technician yet (maroon badge in the menu) |
| En route / on site | Visits in progress |
| Technicians available | Technicians free now, out of the total |
| Low-stock depot lines | Spare lines below their minimum at any depot |

## Field activity – live

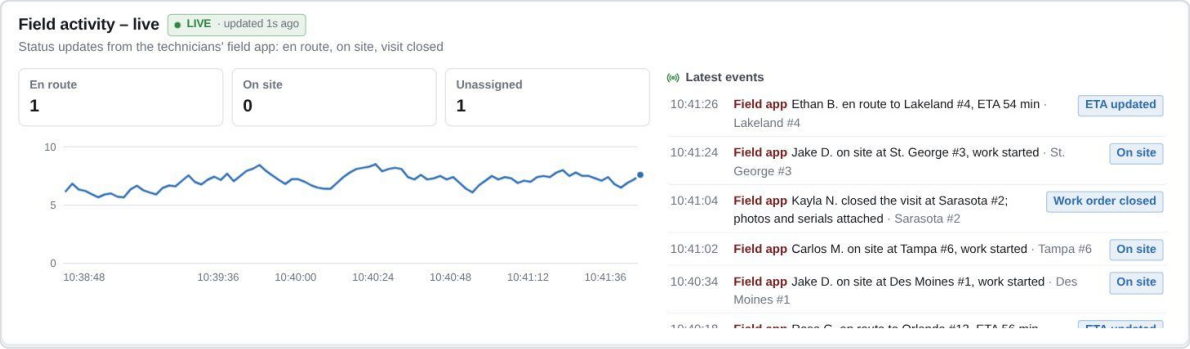



**Figure 133.** Field activity

Status updates from the technicians' field app: en route with ETA, on site, visit closed.

## Forecasts and capacity

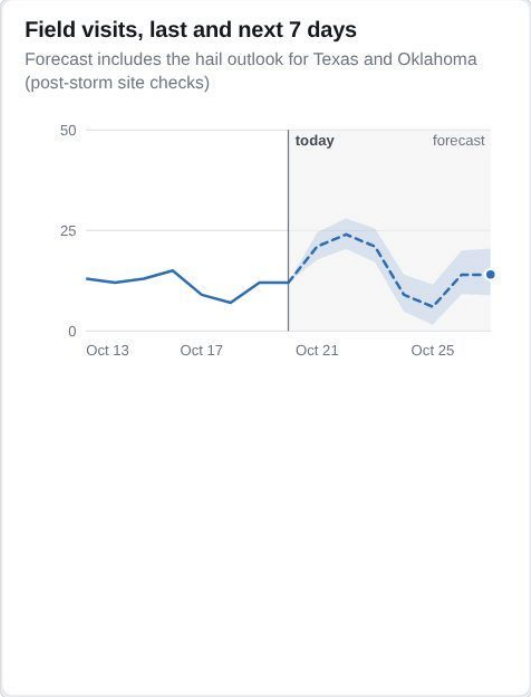



**Figure 134.** Field visits

**Field visits, last and next 7 days:** visits per day, with a forecast that includes the weather outlook, such as post-storm site checks after hail.




**Figure 135.** Spares cover by depot

**Spares cover by depot:** for each depot, the days until its tightest spare runs out at the forecast usage, and which part that is. The black tick is the 21-day target; bars under it need a restock, which **Restock under 21 days** raises.

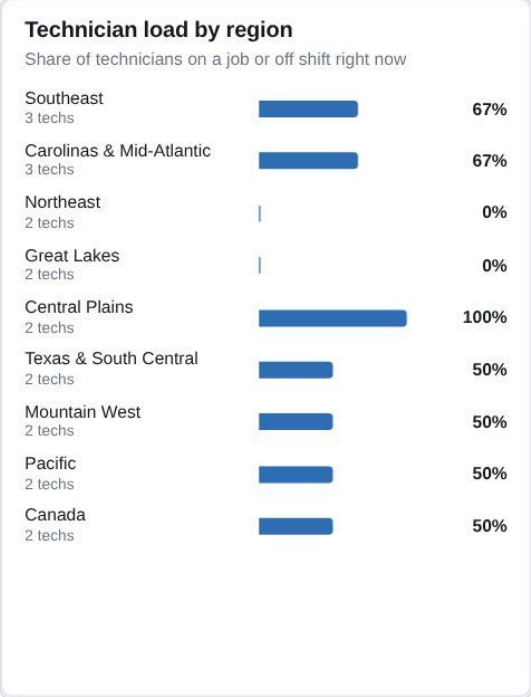



**Figure 136.** Technician load by region

**Technician load by region:** the share of technicians on a job or off shift now, per region, with the head count. A region at 100% has no one free for a new P1.

Field visits are SRS field technical services, booked and directed by the Movate NOC. Most are booked from the root cause: Mova iO's Dispatch Agent prepares the job, skills, spare and technician in **Mova iO: who fixes it** (chapter 3); once an engineer approves, it books the work order here with its incident and reserves the spare.

## Onsite work pack

Every work order carries an onsite work pack that Mova iO assembles, so the technician arrives prepared instead of starting the investigation again. **Work pack** on a work order (or **Open the work pack** in the root-cause view) opens it. The technician sees it on a tablet and starts with the **Summary**; the NOC sees the same pack in the console.

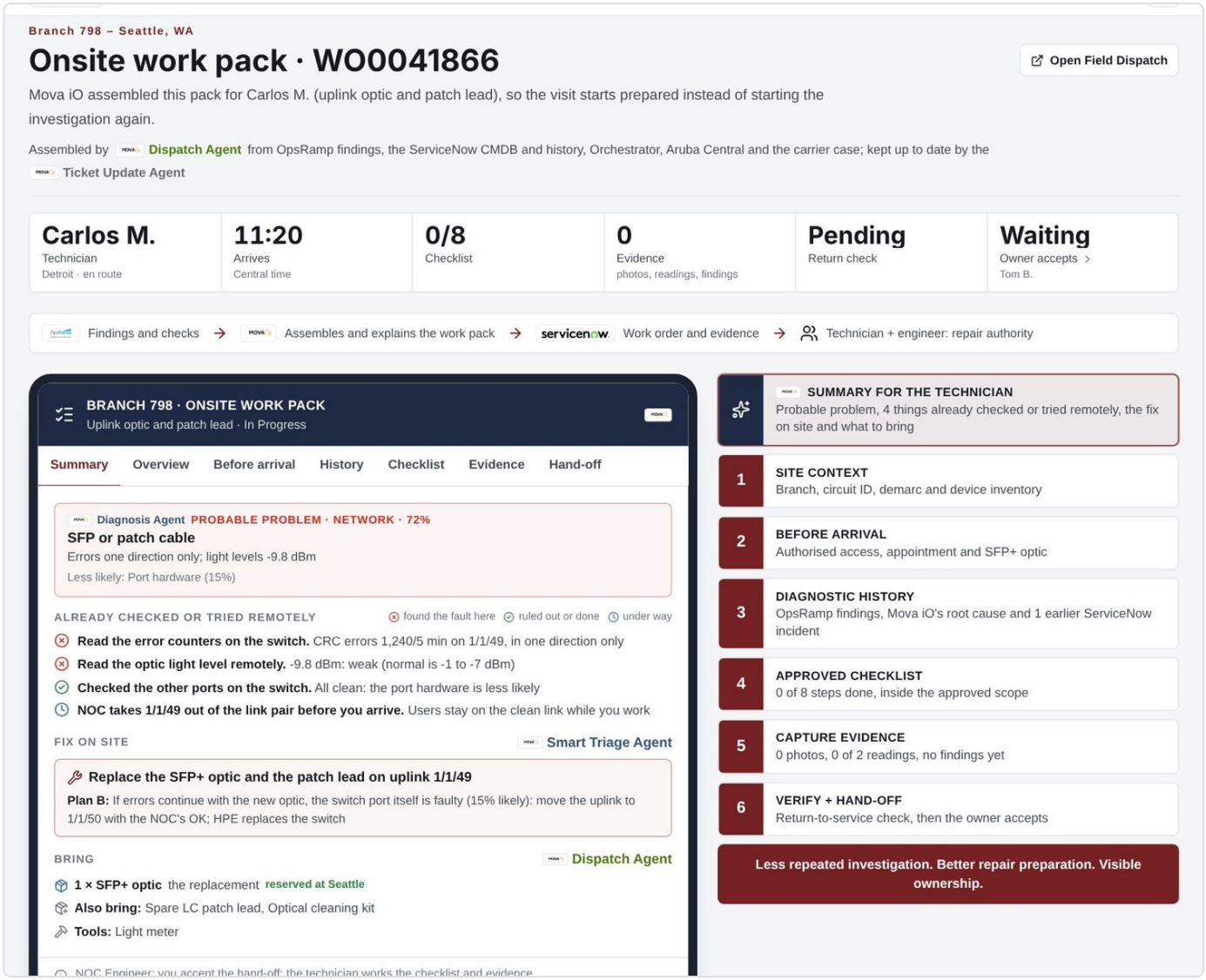



**Figure 137.** The onsite work pack for WO0041866: the tablet view and the six parts Mova iO assembled

The strip under the numbers shows where it comes from: **OpsRamp** findings and checks, **Mova iO** assembles and explains the pack, **ServiceNow** holds the work order and the evidence, and the technician and the NOC engineer share the repair authority.

| Part | Tab | What is in it |
|---|---|---|
| Summary for the technician | Summary | Mova iO's summary, the first thing the technician reads: the probable problem with its confidence and evidence, and what is less likely; what Mova iO and the NOC already checked or tried remotely and what each showed (found the fault here, ruled out or done, under way); the fix to carry out on site with a plan B; and what to bring: the replacement part (reserved at the depot), anything to bring in case, and the tools |
| 1. Site context | Overview | The branch, local time and hours, the circuit IDs from the CMDB, where the demarc is, the service affected and the status; the network path (circuit, appliance, switch, access point) with the fault in red; and the key devices with model, serial and location |
| 2. Before arrival | Before arrival | Authorised access and site rules (sign-in, hi-vis in the yard, telling the counter before unplugging anything), the appointment, the parts checklist with the depot where the Dispatch Agent reserved them, and the tools for the job |
| 3. Diagnostic history | History | Mova iO's root cause with its confidence and evidence, the OpsRamp findings in time order, what Mova iO already tried remotely, the carrier or HPE case, and earlier ServiceNow incidents at the branch in the last 90 days |
| 4. Approved checklist | Checklist | The steps for this kind of fix, approved by the ticket owner. Safe steps stay inside the approved scope; anything else needs the NOC engineer's OK first |
| 5. Capture evidence | Evidence | Photo slots for the job (for example the port before, the new optic seated, the light meter), readings with the range they must fall in (checked by the Verification Agent), and the technician's findings |
| 6. Verify + hand-off | Hand-off | The Verification Agent's return-to-service check, the receiving owner who accepts, what goes back to the depot or HPE, and how to ask for anything outside the approved steps |

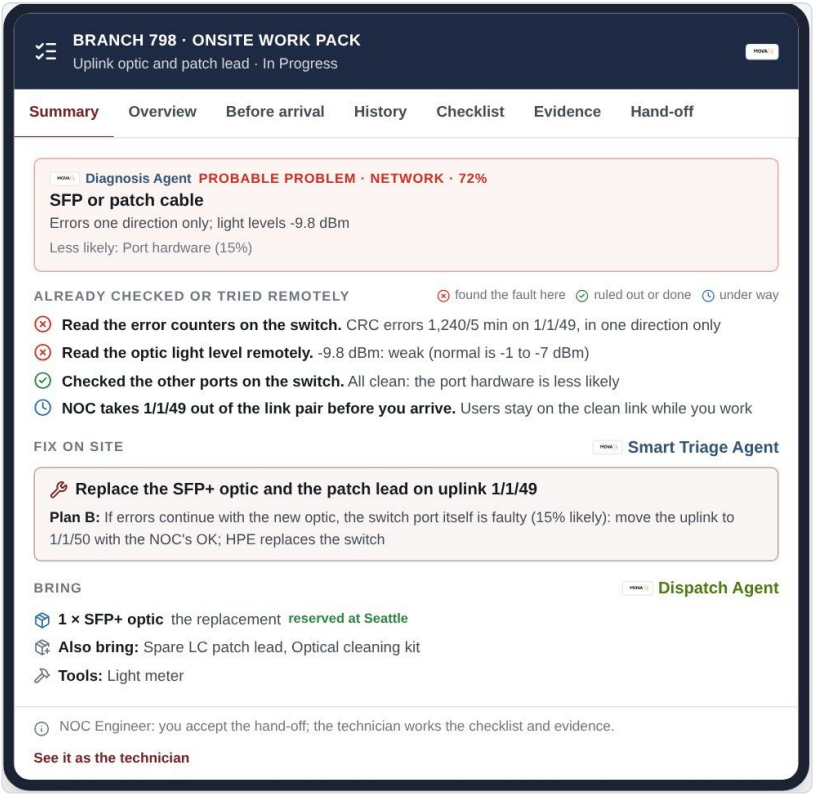



**Figure 138.** Summary for the technician: probable problem, what was already done remotely, the fix on site and what to bring

The same summary shows when the NOC reviews the dispatch (**What Mova iO tells the technician**), so the engineer approving the visit sees exactly what the technician will get.




**Figure 139.** Approved checklist on site

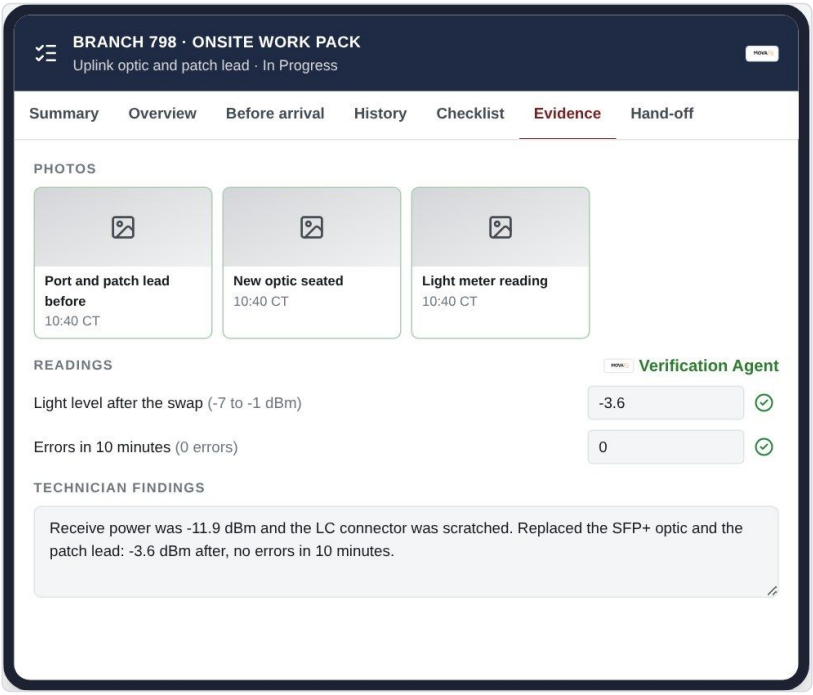



**Figure 140.** Evidence: photos, readings in range and the technician's findings

**How a visit runs.** The technician ticks each step (the work order moves to On site at the first tick), takes the photos, enters the readings and writes the findings. **Run the return-to-service check** asks the Verification Agent to confirm every step is done, every reading is in range and the photos are there; if not, it lists what is missing. Once it passes, the ticket owner (or an NOC engineer for them) presses **Accept the hand-off**: the work order closes with the evidence attached, the technician is free again, and the incident resolves with the right close code (for example Hardware replaced). Every step is in the ticket's work notes and the audit log.

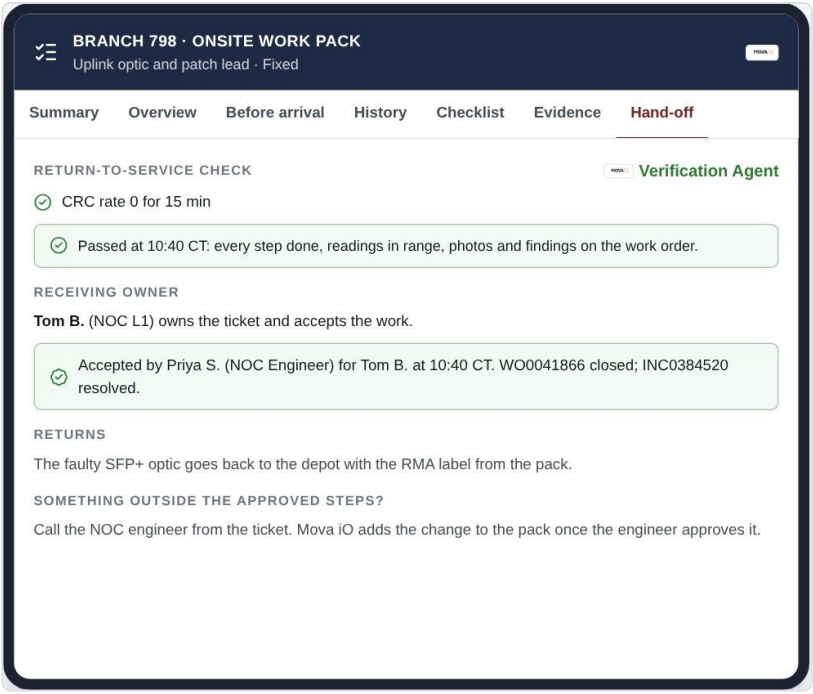



**Figure 141.** The hand-off: return check passed and the owner accepted

The Field Technician (SRS field technical services) records the onsite work: the checklist and evidence. That is the only thing an SRS role does beyond seeing and escalating. The NOC Engineer and Network Engineer L3 accept the hand-off (**See it as the technician** switches the role for a demonstration). For the Field Technician, Field Dispatch opens with **My jobs today**, one card per job with its work pack. On the role dashboards the pack is read-only.




**Figure 142.** My jobs today for the Field Technician

## Work orders tab

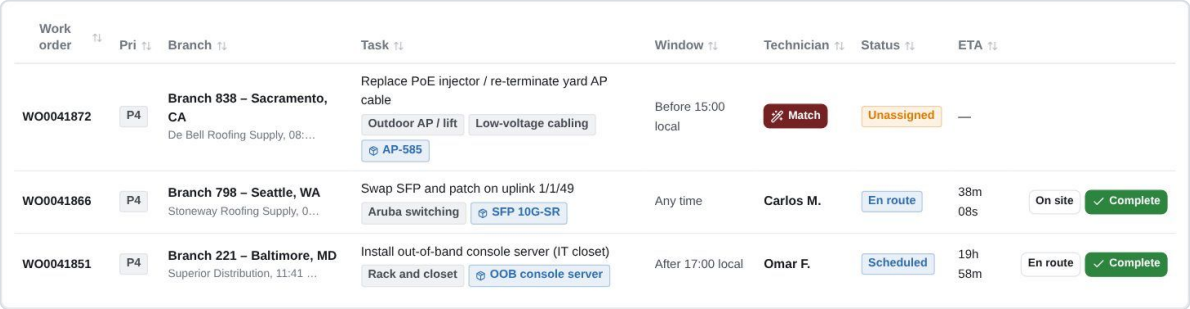



**Figure 143.** Work orders

| Column | Meaning |
|---|---|
| Work order | The WO number |
| Pri | Priority of the linked incident |
| Branch | Branch, brand, local time and whether it is open |
| Task | What to do, the skills needed and the parts |
| Window | When the visit may happen, for example after close, before 15:00 local or "Now (P1)" |
| Technician | The assigned technician, or **Match** to pick the best fit |
| Work pack | Opens the onsite work pack; shows how many steps are done, or Accepted |
| Status | Unassigned, Dispatched, En route, On site, Closed |
| ETA | Time to arrival |

**Match** ranks technicians by skill fit, distance, current load and branch hours.

## Technicians tab

| Technician | Region | Base | Skills | Status | Jobs 30d | Rating |
|---|---|---|---|---|---|---|
| **Luis R.** | Southeast | Tampa | Aruba wireless · Rack and closet · EdgeConnect swap | Off shift | 10 | 4.8 |
| **Brandon K.** | Carolinas & Mid-Atlantic | Charlotte | Aruba switching · Outdoor AP / lift · Rack and closet · Low-voltage cabling | On site | 13 | 4.8 |
| **Jasmine T.** | Northeast | Philadelphia | Aruba wireless · Rack and closet · EdgeConnect swap · LTE / cellular | Available | 28 | 4.8 |
| **Carlos M.** | Great Lakes | Detroit | Low-voltage cabling · LTE / cellular · Rack and closet · EdgeConnect swap | Available | 18 | 4.4 |
| **Derek H.** | Central Plains | Des Moines | Rack and closet · Aruba switching · Aruba wireless · EdgeConnect swap | Off shift | 8 | 4.5 |
| **Amanda P.** | Texas & South Central | Lafayette | Rack and closet · Outdoor AP / lift · LTE / cellular | Available | 20 | 4.8 |
| **Tyrell J.** | Mountain West | Colorado Springs | Rack and closet · LTE / cellular · Aruba wireless · EdgeConnect swap | Available | 26 | 4.6 |
| **Megan S.** | Pacific | Yakima | Aruba switching · Rack and closet | Off shift | 24 | 4.3 |
| **Omar F.** | Canada | Tampa | Low-voltage cabling · EdgeConnect swap | Available | 12 | 4.7 |
| **Hannah W.** | Southeast | Orlando | Aruba wireless · Rack and closet · LTE / cellular · EdgeConnect swap | On site | 26 | 4.8 |
| **Jake D.** | Carolinas & Mid-Atlantic | Greensboro | Aruba switching · EdgeConnect swap | Available | 18 | 4.3 |
| **Nina V.** | Northeast | Amagansett | Outdoor AP / lift · Low-voltage cabling | Available | 26 | 4.4 |
| **Chris O.** | Great Lakes | Columbus | EdgeConnect swap · Aruba wireless · LTE / cellular | Available | 22 | 4.4 |
| **Rosa G.** | Central Plains | St. Louis | EdgeConnect swap · Low-voltage cabling · Aruba switching | On site | 13 | 4.4 |
| **Ethan B.** | Texas & South Central | Fort Worth | Aruba wireless · Low-voltage cabling · Outdoor AP / lift | On site | 16 | 4.5 |
| **Kayla N.** | Mountain West | Tucson | LTE / cellular · Aruba wireless · Outdoor AP / lift · Low-voltage cabling · EdgeConnect swap | Off shift | 19 | 4.4 |
| **Andre L.** | Pacific | Oakland | Rack and closet · EdgeConnect swap · Outdoor AP / lift | Available | 11 | 4.9 |
| **Sophie C.** | Canada | Orlando | Rack and closet · EdgeConnect swap | Off shift | 19 | 4.5 |
| **Victor A.** | Southeast | Miami | Aruba switching · Rack and closet · EdgeConnect swap | Available | 23 | 4.8 |
| **Grace Y.** | Carolinas & Mid-Atlantic | Greenville | Outdoor AP / lift · Aruba wireless · LTE / cellular · Aruba switching | On site | 22 | 4.7 |

**Figure 144.** Technicians

Every technician with region, base, skills (for example outdoor AP and lift, Aruba switching, LTE and cellular, EdgeConnect swap), status, jobs in 30 days and rating.

## Depot spares tab

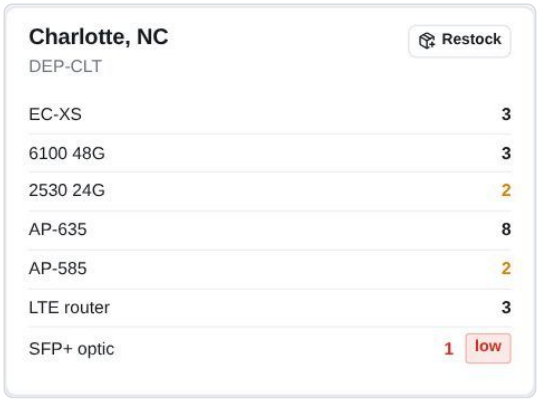



**Figure 145.** Depot card

One card per depot with its code and the count of each spare: EC-XS, CX 6100, 2530, AP-635, AP-585 or AP-565, LTE router and SFP+ optic. "low" marks a line below its minimum. **Restock** raises the order. Spares for a predicted failure are reserved here from the Predictive Watchlist and the inventory.

# 16. Approvals & Changes

Approvals & Changes is where people stay in control: the human-in-the-loop queue, the change calendar with freeze windows, Network-as-Code changes, drift from the golden configuration and firmware compliance. It is the landing page for the Network Engineer L3 and, on the firmware tab, the Security Lead.

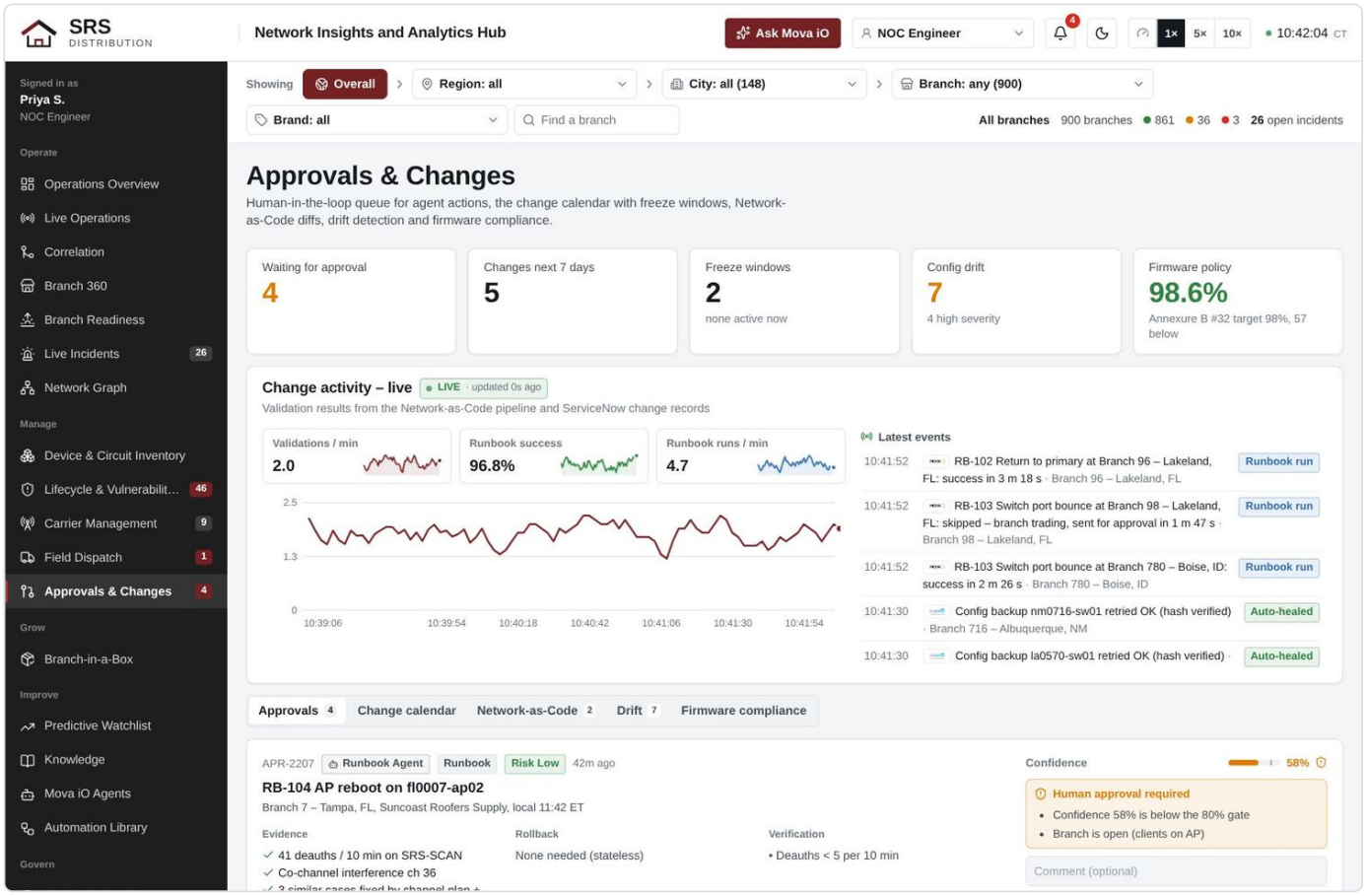



**Figure 146.** Approvals & Changes

## Tiles

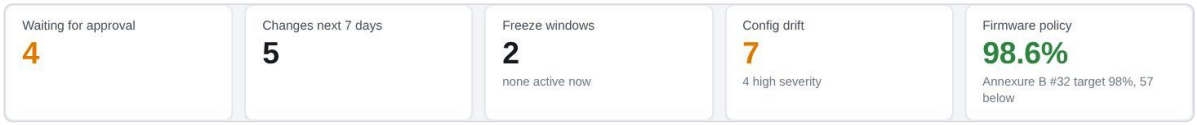



**Figure 147.** Change tiles

| Tile | Meaning |
|---|---|
| Waiting for approval | Mova iO actions held for a person (maroon badge in the menu) |
| Changes next 7 days | Scheduled changes in the coming week |
| Freeze windows | Freeze windows defined, and whether one is active now |
| Config drift | Devices differing from the golden config, with how many are high severity |
| Firmware policy | Share of devices within the firmware policy, against service level #32 (98%), and how many are below it |

## Change activity – live

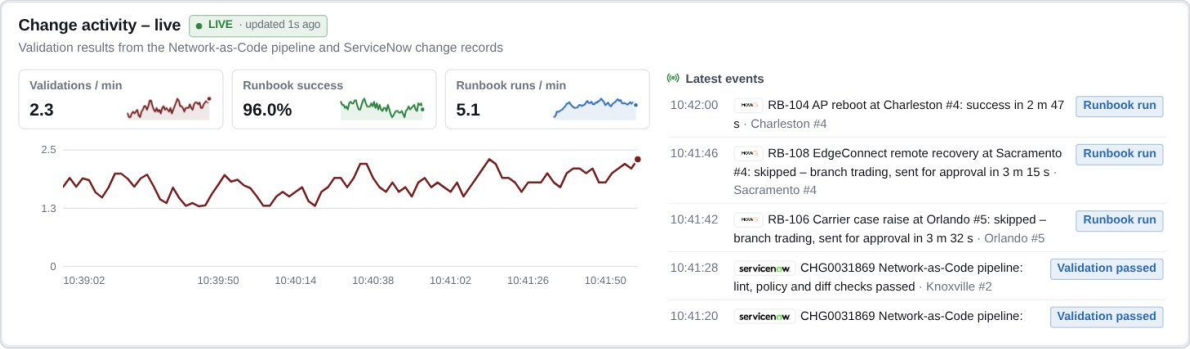



**Figure 148.** Change activity

Network-as-Code validations per minute, runbook success rate and runbook runs per minute, with the latest pipeline and runbook events, such as "skipped – branch trading, sent for approval".

## Approvals tab




**Figure 149.** Approval card

Each approval is a card with everything needed to decide:

| Part | Meaning |
|---|---|
| ID, agent, type, risk, age | Who is asking (for example Runbook Agent, Change Validation Agent, Prediction Agent), what kind of action, its risk and how long it has waited |
| Title and branch | The action and where, with the branch's local time |
| Evidence | Why the agent proposes it |
| Rollback | How it would be undone |
| Verification | How success will be checked |
| Confidence | The agent's confidence |
| Why a person is needed | The gate reason: confidence below 80%, branch open, freeze, or "predictions never act directly" |

**Approve** runs the action within its runbook and records the approver. **Reject** keeps it as a recommendation and returns it to the engineer. The open incident link goes to the incident.

## Change calendar tab

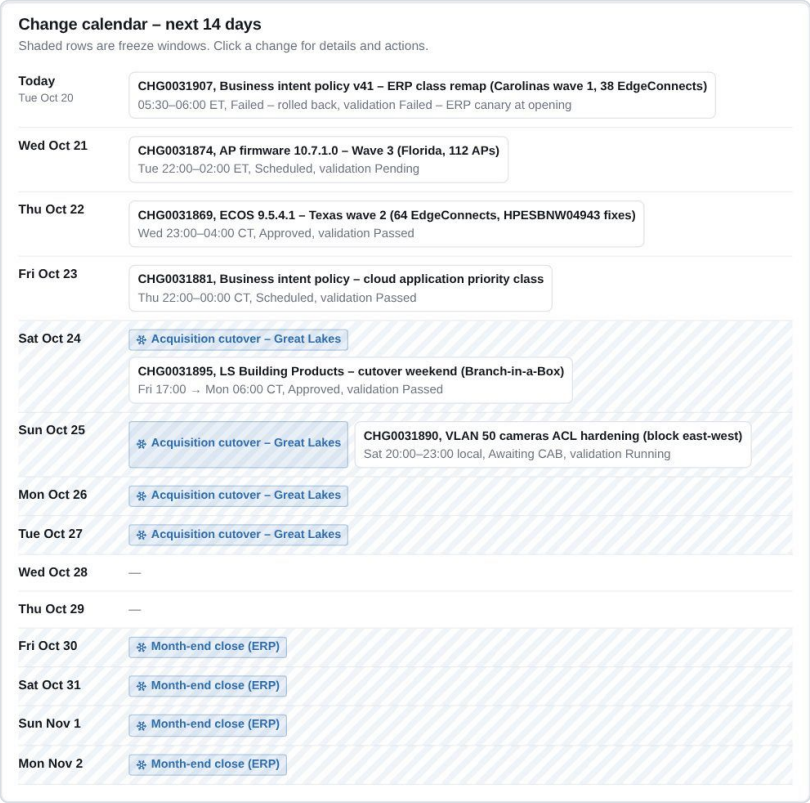



**Figure 150.** Change calendar

**Change calendar, next 14 days:** one row per day with each change, its window in local time, state (Scheduled, Approved, In progress, Paused) and validation result. Shaded rows are freeze windows. Click a change for details and actions, including **Pause** and **Rollback**.

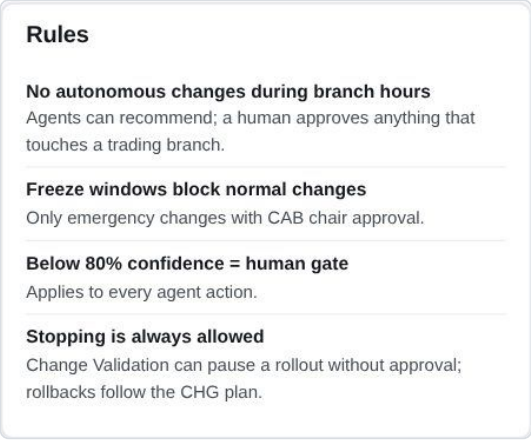



**Figure 151.** Rules

**Rules** lists the four change rules: no autonomous changes during branch hours; freeze windows block normal changes (only emergency changes with CAB chair approval); below 80% confidence means a person approves; stopping is always allowed (Change Validation can pause a rollout without approval, and rollbacks follow the change plan).

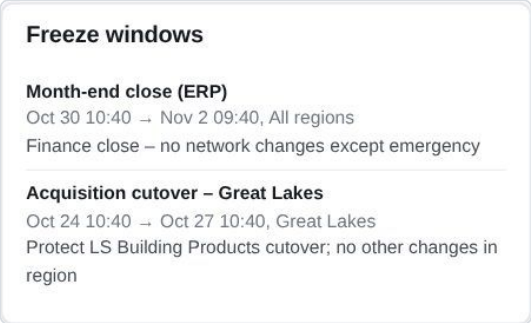



**Figure 152.** Freeze windows

**Freeze windows** lists each freeze with its dates, scope and reason, for example month-end ERP close (all regions) or an acquisition cutover in one region.

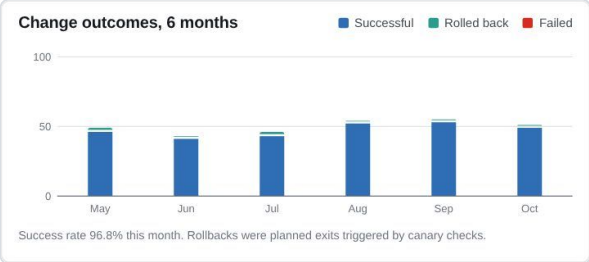



**Figure 153.** Change outcomes

**Change outcomes, 6 months:** changes per month that succeeded, were rolled back or failed, with this month's success rate. Rollbacks triggered by a canary are planned exits, not failures.




**Figure 154.** Change risk scores

**Change risk scores, last 90 days:** how many changes fell in each risk band of the Change Impact Analyzer score (0 to 100). Above 60 needs the Change Review Board and a canary.

### Network-as-Code tab

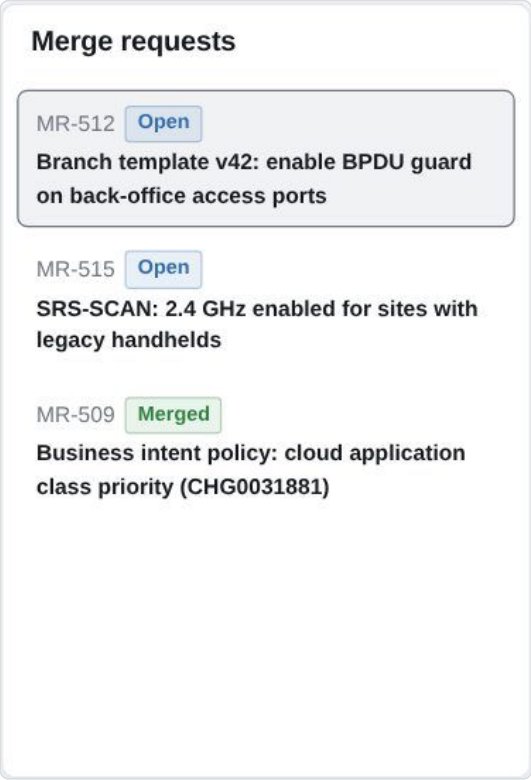



**Figure 155.** Merge requests

**Merge requests** lists proposed changes to the templates and site files in the Network-as-Code repository, open or merged.

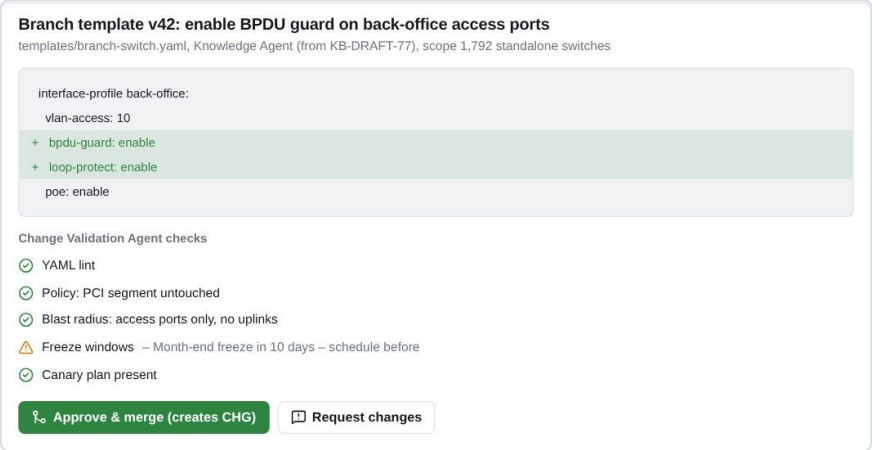
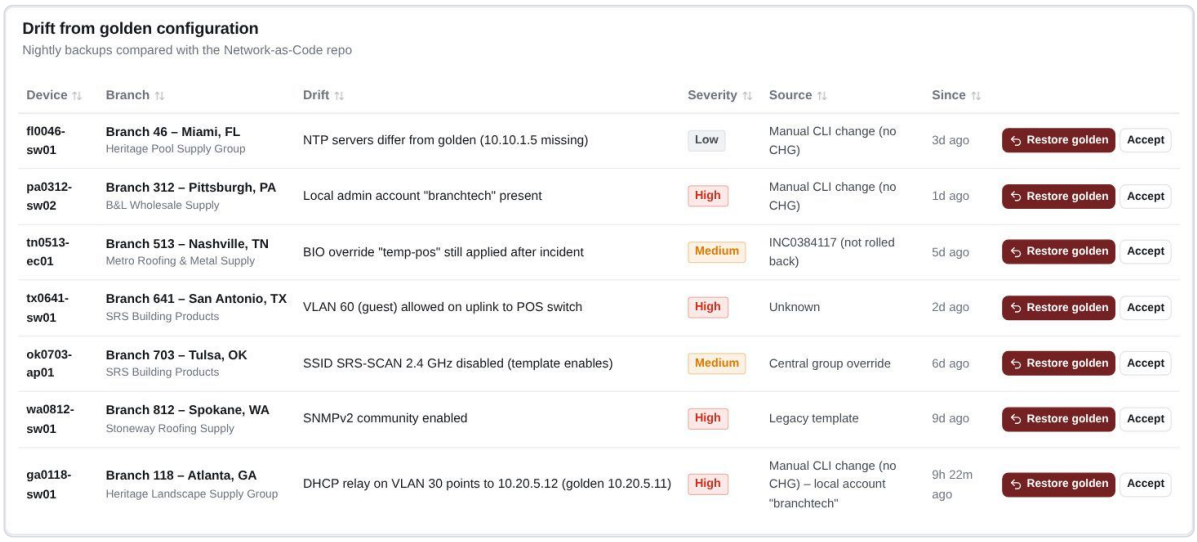



**Figure 156.** Merge request detail

The selected merge request shows the file, who proposed it and from what (for example the Knowledge Agent from a draft article), its scope, the diff (green lines added), and the Change Validation Agent's checks: YAML lint, policy (PCI segment untouched), blast radius, freeze windows and whether a canary plan is present. **Approve & merge** creates the change record; **Request changes** sends it back.

## Drift tab



**Drift from golden configuration**

Nightly backups compared with the Network-as-Code repo

| Device | Branch | Drift | Severity | Source | Since | |
|---|---|---|---|---|---|---|
| fl0046-sw01 | Branch 46 – Miami, FL<br>Heritage Pool Supply Group | NTP servers differ from golden (10.10.1.5 missing) | Low | Manual CLI change (no CHG) | 3d ago | Restore golden Accept |
| pa0312-sw02 | Branch 312 – Pittsburgh, PA<br>B&L Wholesale Supply | Local admin account "branchtech" present | High | Manual CLI change (no CHG) | 1d ago | Restore golden Accept |
| tn0513-ec01 | Branch 513 – Nashville, TN<br>Metro Roofing & Metal Supply | BIO override "temp-pos" still applied after incident | Medium | INC0384117 (not rolled back) | 5d ago | Restore golden Accept |
| tx0641-sw01 | Branch 641 – San Antonio, TX<br>SRS Building Products | VLAN 60 (guest) allowed on uplink to POS switch | High | Unknown | 2d ago | Restore golden Accept |
| ok0703-ap01 | Branch 703 – Tulsa, OK<br>SRS Building Products | SSID SRS-SCAN 2.4 GHz disabled (template enables) | Medium | Central group override | 6d ago | Restore golden Accept |
| wa0812-sw01 | Branch 812 – Spokane, WA<br>Stoneway Roofing Supply | SNMPv2 community enabled | High | Legacy template | 9d ago | Restore golden Accept |
| ga0118-sw01 | Branch 118 – Atlanta, GA<br>Heritage Landscape Supply Group | DHCP relay on VLAN 30 points to 10.20.5.12 (golden 10.20.5.11) | High | Manual CLI change (no CHG) – local account "branchtech" | 9h 22m ago | Restore golden Accept |

**Figure 157.** Drift from golden configuration

Nightly backups compared with the Network-as-Code repository. Each row shows the device, branch, the drift found, its severity, its source (for example a manual CLI change with no change record) and how long ago. **Restore golden** puts the approved config back (a change, so it follows the gate rules); **Accept** makes the drift the new approved state through a change.

## Firmware compliance tab

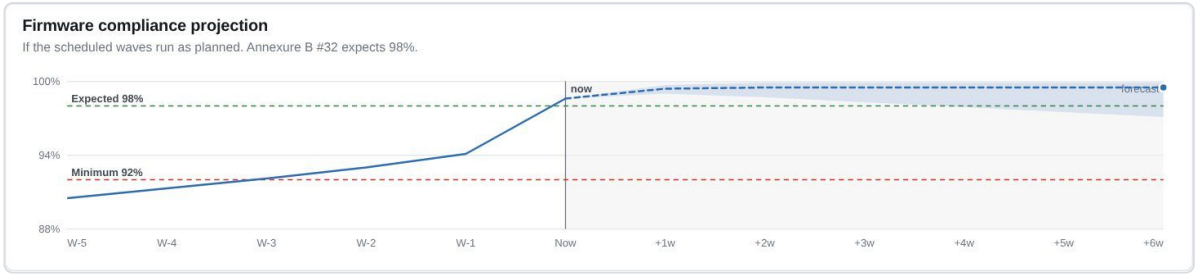



**Figure 158.** Firmware compliance projection

**Firmware compliance projection:** compliance for the last five weeks and the forecast for six weeks if the scheduled waves run, against the expected 98% and minimum 92% lines from service level #32.




**Figure 159.** Firmware compliance by platform

**Firmware compliance by platform:** for each platform, the target release (N), device count, how many are on N, N-1, N-2 and older, the share compliant and how many are below policy. The policy: critical devices (EdgeConnects) at N-1 or newer, others at N-2 or newer, and Day-0 critical patches inside the emergency window.

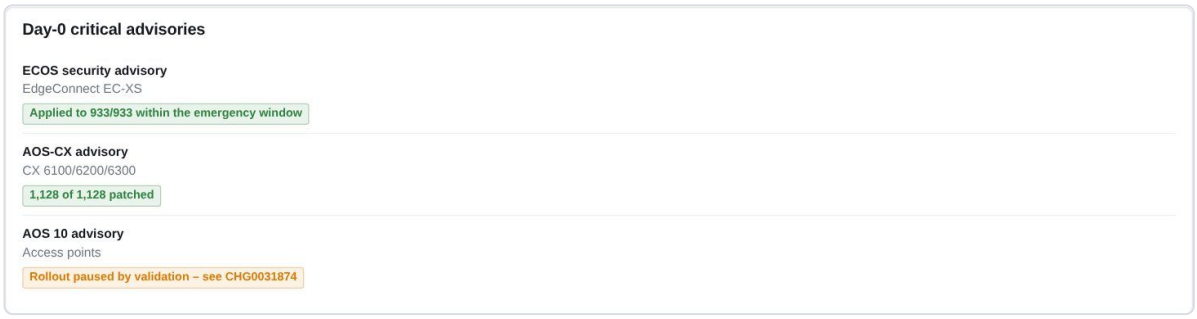



**Figure 160.** Day-0 critical advisories

**Day-0 critical advisories:** each critical advisory and its status: applied to every affected device within the emergency window, partly patched, or paused by validation with the change number.

# 17. Branch-in-a-Box

Branch-in-a-Box is the standard, repeatable kit and process that brings any new or acquired branch onto the SRS standard network, usually over one weekend, with the business trading again on Monday morning. Acquired branches keep their brand name. It is the M&A Integration Lead's landing page. It is a project board, not a live feed: nothing on it changes every few seconds; numbers move only when a gate passes, a stage completes or a project is created. Its tabs are **Portfolio**, **Two tracks**, **Kit & VLANs**, **Floor plan**, **Acquired-branch steps**, **Cutover weekend** and **Day-1 parity**.

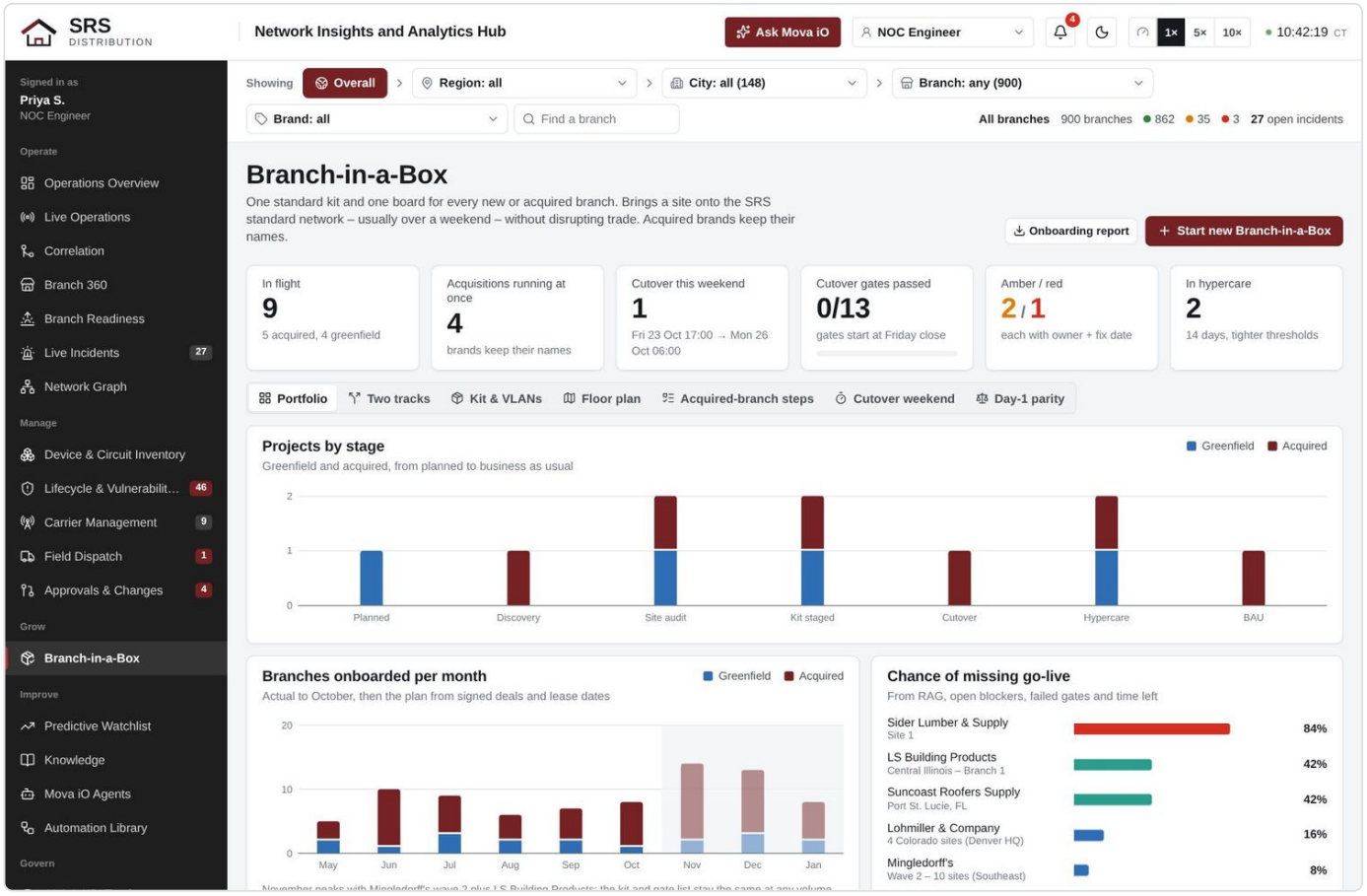



**Figure 161.** Branch-in-a-Box

## Tiles

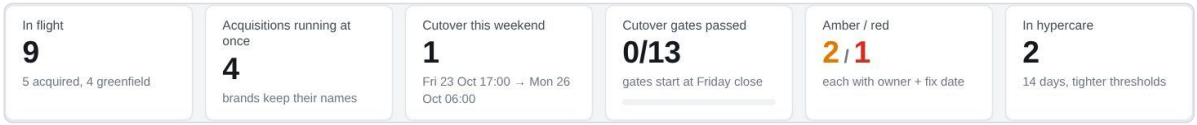



**Figure 162.** Branch-in-a-Box tiles

| Tile | Meaning |
|---|---|
| In flight | Projects running, split into acquired and greenfield |
| Acquisitions running at once | Acquired brands being onboarded in parallel; brands keep their names |

| Tile | Meaning |
|---|---|
| Cutover this weekend | Projects whose cutover runs this coming weekend, with the dates (Friday 17:00 close to Monday 06:00) |
| Cutover gates passed | Gates passed out of 13 for that cutover; the gates start at Friday close, so the count is 0 during the week |
| Amber / red | Projects at risk, each with an owner and a fix date |
| In hypercare | Projects in their 14 days of tighter thresholds after go-live |

**Start new Branch-in-a-Box** opens a wizard that creates a project from the template; **Onboarding report** downloads the report for the selected project.

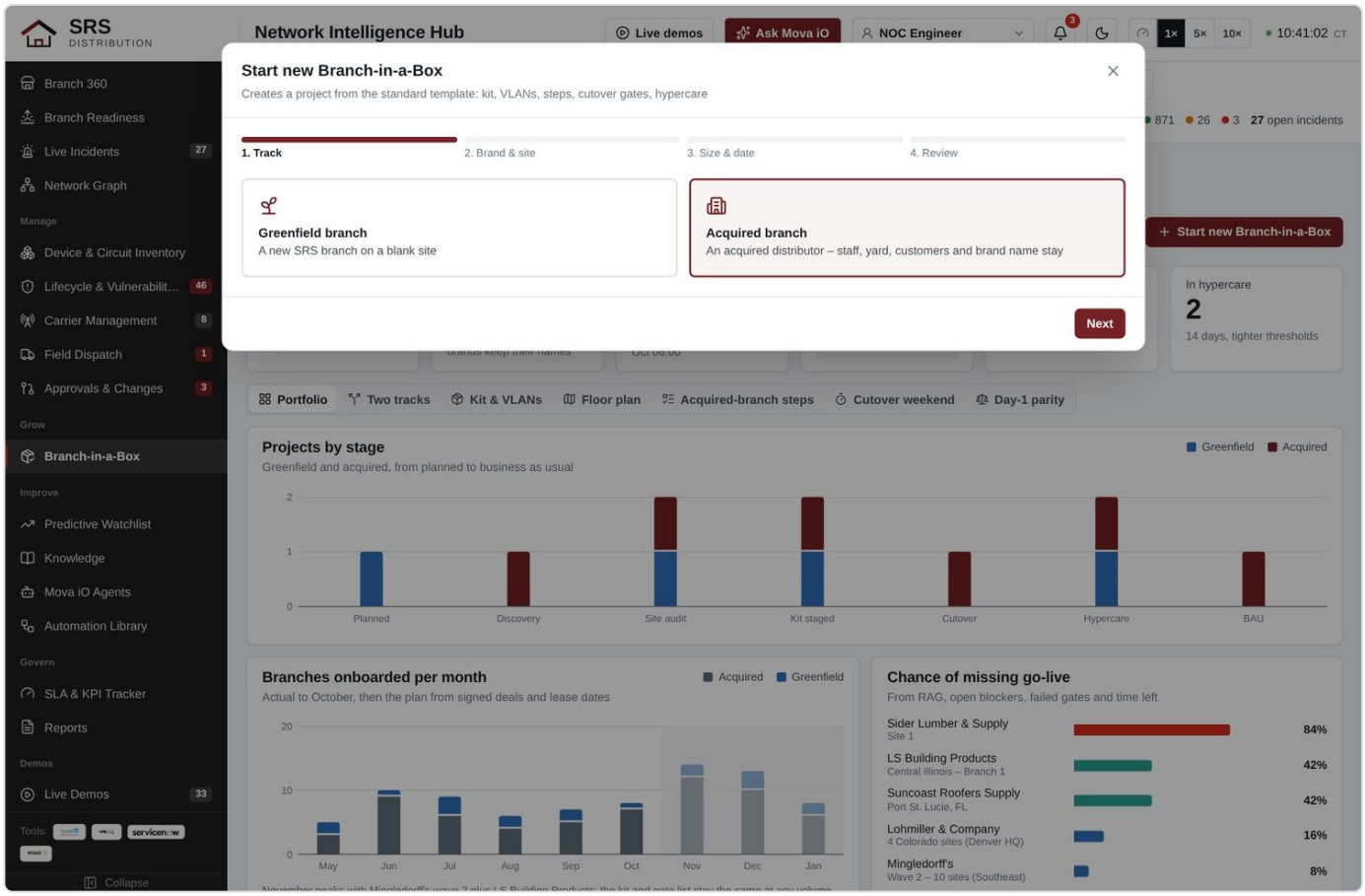



**Figure 163.** New project wizard

## Portfolio tab

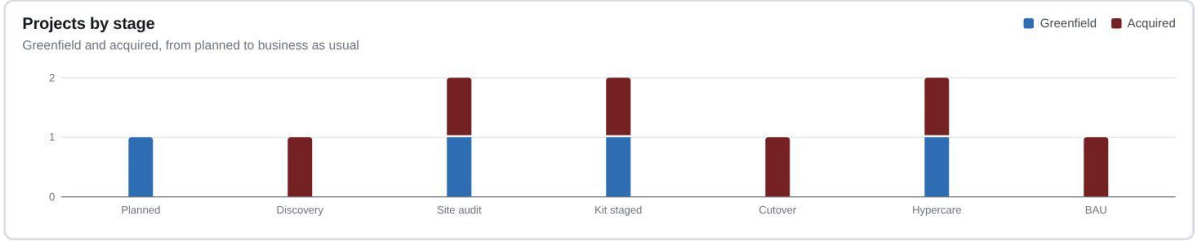



**Figure 164.** Projects by stage

**Projects by stage:** greenfield and acquired projects counted at each stage: Planned, Discovery, Site audit, Kit staged, Cutover, Hypercare, BAU.

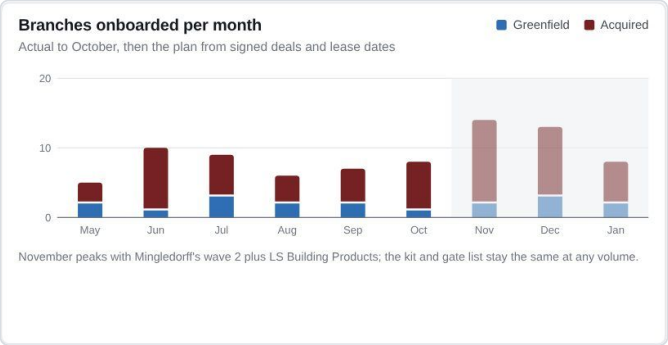



**Figure 165.** Branches onboarded per month

**Branches onboarded per month:** actual onboarding to date, then the plan from signed deals and lease dates, split into acquired and greenfield. The same kit and gate list apply at any volume.




**Figure 166.** Chance of missing go-live

**Chance of missing go-live:** each project's risk of missing its date, worked out from its RAG status, open blockers, failed gates and time left. Above 50% needs the integration lead's attention this week.

The project table on this tab lists every project with brand, site, track, stage, RAG, gates passed, days to go-live and blockers.

## Two tracks tab

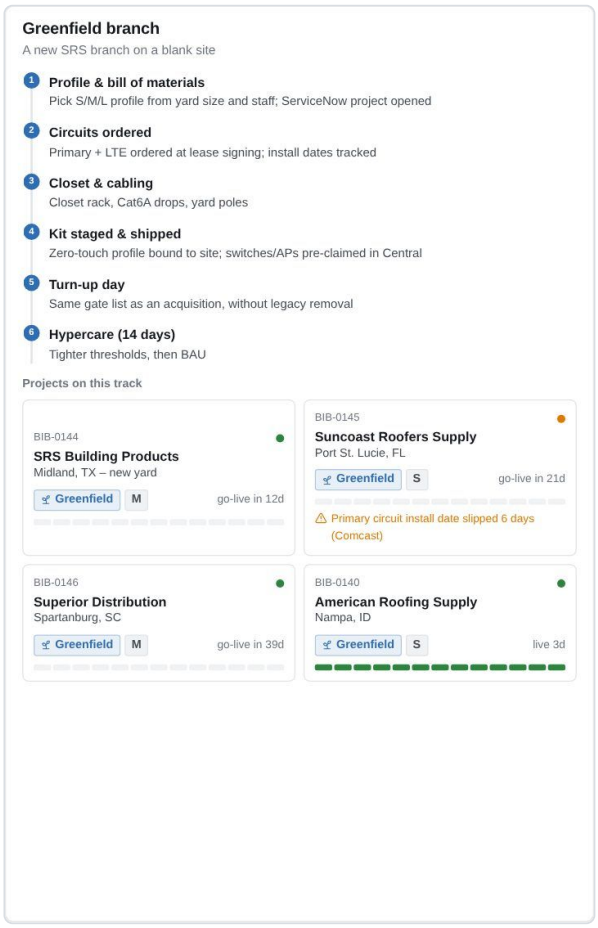



**Figure 167.** Greenfield track

**Greenfield branch** (a new SRS branch on a blank site): profile and bill of materials, circuits ordered at lease signing, closet and cabling, kit staged and shipped with zero-touch profiles, turn-up day using the same gates as an acquisition, and 14 days of hypercare.

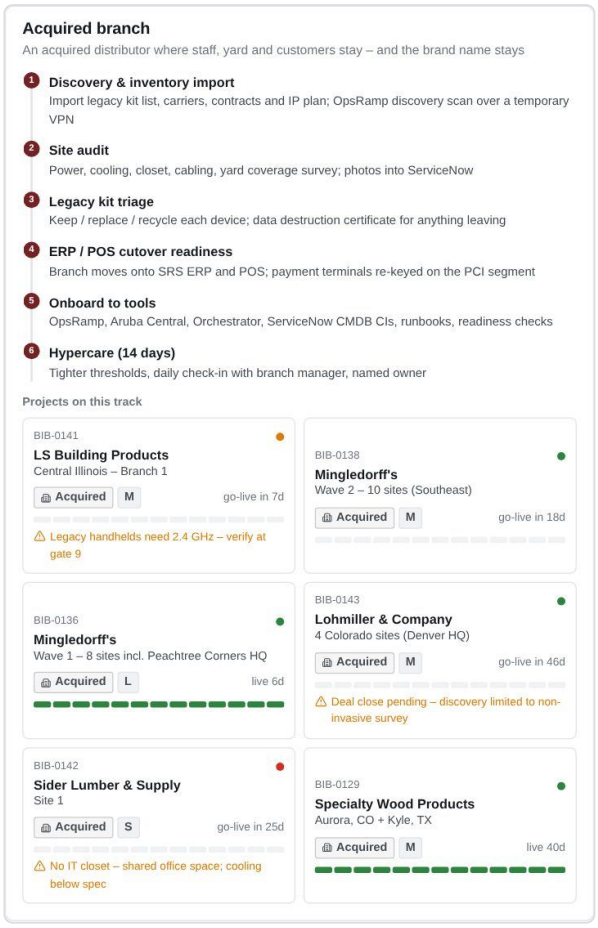



**Figure 168.** Acquired track

**Acquired branch** (staff, yard, customers and brand stay): discovery and inventory import, site audit, legacy kit triage (keep, replace or recycle, with a data destruction certificate), ERP and POS cutover readiness, onboarding to the tools, and hypercare. Each track lists its current projects.

## Kit & VLANs tab

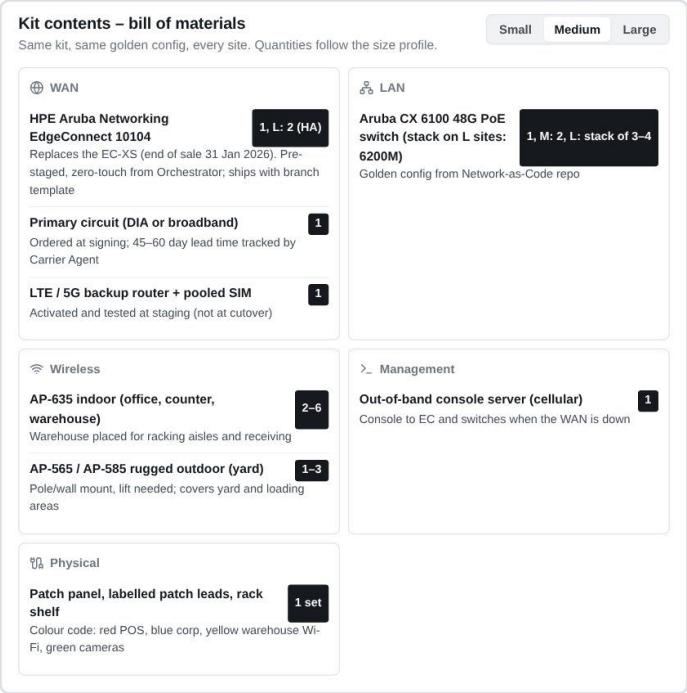



**Figure 169.** Kit contents

**Kit contents, bill of materials:** the same kit and golden config at every site, with quantities by size profile (Small, Medium, Large):

| Group | Items |
|---|---|
| WAN | EdgeConnect 10104 (two as an HA pair on large sites), which replaces the EC-XS after its end of sale on 31 Jan 2026, pre-staged for zero-touch; a primary circuit ordered at signing; an LTE/5G backup router with a pooled SIM, tested at staging |
| LAN | Aruba CX 6100 48-port PoE switches (a 6200M stack on large sites) with the golden config |
| Wireless | AP-635 indoor for office, counter and warehouse; AP-565 or AP-585 rugged outdoor for the yard |
| Management | An out-of-band console server on cellular, for console access when the WAN is down |
| Physical | Patch panel, labelled leads and a colour code: red POS, blue corporate, yellow warehouse Wi-Fi, green cameras |

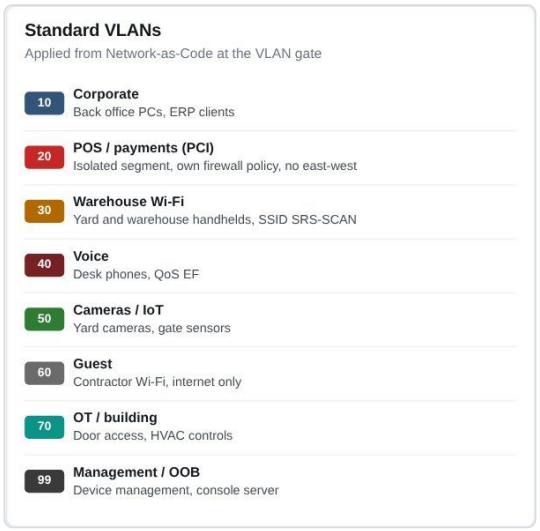



**Figure 170.** Standard VLANs

**Standard VLANs,** applied from Network-as-Code at the VLAN gate: 10 Corporate, 20 POS and payments (PCI), 30 Warehouse Wi-Fi (SSID SRS-SCAN), 40 Voice (QoS EF), 50 Cameras and IoT, 60 Guest (internet only), 70 OT and building, 99 Management and out-of-band.

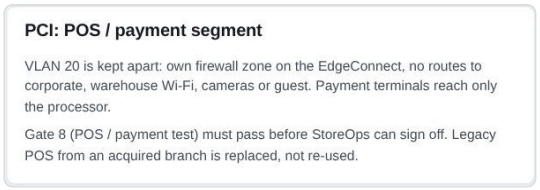



**Figure 171.** PCI segment

**PCI: POS and payment segment.** VLAN 20 has its own firewall zone on the EdgeConnect and no routes to corporate, warehouse Wi-Fi, cameras or guest; payment terminals reach only the processor. Gate 8 must pass before StoreOps can sign off, and legacy POS from an acquired branch is replaced, not re-used.

## Floor plan tab

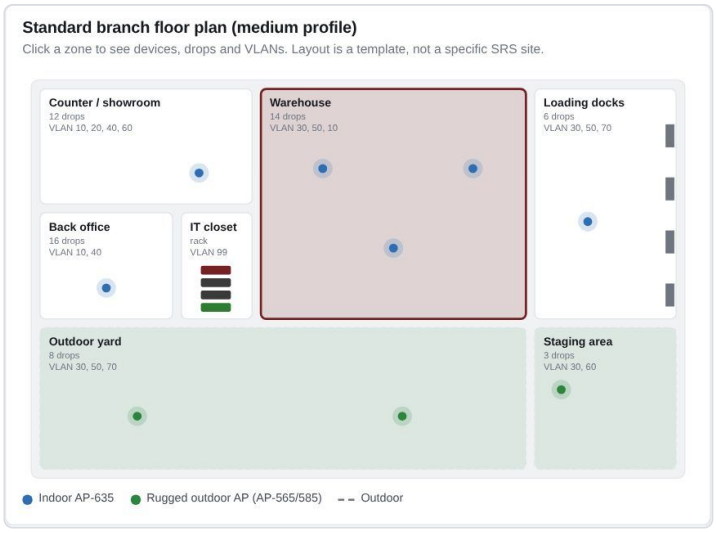



**Figure 172.** Standard floor plan

The standard medium branch layout: counter and showroom, back office, IT closet, warehouse, loading docks, outdoor yard and staging area, each with its drops and VLANs, and the AP positions (indoor and rugged outdoor). Click a zone to see its devices, drops, VLANs and what is tested there at cutover.

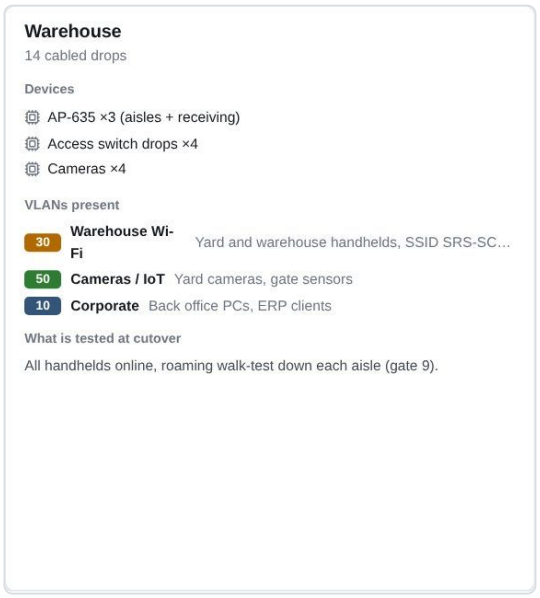



**Figure 173.** Zone detail

## Acquired-branch steps tab

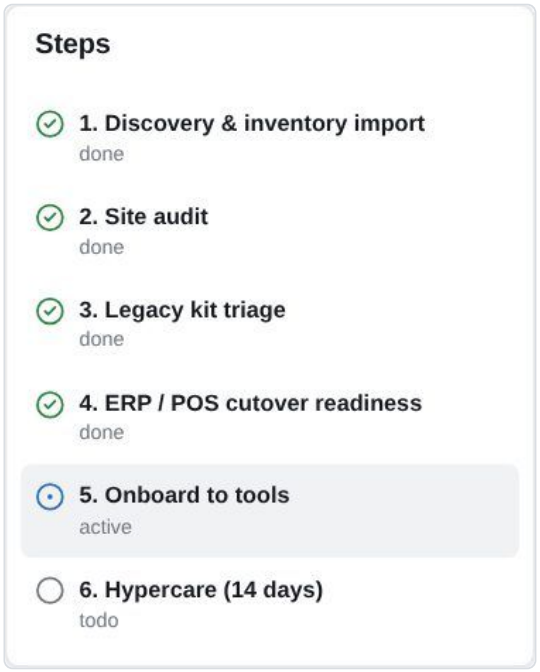



**Figure 174.** Acquired-branch steps

The six steps for the selected acquired project, each done, active or to do. The active step shows its checklist; for **Onboard to tools** that is OpsRamp discovery and monitors, Aruba Central claim and group, Orchestrator template, ServiceNow CIs and relationships, runbooks RB-101 to RB-110 enabled, readiness checks scheduled at 05:30 local and Mova iO agents scoped to the site. **Run onboarding automation** performs the checklist; **Complete step** moves on.

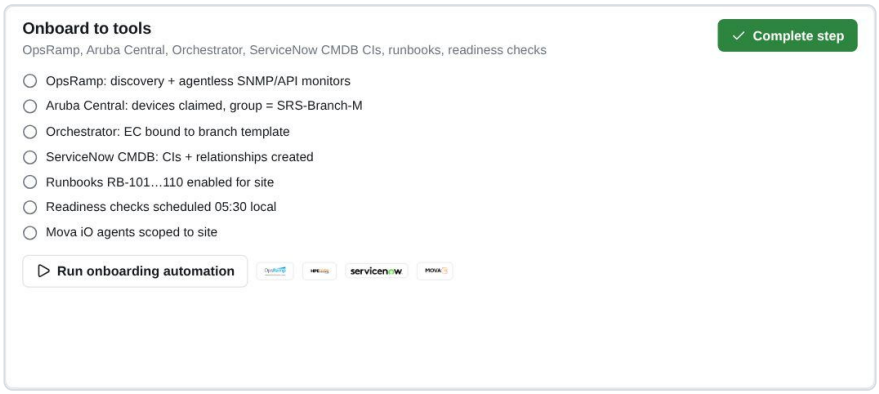



**Figure 175.** Onboard to tools

## Cutover weekend tab

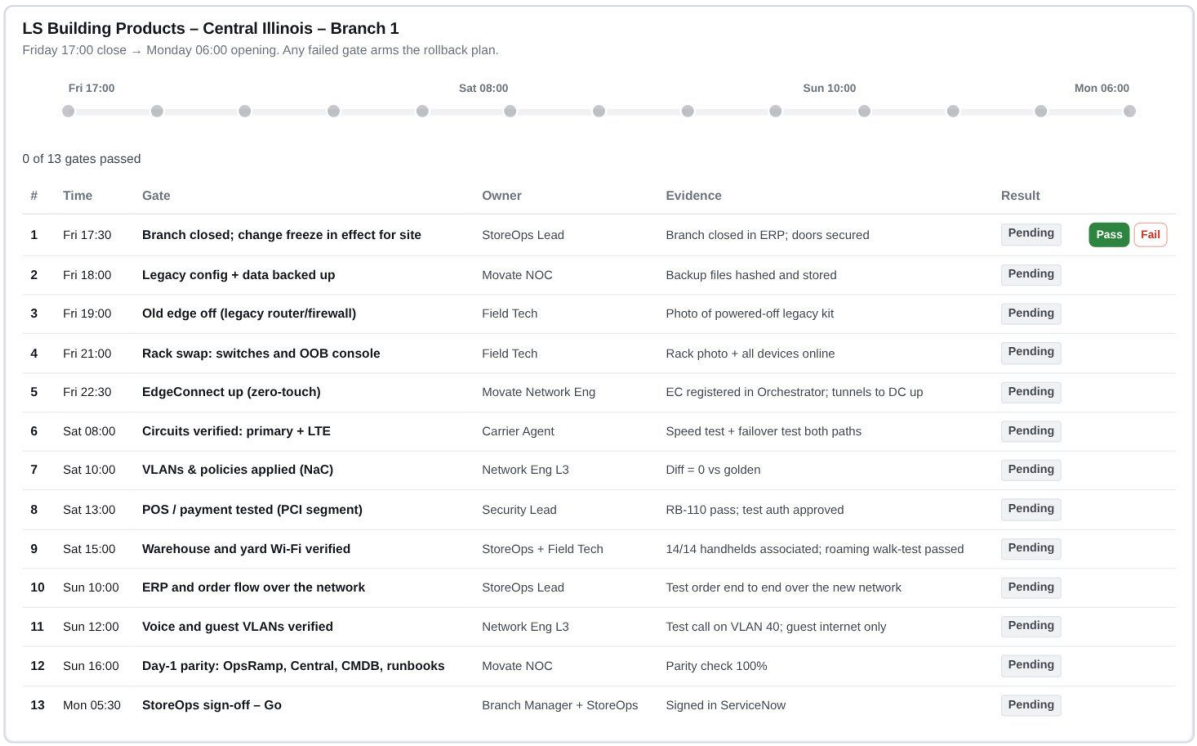



**Figure 176.** Cutover gate tracker

The gate tracker for the cutover weekend, from Friday 17:00 close to Monday 06:00 opening, with a timeline and the gates passed. Each of the 13 gates has a time, an owner, the evidence required and a result (Pending, Pass or Fail, with **Pass** and **Fail** buttons). Any failed gate arms the rollback plan.

| # | Time | Gate | Owner |
|---|---|---|---|
| 1 | Fri 17:30 | Branch closed; change freeze in effect for site | StoreOps Lead |
| 2 | Fri 18:00 | Legacy config and data backed up | Movate NOC |
| 3 | Fri 19:00 | Old edge off (legacy router or firewall) | Field Tech |
| 4 | Fri 21:00 | Rack swap: switches and OOB console | Field Tech |
| 5 | Fri 22:30 | EdgeConnect up (zero-touch) | Movate Network Eng |

| # | Time | Gate | Owner |
|---|---|---|---|
| 6 | Sat 08:00 | Circuits verified: primary and LTE | Carrier Agent |
| 7 | Sat 10:00 | VLANs and policies applied (Network-as-Code) | Network Eng L3 |
| 8 | Sat 13:00 | POS and payment tested (PCI segment) | Security Lead |
| 9 | Sat 15:00 | Warehouse and yard Wi-Fi verified | StoreOps and Field Tech |
| 10 | Sun 10:00 | ERP and order flow over the network | StoreOps Lead |
| 11 | Sun 12:00 | Voice and guest VLANs verified | Network Eng L3 |
| 12 | Sun 16:00 | Day-1 parity: OpsRamp, Central, CMDB, runbooks | Movate NOC |
| 13 | Mon 05:30 | StoreOps sign-off: Go | Branch Manager and StoreOps |

## Day-1 parity tab

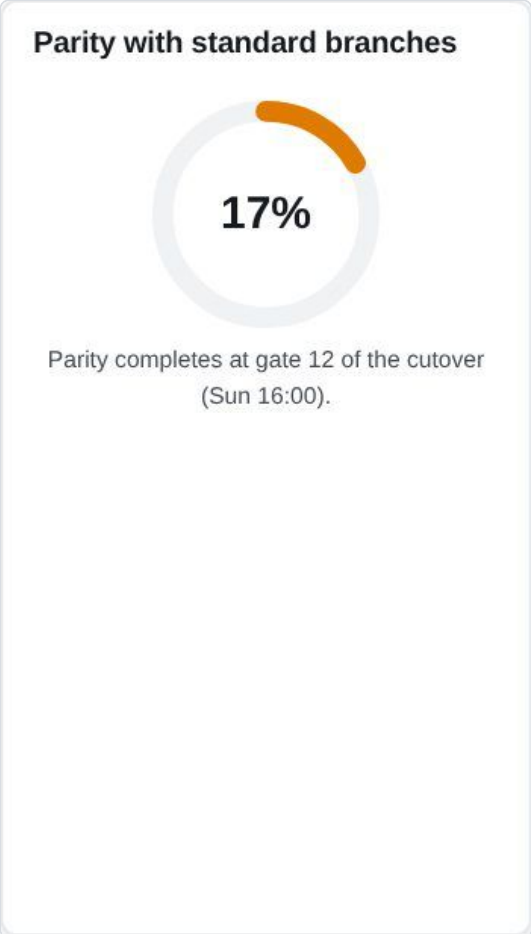



**Figure 177.** Parity with standard branches

Day-1 parity checks that the new branch has the same telemetry, automation, runbooks and AI coverage as every other branch from its first day: monitored in OpsRamp, claimed in Central, bound in Orchestrator, in the CMDB, runbooks enabled, readiness checks scheduled and agents scoped. The percentage reaches 100% at gate 12.

# 18. Predictive Watchlist

The Predictive Watchlist holds every prediction Mova iO has made about the network: degrading circuits, access points and switches likely to fail, capacity running out and backup paths at risk. A prediction never changes anything by itself. It can only create a watch or ask for approval. The tabs filter by kind: **All**, **Degrading circuit**, **AP failure risk**, **Switch failure risk**, **Capacity forecast** and **Backup path risk**.

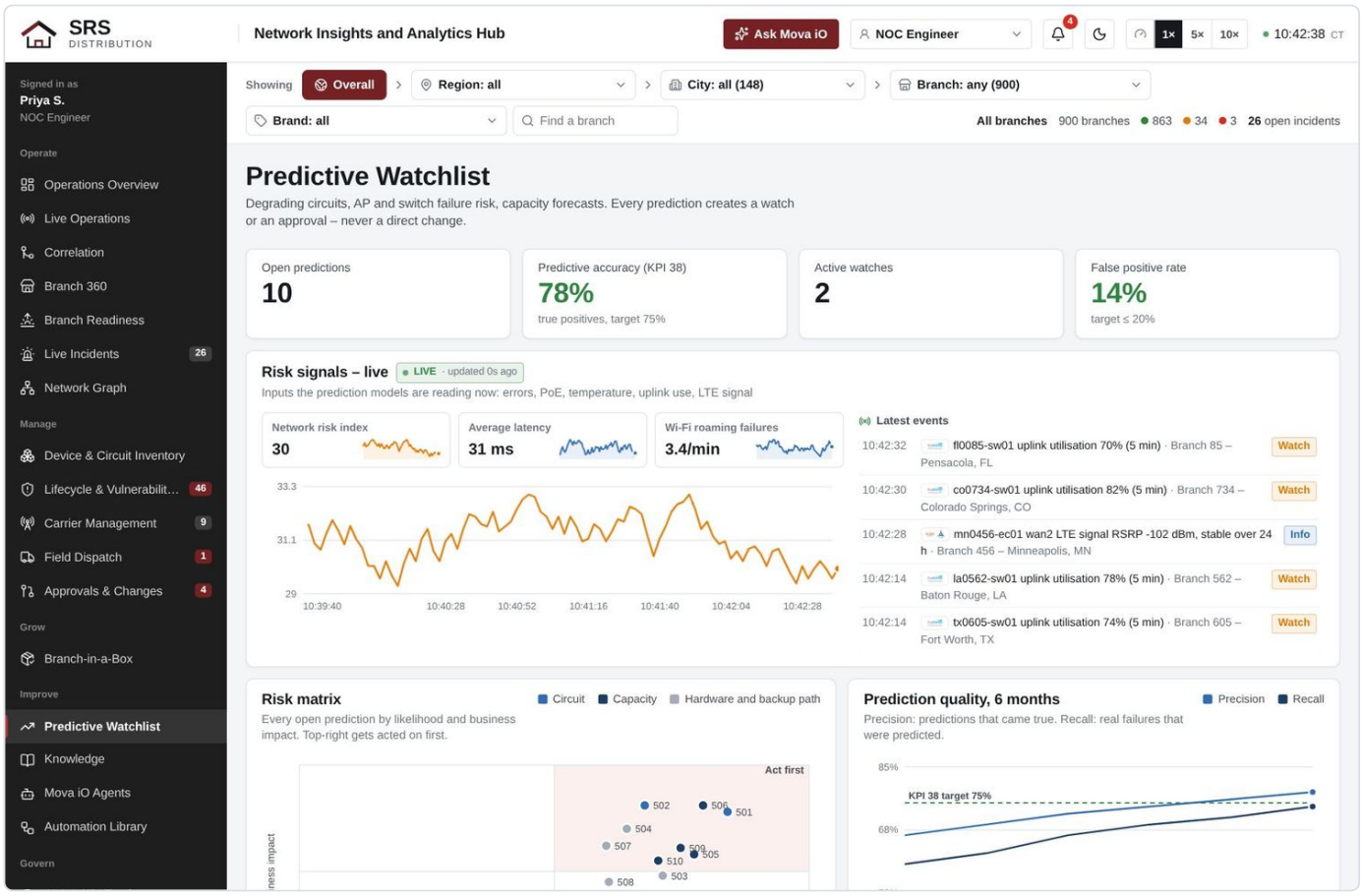



**Figure 178.** Predictive Watchlist

## Tiles




**Figure 179.** Prediction tiles

| Tile | Meaning | Target |
|---|---|---|
| Open predictions | Predictions not yet resolved or dismissed | – |
| Predictive accuracy (KPI 38) | Share of predictions that came true | 75% or more |
| Active watches | Predictions being watched with an alert set | – |
| False positive rate | Share of predictions marked false | 20% or less |

## Risk signals – live




**Figure 180.** Risk signals

The inputs the models are reading now: a network risk index (the average risk score across branches), the average latency to the data centre and Wi-Fi roaming failures, with the latest risk events such as uplink use, PoE renegotiations, temperature and LTE signal.

## Risk matrix




**Figure 181.** Risk matrix

Every open prediction placed by likelihood within its horizon (across) and business impact (up), labelled with its number and coloured by kind: circuit, capacity, or hardware and backup path. The top-right corner is **Act first**; the bottom-left is **Watch**.

## Prediction quality, 6 months

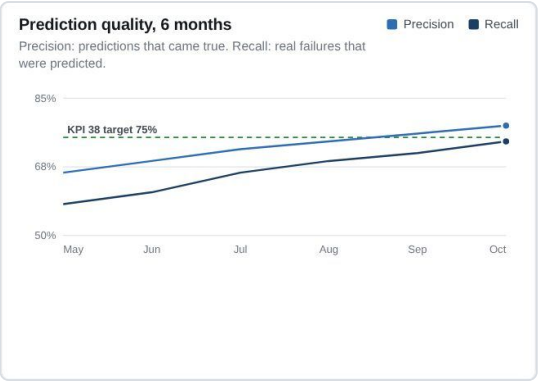



**Figure 182.** Prediction quality

**Precision** is the share of predictions that came true; **recall** is the share of real failures that were predicted. Both should stay above the 75% line of KPI 38. Precision falling means too many false alarms; recall falling means failures are being missed.

## Preventive maintenance, next 9 days

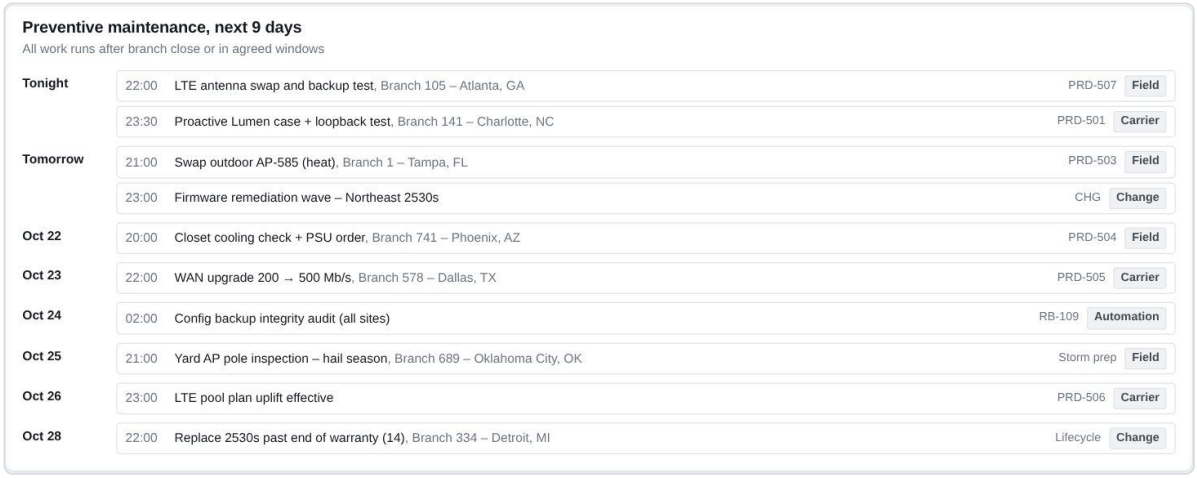



**Figure 183.** Preventive maintenance

Work booked from predictions, by day and time, with the branch, the prediction or change it came from and the owner (Field, Carrier or Change). All of it runs after branch close or in agreed windows.

## Prediction cards




**Figure 184.** Prediction card

| Part | Meaning |
|---|---|
| Number, kind, horizon, state | For example PRD-501, degrading circuit, 48 to 72 hours, New or Watching |
| Target | The circuit, device or pool, with its branch |
| Signal | What the model sees, such as loss rising from 0.1% to 0.9% over five days |
| Chart | The metric for the last nine days and the forecast, against its limit |
| Probability | Chance the failure happens within the horizon |
| Suggests | The preventive action, and whether it needs approval |

**Create watch** sets an alert on the metric. **Request approval** sends the suggested action to Approvals & Changes. **False positive** dismisses the prediction and feeds back into the model's accuracy.

## Capacity forecast




**Figure 185.** Capacity forecast

For the capacity prediction in view: p95 utilisation for ten weeks of actuals and four weeks of forecast, with the likely range and the 90% upgrade threshold. Where the forecast crosses 90% is when the upgrade must be in place.

# 19. Knowledge

Knowledge holds the articles engineers use, finds similar past incidents, reviews articles Mova iO drafts from resolved incidents and tracks problem records. Its tabs are **Search**, **Auto-generated drafts** and **Problem records**.

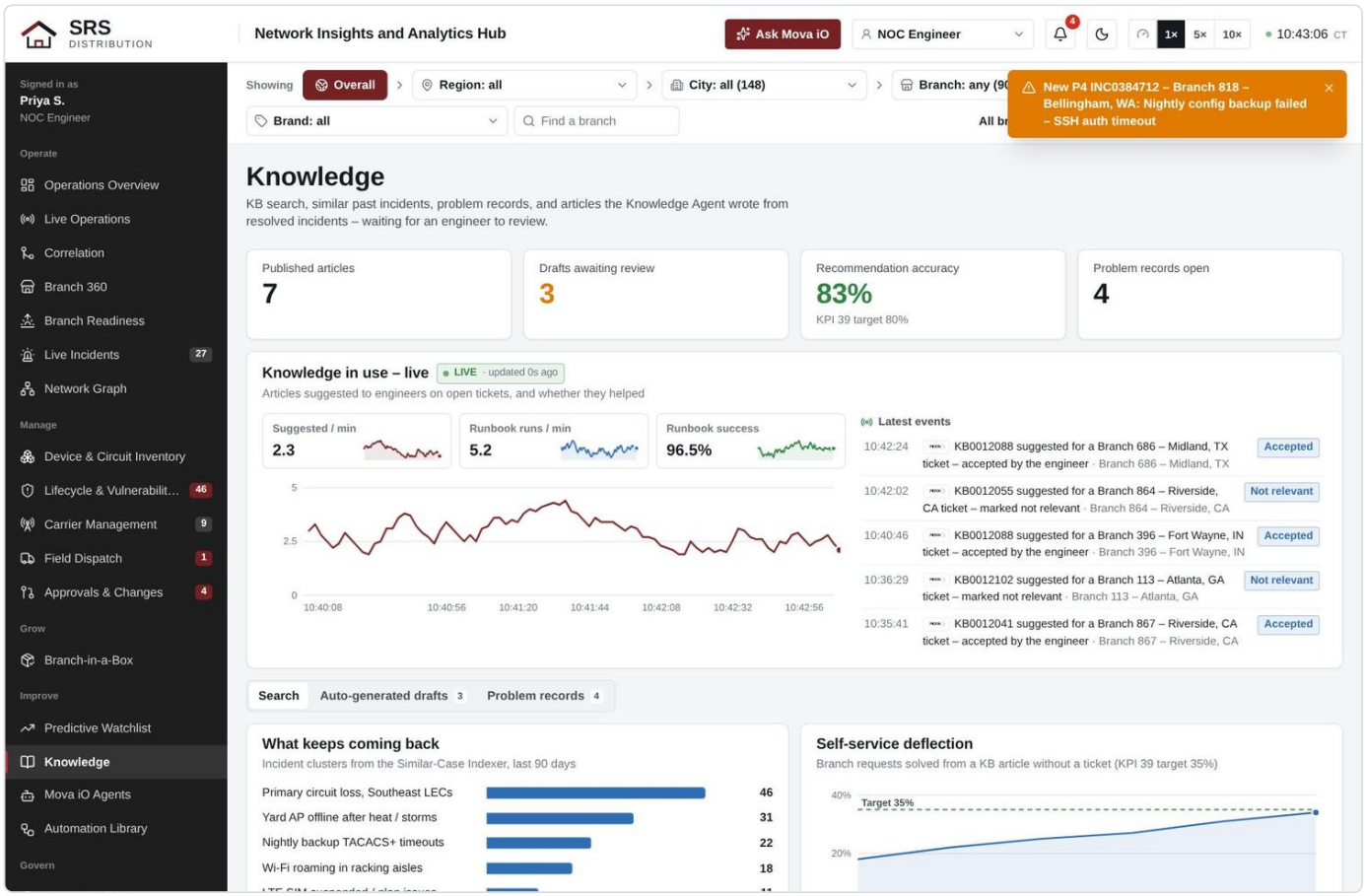



**Figure 186.** Knowledge

## Tiles




**Figure 187.** Knowledge tiles

| Tile | Meaning | Target |
|---|---|---|
| Published articles | Articles live in the knowledge base | – |
| Drafts awaiting review | Articles written by the Knowledge Agent, waiting for an engineer | – |
| Recommendation accuracy (KPI 39) | Share of suggested articles that engineers found relevant | 80% |
| Problem records open | Open problems: the causes behind repeat incidents | – |

## Knowledge in use – live

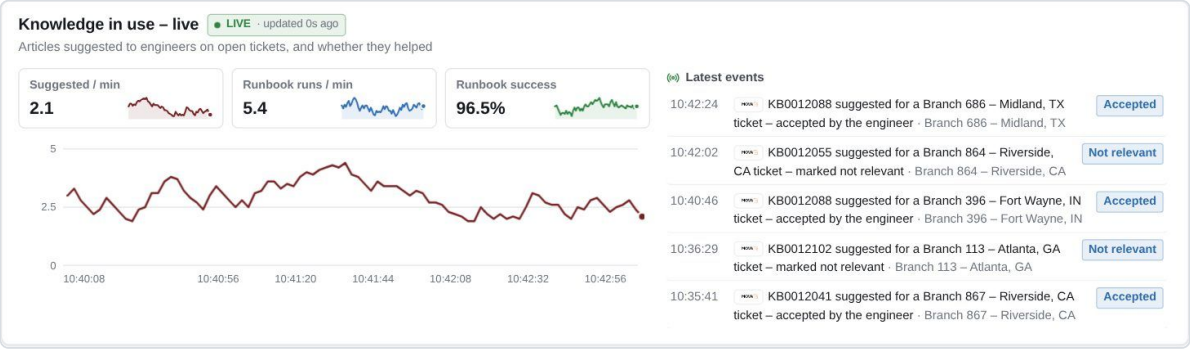



**Figure 188.** Knowledge in use

Articles suggested per minute on open tickets, runbook runs and success, and the latest suggestions with whether the engineer accepted the article or marked it not relevant.

## What keeps coming back

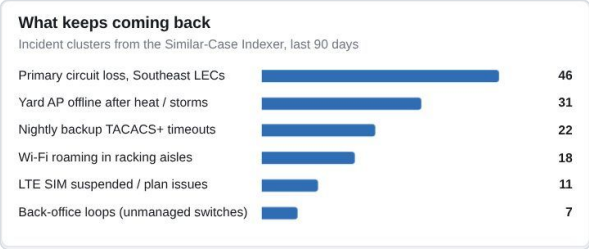



**Figure 189.** What keeps coming back

Incident clusters found by the Similar-Case Indexer over 90 days, longest bar first: for example primary circuit loss on Southeast LECs, yard APs offline after heat or storms, backup timeouts. Each cluster is a candidate for a problem record and a permanent fix.

## Self-service deflection




**Figure 190.** Self-service deflection

Branch requests solved from a knowledge article without a ticket, by month, against the 35% target of KPI 39.

## Most used articles

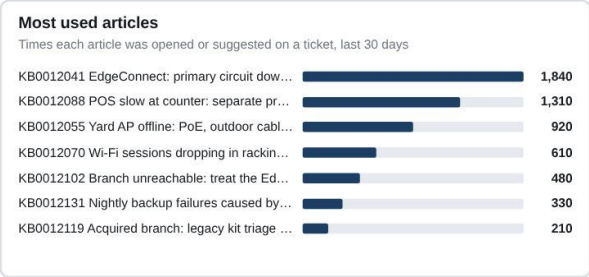



**Figure 191.** Most used articles

The seven articles opened or suggested most often on tickets in the last 30 days, with the count. The top articles are usually the runbook-backed ones for the commonest faults, such as a primary circuit down or a slow payment path.

## Suggestions accepted by engineers




**Figure 192.** Suggestions accepted

Articles Mova iO suggested on tickets each week, split into those the engineer accepted and those marked not relevant. A rising green share means the recommendations are getting better (KPI 39 asks for 80% relevant).

## Knowledge base and similar incidents

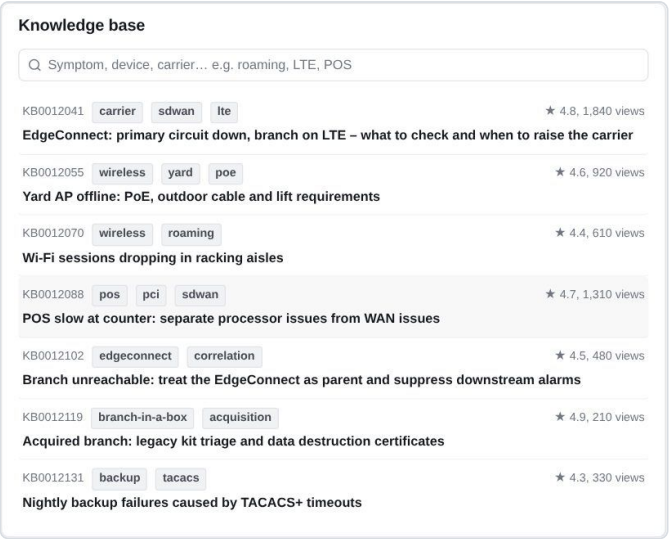



**Figure 193.** Knowledge base

Each article shows its number, tags, rating and views, and its title. Click to read it.

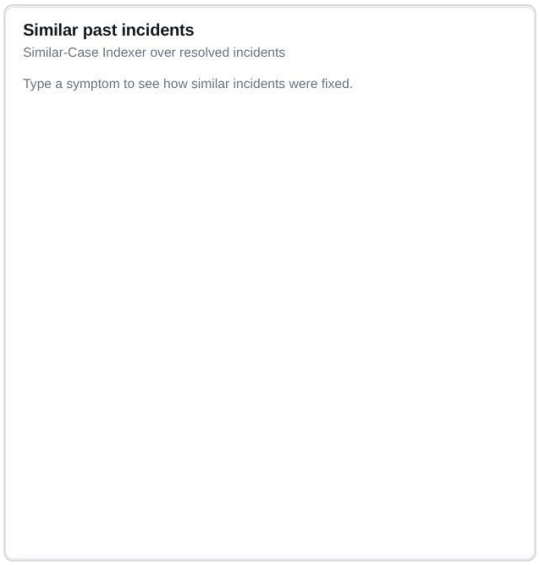



**Figure 194.** Similar past incidents

Type a symptom to see how similar incidents were fixed, ranked by similarity.

## Auto-generated drafts tab




**Figure 195.** Draft article

Each draft shows the agent, the incident it came from, its age, the title, the symptom and fix, and the agent's confidence. **Approve & publish** makes it live, **Edit** changes it first, **Reject** discards it. No draft is published without an engineer.

## Problem records tab



| Problem | Title | Linked incidents | Owner | Opened | State |
|---|---|---|---|---|---|
| PRB0004211 | Repeat Lumen DIA loss at Southeast branches on shared LEC | 14 | Network Eng L3 | 18d ago | Root cause analysis |
| PRB0004218 | Outdoor AP-585 reboots in high heat (Sun Belt yards) | 9 | Wireless team | 26d ago | Known error |
| PRB0004230 | TACACS+ node timeouts during 02:00 backup window | 22 | Security Lead | 9d ago | Fix in progress |
| PRB0004236 | Policy v41 maps ERP DSCP to the bulk class; pre-open validation has no ERP test | 1 | Network Eng L3 | 4h 28m ago | Fix in progress |

**Figure 196.** Problem records

Each problem has a number, title, linked incidents, owner, age and state, moved along New, Root cause analysis, Known error, Fix in progress and Closed. A problem is opened when an incident repeats three or more times in 30 days, after every P1, or when a cluster appears in "What keeps coming back".

# 20. Mova iO Agents

This screen shows the 13 Mova iO agents: what each is doing now, how often it acts and how often it gets it right. Every action follows the gate rules and lands in the audit log. The header shows how many agents are working right now and the actions taken today.

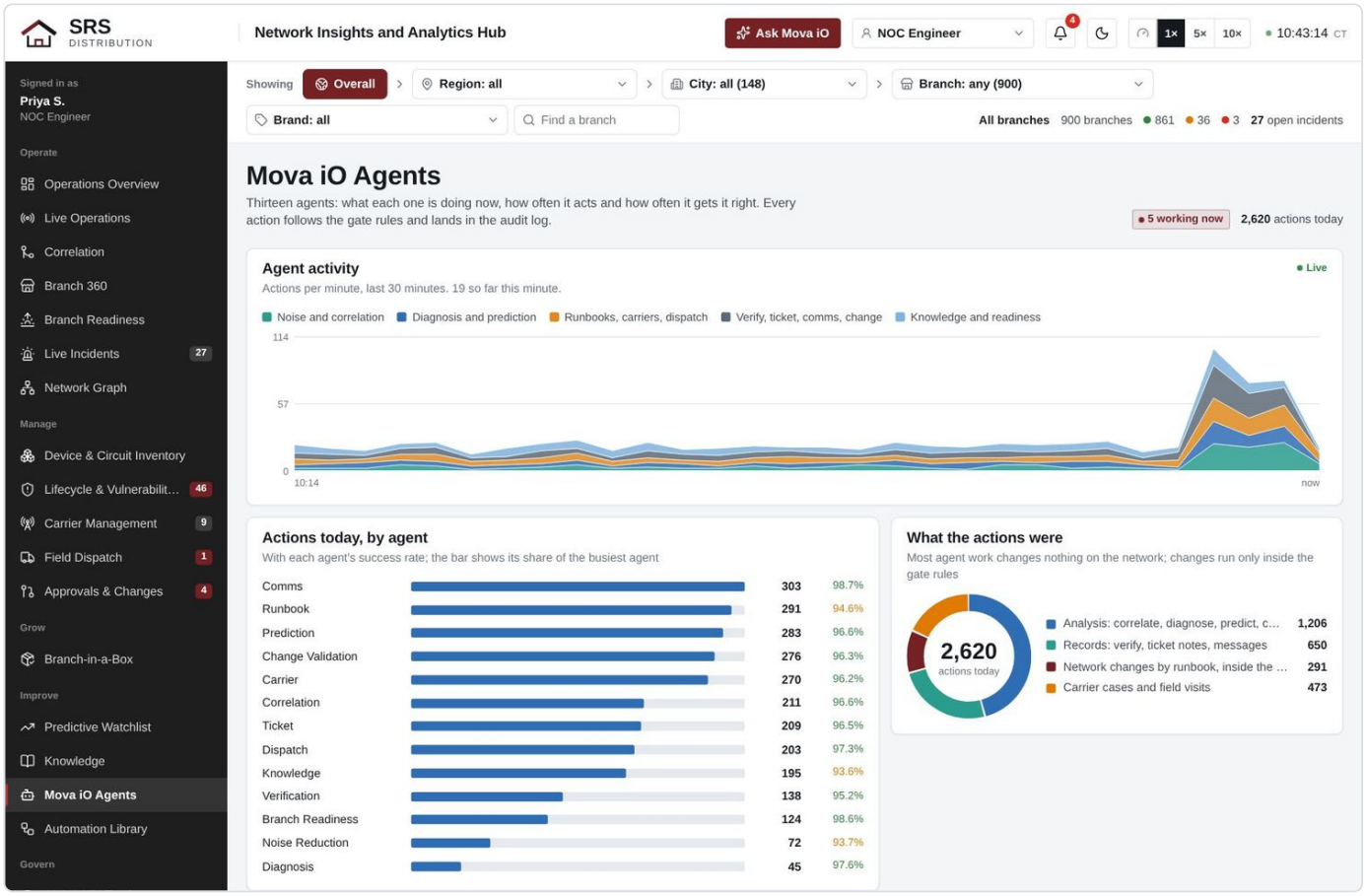



**Figure 197.** Mova iO Agents

## Agent activity

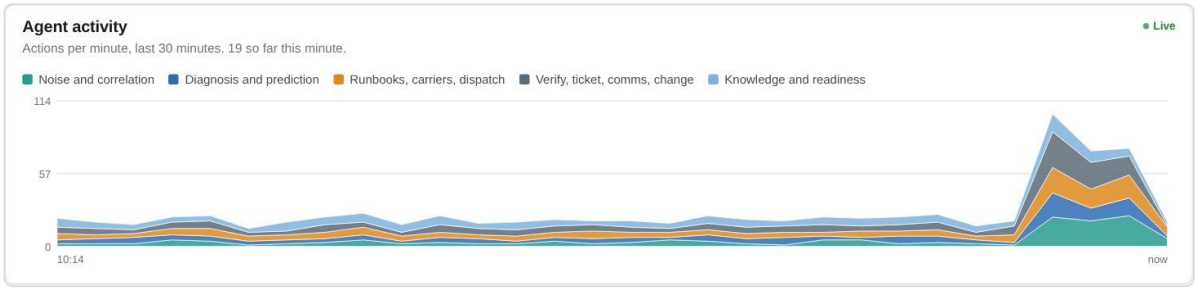



**Figure 198.** Agent activity

Actions per minute for the last 30 minutes, stacked by agent group: noise and correlation; diagnosis and prediction; runbooks, carriers and dispatch; verify, ticket, comms and change; knowledge and readiness. The subtitle gives the count so far this minute. A busy noise-and-correlation band with a thin runbook band is normal: most work is filtering.

## Actions today, by agent

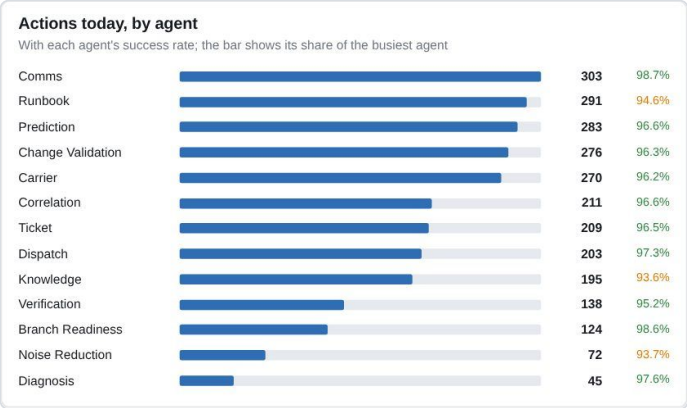



**Figure 199.** Actions today by agent

Every agent ranked by actions today, with its success rate on the right (green from 95%). Comms, runbooks and prediction are usually the busiest; diagnosis acts only when an incident needs a root cause.

## What the actions were

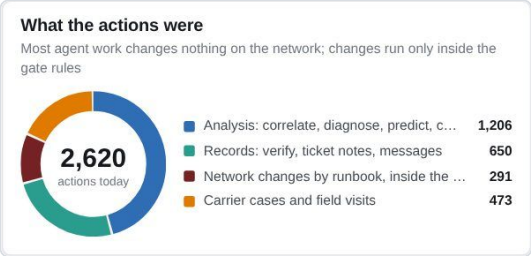



**Figure 200.** What the actions were

Today's actions by kind: analysis (correlate, diagnose, predict, check readiness and changes), records (verification, ticket notes, messages), network changes by runbook inside the gate rules, and carrier cases and field visits. Most agent work changes nothing on the network; the maroon slice is the only part that touches devices.

## Agent cards

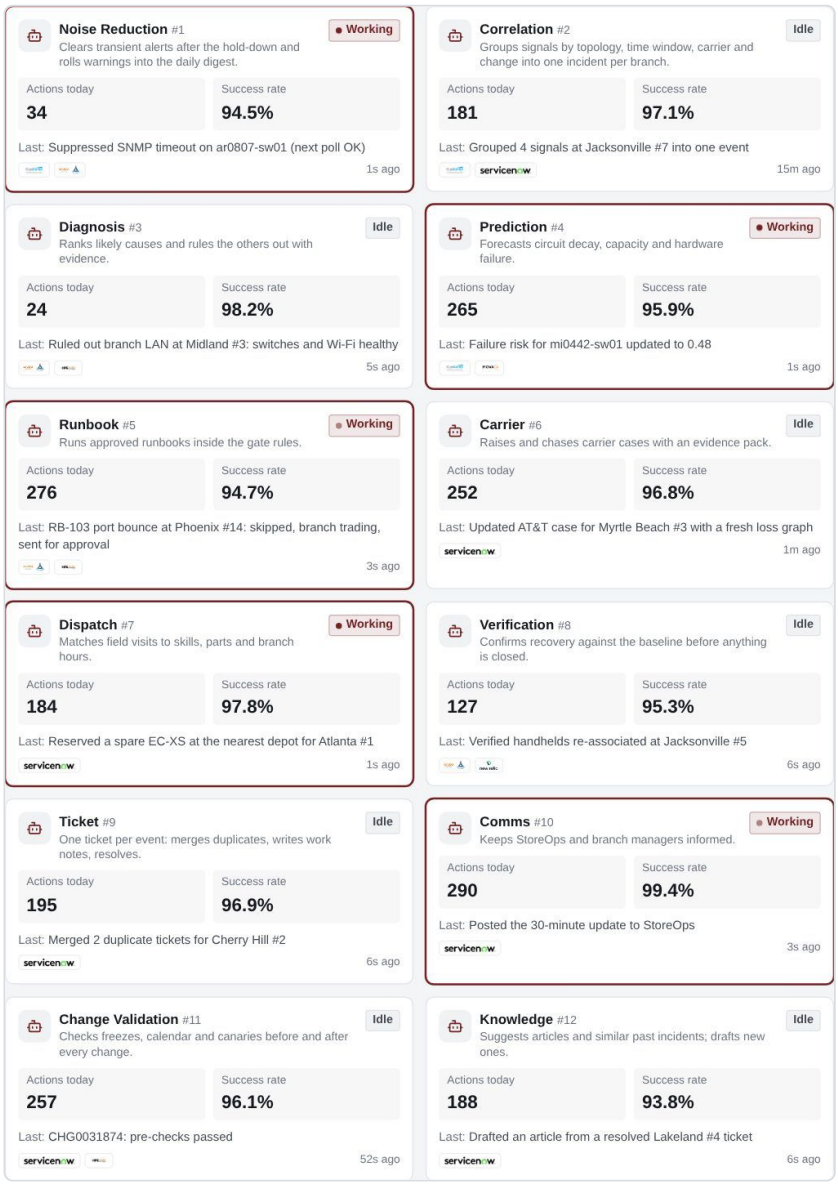



**Figure 201.** Agent cards

One card per agent: its name and number, what it does, **Working** (maroon, it acted in the last few seconds) or **Idle**, actions today, success rate, its last action, the tools it acts through and how long ago.

| Agent | What it does |
|---|---|
| Noise Reduction | Clears transient alerts after the hold-down and rolls warnings into the daily digest |
| Correlation | Groups signals by topology, time window, carrier and change into one incident per branch |
| Diagnosis | Ranks likely causes and rules the others out with evidence |
| Prediction | Forecasts circuit decay, capacity and hardware failure |
| Runbook | Runs approved runbooks inside the gate rules |
| Carrier | Raises and chases carrier cases with an evidence pack |

| Agent | What it does |
|---|---|
| Dispatch | Matches field visits to skills, parts and branch hours |
| Verification | Confirms recovery against the baseline before anything is closed |
| Ticket | One ticket per event: merges duplicates, writes work notes, resolves |
| Comms | Keeps StoreOps and branch managers informed |
| Change Validation | Checks freezes, the calendar and canaries before and after every change |
| Knowledge | Suggests articles and similar past incidents; drafts new ones |
| Branch Readiness | Runs the pre-opening checks at 05:30 local for every branch |

## Agent activity stream

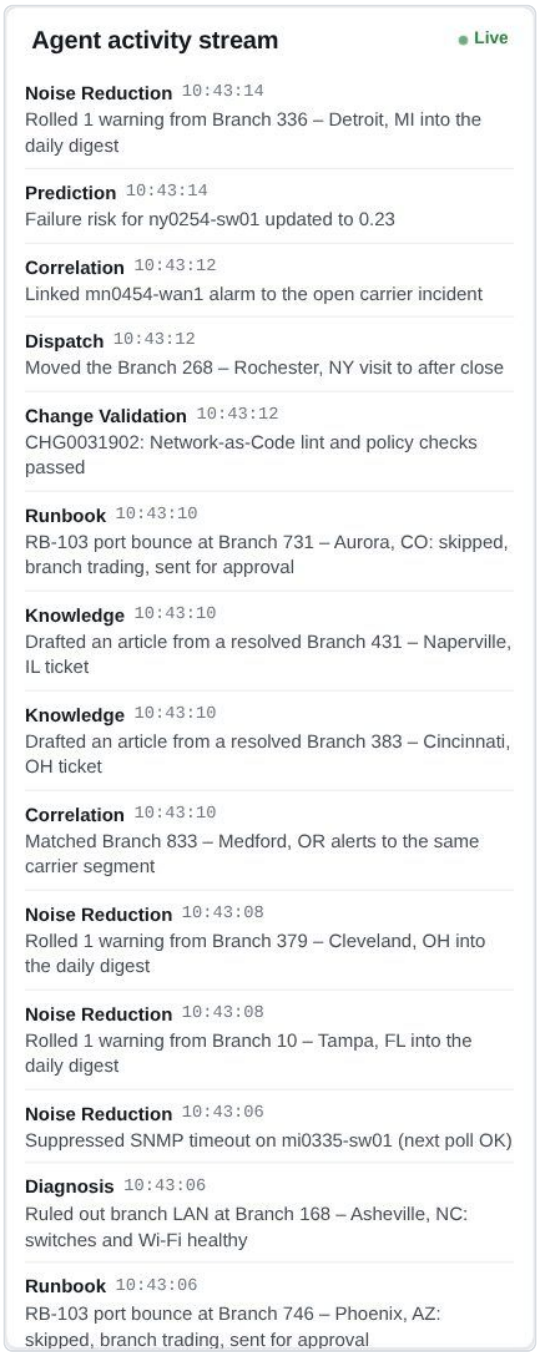



**Figure 202.** Agent activity stream

Every agent action as it happens, with the agent, time and what it did.

# 21. Automation Library

The Automation Library is the catalog of approved runbooks: what each does, how much it may do on its own, how often it runs and how often it succeeds.

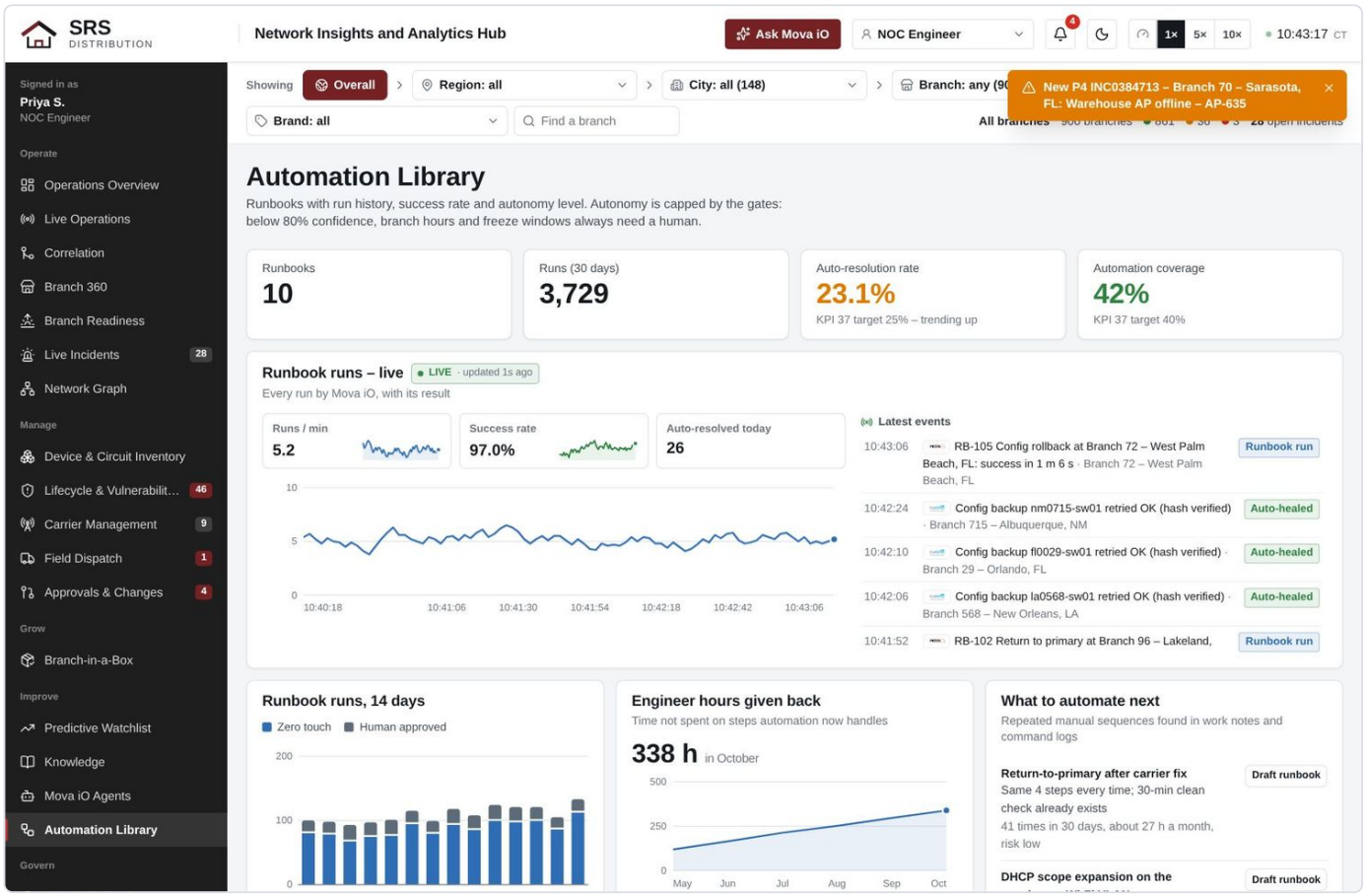



**Figure 203.** Automation Library

## Tiles

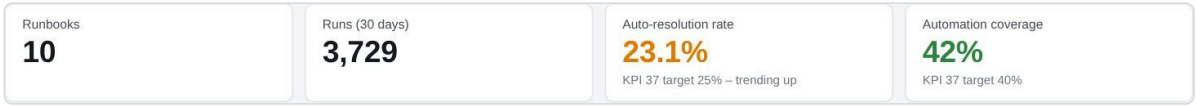



**Figure 204.** Automation tiles

| Tile | Meaning | Target |
|---|---|---|
| Runbooks | Runbooks in the catalog | – |
| Runs (30 days) | All runs in the last 30 days | – |
| Auto-resolution rate (KPI 37) | Incidents closed by a runbook with verification and no human touch | 25% in year 1, rising |
| Automation coverage (KPI 37) | Share of incident types with an approved runbook | 40% |

## Runbook runs – live

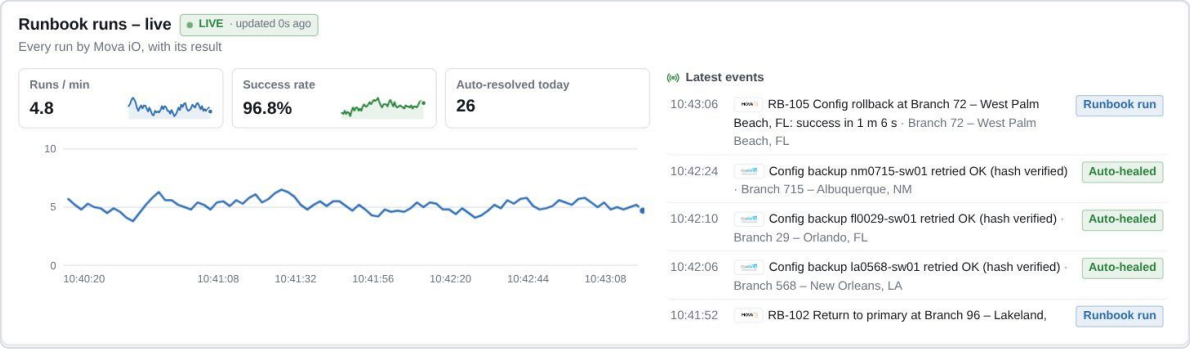



**Figure 205.** Runbook runs

Runs per minute, success rate and incidents auto-resolved today, with the latest runs and their results.

## Runbook runs, 14 days and hours given back

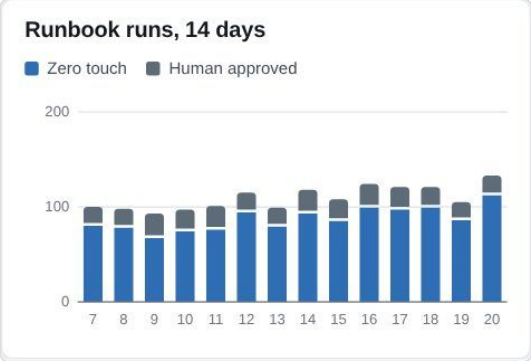



**Figure 206.** Runbook runs, 14 days

Runs per day split into **zero touch** (ran inside its tier with no approval) and **human approved**. Zero touch growing is autonomy growing.

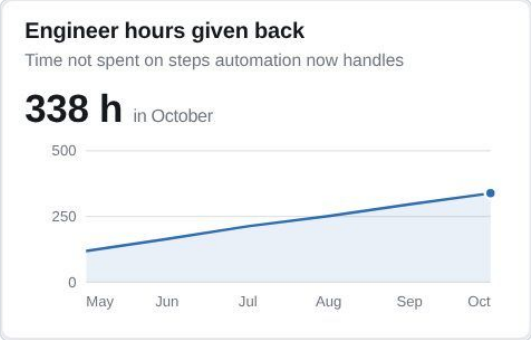



**Figure 207.** Engineer hours given back

Engineer hours not spent on steps automation now handles, this month and by month.

## What to automate next

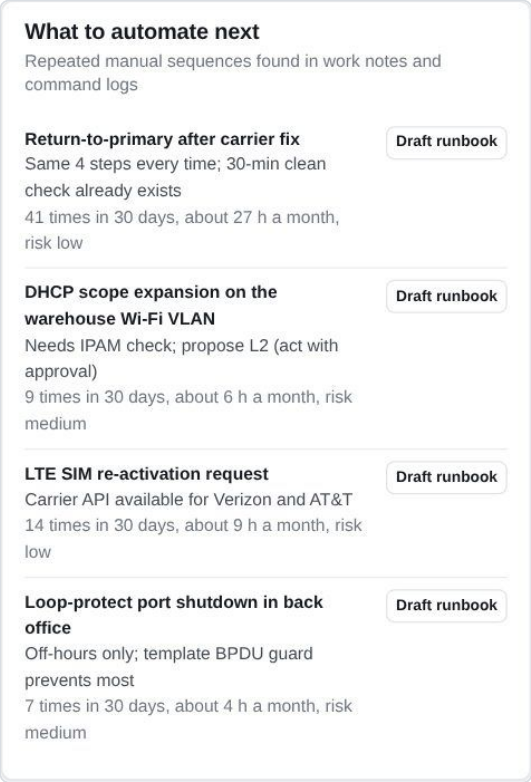
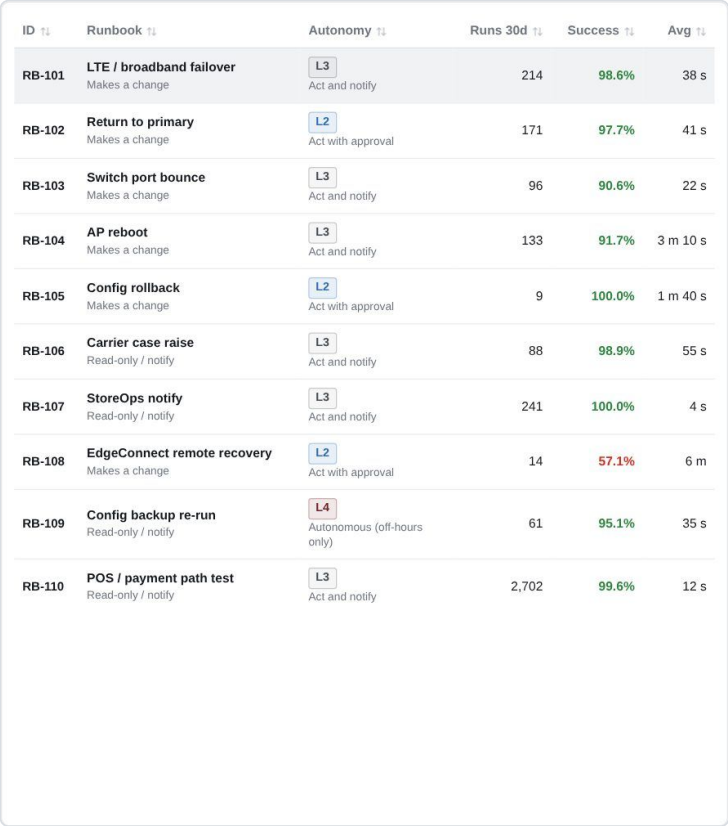



**Figure 208.** What to automate next

Repeated manual sequences found in work notes and command logs, with how often they happen, the hours they cost, their risk and the proposed autonomy level. **Draft runbook** starts the Toil Miner flow shown in the toil demo: a draft, generated checks, a replay against past cases and approval at the Stage tier.

## The catalog



| ID | Runbook | Autonomy | Runs 30d | Success | Avg |
|---|---|---|---|---|---|
| RB-101 | LTE / broadband failover<br>Makes a change | L3<br>Act and notify | 214 | 98.6% | 38 s |
| RB-102 | Return to primary<br>Makes a change | L2<br>Act with approval | 171 | 97.7% | 41 s |
| RB-103 | Switch port bounce<br>Makes a change | L3<br>Act and notify | 96 | 90.6% | 22 s |
| RB-104 | AP reboot<br>Makes a change | L3<br>Act and notify | 133 | 91.7% | 3 m 10 s |
| RB-105 | Config rollback<br>Makes a change | L2<br>Act with approval | 9 | 100.0% | 1 m 40 s |
| RB-106 | Carrier case raise<br>Read-only / notify | L3<br>Act and notify | 88 | 98.9% | 55 s |
| RB-107 | StoreOps notify<br>Read-only / notify | L3<br>Act and notify | 241 | 100.0% | 4 s |
| RB-108 | EdgeConnect remote recovery<br>Makes a change | L2<br>Act with approval | 14 | 57.1% | 6 m |
| RB-109 | Config backup re-run<br>Read-only / notify | L4<br>Autonomous (off-hours only) | 61 | 95.1% | 35 s |
| RB-110 | POS / payment path test<br>Read-only / notify | L3<br>Act and notify | 2,702 | 99.6% | 12 s |

**Figure 209.** Runbook catalog

| Column | Meaning |
|---|---|
| ID and runbook | RB number, name, and whether it makes a change or is read-only and notify |
| Autonomy | L1 to L4 and what it allows |
| Runs 30d, success, avg | Volume, success rate and average run time |

| Level | Name | Meaning |
|---|---|---|
| L1 | Recommend only | Mova iO suggests it; a person runs it |
| L2 | Act with approval | Stage tier: prepared, then waits for a named person |
| L3 | Act and notify | Runs inside the gate rules and notifies at once |
| L4 | Autonomous (off-hours only) | Runs on its own outside trading hours, for read-only or no-impact work |

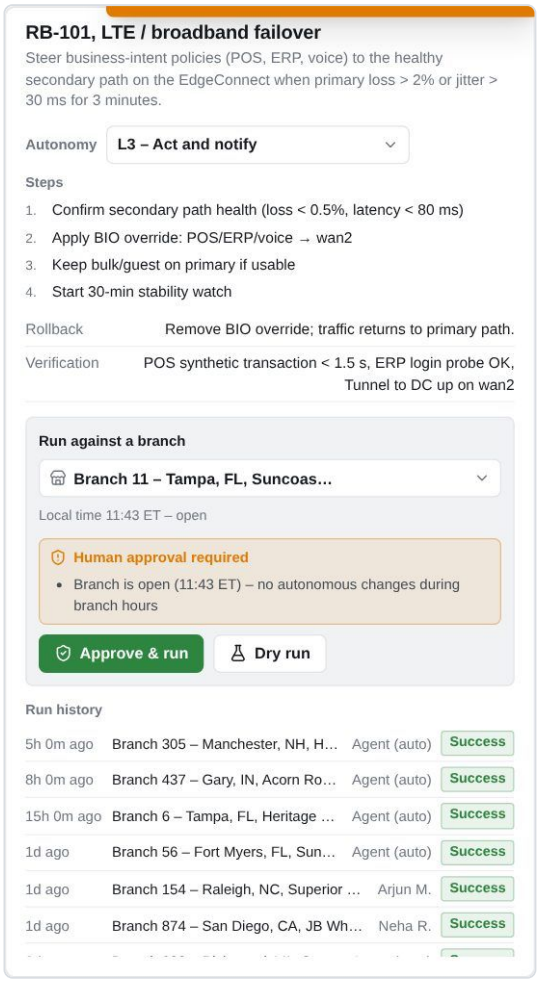



**Figure 210.** Runbook detail

The selected runbook shows its trigger, its autonomy level (with the levels it could move to), its steps, rollback, verification checks and recent runs. Changing the autonomy level is itself a change that SRS approves.

# 22. SLA & KPI Tracker

The SLA & KPI Tracker measures every service level and KPI agreed for the contract (Annexure B) against its minimum service level (MSL) and expected service level (ESL), with trends and breach risk.

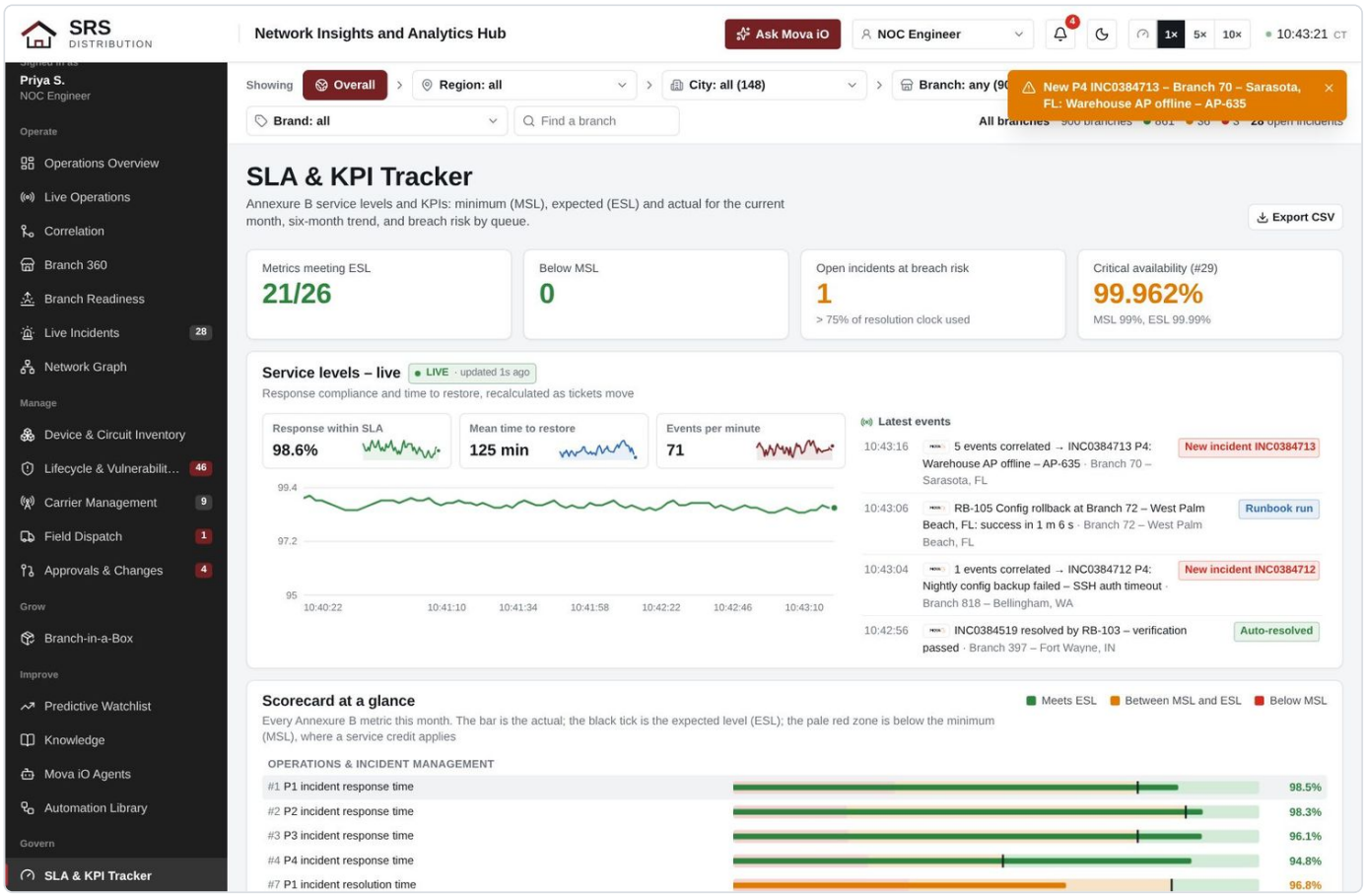



**Figure 211.** SLA & KPI Tracker

## Tiles

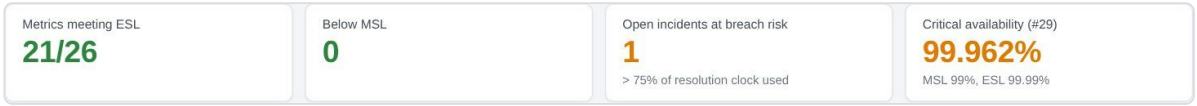



**Figure 212.** SLA tiles

| Tile | Meaning |
|---|---|
| Metrics meeting ESL | Metrics at or above their expected level, out of all measured |
| Below MSL | Metrics under their minimum; below MSL in a month triggers a service credit |
| Open incidents at breach risk | Open incidents with more than 75% of their resolution clock used |
| Critical availability (#29) | Availability of in-scope critical network devices and services, against MSL 99% and ESL 99.99% |

## Service levels – live

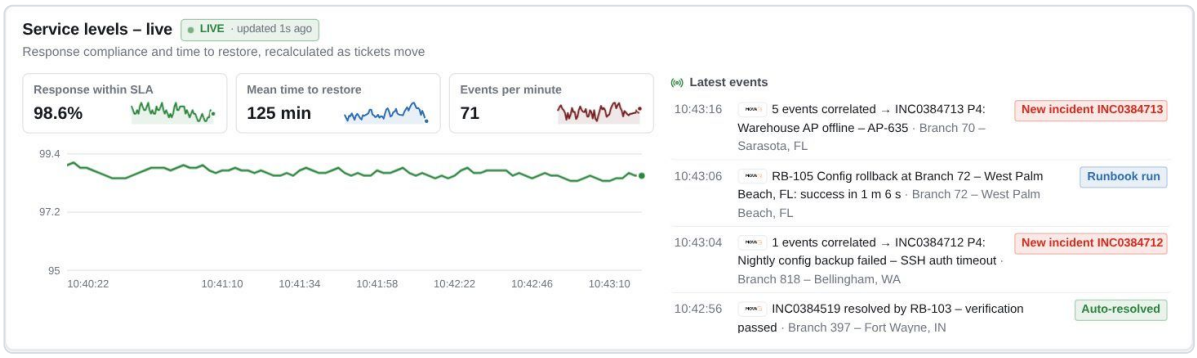



**Figure 213.** Service levels live

Response within SLA, mean time to restore and events per minute, recalculated as tickets move, with the latest ticket events.

## Scorecard at a glance




**Figure 214.** Scorecard at a glance

Every Annexure B metric this month on one card, grouped by service line. Each row is a bullet bar:

| Part | Meaning |
|---|---|
| Coloured bar | This month's actual: green meets the expected level, amber is between the minimum and the expected level, red is below the minimum |
| Black tick | The expected service level (ESL), the target |
| Pale red zone | Below the minimum service level (MSL), where a service credit applies |
| Pale amber and green zones | Between MSL and ESL, and above ESL |
| Figure on the right | The actual, in the colour of its status |

Click a row to show that metric's projection at the bottom of the screen. For a "lower is better" metric, such as the alert false positive rate, the bar is green while it stays under the tick.

## Where the service stands




**Figure 215.** Where the service stands

A donut of all 26 metrics by status (meets ESL, between MSL and ESL, below target for KPIs, below MSL), and a bar per service line showing how many of its metrics meet the expected level.

## Respond and resolve on time, by priority

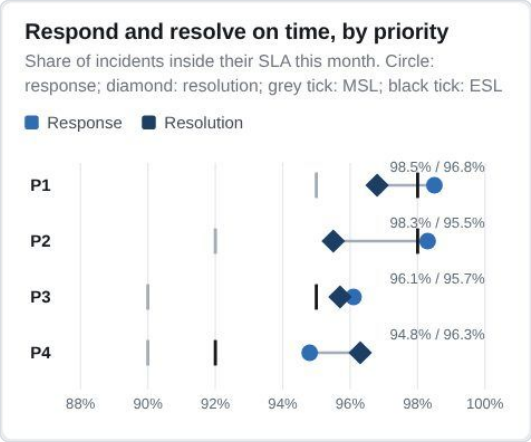



**Figure 216.** Respond and resolve by priority

One line per priority on a scale from 88% to 100%: the circle is the share of incidents answered within the response time, the diamond the share fixed within the resolution time. The grey tick is the MSL and the black tick the ESL. A circle or diamond left of its black tick is the place to focus.

## Service credit exposure

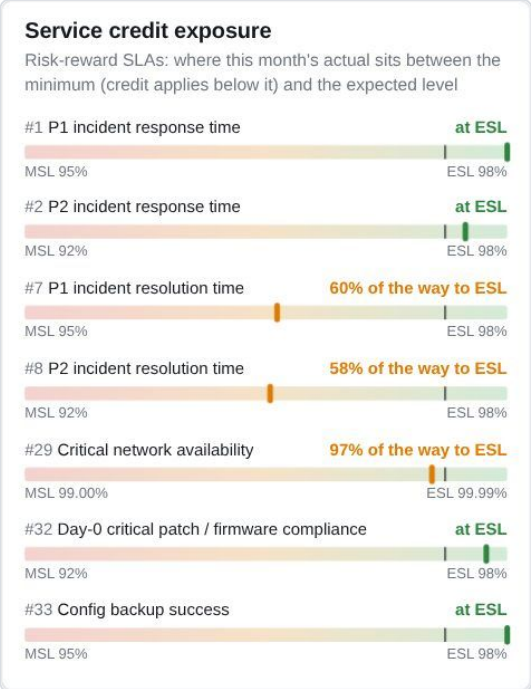



**Figure 217.** Service credit exposure

The risk-reward SLAs, which carry a service credit. Each bar runs from the minimum (left, red: a credit applies below it) to the expected level (the black line); the marker shows where this month's actual sits, labelled "at ESL" or how far of the way to ESL it is. A marker in the red end is a credit risk.

## Service levels and error budgets

The same four error budgets as on the Operations Overview (chapter 5): branches online, jitter under 30 ms, loss under 1%, and P1/P2 answered in 15 minutes.

## Mean time to restore, and tickets per month

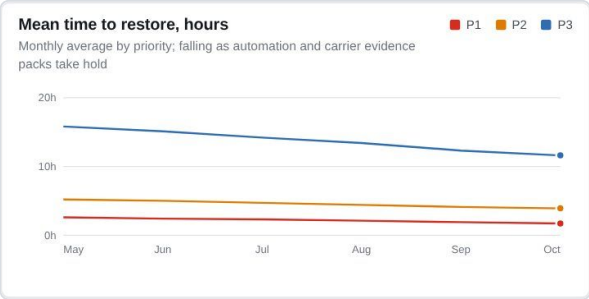



**Figure 218.** Mean time to restore

Average hours to restore service each month, for P1, P2 and P3. Falling lines show automation and carrier evidence packs shortening repairs.

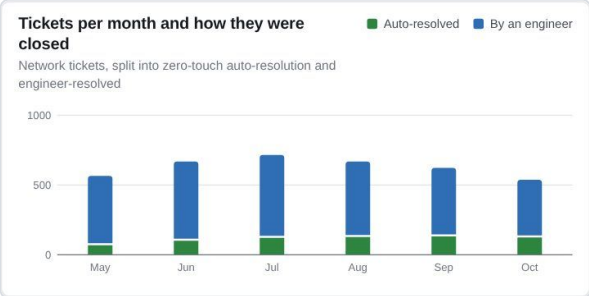



**Figure 219.** Tickets per month

Network tickets per month, split into those closed by zero-touch automation (green) and those resolved by an engineer (blue). The green share is the auto-resolution rate (KPI 37).

## Six months at a glance




**Figure 220.** Six months at a glance

One row per metric and one column per month, each cell coloured by that month's status. Read across a row to see a metric's history and down a column to see which month slipped; an exclamation mark is a month below MSL. Click a row to show its projection.

## Incidents per day, last 12 weeks

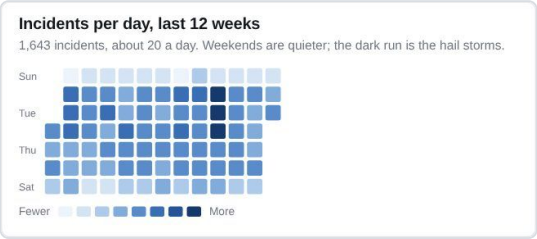



**Figure 221.** Incidents per day

A calendar of incidents per day for 12 weeks, one square per day, rows Sunday to Saturday; darker means more. Weekends are quieter; a dark run usually matches a storm.

## Breach risk by queue

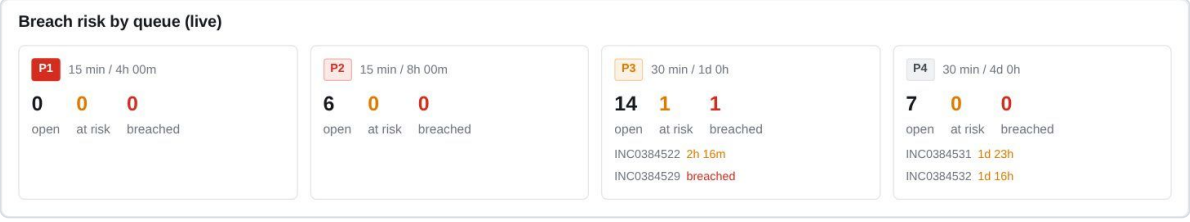
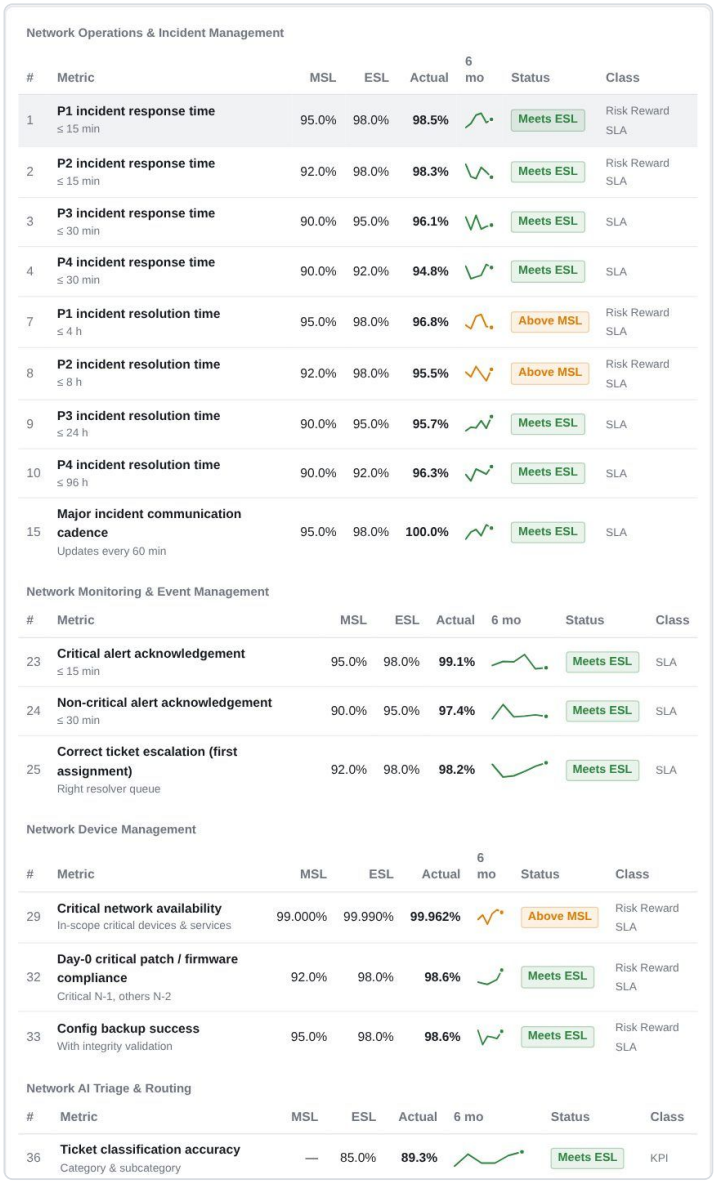



**Figure 222.** Breach risk by queue

One block per priority with its response and resolution targets, the number open, at risk and breached, and the incidents closest to their limit.

## The metrics table



Network Operations & Incident Management

| # | Metric | MSL | ESL | Actual | 6 mo | Status | Class |
|---|---|---|---|---|---|---|---|
| 1 | **P1 incident response time** ≤ 15 min | 95.0% | 98.0% | **98.5%** | | Meets ESL | Risk Reward SLA |
| 2 | **P2 incident response time** ≤ 15 min | 92.0% | 98.0% | **98.3%** | | Meets ESL | Risk Reward SLA |
| 3 | **P3 incident response time** ≤ 30 min | 90.0% | 95.0% | **96.1%** | | Meets ESL | SLA |
| 4 | **P4 incident response time** ≤ 30 min | 90.0% | 92.0% | **94.8%** | | Meets ESL | SLA |
| 7 | **P1 incident resolution time** ≤ 4 h | 95.0% | 98.0% | **96.8%** | | Above MSL | Risk Reward SLA |
| 8 | **P2 incident resolution time** ≤ 8 h | 92.0% | 98.0% | **95.5%** | | Above MSL | Risk Reward SLA |
| 9 | **P3 incident resolution time** ≤ 24 h | 90.0% | 95.0% | **95.7%** | | Meets ESL | SLA |
| 10 | **P4 incident resolution time** ≤ 96 h | 90.0% | 92.0% | **96.3%** | | Meets ESL | SLA |
| 15 | **Major incident communication cadence** Updates every 60 min | 95.0% | 98.0% | **100.0%** | | Meets ESL | SLA |

Network Monitoring & Event Management

| # | Metric | MSL | ESL | Actual | 6 mo | Status | Class |
|---|---|---|---|---|---|---|---|
| 23 | **Critical alert acknowledgement** ≤ 15 min | 95.0% | 98.0% | **99.1%** | | Meets ESL | SLA |
| 24 | **Non-critical alert acknowledgement** ≤ 30 min | 90.0% | 95.0% | **97.4%** | | Meets ESL | SLA |
| 25 | **Correct ticket escalation (first assignment)** Right resolver queue | 92.0% | 98.0% | **98.2%** | | Meets ESL | SLA |

Network Device Management

| # | Metric | MSL | ESL | Actual | 6 mo | Status | Class |
|---|---|---|---|---|---|---|---|
| 29 | **Critical network availability** In-scope critical devices & services | 99.000% | 99.990% | **99.962%** | | Above MSL | Risk Reward SLA |
| 32 | **Day-0 critical patch / firmware compliance** Critical N-1, others N-2 | 92.0% | 98.0% | **98.6%** | | Meets ESL | Risk Reward SLA |
| 33 | **Config backup success** With integrity validation | 95.0% | 98.0% | **98.6%** | | Meets ESL | Risk Reward SLA |

Network AI Triage & Routing

| # | Metric | MSL | ESL | Actual | 6 mo | Status | Class |
|---|---|---|---|---|---|---|---|
| 36 | **Ticket classification accuracy** Category & subcategory | — | 85.0% | **89.3%** | | Meets ESL | KPI |

**Figure 223.** Metrics table

Metrics are grouped by area, for example Network Operations & Incident Management.

| Column | Meaning |
|---|---|
| # and metric | The Annexure B number and name, with its definition (for example P1 response within 15 minutes) |
| MSL | Minimum service level; below it a credit applies |
| ESL | Expected service level, the target |
| Actual | This month's result |
| 6 mo | The trend over six months |

| Column | Meaning |
|---|---|
| Status | Meets ESL (green), Above MSL (amber), Below MSL (red) |
| Class | **Risk Reward SLA** (carries a service credit and earn-back), **SLA** or **KPI** |

The SLAs that carry credits are P1 and P2 response, P1 and P2 resolution, critical network availability (#29), Day-0 critical patch and firmware compliance (#32) and config backup success (#33). AI and automation KPIs, such as auto-resolution, automation coverage, predictive accuracy and alert noise reduction, have expected levels only.

## Metric projection




**Figure 224.** Metric projection

The selected metric's monthly results and a two-month projection with its likely range, against its minimum and expected lines, with a sentence on whether the projection meets the expected level. Click any row in the table to see its projection.

# 23. Reports

Every report is built from live console data at the moment it is opened, with who generated it and when. **Download HTML** saves it with its charts (it opens in Word or a browser); **Print / PDF** prints it. Each report opens with a row of charts, then the tables:

| Report | Charts at the top |
|---|---|
| Shift handover | Open incidents by priority; SLA clock used for each open P1 and P2, with the 75% at-risk line; incidents opened per hour over the last 12 hours, by priority |
| Daily operations summary | Branch health now; incidents opened per hour over 24 hours, resolved or still open; incidents by category; WAN lines down or degraded, by carrier |
| Weekly service review pack | Annexure B metrics by status; the risk-reward SLAs this month; network tickets per month by close code; each carrier's restore commitment met, against 90%; each carrier's mean time to restore |
| Branch-in-a-Box onboarding report | Projects by stage and track; project health (RAG); cutover gates passed for each project, out of 13 |
| Audit log | Who acted (Mova iO agents, people, system); actions per hour over the last 12 hours |

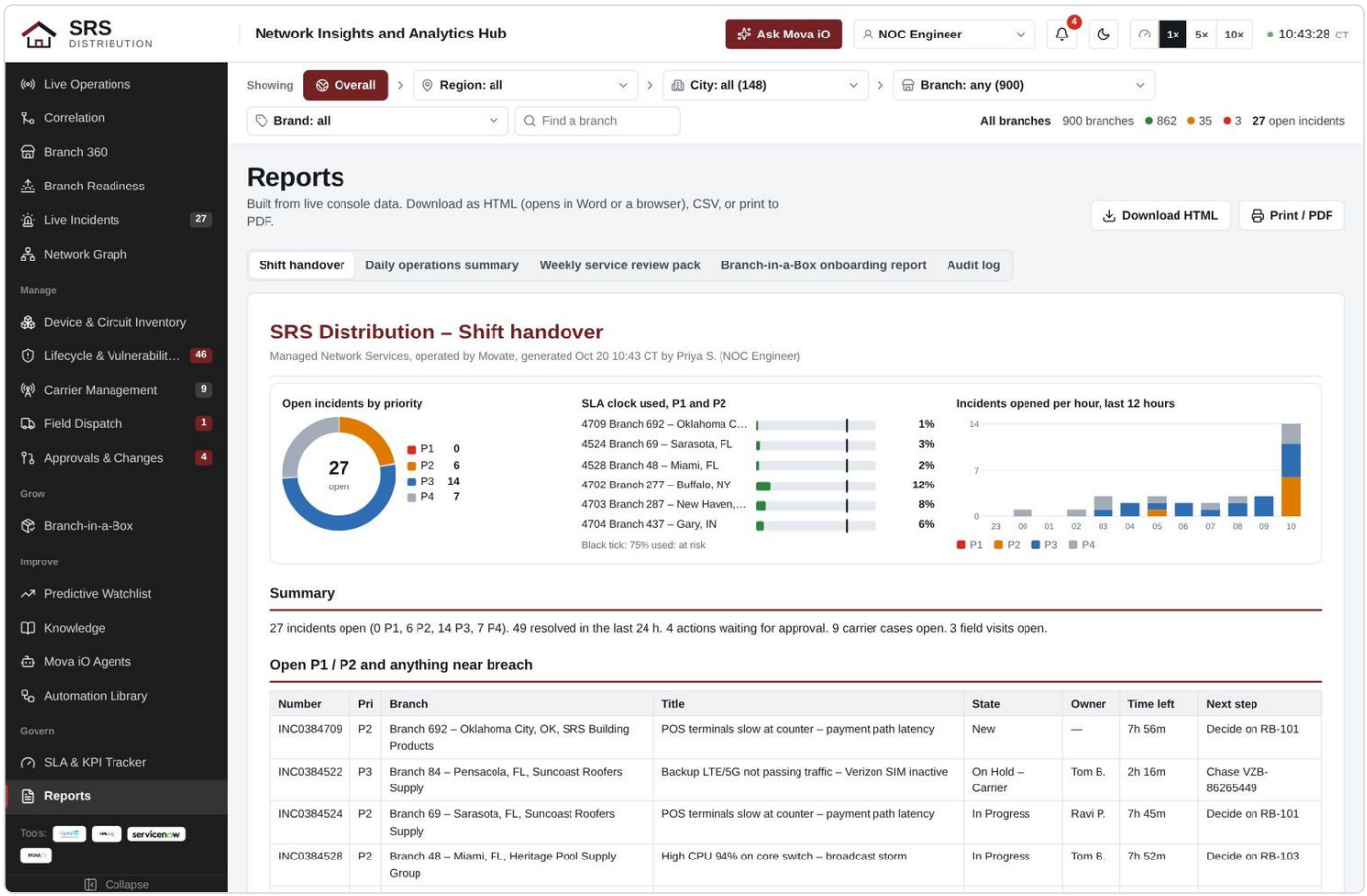



**Figure 225.** Reports: shift handover

| Report | For | What it contains |
|---|---|---|
| Shift handover | The incoming NOC shift | Summary line; open P1 and P2 and anything near breach, with owner, time left and next step; actions waiting on a human; carrier cases |
| Daily operations summary | StoreOps and IT leadership | Network health; incidents in the last 24 hours; readiness this morning; carriers |
| Weekly service review pack | The weekly service review | Service levels against Annexure B; ticket mix for six months; problems and improvements; carriers |
| Branch-in-a-Box onboarding report | The M&A integration lead | The project portfolio and this weekend's cutover gates |
| Audit log | Security and audit | Every human and agent action this session: approvals, runbook runs, changes, demos and settings, with who and when |

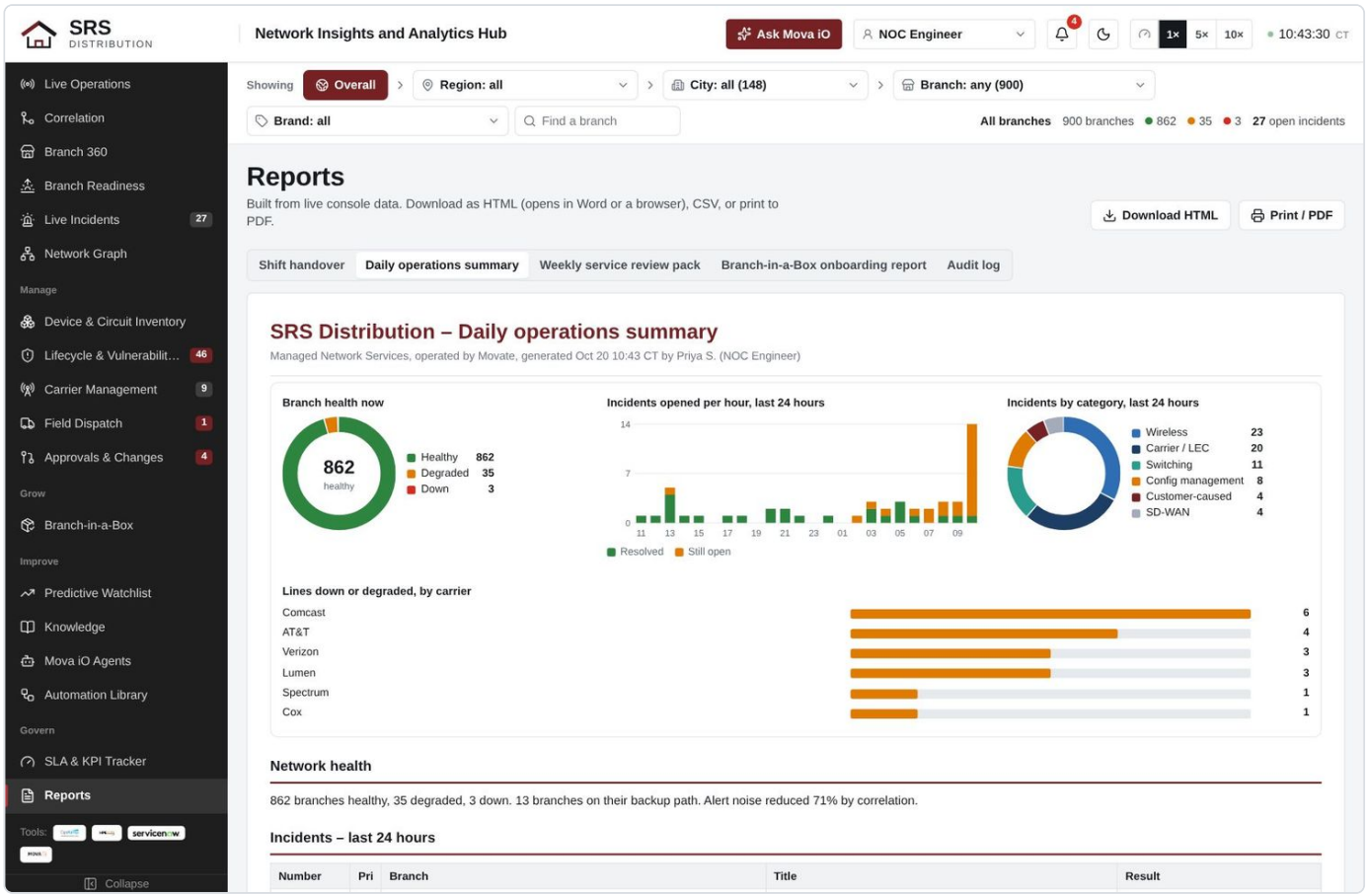



**Figure 226.** Daily operations summary

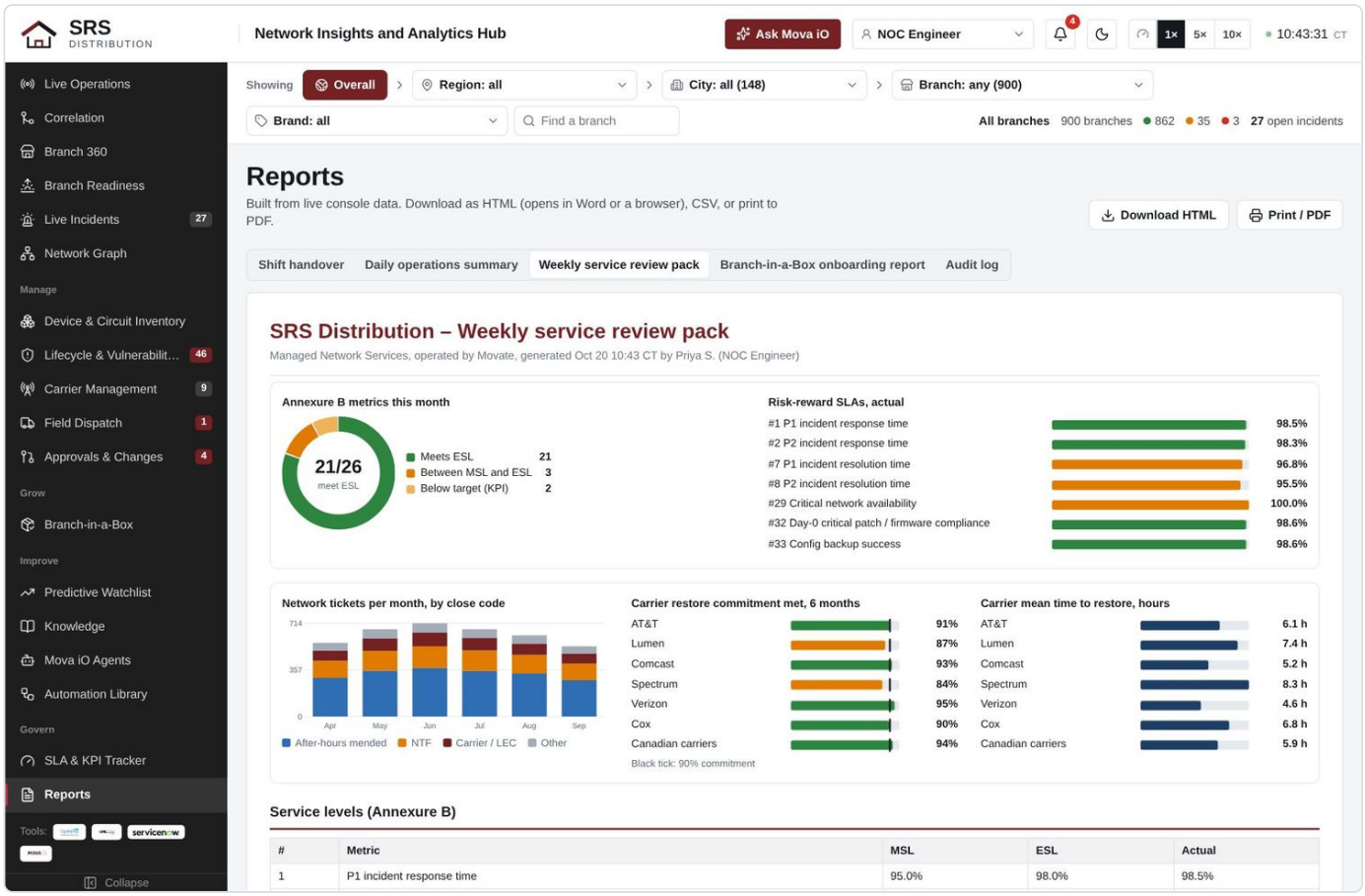



**Figure 227.** Weekly service review pack

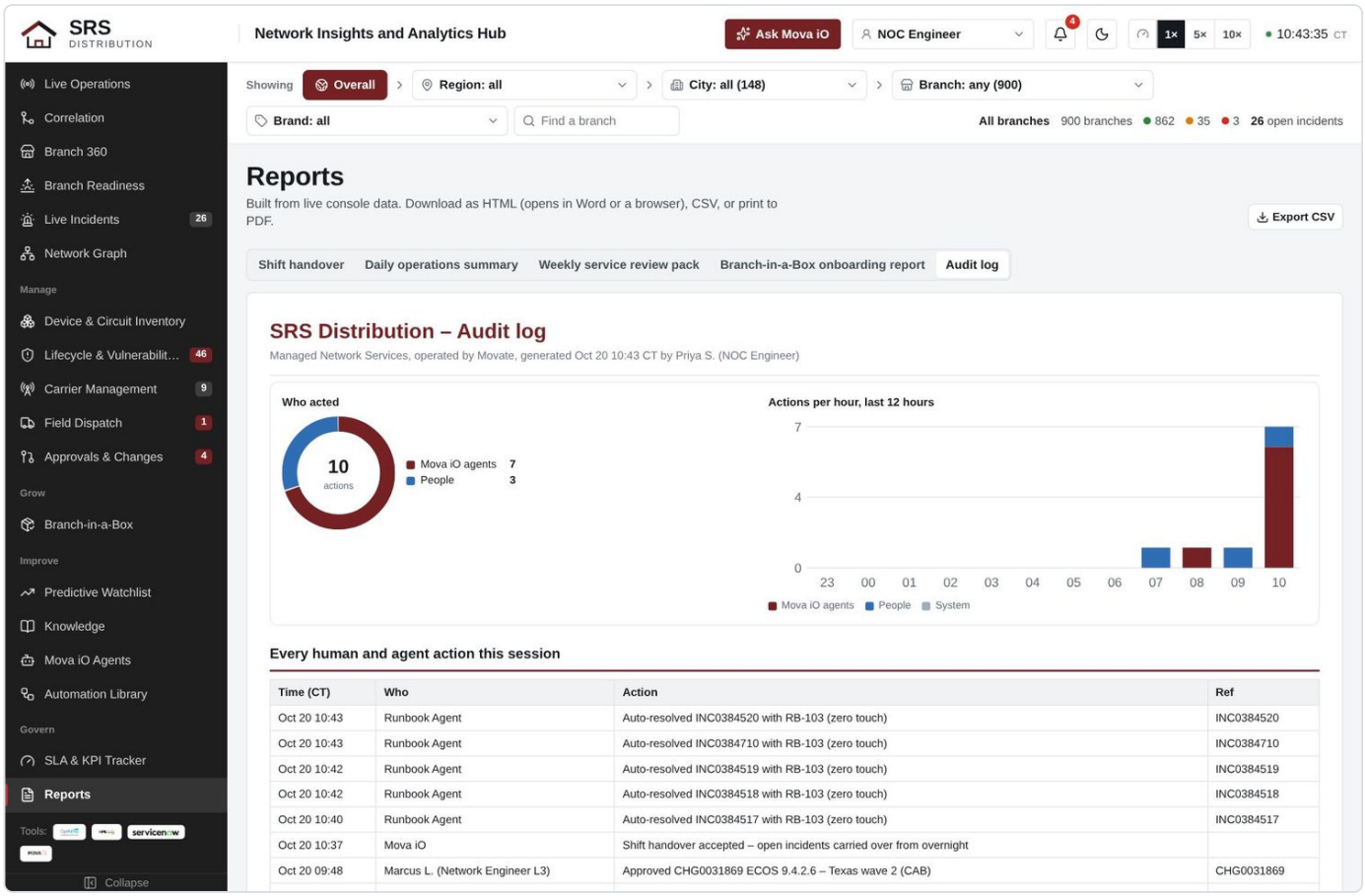



**Figure 228.** Audit log

# 24. Live Demos

Live Demos is the library of 33 scripted SRS branch situations. It is a page of its own ( `demos/` on the site) with its own link; the console and the demos page do not link to each other, so the console can be shown to anyone without the scripted stories. Each demo plays on a full-screen stage that shows every system involved at once, and shows the incident, approvals, carrier cases and work orders it creates exactly as they would appear in the console. Part III describes each demo.

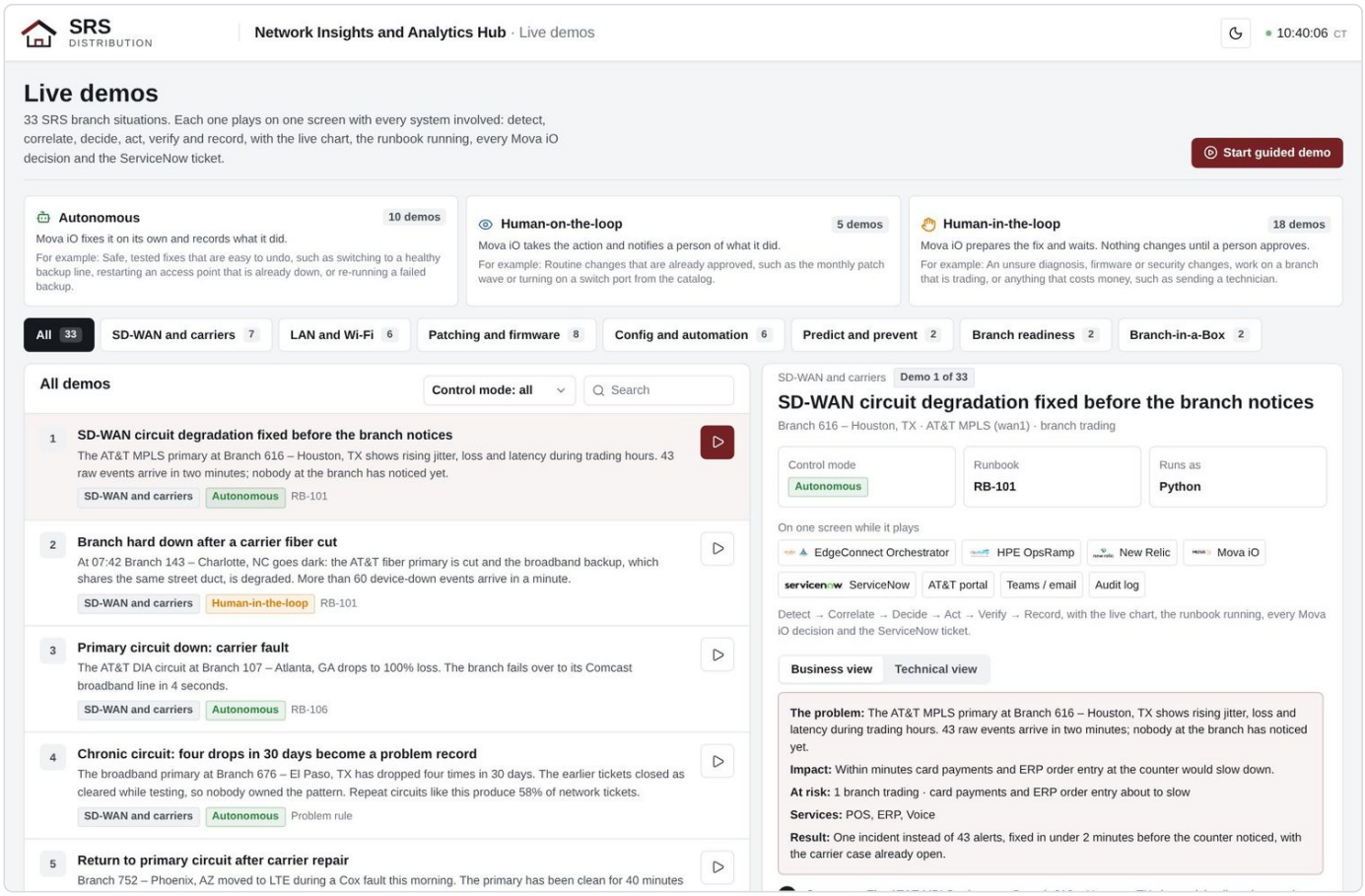



**Figure 229.** Live Demos screen

## The demo library

| Part of the screen | What it shows | How to use it |
|---|---|---|
| Start guided demo | Plays demo 1 and then moves through the list | Use it for a non-stop walkthrough |
| Control mode cards | Autonomous, Human-on-the-loop and Human-in-the-loop, each with its rule and how many demos use it | Click a card to show only that mode; click again to clear |
| Area chips | SD-WAN and carriers, LAN and Wi-Fi, Patching and firmware, Config and automation, Predict and prevent, Branch readiness, Branch-in-a-Box, with counts | Click to filter the list by area |

| Part of the screen | What it shows | How to use it |
|---|---|---|
| Search and control-mode list | Text search over titles and problems | Find a demo by branch, device or keyword |
| Demo list | Number, title, the problem in two lines, area, control mode and runbook | Click a row to see its details; click the play button to run it |
| Detail card | Control mode, runbook, whether it runs as Ansible or Python, the systems shown on stage, and the business or technical explanation | Read it before you play, to know what the audience will see |

## The demo stage

The stage is one screen with every system involved, in the order a real event moves through them.

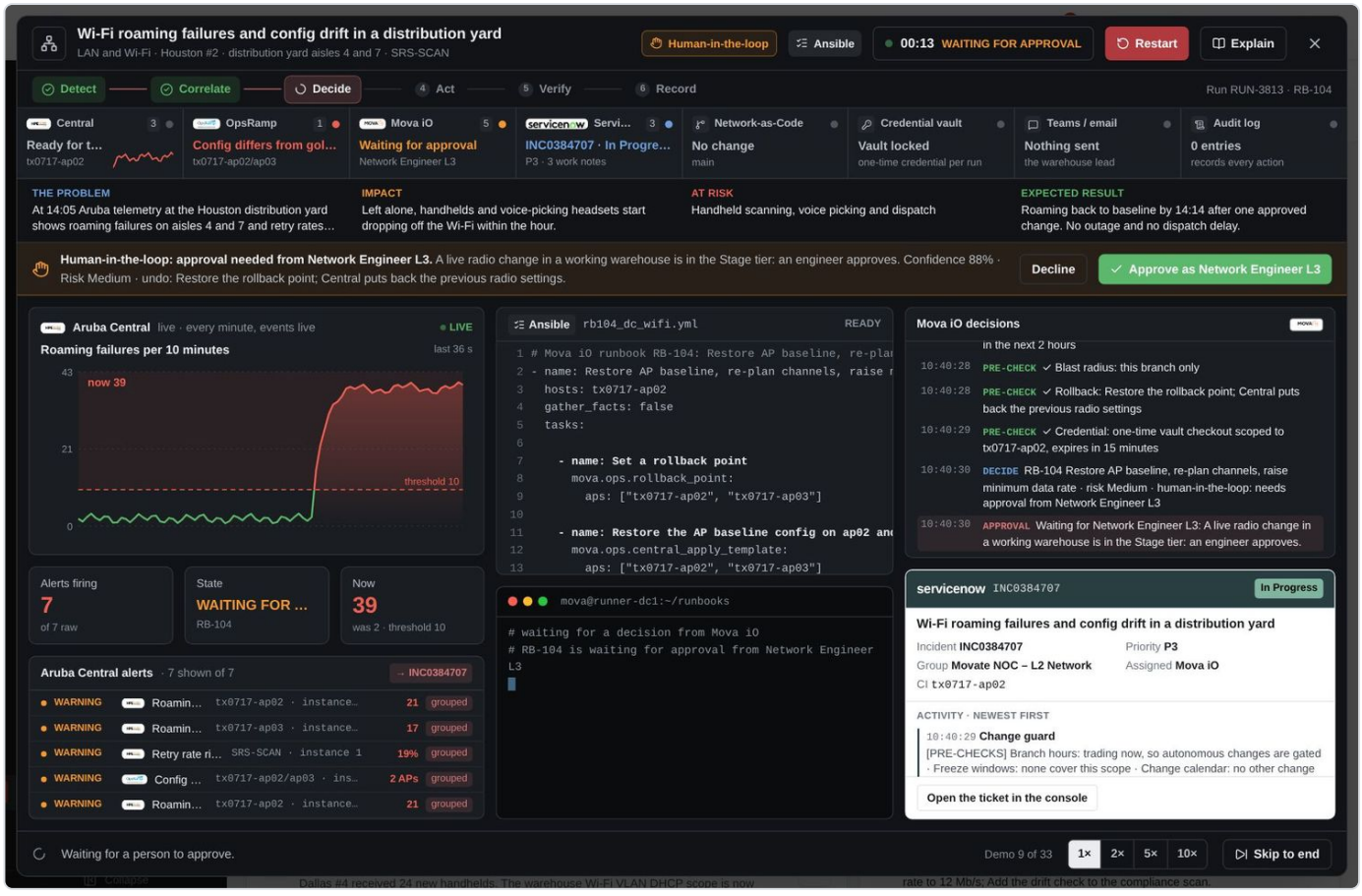



**Figure 230.** Stage waiting for an approval (Human-in-the-loop)

| Area of the stage | What it shows |
|---|---|
| Title bar | Demo title, area, branch and device; the control mode badge; whether the runbook is Ansible or Python; the run clock and state; Restart (or Replay), Explain and close |
| Phase bar | Detect, Correlate, Decide, Act, Verify, Record. Each phase turns green when it is done; the current one is highlighted |

| Area of the stage | What it shows |
|---|---|
| System strip | One tile per system on stage (for example Central, OpsRamp, Mova iO, ServiceNow, Network-as-Code, credential vault, Teams, audit log). Each shows its state now, a count of updates and a status dot |
| Story strip | The problem, the impact, what is at risk (later "Protected") and the expected result (later "Result") in business words |
| Approval or stop bar | Appears at the decision. Human-in-the-loop shows **Approve as (role)** and **Decline**; Human-on-the-loop shows a countdown and **Stop** |
| Live chart | The key metric from the source system, with its threshold line, a red zone above it and a "Mova iO acted" marker when the runbook changes something |
| Three numbers | Alerts firing (raw count, then grouped, then cleared), state of the run, and the metric now against its normal value and threshold |
| Source alerts | The raw alerts from the source system, then "grouped" into the ticket, then "cleared" |
| Runbook and terminal | The Ansible playbook or Python job that runs, step by step, and the runner's terminal output including verification lines |
| Mova iO decisions | The decision log: intake, correlation, evidence, ruled-out causes, diagnosis with confidence, pre-checks, decision, approval, actions, verification, ticket, notice and learning |
| ServiceNow ticket | The real incident record: number, priority, group, assignee, CI and activity, newest first |
| Bottom bar | Status line, demo number, speed (1×, 2×, 5× or 10×) and **Skip to end** |

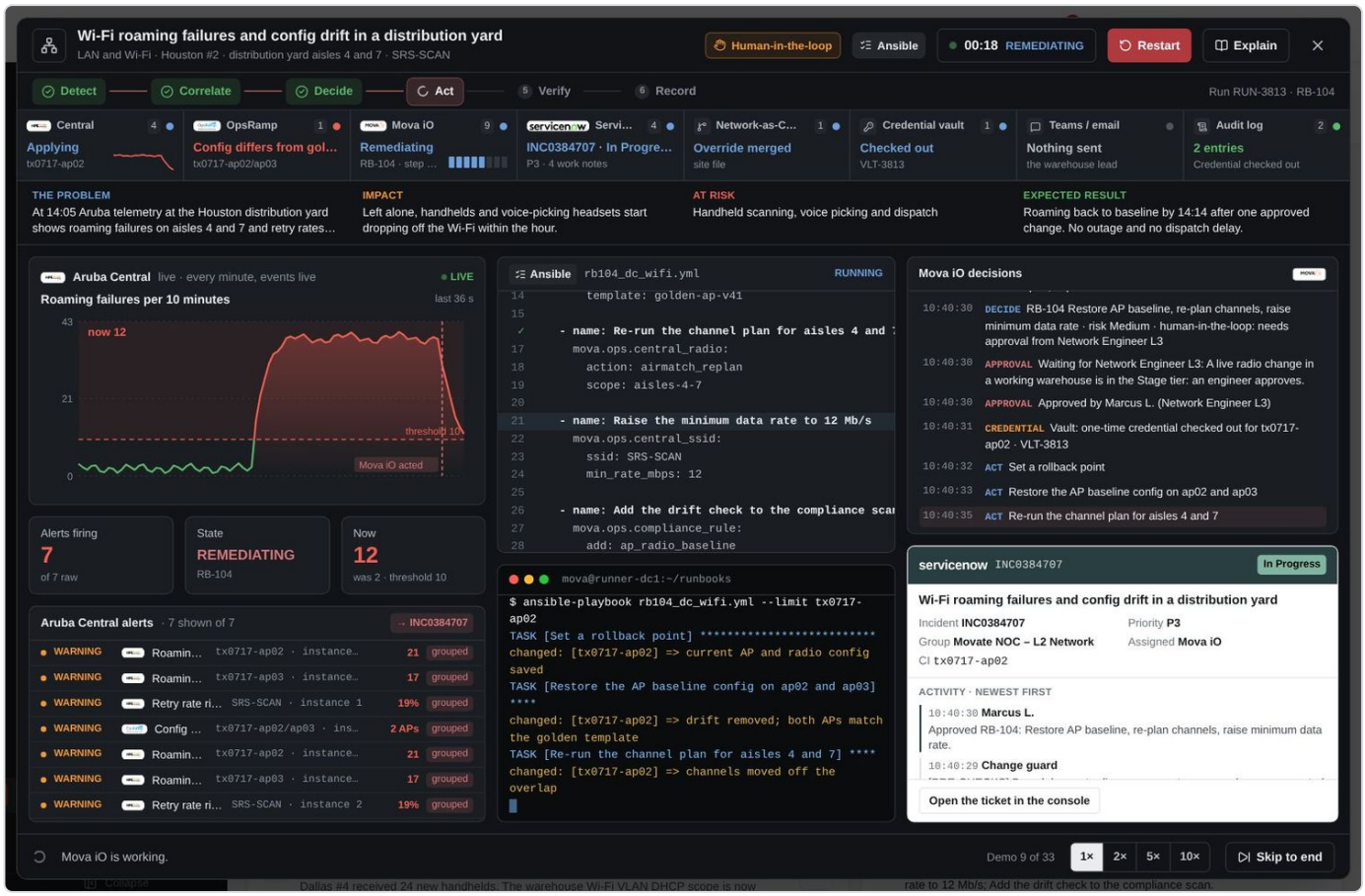



**Figure 231.** Stage running after approval

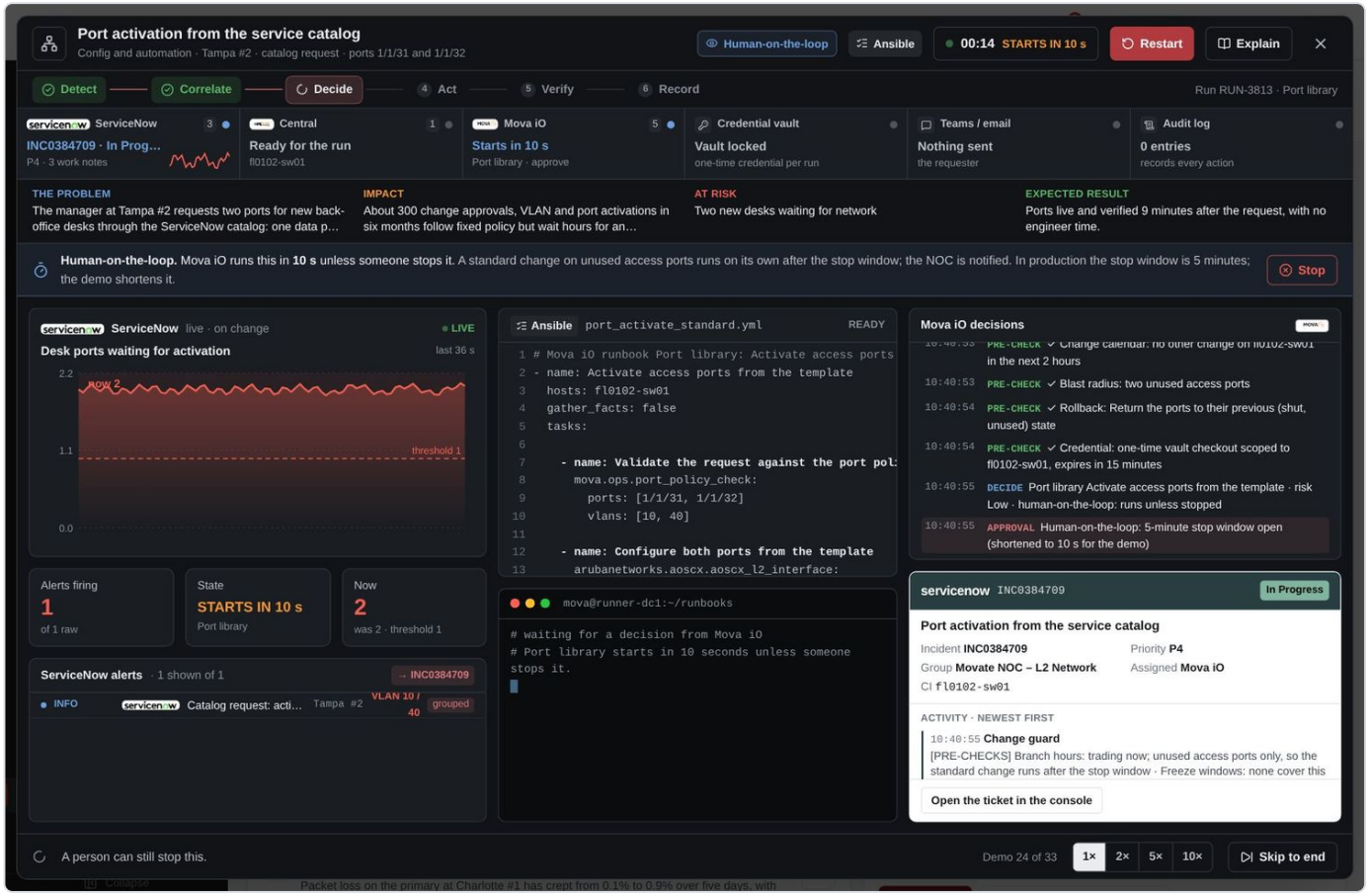



**Figure 232.** Stage on Human-on-the-loop: the stop window is counting down

When the run finishes, the bottom bar shows the outcome: the metric before and after, how many alerts became one ticket, how many people were needed, and how long the demo ran. **Next demo** plays the next one.

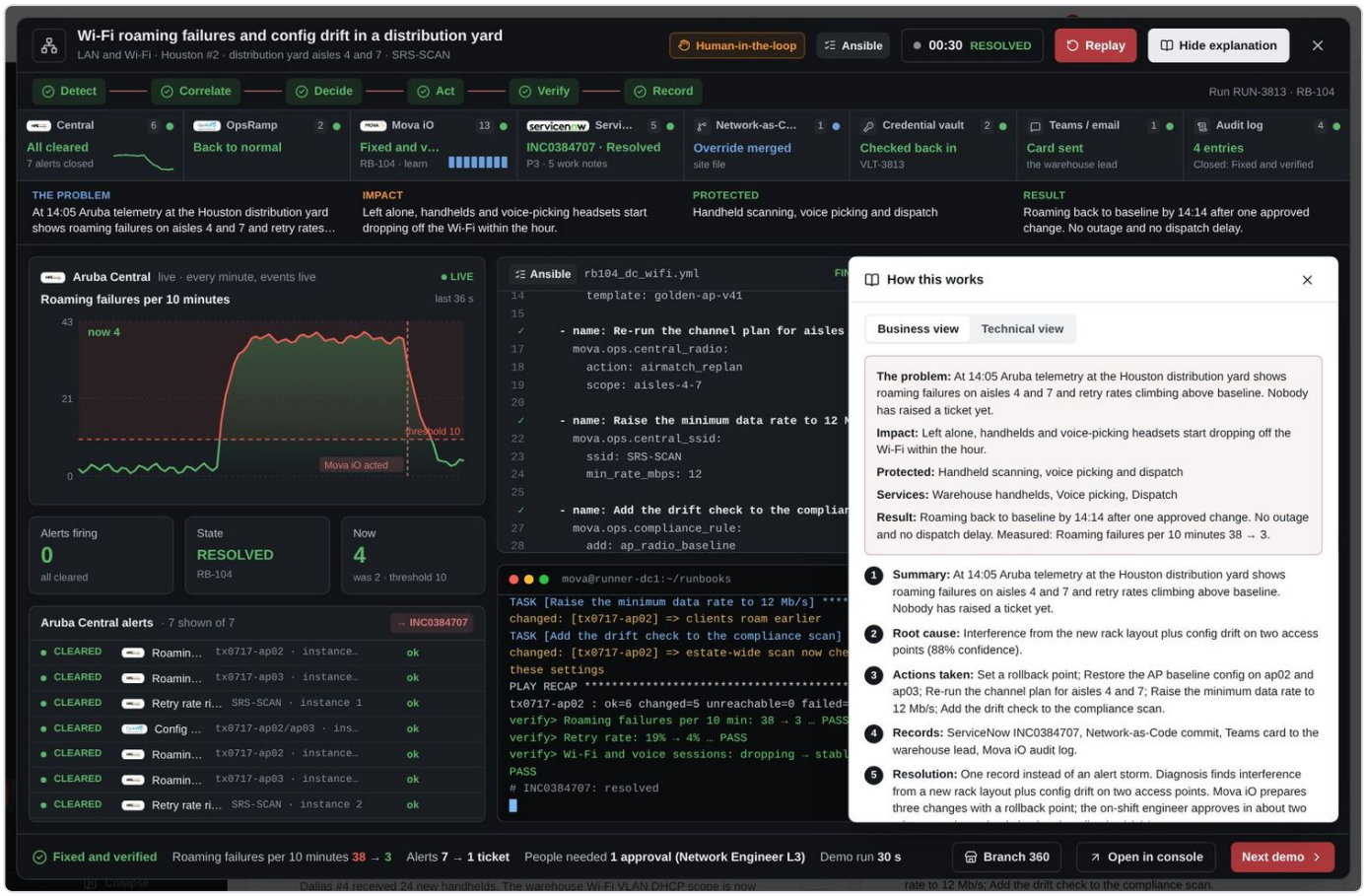



**Figure 233.** Finished run with the explanation open

## The explanation panel

**Explain** opens a panel with two views. The **Business view** gives the problem, impact, what was protected, services, the result with the measured change, then five numbered points: summary, root cause with confidence, actions taken, records written and resolution. Two boxes list how Mova iO knew and what it ruled out.

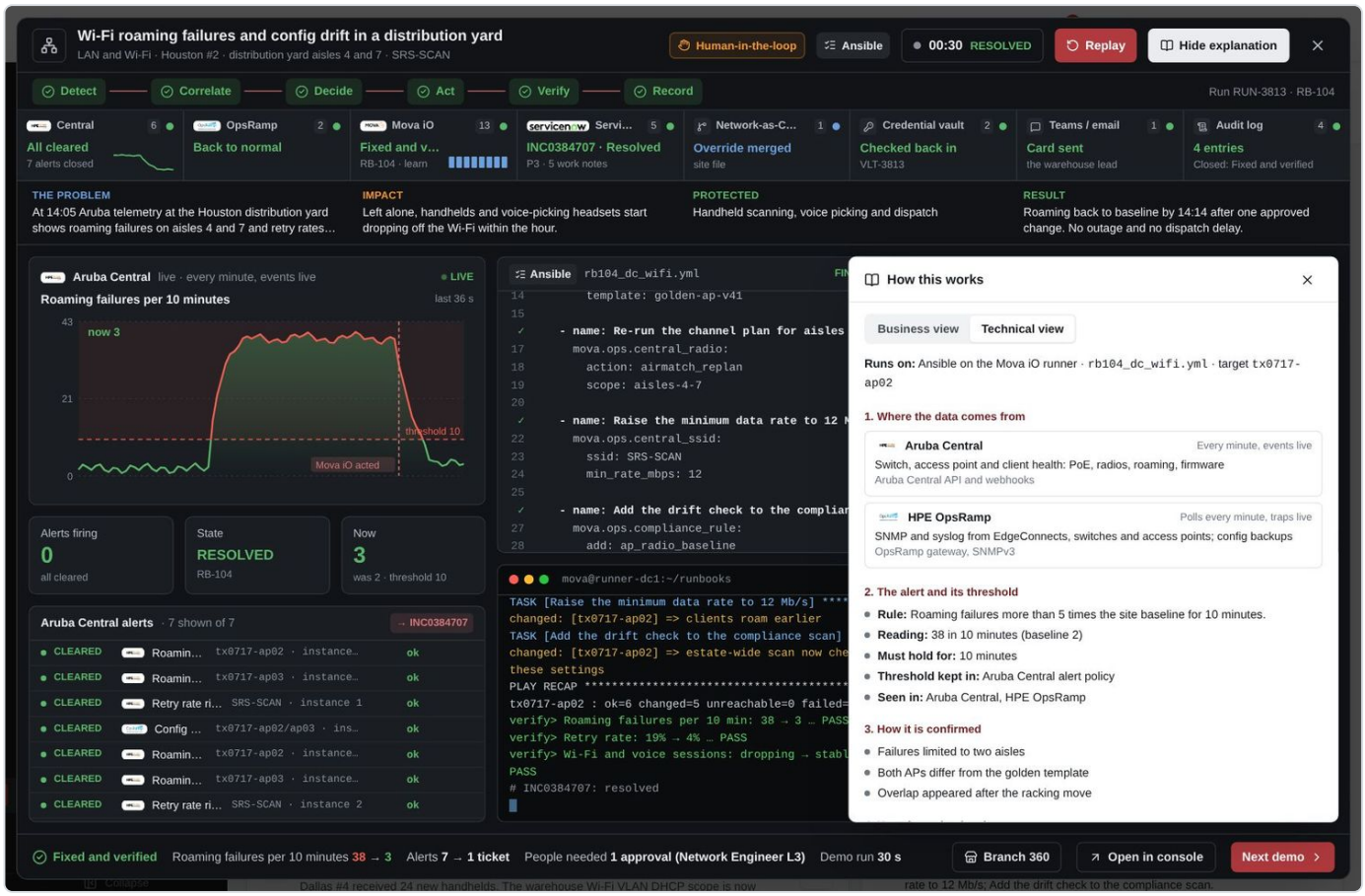



**Figure 234.** Technical view of the explanation

The **Technical view** answers the questions an engineer asks, in six sections:

1. **Where the data comes from:** each source system, how often it reports, what it measures and how it connects.
2. **The alert and its threshold:** the rule, the reading, how long it must hold and where the threshold is kept.
3. **How it is confirmed:** the cross-checks that rule out a false alarm.
4. **How the action is taken:** the engine, file, credential and each step, then the undo.
5. **How it is checked:** the verification tests.
6. **Control mode:** why this run is autonomous, on the loop or in the loop, with the rules.

## Presenting a demo

- On a Human-in-the-loop demo, pause at the approval bar and read the evidence aloud before clicking **Approve**. **Decline** shows the other path: no change is made and the incident goes to the engineer.
- On a Human-on-the-loop demo, point at the countdown: the run will go ahead unless someone presses **Stop**.
- Use **Skip to end** for a long demo and then open the explanation.
- After the demo, open the console in its own tab to show the same kind of record in its screen (Live Incidents, Carrier Management or Approvals & Changes).

# 25. Role dashboards

The role dashboards are a third page, separate from the console and the demos, with their own link (`dashboards/` on the site; the older `briefs/` link forwards there). They are real-time dashboards with role-based access (RBAC): six leaders each sign in and see only their own dashboard, their own scope and the detail their access allows. Every number refreshes from the same live data as the console, and every number, chart, insight and event opens the detail behind it, as deep as the console goes: from the whole estate down to one internet line at one branch.

The dashboards are SRS leaders' views: they see everything and escalate to Movate; they do not trigger actions. **Escalate to Movate** sits in the header and at the top of every drill-down level (chapter 3).

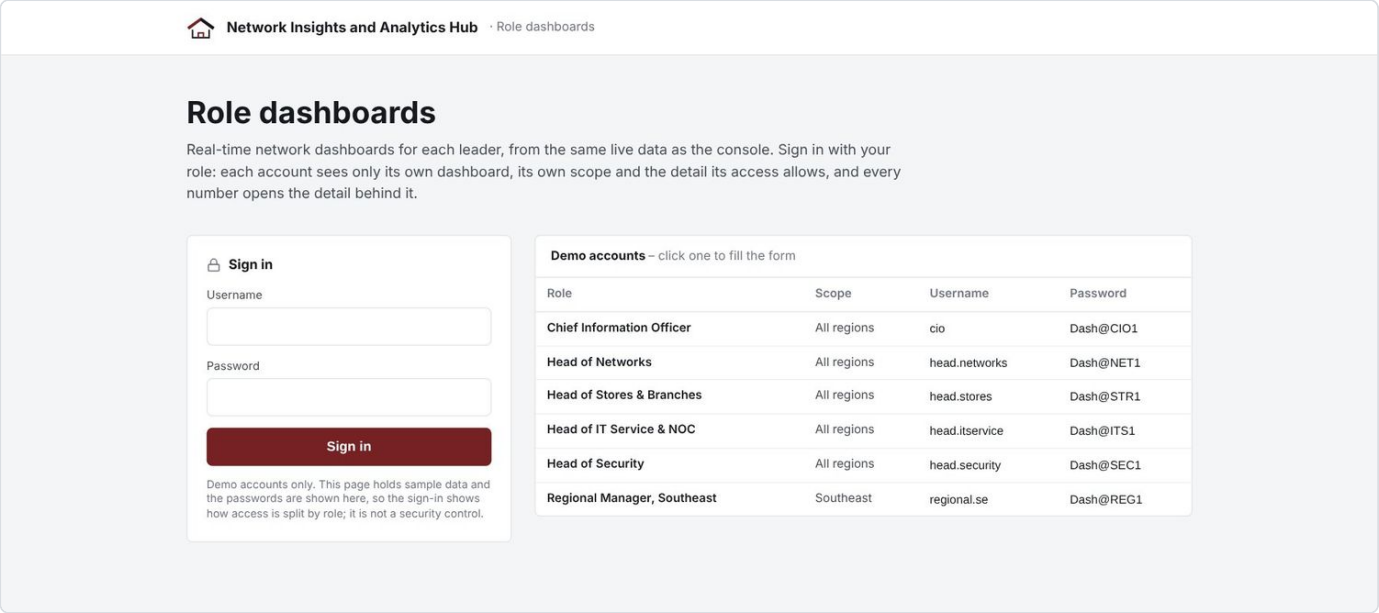



**Figure 235.** Sign-in with the six demo accounts

## Roles and access

The accounts are demo accounts and their passwords are shown on the sign-in screen: the page holds sample data, so the sign-in shows how access is split by role; it is not a security control.

| Role | Scope | Username | What the dashboard answers |
|---|---|---|---|
| Chief Information Officer | All regions | `cio` | Is the business trading, what is it costing, and what needs my decision? |
| Head of Networks | All regions | `head.networks` | What is broken, why, what changed, and what will break next? |
| Head of Stores & Branches | All regions | `head.stores` | Can every branch trade today, and what will my branch managers feel? |
| Head of IT Service & NOC | All regions | `head.itservice` | Is the queue under control, are we inside SLA, and who is waiting on whom? |
| Head of Security | All regions | `head.security` | Where are we exposed, what changed without a record, and who approved what? |

| Role | Scope | Username | What the dashboard answers |
|---|---|---|---|
| Regional Manager, Southeast | Southeast | `regional.se` | Which of my branches need help today, and who is on the way? |

The role button in the header (a green shield with the role and scope) opens **Your access**: the role, its scope, the question the dashboard answers and the areas its access covers. Another role's dashboard is never shown to this account, and the regional manager's lists, charts, live activity and drill-down stay inside the Southeast.

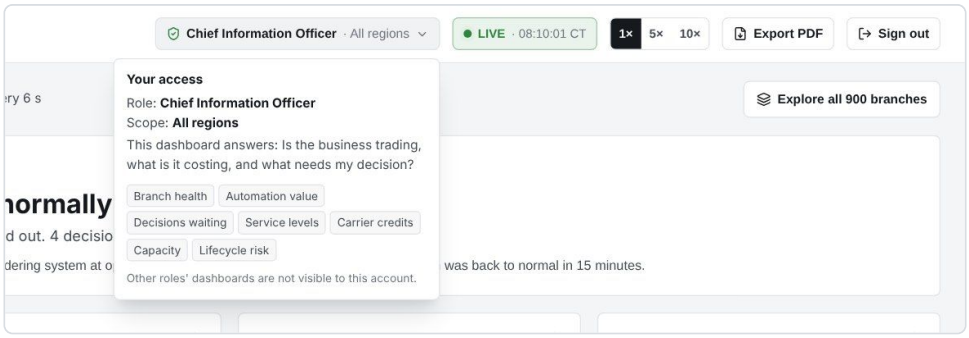



**Figure 236.** Your access

## Real time

The dashboard is worked out again from live data every third live tick: about every 6 seconds at 1×, faster at 5× and 10×. Numbers that change flash as they update, and the live activity stream adds new events at the top.

| Header control | What it does |
|---|---|
| **LIVE** with the time | The time of the last update, in Central Time. Click it to pause (it turns amber and reads PAUSED) and again to resume |
| 1×, 5×, 10× | How fast the live data moves |
| Export PDF | Saves the current view as a PDF, with the same four charts drawn in the PDF |
| Sign out | Returns to the sign-in screen |

## Three views

| View | Looks at | Its four charts |
|---|---|---|
| Live now | What is happening right now: who is down, what is due in the next 24 hours, the decisions waiting | Every branch right now; where the problems are (map); trading normally by region (by state for the regional manager); one chart for the role |
| Last 7 days | Problems fixed, how fast, what automation saved, changes made, and the next 7 days | Problems fixed day by day; what went wrong; how long fixes took; trouble-free branches by region, or changes made and planned |
| Last month | The last full month against target, the six-month trend and the next 30 days | Network uptime for six months; time to fix for six months; service levels; lines below their carrier promise (or capacity, or switches reaching end of support) |

The role's own chart on **Live now**: the CIO sees problems fixed automatically day by day; the Head of Networks sees the delay to the data centre by region; the Head of Stores and the regional manager see readiness to open; the Head of IT Service sees open incidents by priority, on track or to watch; the Head of Security sees device health (all clear, critical flaw to patch, behind on software, differs from the standard setup).

## The dashboard

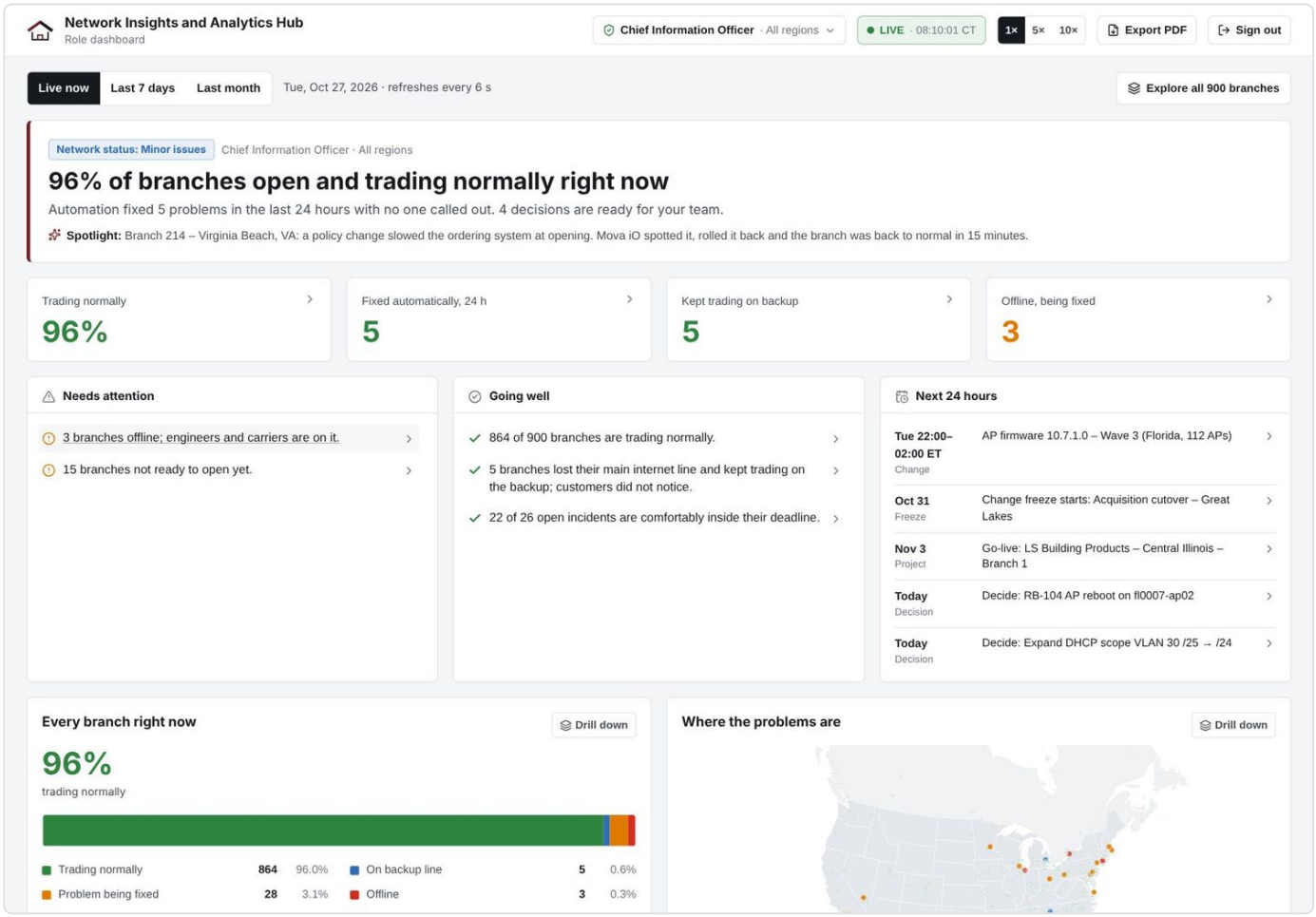



**Figure 237.** The CIO's dashboard, Live now

| Part | What it shows | Click it to see |
|---|---|---|
| Summary | The network status (healthy, minor issues, degraded or major disruption, from the share of branches with a problem), one line that leads with the good news, and a **Spotlight** on one branch event | The spotlight opens that branch or trend |
| Key numbers | Four numbers, good news first | Each opens its list or trend |
| Needs attention | Up to three things to fix, one sentence each | The list behind each, for example the branches offline or the devices to patch |
| Going well | Up to three things going well | The list behind each |
| Next 24 hours (Next 7 days, Next 30 days) | Up to five changes, decisions, technician visits, carrier restore times, go-lives and freezes | The change, the decision, the incident, the line or the project |

| Part | What it shows | Click it to see |
|---|---|---|
| Charts | Four charts, below | Every bar, slice, column and dot; **Drill down** on each card |
| Insights | Four answers from Ask Mova iO, each with a headline that gives the healthy share first, a sentence, up to three numbers and up to three points | **Details** opens the whole answer; **Drill down** opens the detail at the right level |
| More insights | The remaining answers | The whole answer |
| Live activity | The role's own stream of network events, newest first, with events per minute | The incident or the branch |

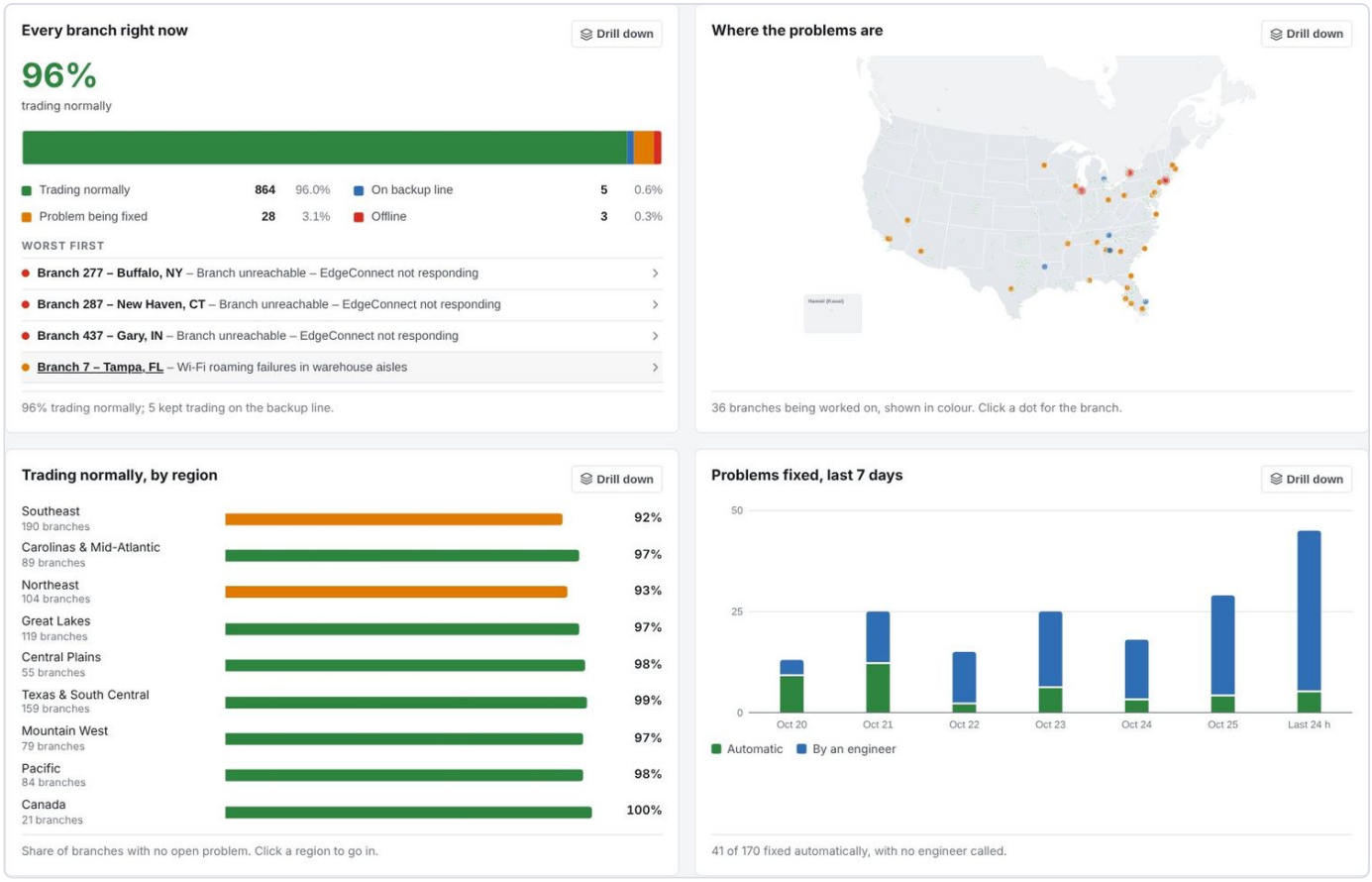



**Figure 238.** Charts on the CIO's dashboard, Live now

| Chart | How to read it | Click |
|---|---|---|
| Every branch right now | The big number is the share trading normally; the bar splits all branches into trading normally (green), on the backup line (blue), problem being fixed (amber) and offline (red). **Worst first** lists the four branches that need help most | A part of the bar or a legend line opens those branches; a branch opens its full detail |
| Where the problems are | Every branch as a dot: healthy ones small and pale, the rest in colour; an offline branch pulses | A dot opens that branch |

| Chart | How to read it | Click |
|---|---|---|
| Trading normally, by region | The share of each region's branches with no open problem. Green is 97% or more, blue 94% to 96%, amber below 94% | A region opens its own page |
| Ready to open | The share that can open; the ring splits all checks passed, can open with one check to watch, and not ready. The list shows each not-ready branch with its first failed check | A slice opens those branches |
| Problems fixed, last 7 days | One column for each 24 hours, split into fixed automatically (green) and fixed by an engineer (blue). The seven columns add up to the week | A column opens that day's fixes |
| Delay to the data centre, by region | Average round trip on the line each branch is using. Under 60 ms keeps card payments and the ordering system quick | A region opens its branches, slowest first |
| Open incidents by priority | P1 to P4, split into on track and to watch (more than half the time to fix used) | A column opens that priority |
| Devices | All clear, critical flaw to patch, behind on software and differs from the standard setup | A slice opens those devices |

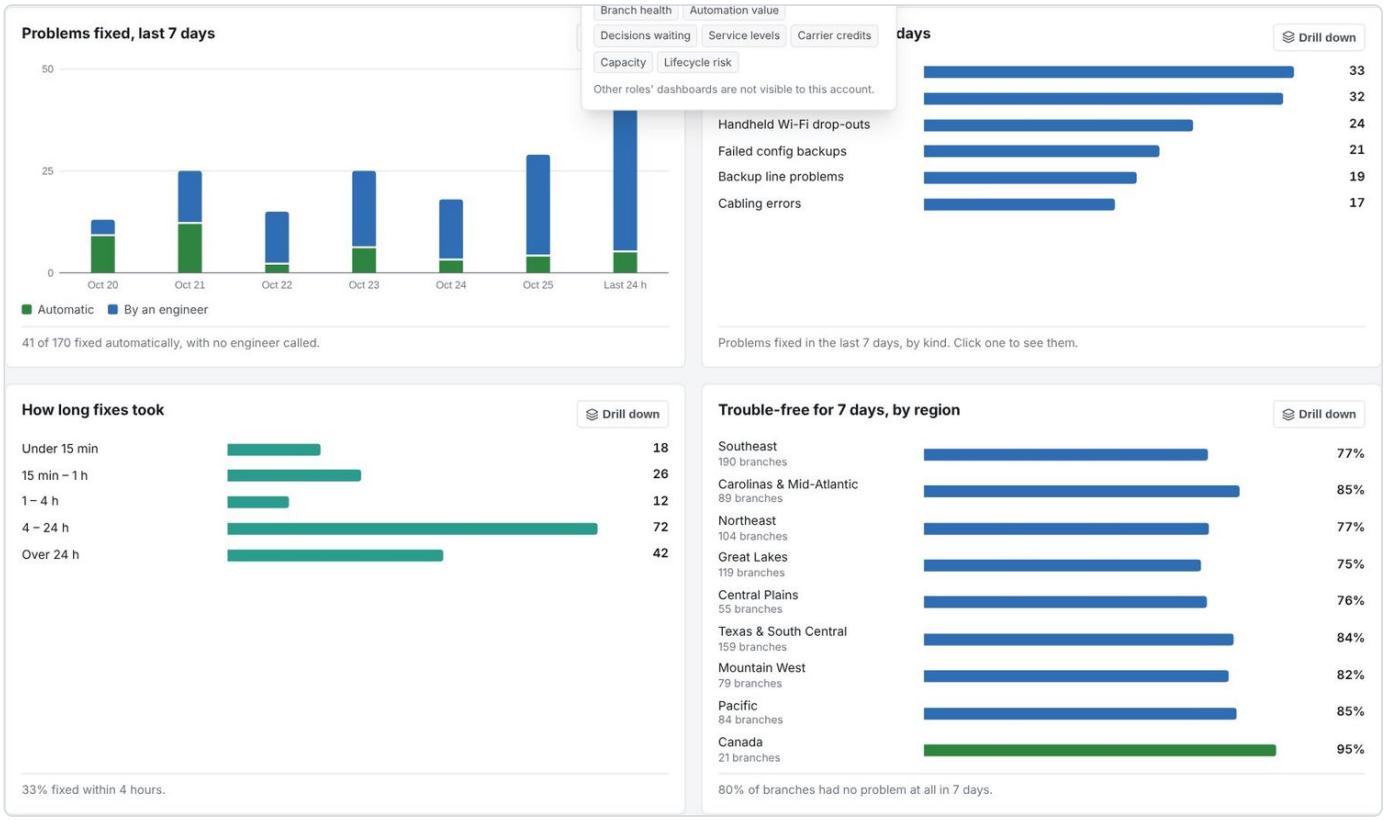



**Figure 239.** Charts, Last 7 days

| Chart (Last 7 days) | How to read it |
|---|---|
| What went wrong, last 7 days | Problems fixed in the last 7 days by kind, most common first. A bar opens those incidents |
| How long fixes took | Fixes in five bands from under 15 minutes to over 24 hours. A band opens those incidents |

| Chart (Last 7 days) | How to read it |
|---|---|
| Trouble-free for 7 days, by region | The share of branches with no incident at all in 7 days |
| Changes, last 7 days and next 7 | Changes made (dark blue) and planned (light blue), day by day; the caption counts any rolled back |

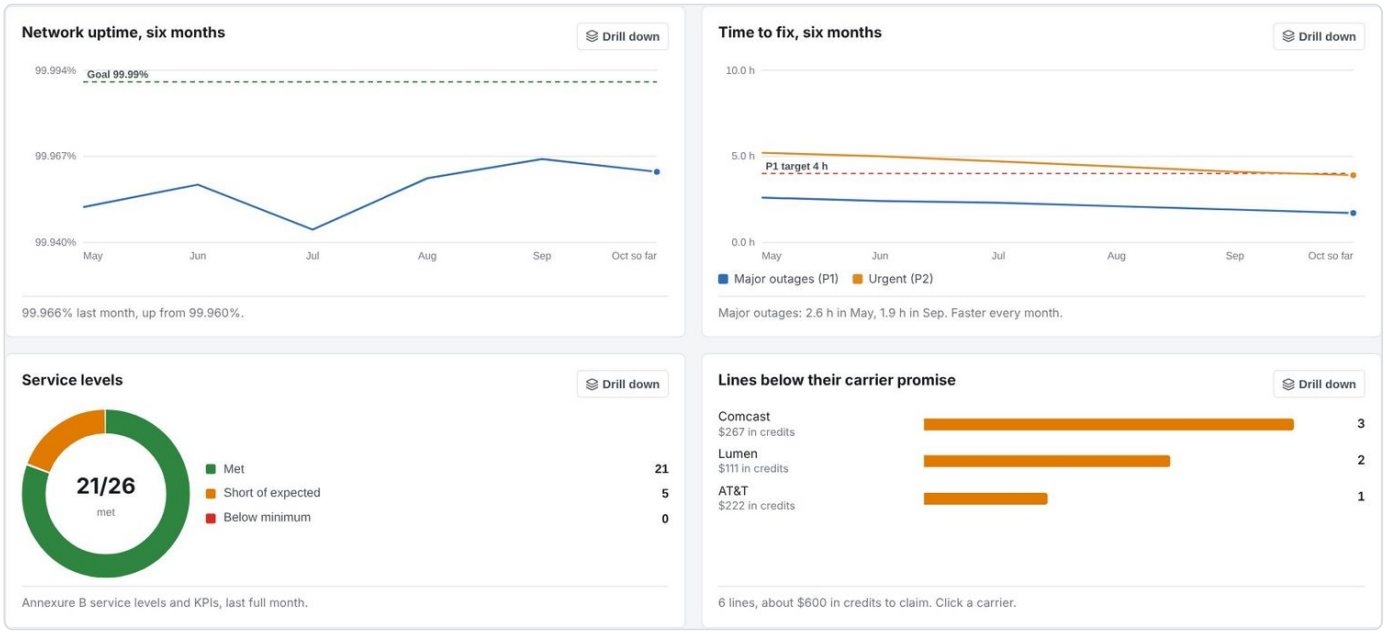



**Figure 240.** Charts, Last month

| Chart (Last month) | How to read it |
|---|---|
| Network uptime, six months | Critical network availability month by month against the 99.99% goal. The last point is the month so far |
| Time to fix, six months | Mean time to restore for major outages (P1) and urgent incidents (P2) against the 4-hour P1 target |
| Service levels | The Annexure B service levels and KPIs last month: met, short of expected, below minimum |
| Lines below their carrier promise | Lines that missed their carrier's uptime promise this month, by carrier, with the service credits to claim |

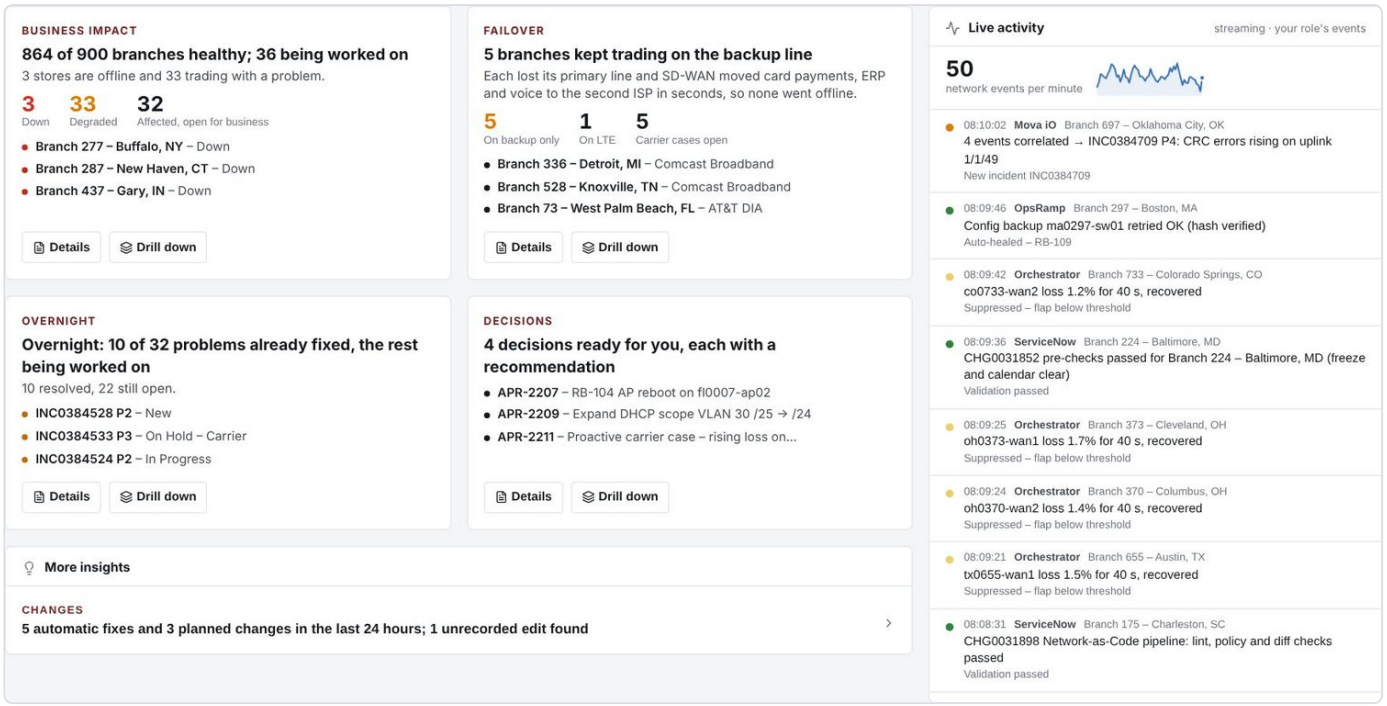



**Figure 241.** Insights and live activity

## Drill down to any level

A click opens a panel from the right. Each level has the same shape: a headline in plain words, a row of numbers, two or more charts and a list. Every number, bar, dot and row opens the next level down. A breadcrumb at the top shows the path; **Back** or any crumb steps out, **Escape** or the cross closes the panel, and the level number shows how deep you are. The panel follows the live data while it is open.




**Figure 242.** Level 1: all 900 branches

| Level | What it shows | Goes on to |
|---|---|---|
| The estate | Branches trading normally, on backup, being fixed and offline; the share trading normally by region; the map; every branch, problems first, with the line it is using, its delay and why it is not normal | A region, any branch |
| A region | The same for the region, broken down by state, with the map zoomed in | A state, any branch |
| A state | The same, broken down by city | A city, any branch |
| A list of branches | Any set from the dashboard: offline, on the backup line, not ready to open, needing more capacity, trouble-free for 7 days, slowest to the data centre, with switches reaching end of support | A region in that set, any branch |

| Level | What it shows | Goes on to |
|---|---|---|
| One branch | Its status and local hours; delay, packet loss and jitter on the line in use; open incidents; opening checks passed; room to grow; both internet lines; the last 24 hours of delay; the ten opening checks; incidents open and fixed in 30 days; every device; what uses the line; technician visits, carrier repairs, decisions, differences from the standard setup, predictions and planned changes | A line, a device, an incident, a decision, a change, a prediction |
| One internet line | Delay, loss and jitter now; uptime over 30 days against the carrier's promise; drops and fault tickets; 24 hours of delay, packet loss and how full the line is; the circuit ID, local exchange company, monthly cost and any service credit; open faults and carrier cases | The carrier, the incident, unstable or repeat-offender lines |
| A carrier | Its lines, how many are down, below promise or unstable, the credits due and its six-month scorecard; uptime by region and by line type; every line, least reliable first | A region, a line type, a line |
| One device | Software against the approved release, critical flaws, lifecycle (end of sale, end of support), warranty or support contract, last config backup; the HPE Aruba Networking advisories that affect it; differences, predictions and incidents | Devices with the same flaw or model |
| A list of devices | Devices with a critical flaw, behind on software, reaching end of support, different from the standard setup or with a failed backup, by model and by region | A model, a region, a device |
| One incident | How long it has been open or took, its deadline and time used, the owner, how many alerts were grouped into it, the likely cause with its confidence; the timeline from the first alert to the fix, Mova iO's steps, the ranked causes, the runbook and the checks before closing | The branch, the line or device, similar incidents |
| The root cause | Where the fault is (network inside SRS, the ISP or the application) on the path from the tills to the data centre; each layer checked; the ranked causes with their evidence; how it was found; the fix, the carrier case, the change behind it, wider carrier problems and repeats. The same view as in the console (chapter 3) | The line, device or change at fault, the branch, the incident |
| Mova iO: who fixes it | What Mova iO's Smart Triage Agent recommends, whether Mova iO can fix it remotely, every route with the agent behind it (runbook, NOC, SME, field engineer, HPE, carrier) and the ticket so far. Read-only here: engineers approve Mova iO's steps in the console | The root cause, the branch |
| A list of incidents | Open, to watch, waiting for an owner, fixed automatically, fixed in 7 days, overnight, needing a carrier; by day, by priority, by kind, by how long the fix took and by region | Any of those, then an incident |
| Changes, decisions, service levels, time to fix, new branches, predictions, carrier repairs | The change calendar and freezes; each decision with its confidence, reasons and evidence; every service level against its target and its six-month trend; time to fix by priority for six months; Branch-in-a-Box projects with stage, blockers and gates; each prediction with its trend | The record behind each line |

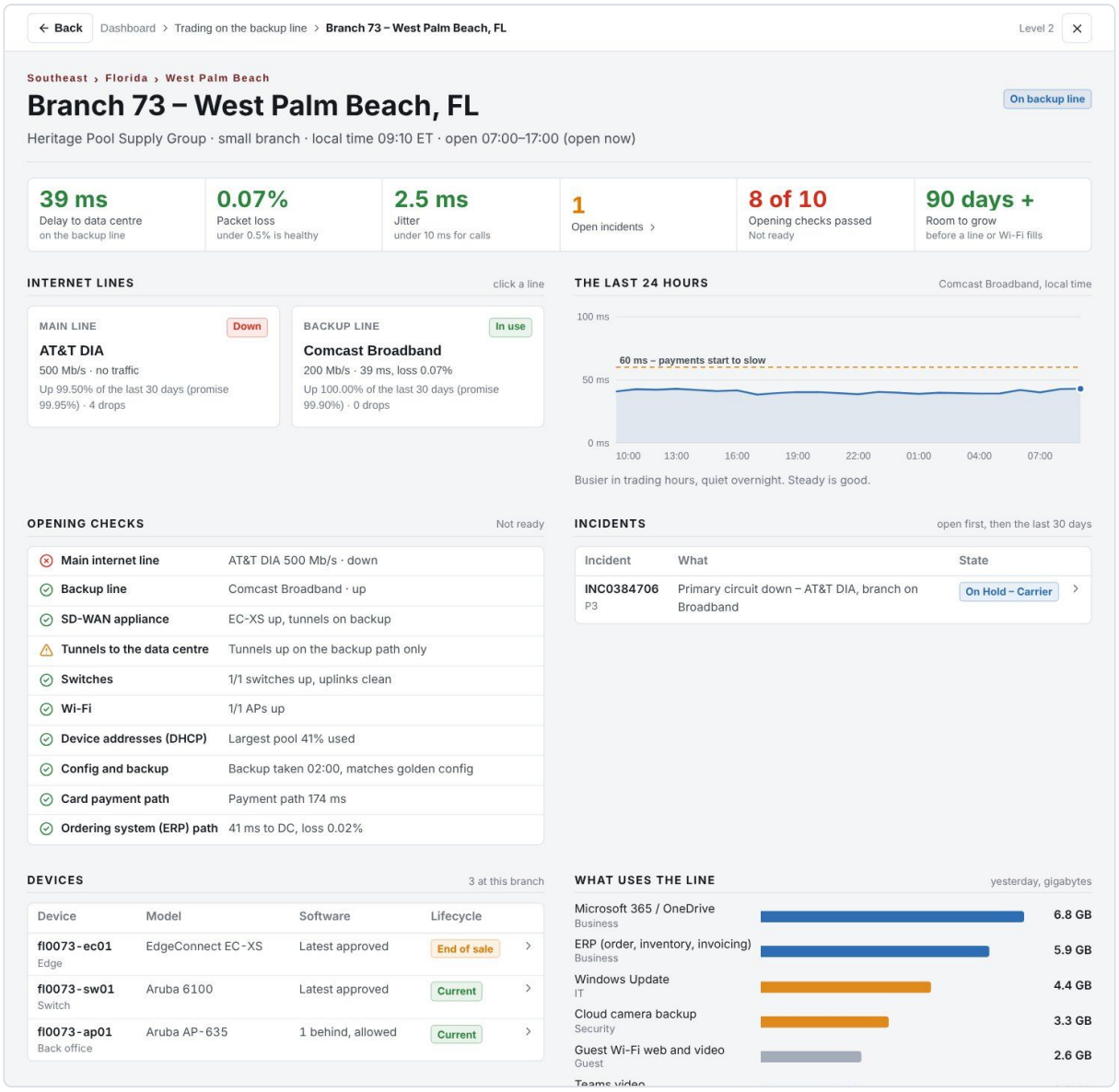



**Figure 243.** Level 2: a branch from the list of branches on the backup line

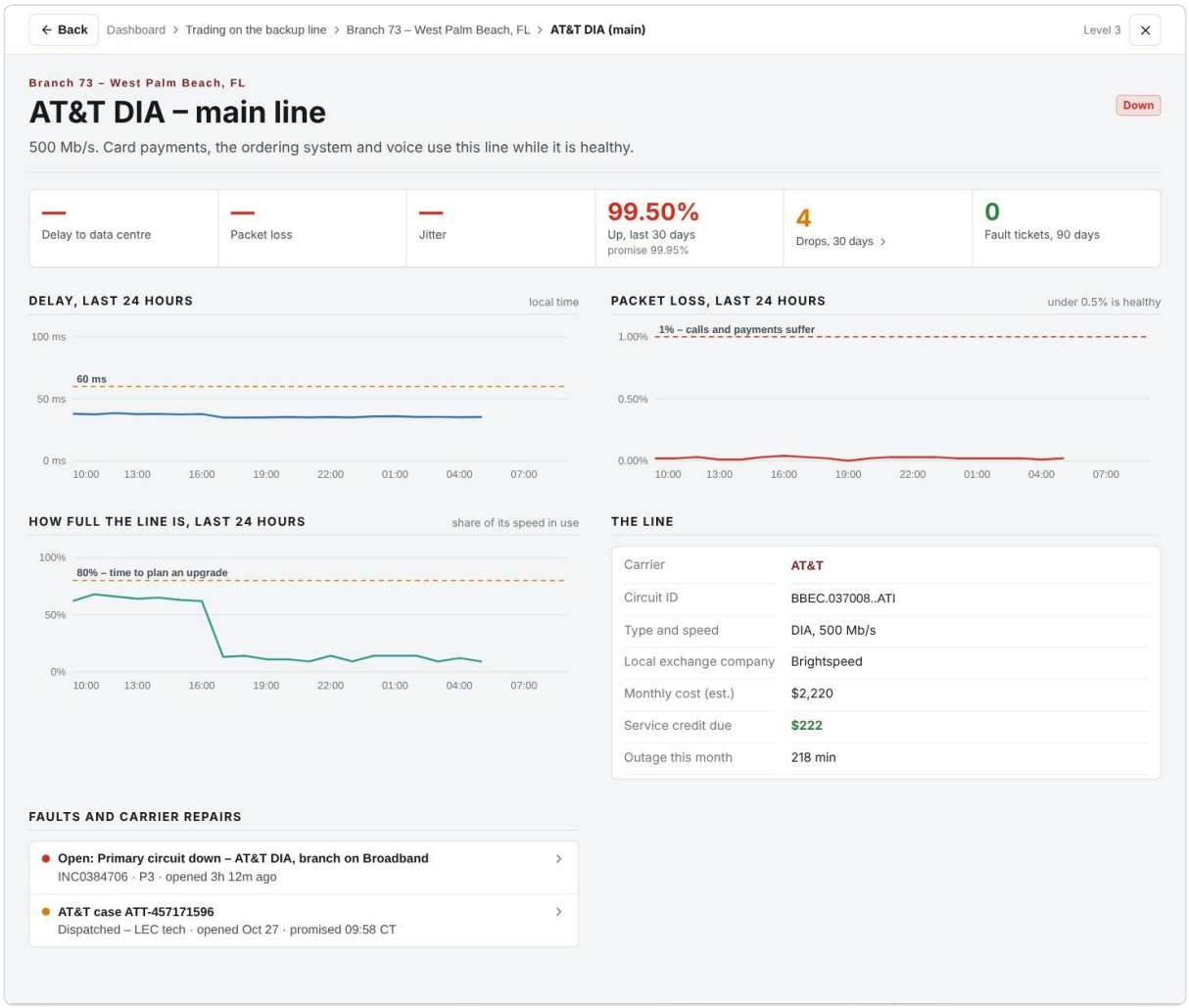



**Figure 244.** Level 3: the branch's main internet line, down with a carrier repair under way

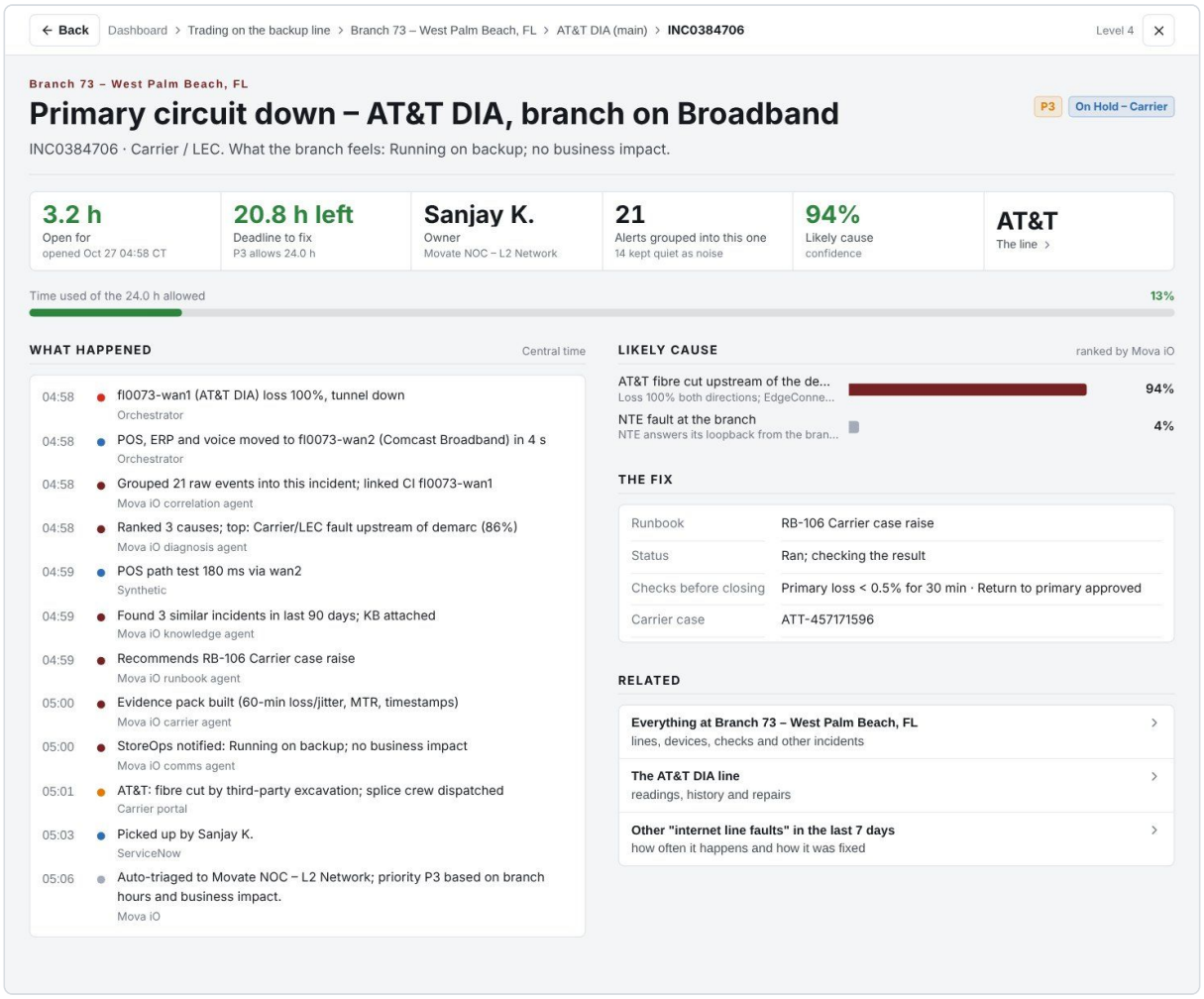



**Figure 245.** An incident: timeline, likely cause and the fix

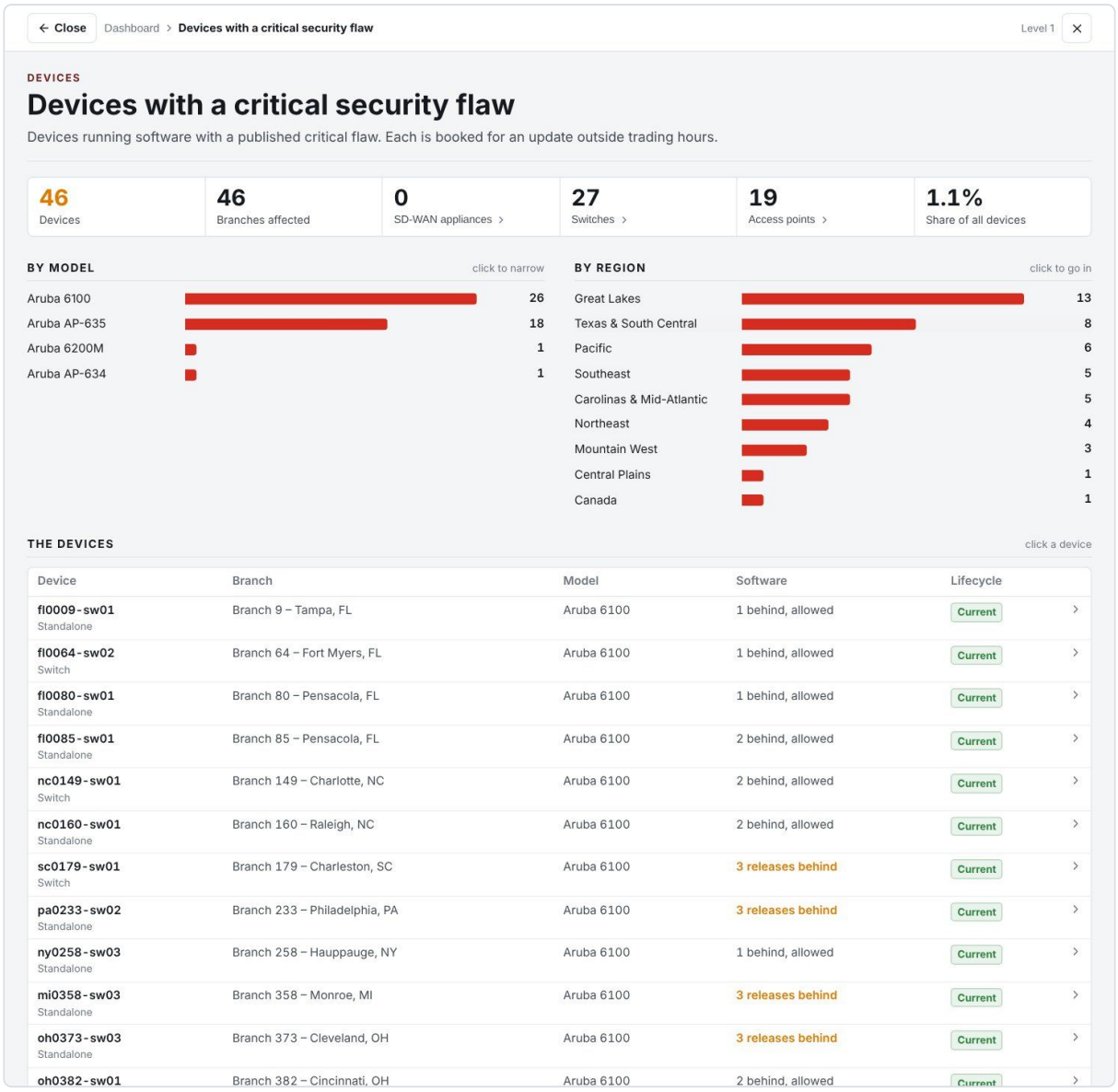



**Figure 246.** Devices with a critical security flaw (Head of Security)

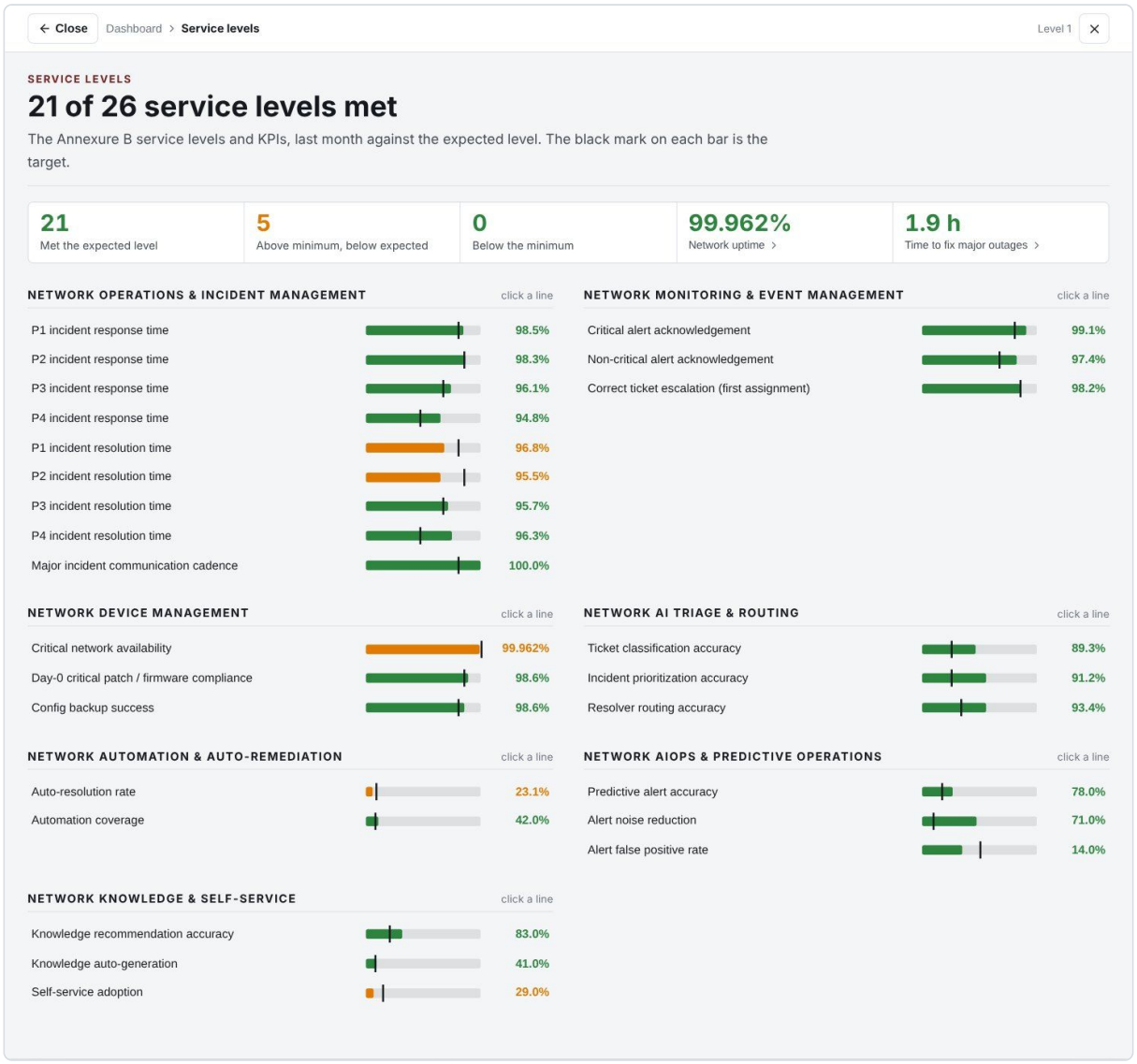



**Figure 247.** Service levels, last month

## Export PDF

**Export PDF** saves the view on screen as a PDF generated in the browser: the header with the role, scope, view and snapshot time, the summary, needs attention and going well, the spotlight, the key numbers, what is coming up, the same four charts drawn in the PDF (the map becomes a list of the branches being worked on), the insights and more insights. It is set in Helvetica, since Aptos cannot be embedded in a PDF.

## Good and bad, and what to do

| Reading | Good | Needs attention | What to do |
|---|---|---|---|
| Network status | Healthy | Degraded or major disruption | Open **Every branch right now**, then the branches with a problem |
| Trading normally | 96% or more of branches | Below 94%, or any region amber | Click the region, then the branches with a problem |

| Reading | Good | Needs attention | What to do |
|---|---|---|---|
| Offline | 0 | Any | Open the branch: the incident shows the cause and the recovery under way |
| Fixed automatically | A quarter or more of fixes | Falling week on week | Open the week's fixes by kind to see which problems still need an engineer |
| Delay to the data centre | Under 45 ms on average, under 60 ms for every region | A region over 60 ms | Open the region, slowest branches first, then the line |
| Service levels | All met | Any short of expected or below minimum | Open the service level for its six-month trend and the incidents behind it |

# Part III. The live demos

## 26. The 33 demos

Every demo is an SRS branch situation built from the proposal documents: the Movate solution response (including its three "day in the life" scenarios), the RFP response, Annexure B (SLA and KPIs) and the console brief. Each one plays on the stage described in chapter 24 and leaves its records in the console.

## Where each demo comes from

| # | Demo | Mode | Source in the proposal |
|---|---|---|---|
| 1 | SD-WAN circuit degradation fixed before the branch notices | Autonomous | Solution response, day in the life #1 |
| 2 | Branch hard down after a carrier fiber cut | Human-in-the-loop | Solution response, day in the life #3 |
| 3 | Primary circuit down: carrier fault | Autonomous | RFP response: carrier isolation sequence and end-to-end carrier ownership |
| 4 | Chronic circuit: four drops in 30 days become a problem record | Autonomous | RFP response: problem rule (3 in 30 days); ticket data: 58% from repeat circuits |
| 5 | Return to primary circuit after carrier repair | Human-on-the-loop | Day in the life #3 (controlled failback); runbook catalog |
| 6 | EdgeConnect hardware failure | Human-in-the-loop | Console brief; RFP response: break-fix and field dispatch |
| 7 | Severe weather across Dallas–Fort Worth | Human-in-the-loop | Solution response: peak-season and storm war room |
| 8 | Yard access point offline | Autonomous | RFP response: closed-loop example (AP unresponsive, Act tier) |
| 9 | Wi-Fi roaming failures and config drift in a distribution yard | Human-in-the-loop | Solution response, day in the life #2 |
| 10 | CRC errors climbing on a switch uplink | Human-in-the-loop | RFP response: Stage example (switch CRC errors) |
| 11 | Automation does not fix it: engineer takes over with full context | Human-in-the-loop | RFP response: handoff when automated remediation fails |
| 12 | Broadcast storm from an unmanaged desk switch | Human-on-the-loop | Solution response: blameless review ends in an engineering action |
| 13 | DHCP scope full after a handheld rollout | Human-in-the-loop | RFP response: deterministic workflows (DHCP scope check) |
| 14 | Day-0 critical patch for all EdgeConnects | Human-in-the-loop | Annexure B #32; RFP response: emergency patch cadence |
| 15 | Monthly switch patch wave with automatic checks | Human-on-the-loop | RFP response: monthly minor patches; Firmware Upgrade Library |
| 16 | Critical access point vulnerability (HPESBNW04712) | Human-in-the-loop | RFP response: vulnerability management; CVE-to-device mapping |
| 17 | Access point firmware wave stopped at canary | Human-on-the-loop | Solution response: progressive rollout with automatic halt; console brief |
| 18 | Annual major switch upgrade with early life support | Human-in-the-loop | RFP response: major upgrades up to once a year, pilot, waves, early life support |
| 19 | Firmware compliance catch-up after hardware swaps | Human-in-the-loop | Annexure B #32; firmware policy (N-1, N-2) |

| # | Demo | Mode | Source in the proposal |
|---|---|---|---|
| 20 | Error budget burning: tonight's patch wave deferred | Autonomous | Solution response: error budget as a control on change |
| 21 | Aruba 2530 end of support: refresh wave 1 | Human-in-the-loop | RFP response: 728 Aruba 2530 switches reach end of support in early 2027 (31 Jan for 48-port, 31 Mar for 24-port) |
| 22 | Nightly config backups fail and are re-run | Autonomous | Annexure B #33; RFP response: configuration management |
| 23 | Unapproved local admin account on a switch | Human-in-the-loop | RFP response: drift monitoring and access management |
| 24 | Port activation from the service catalog | Human-on-the-loop | Solution response: about 300 catalog changes follow fixed policy |
| 25 | Standalone access points brought under Aruba Central | Human-in-the-loop | RFP response: 2,080 devices (51%) outside Aruba Central |
| 26 | A repeated manual fix becomes a tested runbook | Human-in-the-loop | Solution response: toil budget and Toil Miner; RFP response: Eval Runner |
| 27 | Backup LTE fails the weekly failover test | Autonomous | Solution response: dual-ISP failover test library |
| 28 | Circuit degradation predicted: proactive carrier case | Autonomous | RFP response: predictive report example |
| 29 | WAN capacity forecast for hail season | Human-in-the-loop | RFP response: capacity management; peak-season planning |
| 30 | Branch fails its 05:30 readiness check | Autonomous | Console brief: Branch Readiness |
| 31 | After-hours link down held and checked at opening | Autonomous | Solution response: site-closed hold; single link down reclassified from P2 to P3 |
| 32 | Acquired branches monitored from day one, before any kit changes | Human-in-the-loop | Solution response: Day-1 M&A playbook; RFP response: M&A transition service |
| 33 | Acquired branch cutover weekend | Human-in-the-loop | Console brief: Branch-in-a-Box; solution response: weekend cutovers |

Every demo is a core network situation: WAN circuits, SD-WAN, switching, Wi-Fi, firmware and configuration.

## SD-WAN and carriers

**1. SD-WAN circuit degradation fixed before the branch notices** (Houston, RB-101, Autonomous). At 09:12:00 jitter, loss and latency rise on the AT&T MPLS primary while the branch is trading. Forty-three events become one incident at 09:12:15. Diagnosis runs path tests, checks the carrier portal and compares the config with the golden template, and points to the carrier (87%). At 09:13:00 the pre-approved runbook steers POS, ERP and voice to broadband and re-applies QoS; at 09:13:25 synthetic tests pass and the AT&T case opens with the evidence; at 09:13:40 the incident closes on its own. **Point at:** the chart falling back under the threshold at "Mova iO acted", and the closing time under two minutes with no human touch.

**2. Branch hard down after a carrier fiber cut** (Charlotte, RB-101, Human-in-the-loop). At 07:42 the fiber primary is cut and the broadband backup, in the same duct, degrades; 62 events become one P1. Diagnosis rules out the branch using last-gasp messages and the out-of-band router. The carrier case opens with evidence, the branch manager's call arrives with the incident already on screen, and the NOC engineer approves bringing up the LTE/5G path with

POS, ERP and voice first. The branch trades again about 13 minutes after going dark; after the repair, failback is verified, the RCA is published and branches with a shared duct are flagged. **Point at:** the approval, then the probes going from 0% to 100%.

**3. Primary circuit down: carrier fault** (Atlanta, RB-106, Autonomous). The AT&T primary drops and the branch fails over in 4 seconds. The Carrier Agent proves the fault is upstream of the demarc with a loopback test and opens the case with loss graphs, MTR and the loopback result, so the carrier dispatches on the first call instead of closing it as no trouble found.

**4. Chronic circuit becomes a problem record** (El Paso, Autonomous). A fourth drop in 30 days on the broadband primary. Mova iO links the four incidents into a problem record, sends a 30-day evidence pack to the carrier account team as a chronic-circuit review, and proposes a diverse second carrier for approval. **Point at:** the pattern finally has one owner.

**5. Return to primary after carrier repair** (Phoenix, RB-102, Human-on-the-loop). The primary has been clean for 40 minutes but the branch is still on LTE. The runbook proves the line is stable and the case closed, and runs unless the NOC engineer stops it in the stop window.

**6. EdgeConnect hardware failure** (Phoenix, RB-108, Human-in-the-loop). The EdgeConnect stops answering on both paths. Remote recovery is tried first: soft reboot, console read, reload from the backup partition. When it fails, a technician is dispatched with a pre-staged spare, zero-touch configures it and the CMDB updates itself.

**7. Severe weather across Dallas–Fort Worth** (Major incident, Human-in-the-loop). Hail hits seven branches in nine minutes. Sixty-four alerts become one major incident with one master carrier case, a regional change freeze, hourly leadership updates and an LTE plan uplift sent for approval.

## LAN and Wi-Fi

**8. Yard access point offline** (Houston, RB-103, Autonomous). The yard AP is down with its PoE port drawing 0 W. Because the device is already down, the port bounce cannot make anything worse, so it runs at once during trading hours; the AP is back in three minutes.

**9. Wi-Fi roaming failures and config drift in a distribution yard** (Houston, RB-104, Human-in-the-loop). At 14:05 roaming failures rise on two aisles before anyone calls. Diagnosis finds interference from a new rack layout plus drift on two APs. Mova iO prepares three changes with a rollback point; the engineer approves in about two minutes; roaming is back to baseline by 14:14 and a drift check joins the estate-wide scan.

**10. CRC errors climbing on a switch uplink** (Memphis, RB-103, Human-in-the-loop). Errors rise over three days; diagnosis ranks a failing optic. The engineer approves taking that link out of the pair now and swapping the optic after close.

**11. Automation does not fix it** (San Antonio, Human-in-the-loop). An AP reboot runs but verification fails, so the runbook stops and the incident reaches the L2 engineer in the same minute with the SLA clock unchanged and the full handoff package. The package shows the switch PoE budget at 98% after new cameras; the engineer approves new PoE priorities and the AP returns. **Point at:** humans receive only the exceptions, and they receive them with context.

**12. Broadcast storm from an unmanaged desk switch** (Miami, RB-103, Human-on-the-loop). Loop-protect blocks the port, the branch is asked to unplug the desk switch, and a template change for BPDU guard on every branch follows.

**13. DHCP scope full after a handheld rollout** (Dallas, Human-in-the-loop). Twenty-four new handhelds fill the warehouse VLAN pool. The fix is proposed at 79% confidence, just under the gate, so an engineer approves it.

## Patching and firmware

**14. Day-0 critical patch for all EdgeConnects** (Human-in-the-loop). A critical ECOS advisory is matched to all 933 EdgeConnects. Mova iO plans rings: a canary of 10 closed branches, then regions after close, HA pairs one unit at a time. The emergency change board approves, the canary passes its health gate and the waves are released with automatic rollback.

**15. Monthly switch patch wave** (Texas, Human-on-the-loop). An approved standard change patches 214 switches after close: backup, pre-checks, batches of 20 with stack members one at a time, and post-checks compared with pre-checks. One switch fails its post-check and rolls back on its own; 213 finish.

**16. Critical access point vulnerability** (Human-in-the-loop). Exposed APs are counted by branch, Security approves a mitigation for all access points, and patch waves run region by region after close with a canary first.

**17. Access point firmware wave stopped at canary** (Florida, Human-on-the-loop). The canary group shows older handhelds failing fast roaming; Change Validation stops the rollout on its own, the canary rolls back and a revised plan goes for approval.

**18. Annual major switch upgrade with early life support** (Human-in-the-loop). The once-a-year AOS-CX major upgrade is packaged with the lab result, 12 pilot branches, regional waves and the previous image for rollback. The change board approves; the pilot runs after close and enters 14 days of early life support, which spots a slow memory climb on one model. Those waves are held and a vendor case opened while the others continue.

**19. Firmware compliance catch-up** (Human-in-the-loop). The weekly scan finds 37 devices outside policy: replacement units on factory firmware and APs that were offline during the last wave. A catch-up wave is staged as a standard change, approved and run after close; the replacement process now upgrades units before shipping.

**20. Error budget burning: tonight's patch wave deferred** (Great Lakes, Autonomous). The region's availability budget is burning 3.1 times faster than planned. The change policy defers the non-essential AP wave by a week, keeps emergency security changes allowed and narrows the next wave to canary branches first.

**21. Aruba 2530 end of support** (Human-in-the-loop). The 728 switches reaching end of support on 1 February 2027 are ranked by failure risk and traffic, and the engineer approves the first refresh wave with the standard CX 6100 kit and spares pre-staged.

## Config and automation

**22. Nightly config backups fail and are re-run** (Southeast, RB-109, Autonomous). Forty-six backups fail in the same minute; all used one TACACS+ server. Mova iO retries through the second server, verifies each file's hash and opens a problem record. Success is back above 99% before 03:00.

**23. Unapproved local admin account on a switch** (Charlotte, Human-in-the-loop). The nightly backup shows a local admin account outside the golden config. Security approves restoring the golden config after close.

**24. Port activation from the service catalog** (Tampa, Human-on-the-loop). A branch requests two desk ports. The request is checked against policy, built from the template after the stop window and verified with link, LLDP and DHCP checks; it closes in nine minutes.

**25. Standalone access points brought under Aruba Central** (Human-in-the-loop). Wave 1 of the 2,080 devices outside Central: 120 APs are checked, their configs captured and mapped to the golden group, then converted after close in batches. Telemetry and the AP runbooks now cover them.

**26. A repeated manual fix becomes a tested runbook** (Human-in-the-loop). The Toil Miner finds the same manual tunnel reset 23 times in 30 days. A runbook is drafted, checks are generated, and a replay against the 23 past cases passes 21 and correctly refuses 2. It joins the catalog at the Stage tier.

**27. Backup LTE fails the weekly failover test** (Tulsa, RB-106, Autonomous). The Sunday test finds the LTE backup passes no traffic because the SIM was suspended. An account case is opened, the SIM is reactivated and the re-test passes before Monday opening.

## Predict and prevent

**28. Circuit degradation predicted** (Charlotte, Autonomous). Loss creeps from 0.1% to 0.9% over five days. A proactive carrier case opens with the trend, a failover test proves the backup, and the carrier replaces a failing line card before the circuit drops.

**29. WAN capacity forecast for hail season** (Dallas, Human-in-the-loop). A branch's WAN heads for 90% in about three weeks. An engineer approves a bandwidth upgrade and, until it lands, a policy that protects POS and ERP.

## Branch readiness

**30. Branch fails its 05:30 readiness check** (Denver, Autonomous). Two checks fail before opening: a tunnel and a config backup. The branch is closed, so both are fixed at once and the 05:52 re-run passes all ten checks.

**31. After-hours link down held and checked at opening** (Lubbock, Autonomous). The backup line goes down at 22:15 at a closed branch with a clean primary. Mova iO sets P3 instead of P2, matches the carrier's maintenance window, holds the page and re-checks at opening; nobody is woken.

## Branch-in-a-Box

**32. Acquired branches monitored from day one** (Mingledorff's wave 2, Human-in-the-loop). Ten acquired branches still on their own kit are discovered over a temporary VPN, monitored as they are, entered in the CMDB and given interim service levels and notify-only runbooks, weeks before their cutovers.

**33. Acquired branch cutover weekend** (LS Building Products, Human-in-the-loop). Thirteen timed gates from Friday close to Monday opening. Gate 9 fails because legacy handhelds are 2.4 GHz only; a site override is approved, the re-test passes and StoreOps signs off.

# Part IV. Presenting and reference

## 27. Presenting the console

A 33-minute walkthrough that covers the story end to end. Open three tabs before you start: the console, the live demos page and the role dashboards.

1. **Operations Overview (3 min).** The estate on one screen: the tiles, the map, the error budgets. Pause updates with **LIVE** while you talk.

2. **Demo 1, SD-WAN degradation (4 min).** Autonomous: 43 alerts, one incident, fixed and closed in under two minutes.
3. **Demo 9, Wi-Fi drift (4 min).** Human-in-the-loop: read the evidence aloud, then approve.
4. **Demo 2, fiber cut (4 min).** A P1 handled end to end, including the branch manager's call and the RCA.
5. **Correlation (2 min).** The funnel and the flow: why engineers see only a small share of the alerts.
6. **Patching (5 min).** Demo 14 (Day-0 patch) or 18 (major upgrade), then Approvals & Changes on the firmware tab.
7. **Branch Readiness (2 min).** Every branch checked at 05:30 local, problems fixed before opening.
8. **Branch-in-a-Box (4 min).** Demo 32 (day-1 monitoring) and demo 33 (cutover weekend).
9. **SLA & KPI Tracker (2 min).** Every Annexure B metric, live.
10. **Role dashboards (3 min).** Sign in as the CIO: the dashboard refreshes live while you talk. Click **Trading normally** and drill down: a region, a state, one branch, its main line and the incident on it. Finish on **Last month** and click the time-to-fix chart.

## Phrases that help

- "One incident per branch per event, not one per alert."
- "Mova iO prepares the evidence and the rollback; a person decides whenever the rules say so."
- "Firmware never changes without a person, but the waves run themselves once approved."
- "A failed check at 05:30 is fixed or owned before the doors open."
- "The branch keeps its name; the network becomes standard over one weekend."
- "Every number on the dashboard opens the detail behind it, down to one line at one branch."

# 28. Common questions

| Question | Answer | Where to show it |
|---|---|---|
| Can the AI change things on its own? | Only actions in the Act tier, inside the gate rules: confidence 80% or more, branch closed for changes, no freeze. Firmware, routing, security policy and QoS changes always need a person. | Chapter 3; Approvals & Changes, Rules |
| What happens when automation fails? | Verification catches it, the runbook stops and the incident goes to an engineer in the same minute with the full context and the SLA clock unchanged. | Demo 11 |
| How do you stop alert storms? | Correlation by topology, time, change, carrier and similarity; transients are cleared after a hold-down and warnings go to a digest. | Correlation |
| How are carriers held to account? | Every case carries evidence, the scorecard compares carriers fairly and repeat circuits become problem records. | Carrier Management; demo 4 |
| How is patching kept safe? | Rings with a canary, pre and post checks, HA pairs one unit at a time, automatic rollback, early life support for major releases and an error budget that can hold non-essential changes. | Demos 14, 15, 18, 19, 20 |
| How do acquisitions join? | Day-1 monitoring as they are, then the standard kit over one weekend with 13 gates, then 14 days of hypercare. | Branch-in-a-Box; demos 32, 33 |
| How is the service measured? | Every Annexure B SLA and KPI with its MSL and ESL, updated live, with service credits on the risk-reward SLAs. | SLA & KPI Tracker |

# 29. Glossary

Short definitions; chapter 3 explains each network term in more detail, with normal values and what it means for the business.

| Term | Meaning |
|---|---|
| Act / Act and notify / Stage | The three catalog tiers: acts on its own; acts and tells a person at once; prepares and waits for approval |
| Advisory | A vendor security bulletin, such as HPESBNW04712 |
| Alert | An event worth acting on, left after noise removal |
| Annexure B | The contract schedule of SLAs and KPIs, with MSL and ESL for each |
| AP | Access point: the Wi-Fi radio |
| Aruba Central | HPE Aruba's cloud management for switches and APs; 1,062 SRS devices are in it |
| Auto-cleared | An alert that recovered by itself and was checked by ping and SNMP |
| Availability | The share of time a service works |
| Backup line | The second WAN line a branch uses when its primary fails |
| Bandwidth | The capacity of a line, in megabits per second (Mb/s) |
| Blast radius | Everything affected if one element fails or changes |
| BPDU guard | A switch setting that shuts a port when another switch is plugged into it, preventing loops |
| Broadband | A cable or fibre internet line shared with other customers |
| Broadcast storm | Traffic copied endlessly round a cabling loop, slowing or stopping the branch network |
| Canary | A small first group in a rollout, checked against a control group before the rest proceed |
| Carrier | The company that sells a WAN line |
| CI / CMDB | Configuration item / the ServiceNow database of all devices, circuits and their relationships |
| Circuit | One WAN line from a carrier |
| Confidence | Mova iO's certainty in a diagnosis or action, 0 to 100%; below 80% a person decides |
| Config backup | A nightly copy of each device's configuration, taken at 02:00 CT |
| Correlation | Grouping related alerts into one incident |
| CRC errors | Corrupted frames on a link, often a failing optic or cable |
| CVE / CVSS | A published vulnerability / its severity score from 0 to 10; 9.0 and above is critical |
| Degraded line | A line that works but with high latency, jitter or loss |
| Demarc / NTE | The point where the carrier's equipment ends and the branch's begins |
| DHCP / scope | The service that gives devices addresses / the pool of addresses for a VLAN |
| DIA | Dedicated internet access: a business internet line with guaranteed speed |

| Term | Meaning |
|---|---|
| Digest | A daily summary of warnings that do not need a ticket |
| Drill-down | On the role dashboards, the panel that opens behind any number, chart, insight or event and goes level by level from the estate to one branch, line, device or incident |
| Drift | A difference between a device's running config and the approved golden config |
| Early life support | A period of closer monitoring after a major upgrade, before further waves |
| EdgeConnect / ECOS | The HPE Aruba SD-WAN appliance at each branch / its operating system. The EC-XS is replaced by the EdgeConnect 10104 |
| End of sale | The last day a vendor sells a model; it is still supported |
| End of support | The last day of vendor support and security fixes; after it the device should be replaced |
| Error budget / burn rate | The failure a target allows over a period / how fast it is used compared with an even spend |
| ESL / MSL | Expected service level (target) / minimum service level (below it a service credit applies) |
| Event | Any single message or threshold crossing from a monitoring tool |
| Evidence pack | Graphs, traces, test results and timestamps attached to a carrier case |
| Failover / failback | Moving traffic to the other line automatically when one fails / moving it back once the main line is stable |
| Firmware / N, N-1, N-2 | Device software / the target release, one behind and two behind |
| Flap / hold-down | A line or port that goes down and up quickly / the wait before acting, in case it recovers |
| Freeze window | A period when only emergency changes are allowed |
| Golden config | The approved configuration in the Network-as-Code repository |
| HA pair | Two EdgeConnects at one branch, one taking over if the other fails |
| Hypercare | 14 days of tighter thresholds after a branch goes live |
| Incident | A ServiceNow record of a problem, with priority and SLA clocks |
| Jitter | How much the delay varies from packet to packet; above 30 ms calls become choppy |
| LAN | The branch network inside the building: switches, APs and cabling |
| Latency | Time for a request to reach the data centre and come back, in milliseconds |
| LEC | Local exchange carrier, which owns the last mile to the branch |
| Limited lifetime warranty | HPE's warranty for Aruba switches and APs, running until five years after the model's end of sale |
| LTE/5G backup | A cellular backup line; slower and metered, but independent of cables |
| MOS | Voice call quality on a 1 to 4.4 scale; good above 4.0 |
| Mova iO | Movate's AI layer: agents that correlate, diagnose, act, verify and record |
| MPLS | A private line inside a carrier's network, steady and expensive |

| Term | Meaning |
|---|---|
| MTTR | Mean time to restore |
| Network status | On the role dashboards, the one-word state of the network right now: healthy, minor issues, degraded or major disruption, from the share of branches with a problem |
| Network-as-Code | Configurations kept as files in a repository, changed by reviewed merge requests |
| Noise removal | Dropping duplicates, flaps and information-only events before they become alerts |
| NTF | No trouble found: a carrier closing a ticket without finding a fault |
| OOB | Out-of-band: a cellular console path to devices when the WAN is down |
| Orchestrator | The central system that configures every EdgeConnect and collects its statistics |
| Overlay | An SD-WAN rule that says which line each kind of traffic uses |
| p95 | The 95th percentile: 95% of measurements are at or below this value |
| Packet loss | The share of data that never arrives; above 1% payments retry and calls break up |
| Payment path round trip | A synthetic test payment from the counter VLAN to the payment processor and back; normally 100 to 200 ms, slow above 400 ms |
| PCI segment | VLAN 20, the isolated network for card terminals |
| PoE | Power over Ethernet, which powers APs, phones and cameras from the switch |
| Primary line | The main WAN line a branch normally uses |
| Problem record | A ServiceNow record for the cause behind repeat incidents |
| RBAC | Role-based access control: each account sees only what its role allows. On the role dashboards, each role sees its own dashboard and scope |
| Readiness check | The ten network checks at 05:30 local before each branch opens |
| Repeat offender | A circuit with three or more tickets in 90 days |
| Roaming | A handheld moving from one AP to the next |
| RB-1xx | A runbook number in the Automation Library |
| Role dashboard | The real-time, role-based page for one leader: live now, last 7 days and last month views, four charts each, insights, live activity and PDF export |
| Runbook | An approved, tested automation for one kind of fix |
| SD-WAN | Software that uses both lines together and picks the best one for each application |
| SLA clock | Time left to respond to or resolve an incident |
| Standalone | A switch or AP managed one by one, outside Aruba Central, and monitored by OpsRamp; 2,080 at SRS |
| Support contract | The yearly HPE support agreement covering the EdgeConnects |
| Synthetic probe | A scripted test run every minute from each EdgeConnect, such as a test payment or ERP login |
| Toil | Repeated manual work that should be automated |
| Tunnel | An encrypted path over one line from the EdgeConnect to the data centre; 1,866 in total |

| Term | Meaning |
|---|---|
| VLAN | A separate network segment, such as 20 for payments or 30 for warehouse Wi-Fi |
| WAN | The lines that connect a branch to the data centre, Azure and the internet |
| WAN use | The share of a line's bandwidth in use; above 85% traffic queues |
| Wi-Fi experience | Aruba Central's 0 to 100 score for Wi-Fi inside the branch; independent of the WAN |
| Zero-touch | An EdgeConnect or switch that configures itself from its template when plugged in |